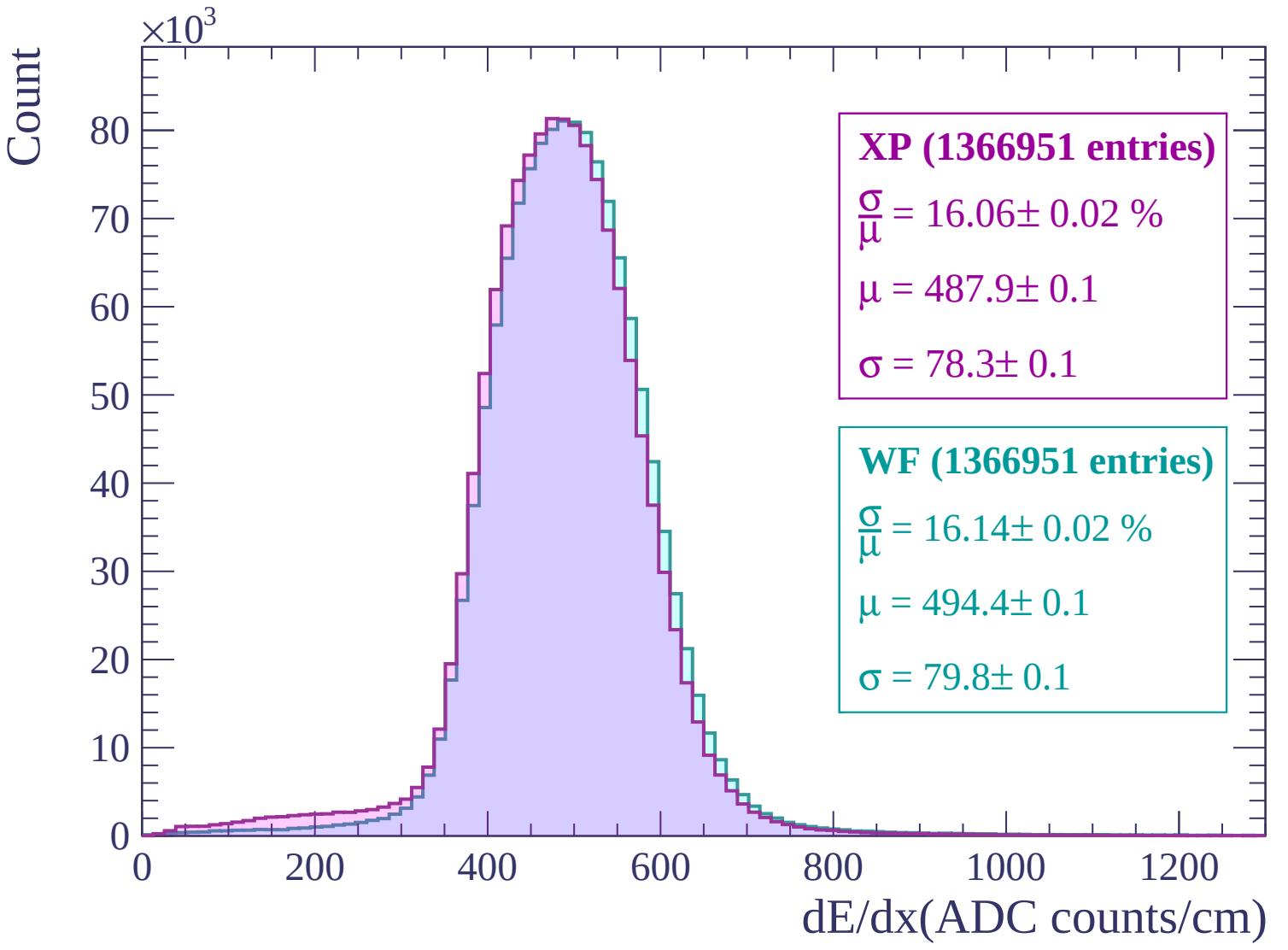
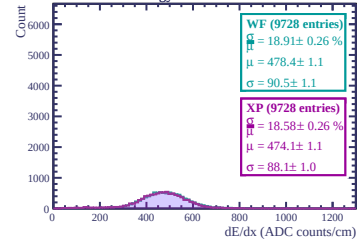
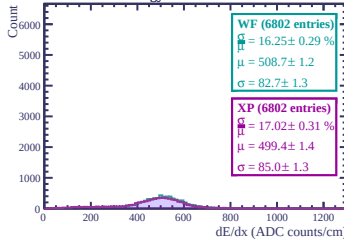




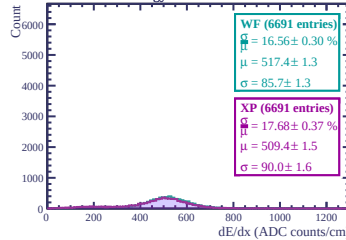
Energy loss in ERAM 14



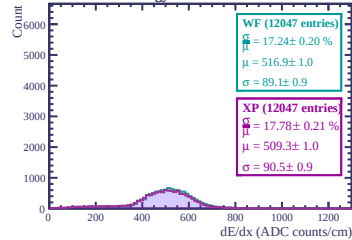
Energy loss in ERAM 30



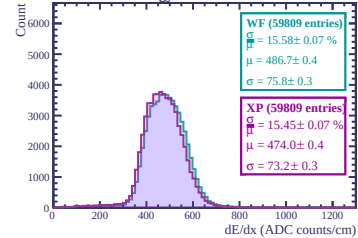
Energy loss in ERAM 28



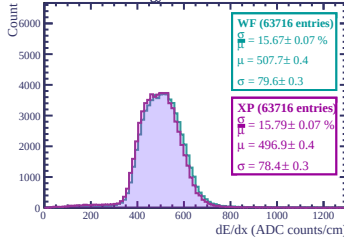
Energy loss in ERAM 19



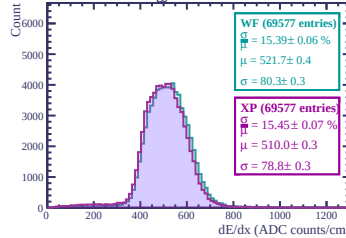
Energy loss in ERAM 21



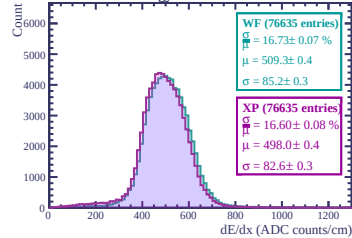
Energy loss in ERAM 13



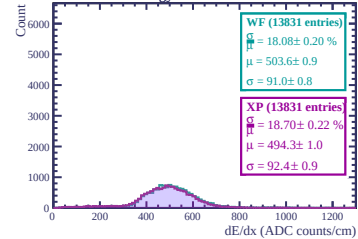
Energy loss in ERAM 9



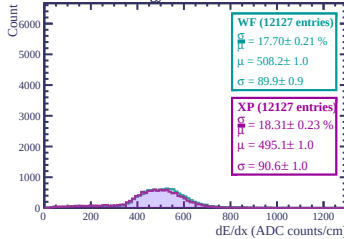
Energy loss in ERAM 2



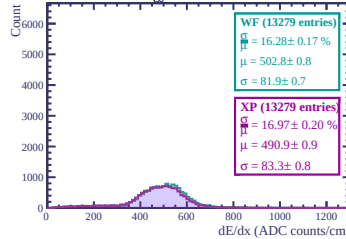
Energy loss in ERAM 26



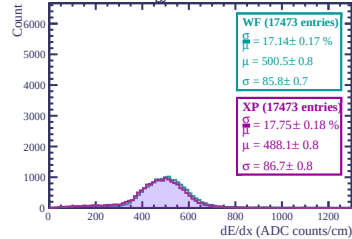
Energy loss in ERAM 17



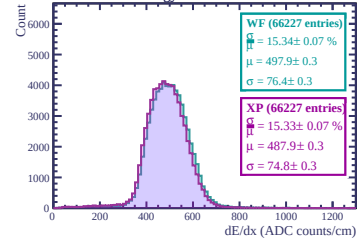
Energy loss in ERAM 23



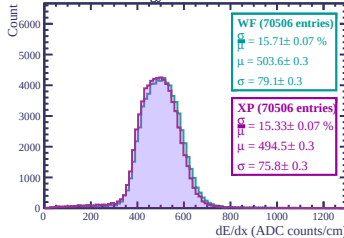
Energy loss in ERAM 29



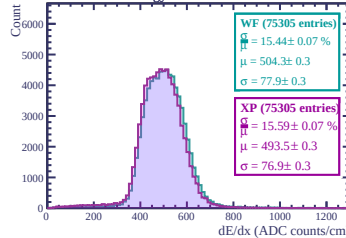
Energy loss in ERAM 1



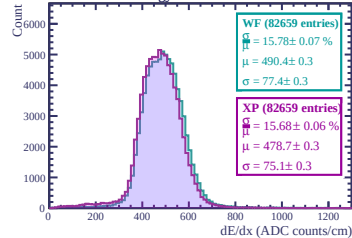
Energy loss in ERAM 10



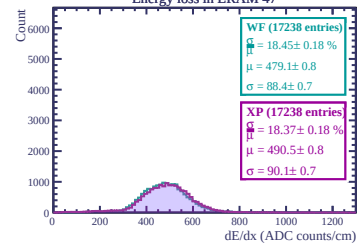
Energy loss in ERAM 11



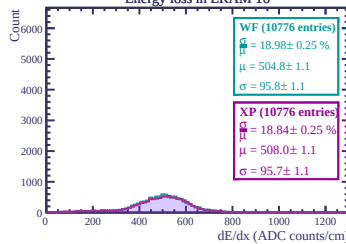
Energy loss in ERAM 3



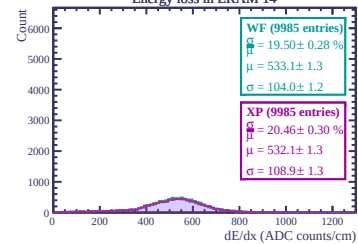
Energy loss in ERAM 47



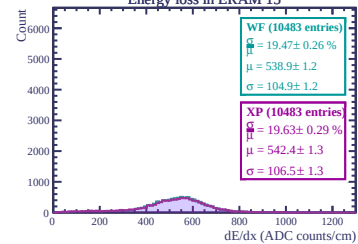
Energy loss in ERAM 16



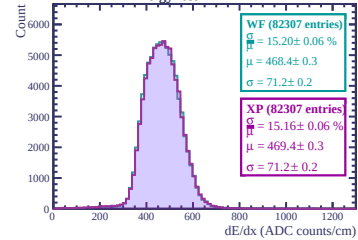
Energy loss in ERAM 14



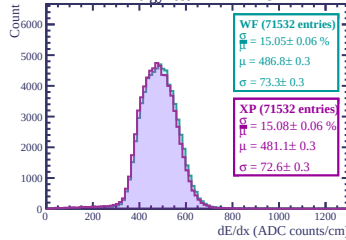
Energy loss in ERAM 15



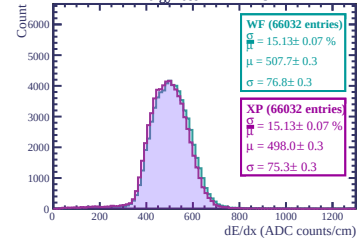
Energy loss in ERAM 42



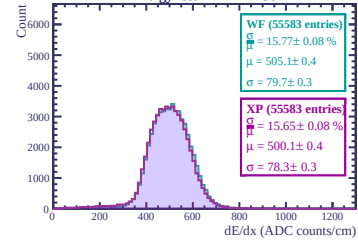
Energy loss in ERAM 45



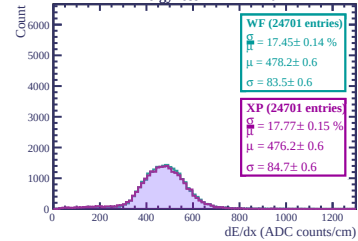
Energy loss in ERAM 37



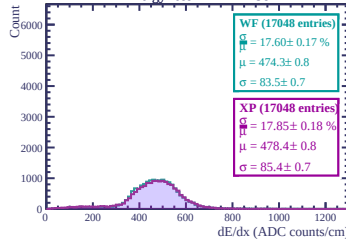
Energy loss in ERAM 36



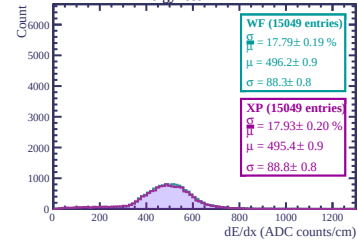
Energy loss in ERAM 20



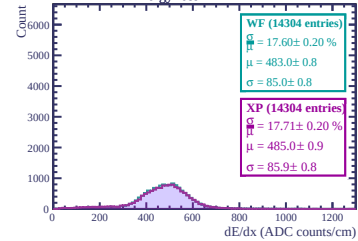
Energy loss in ERAM 38



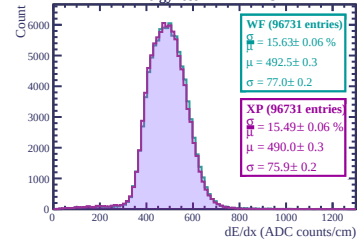
Energy loss in ERAM 7



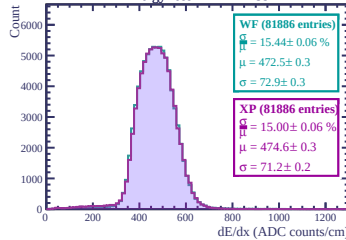
Energy loss in ERAM 44



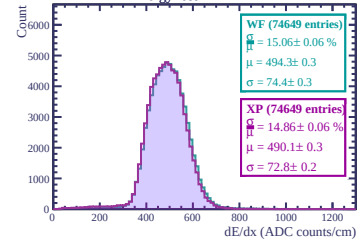
Energy loss in ERAM 43



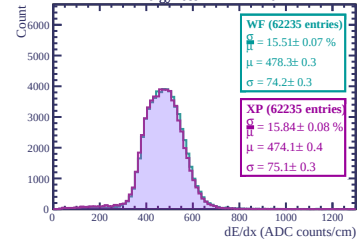
Energy loss in ERAM 39

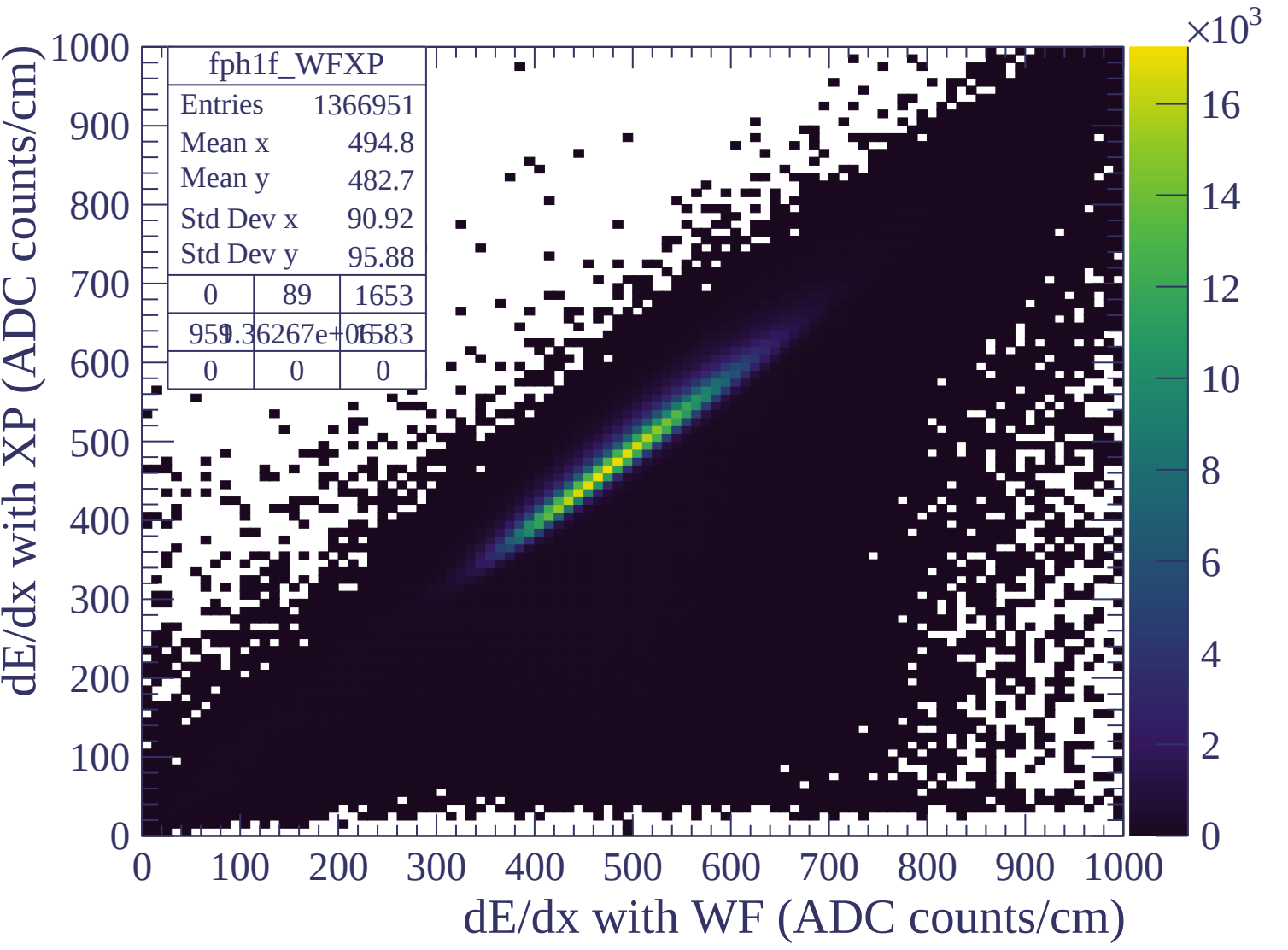


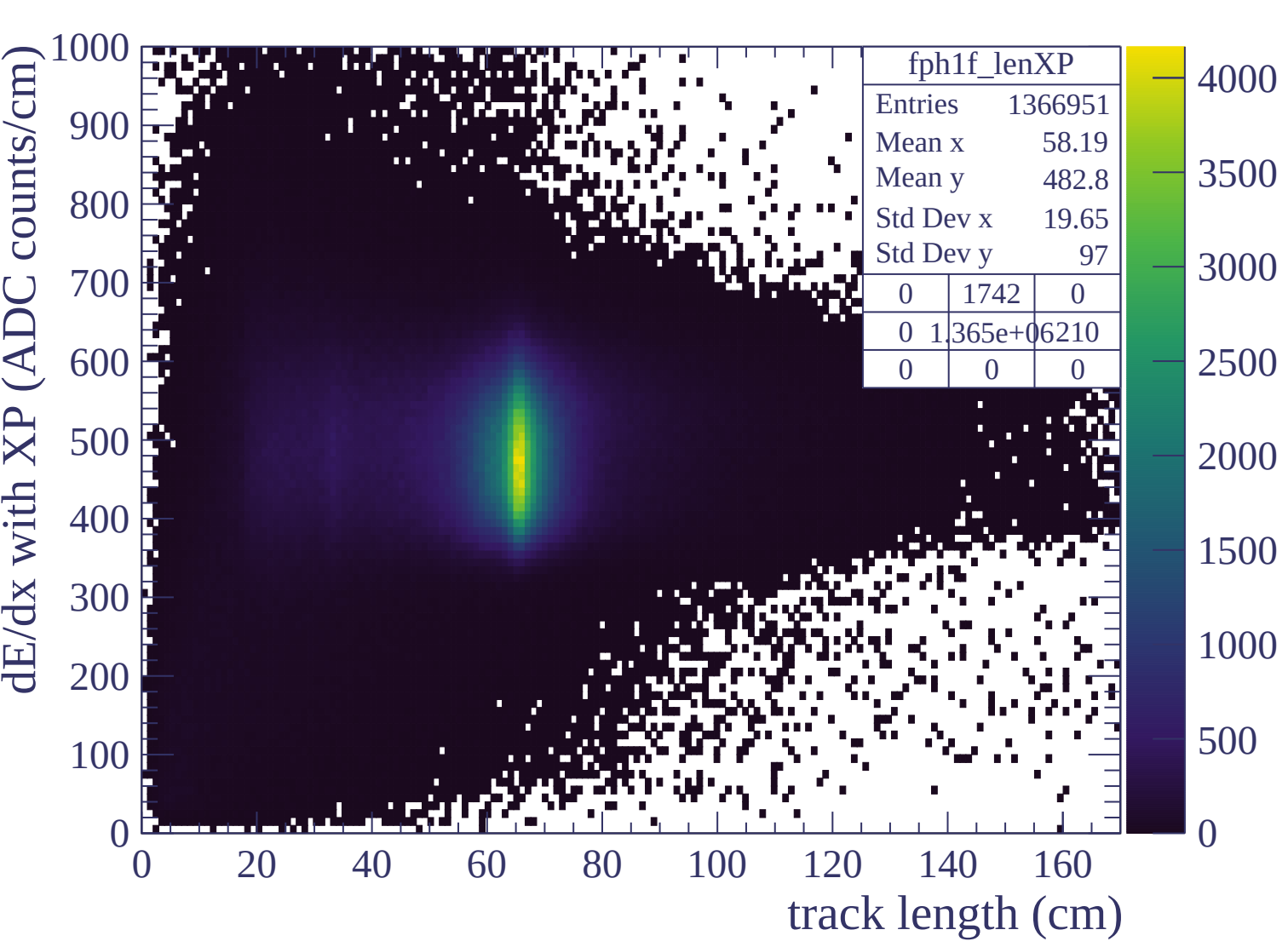
Energy loss in ERAM 41

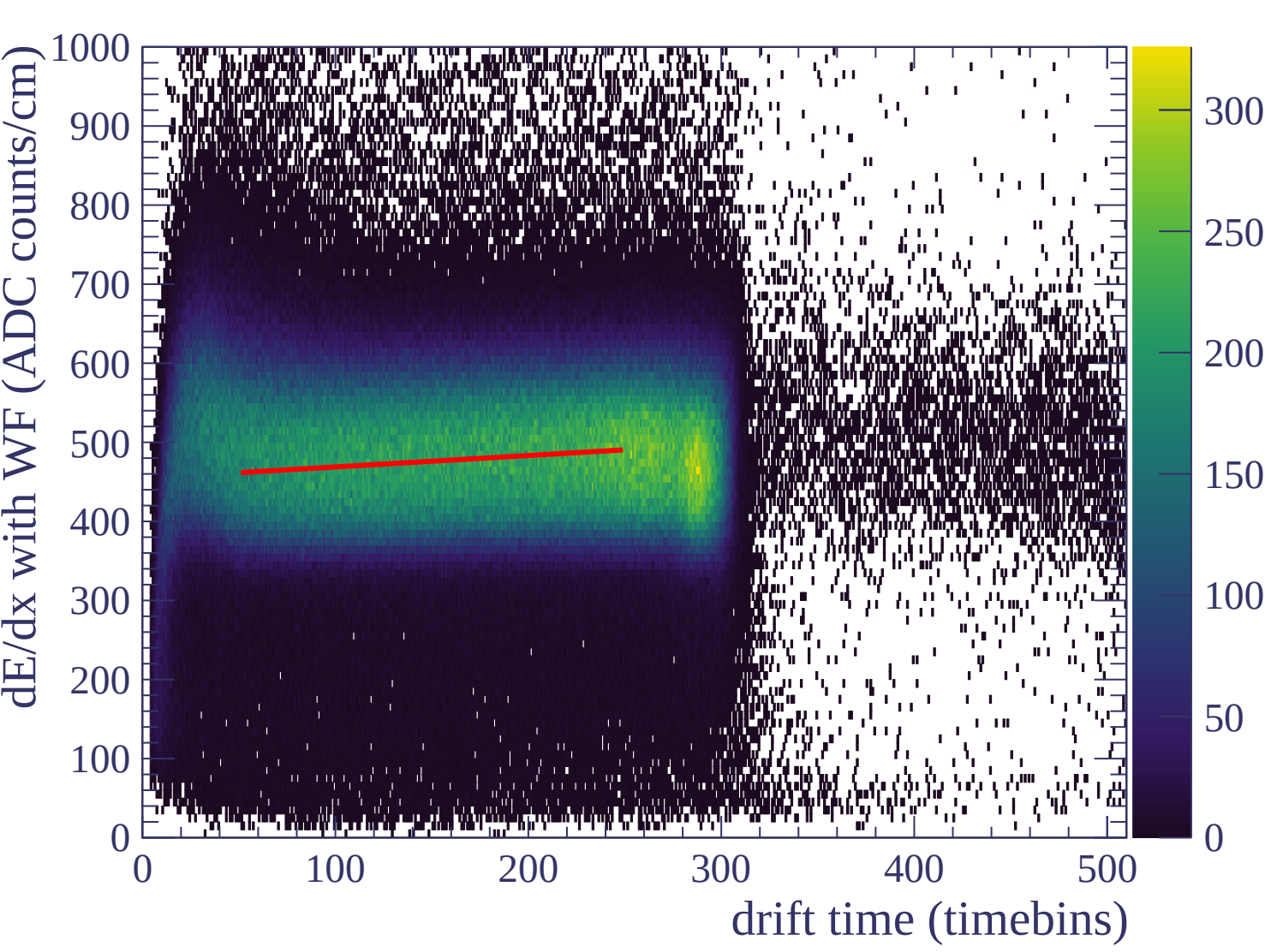


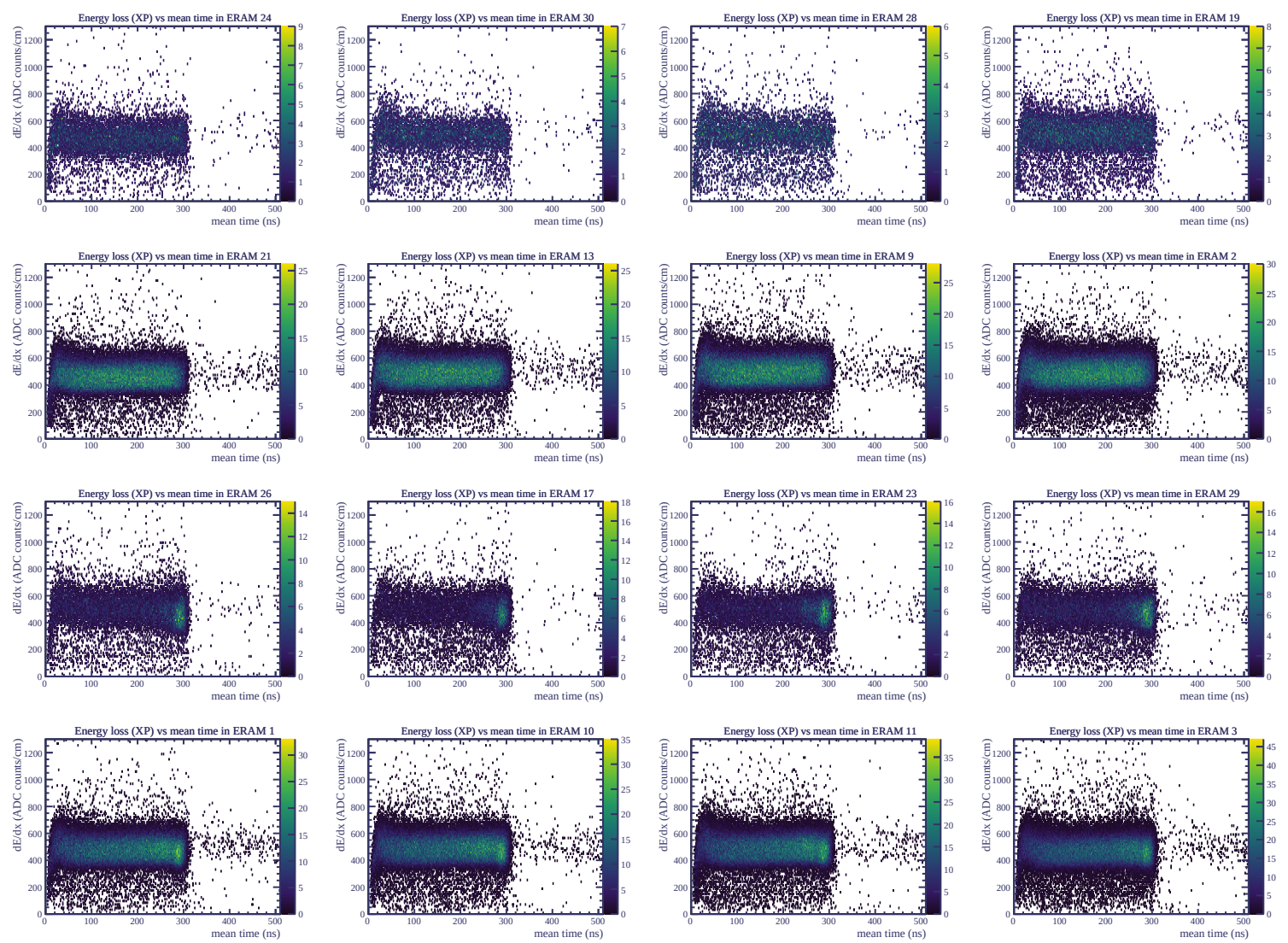
Energy loss in ERAM 46

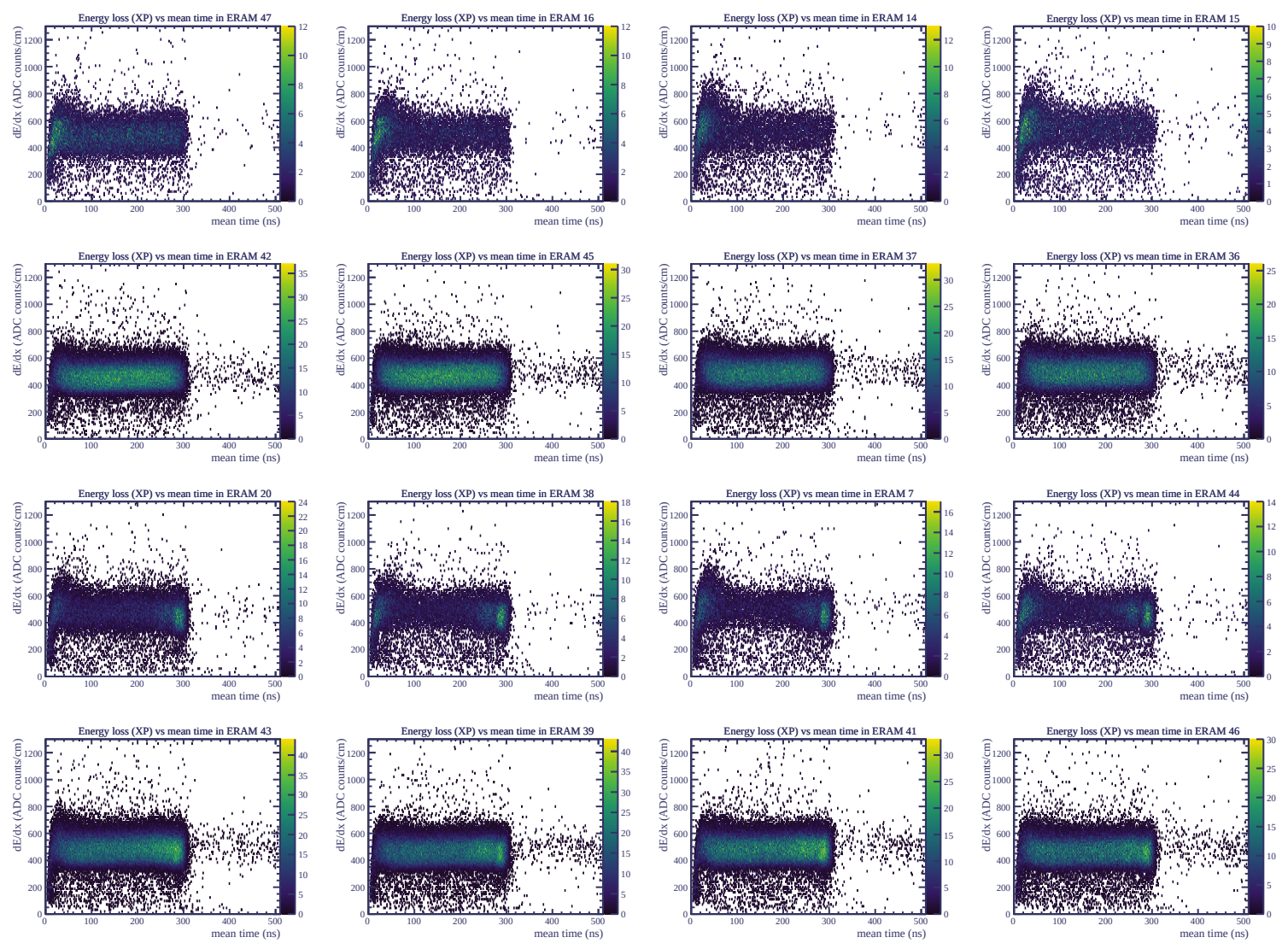




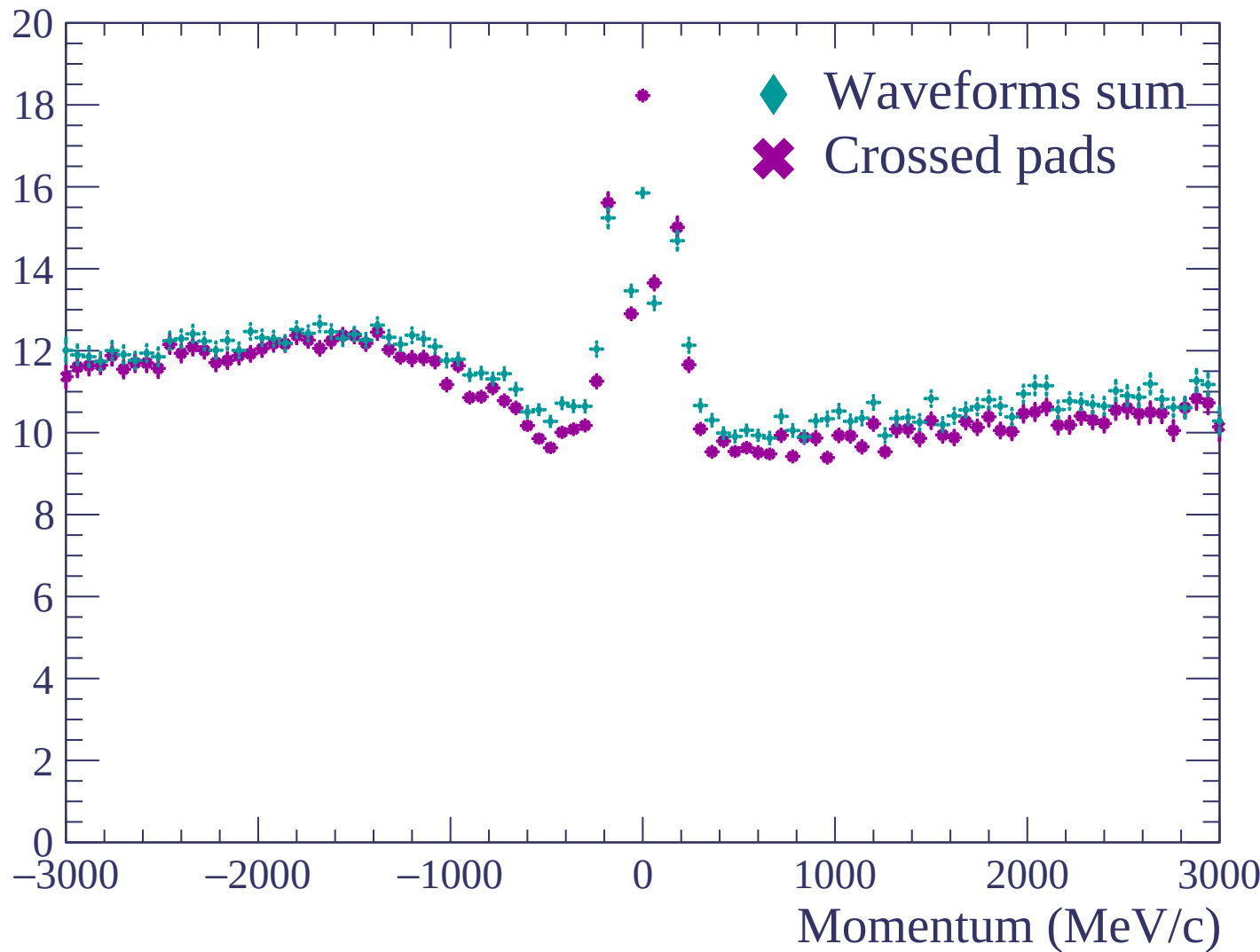


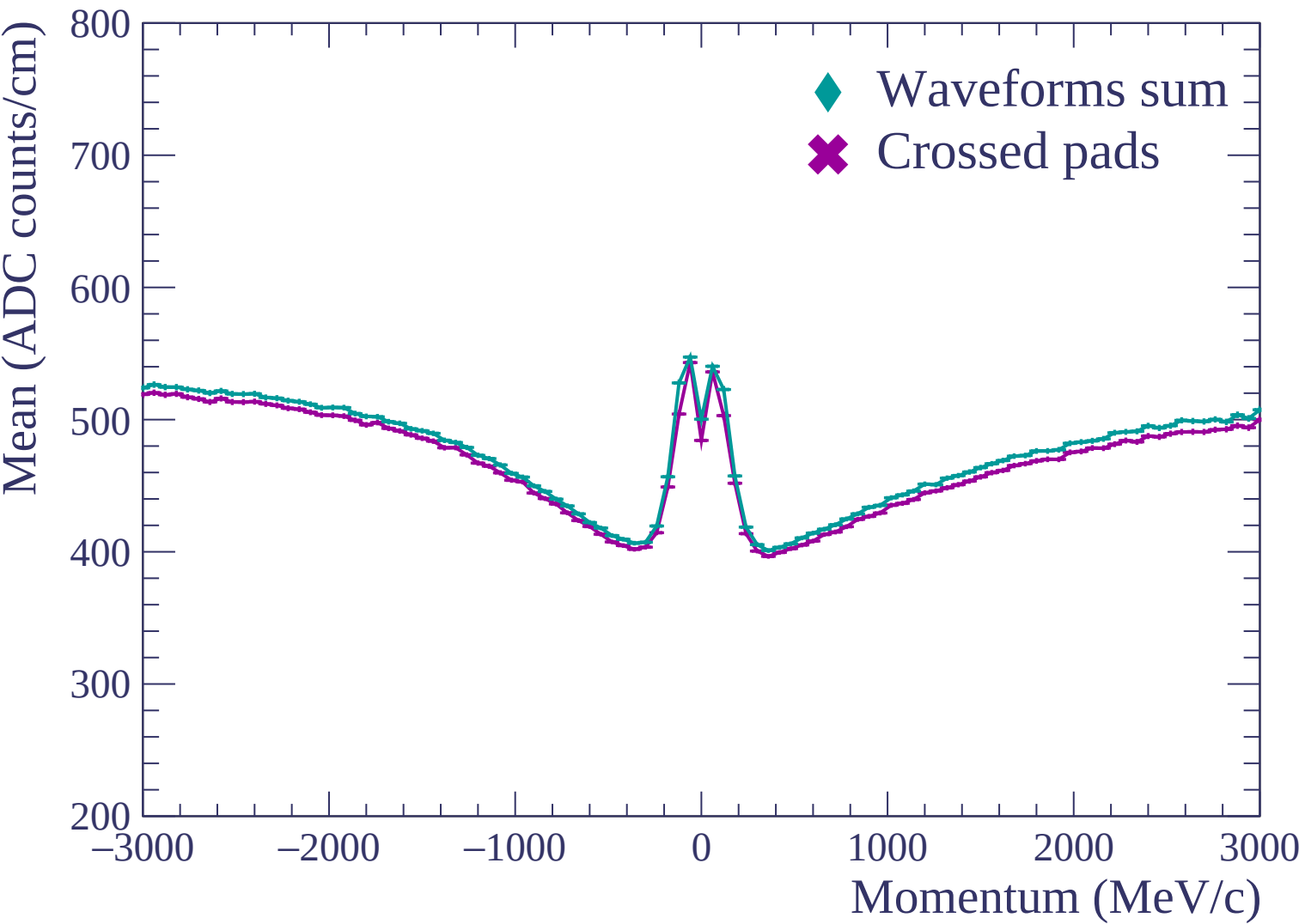


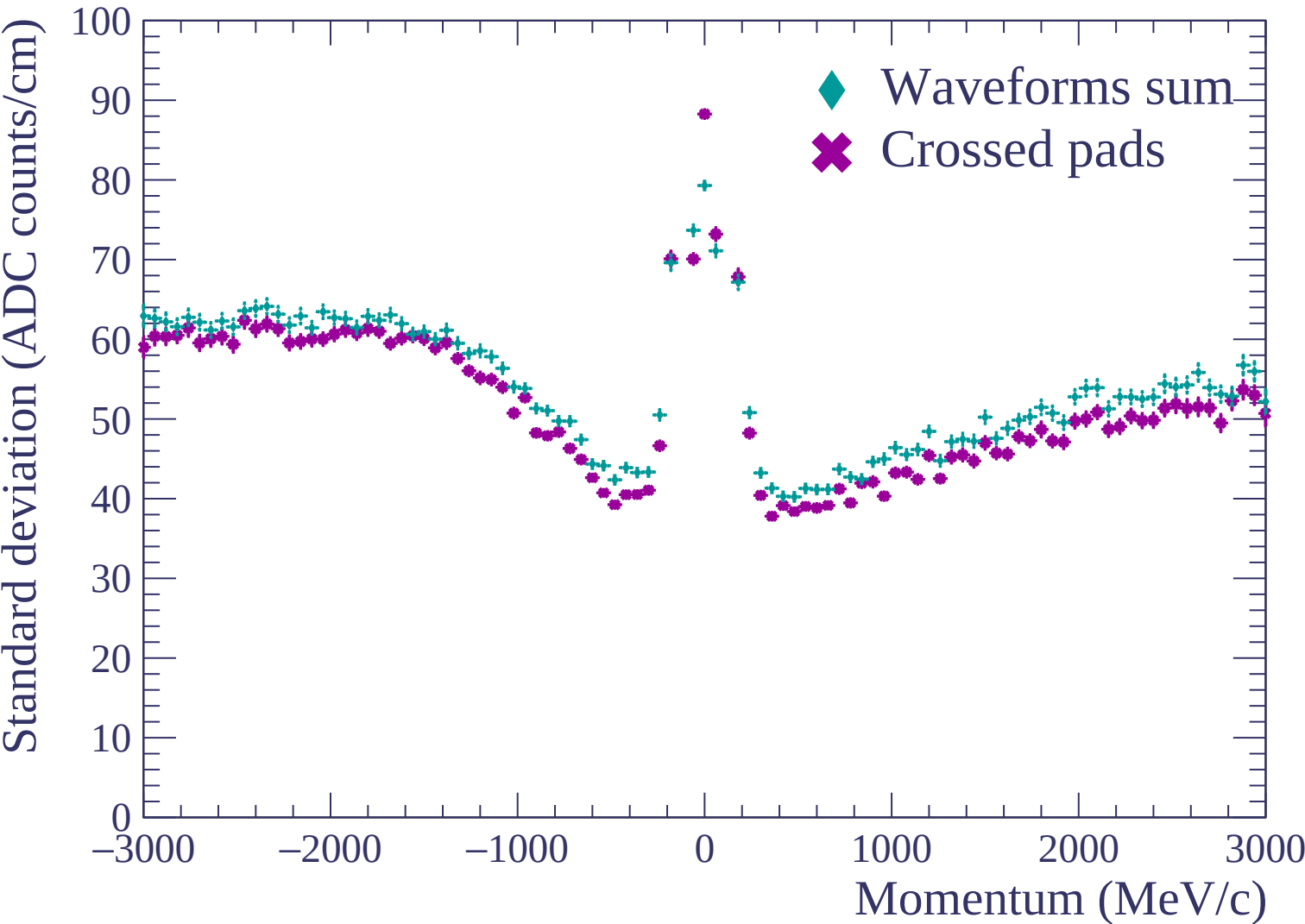


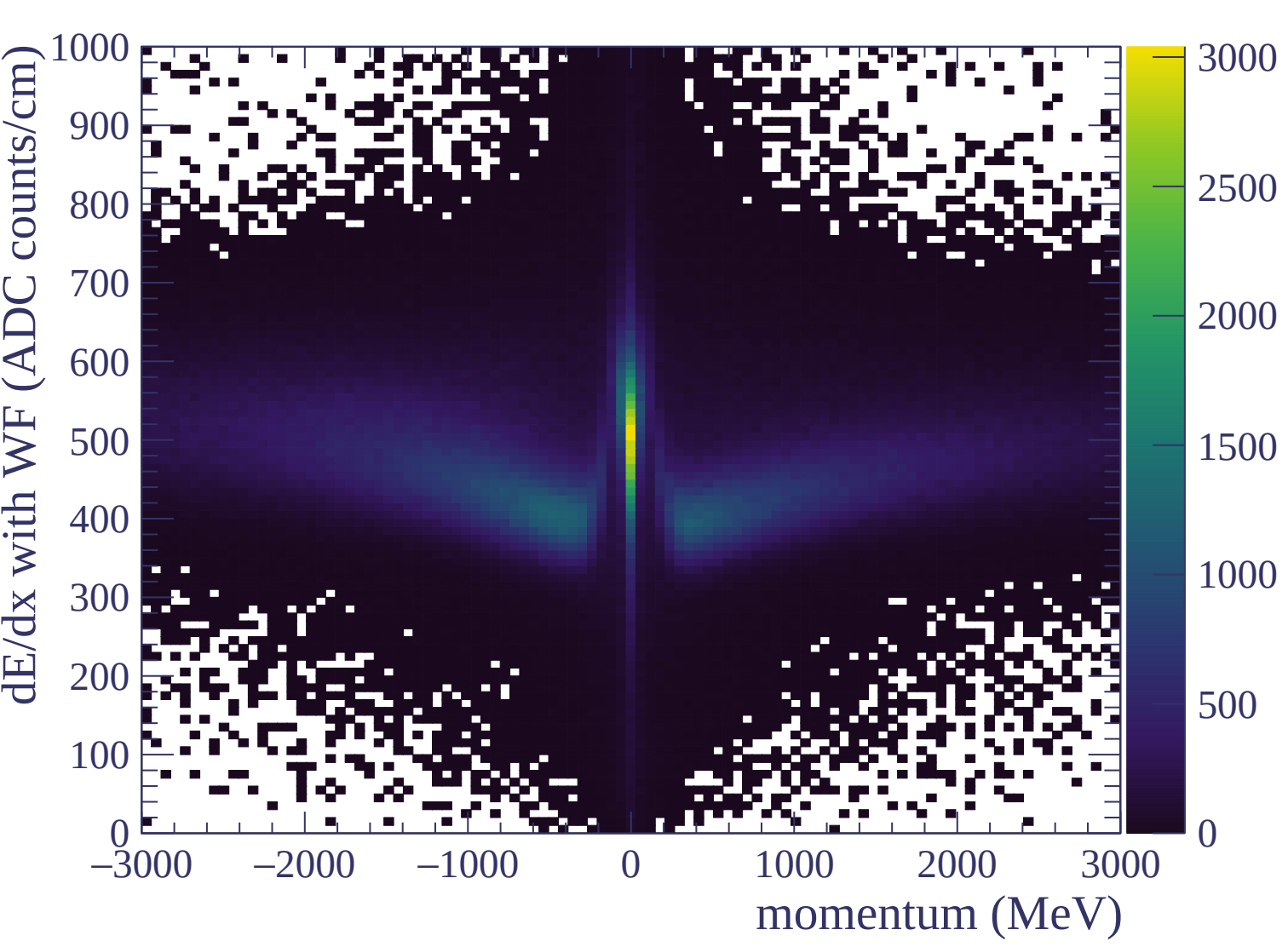


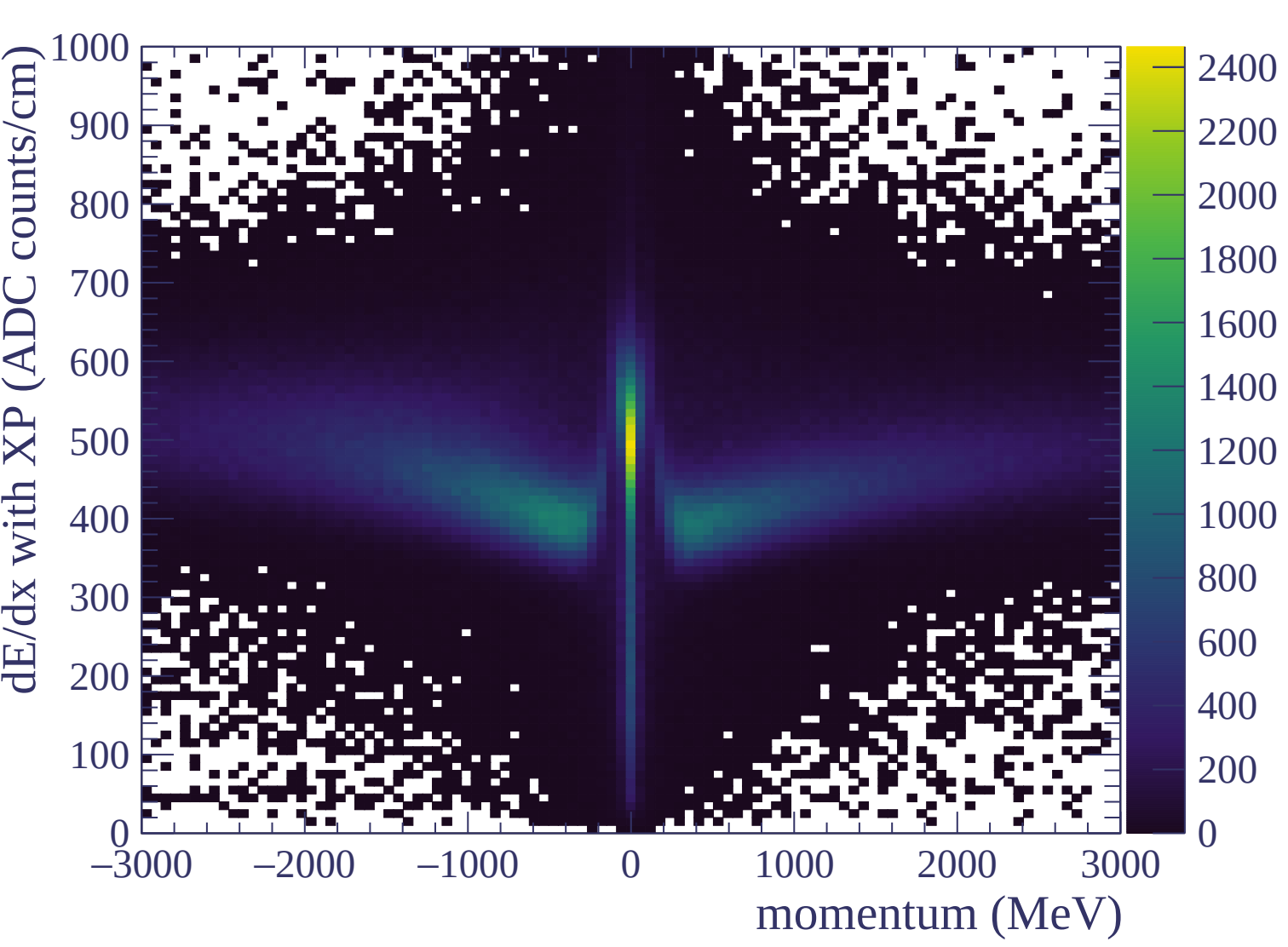
Resolution (%)

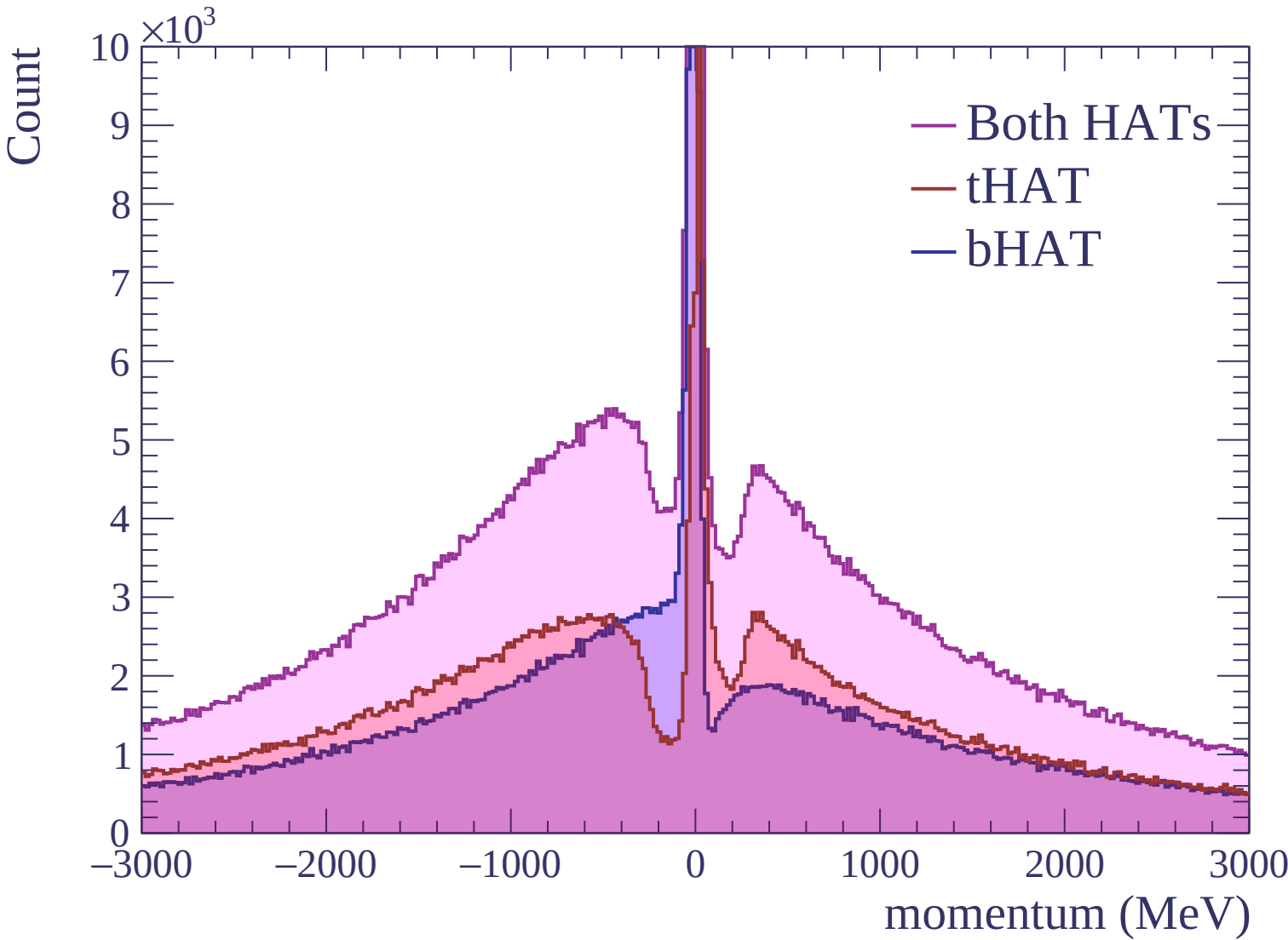


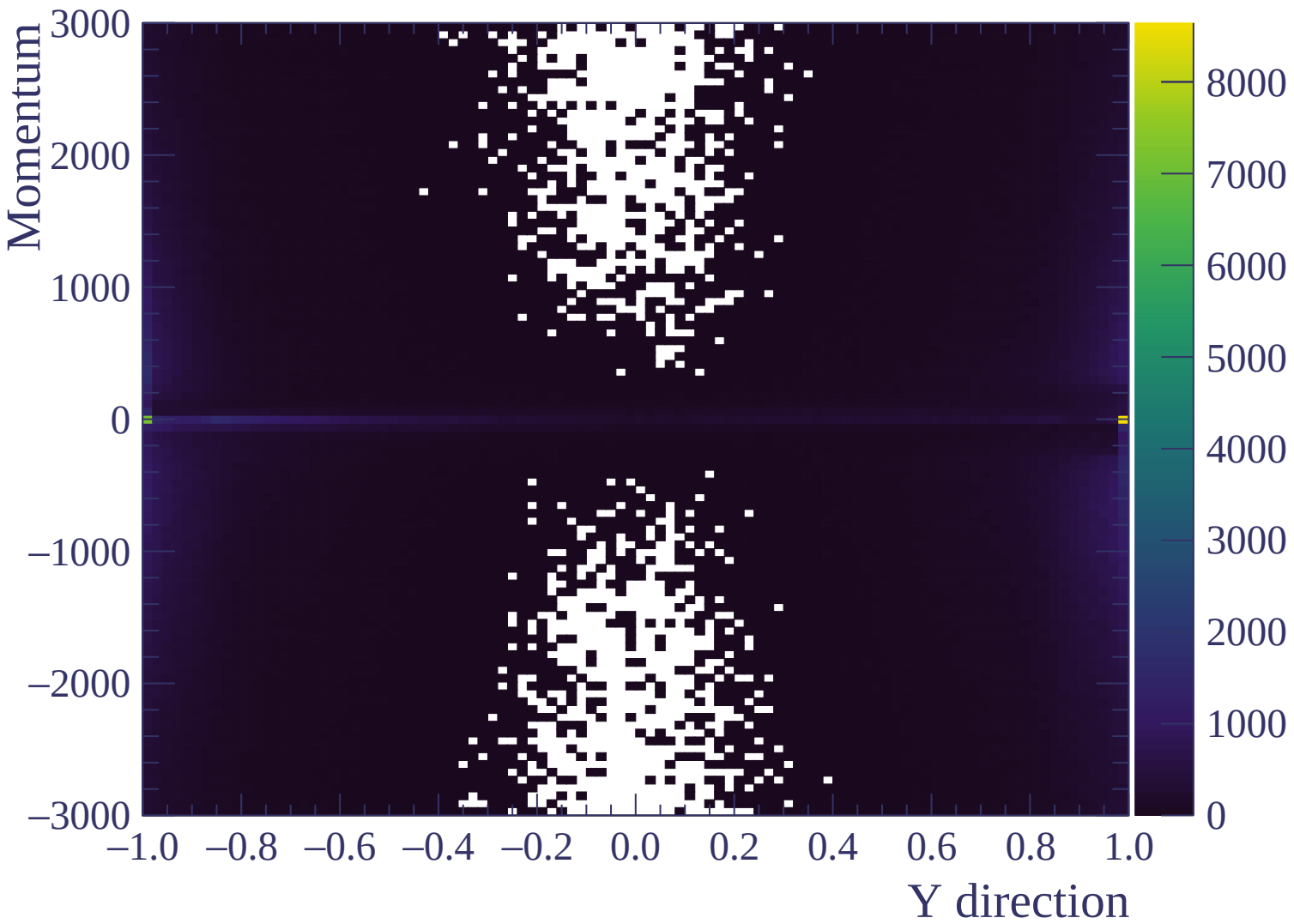


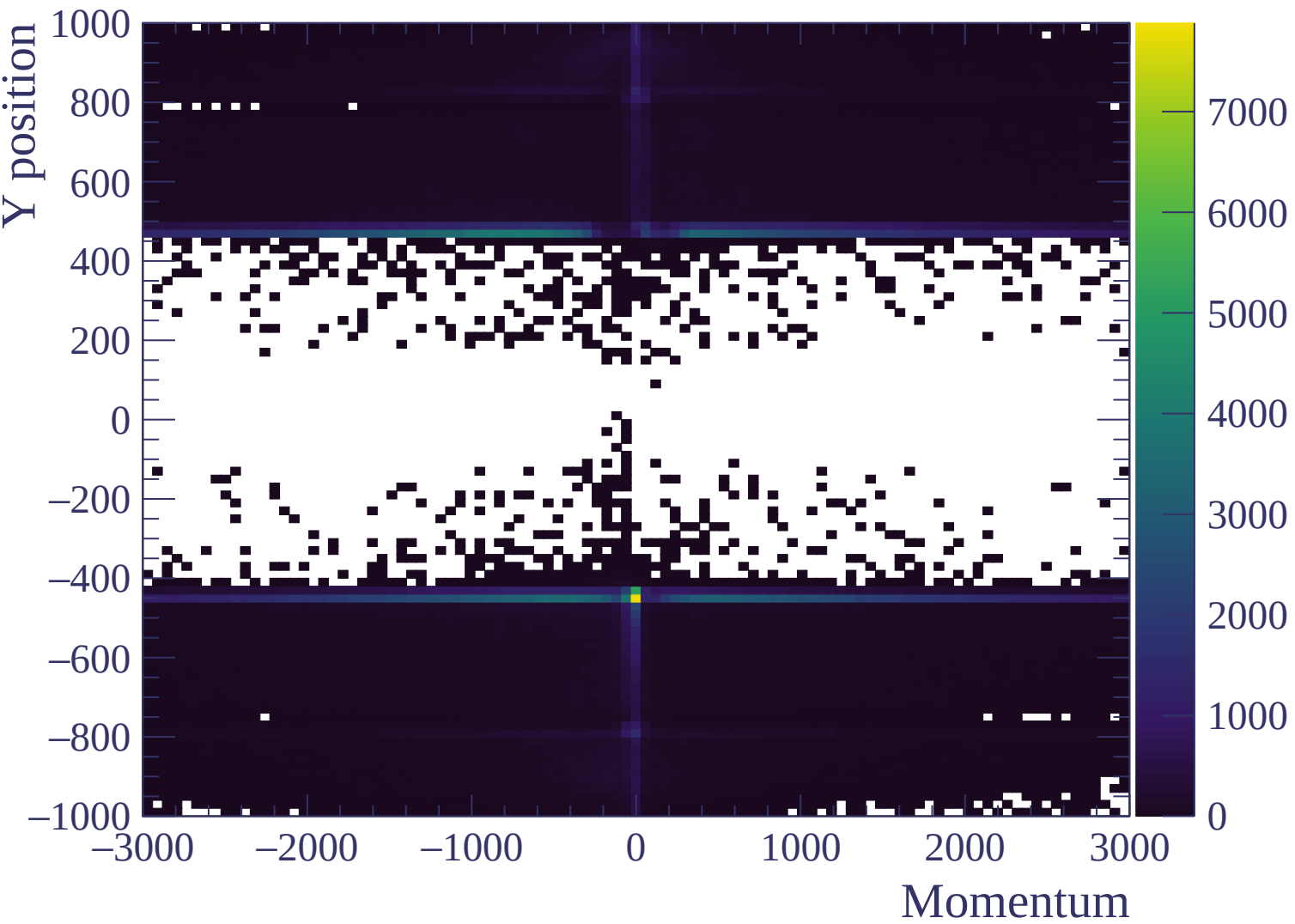


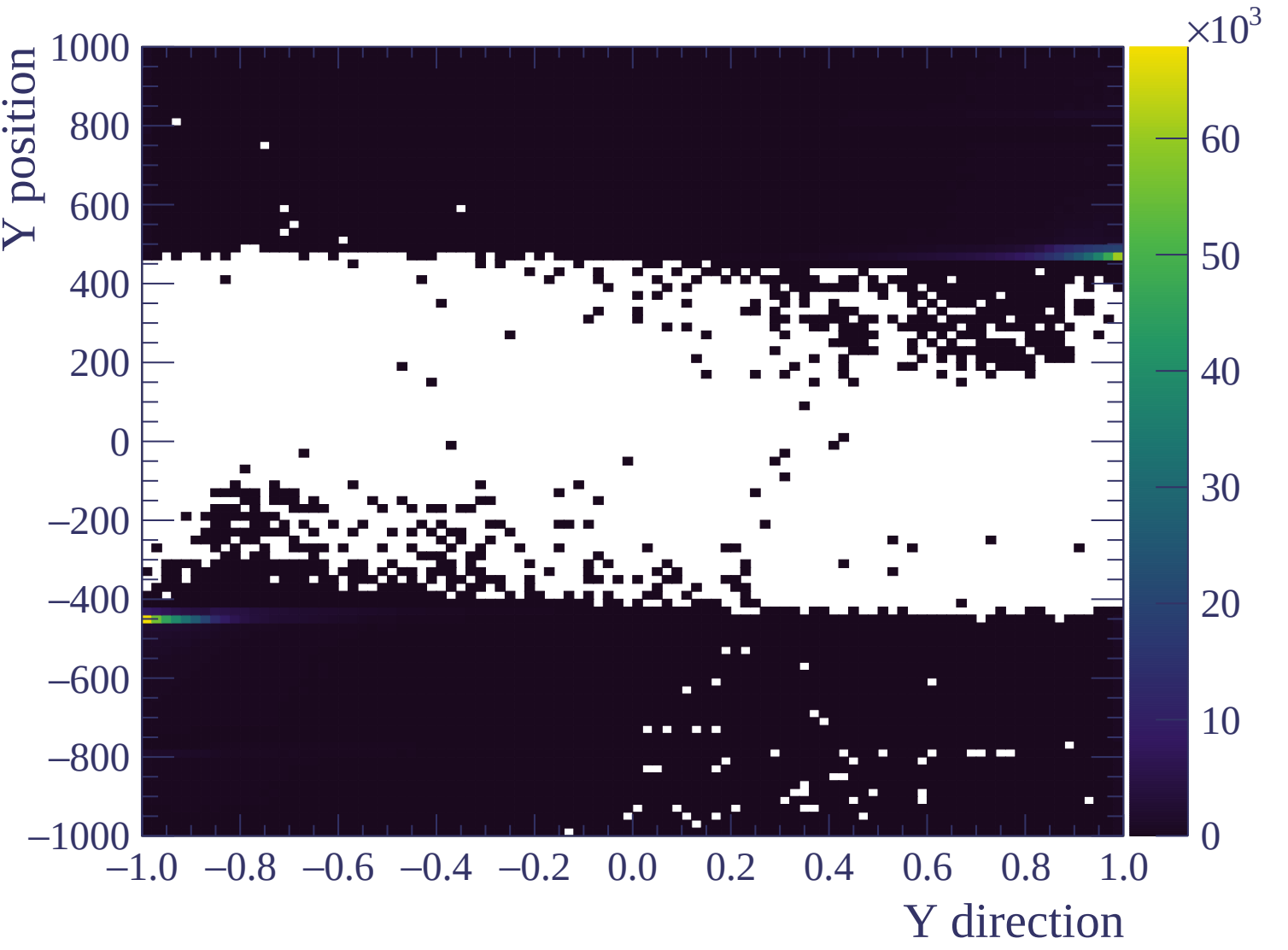






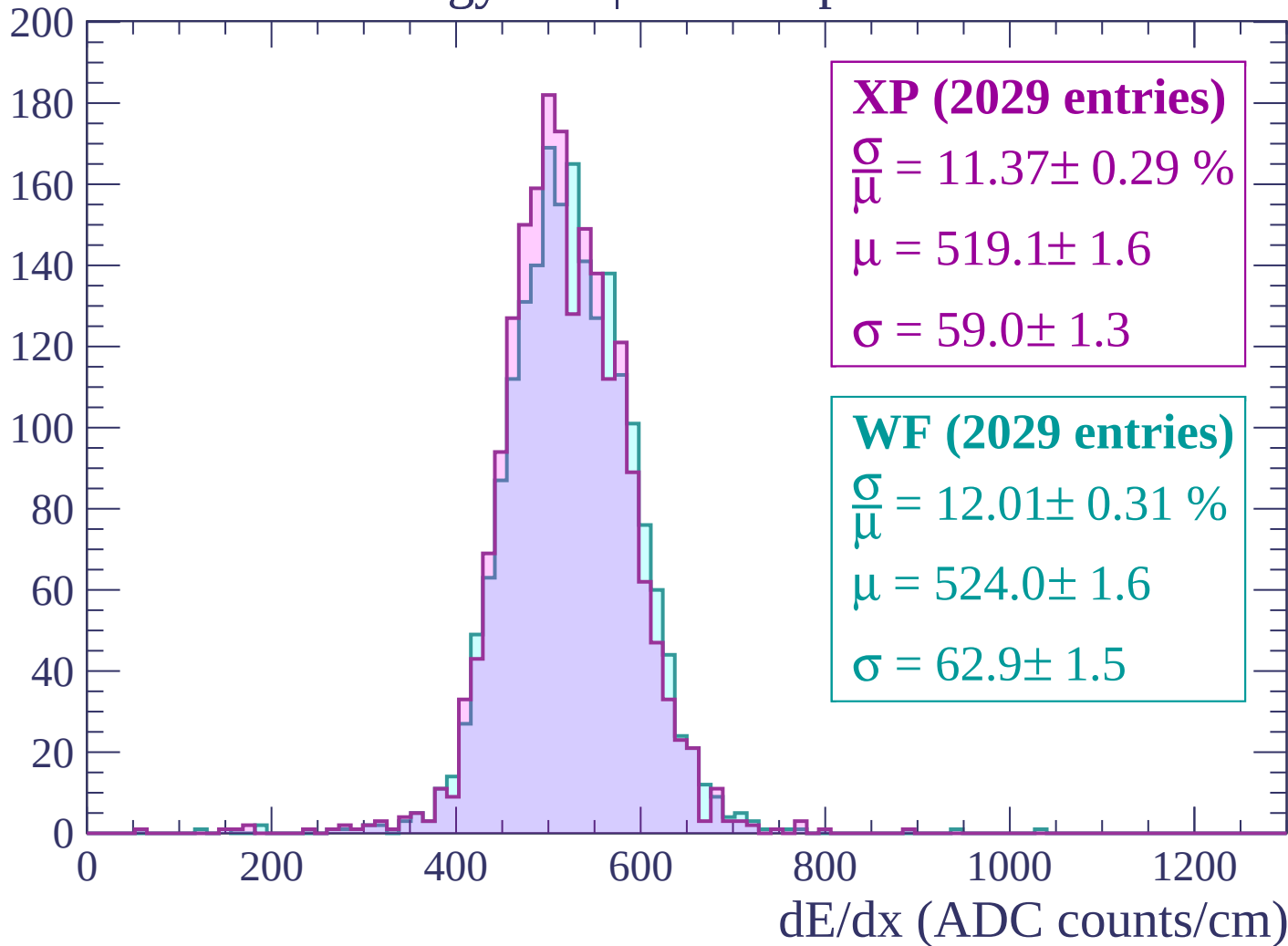






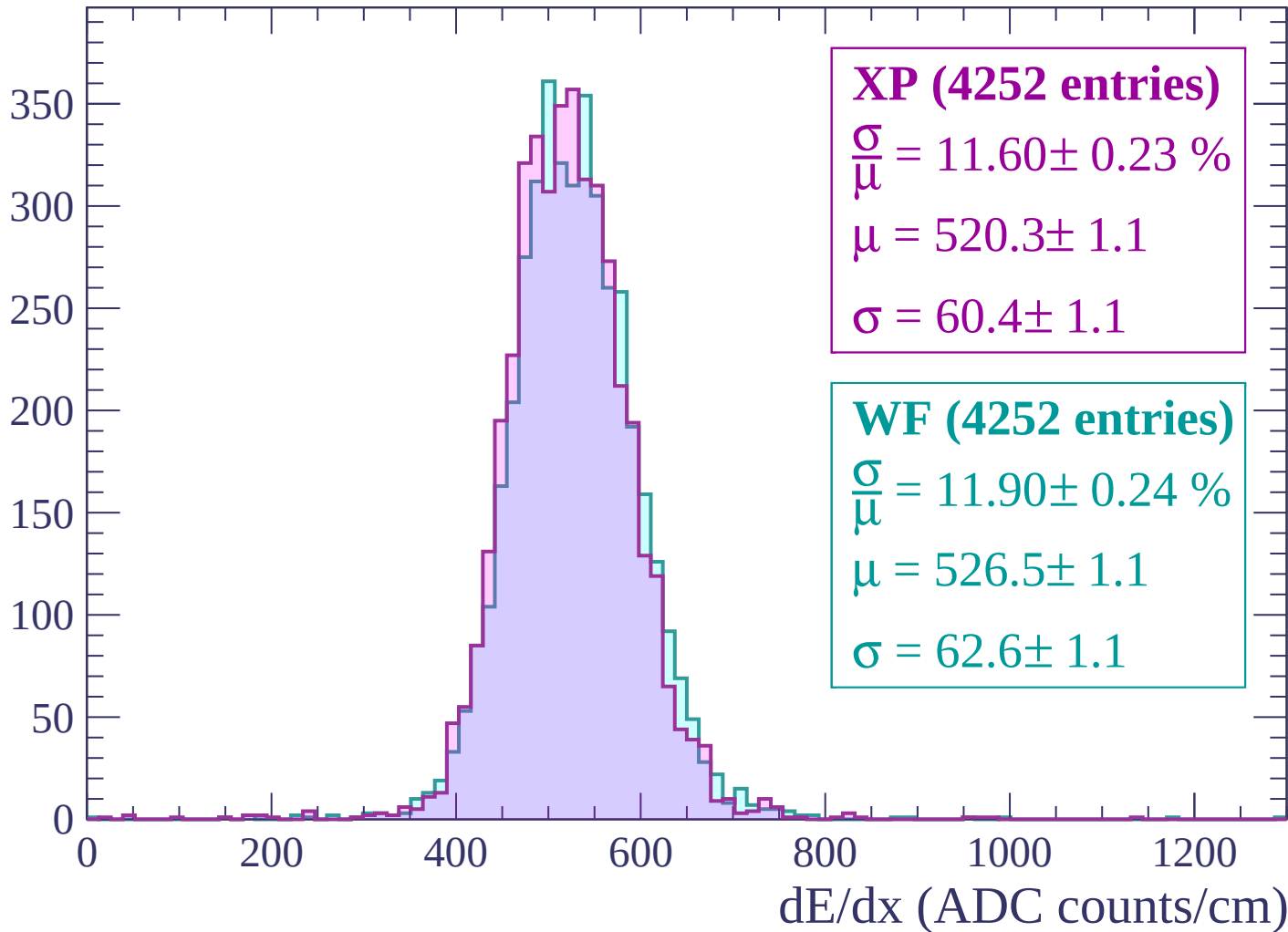
Energy loss | -3000 < p < -2940

Count

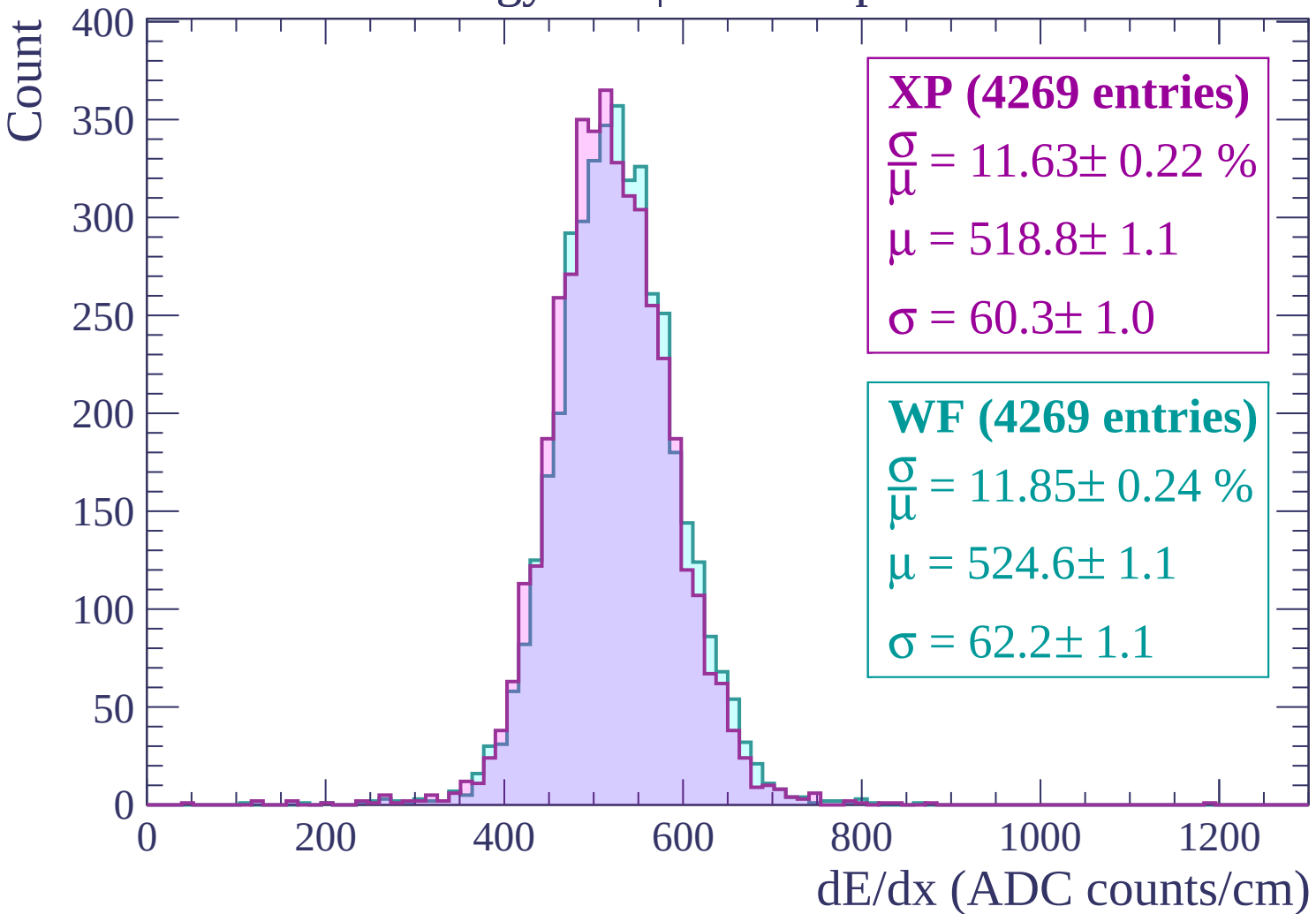


Energy loss | $-2940 < p < -2880$

Count

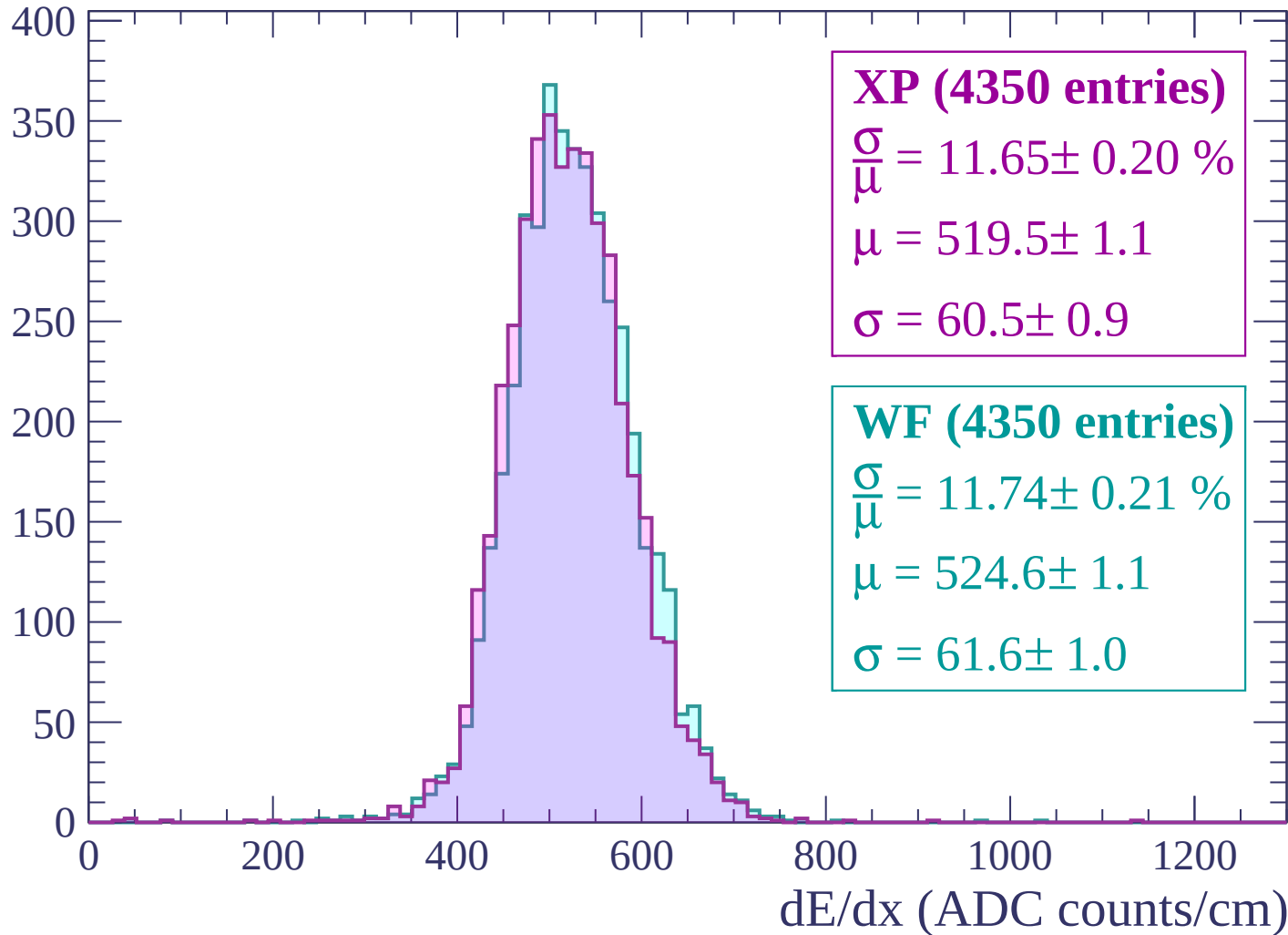


Energy loss | $-2880 < p < -2820$



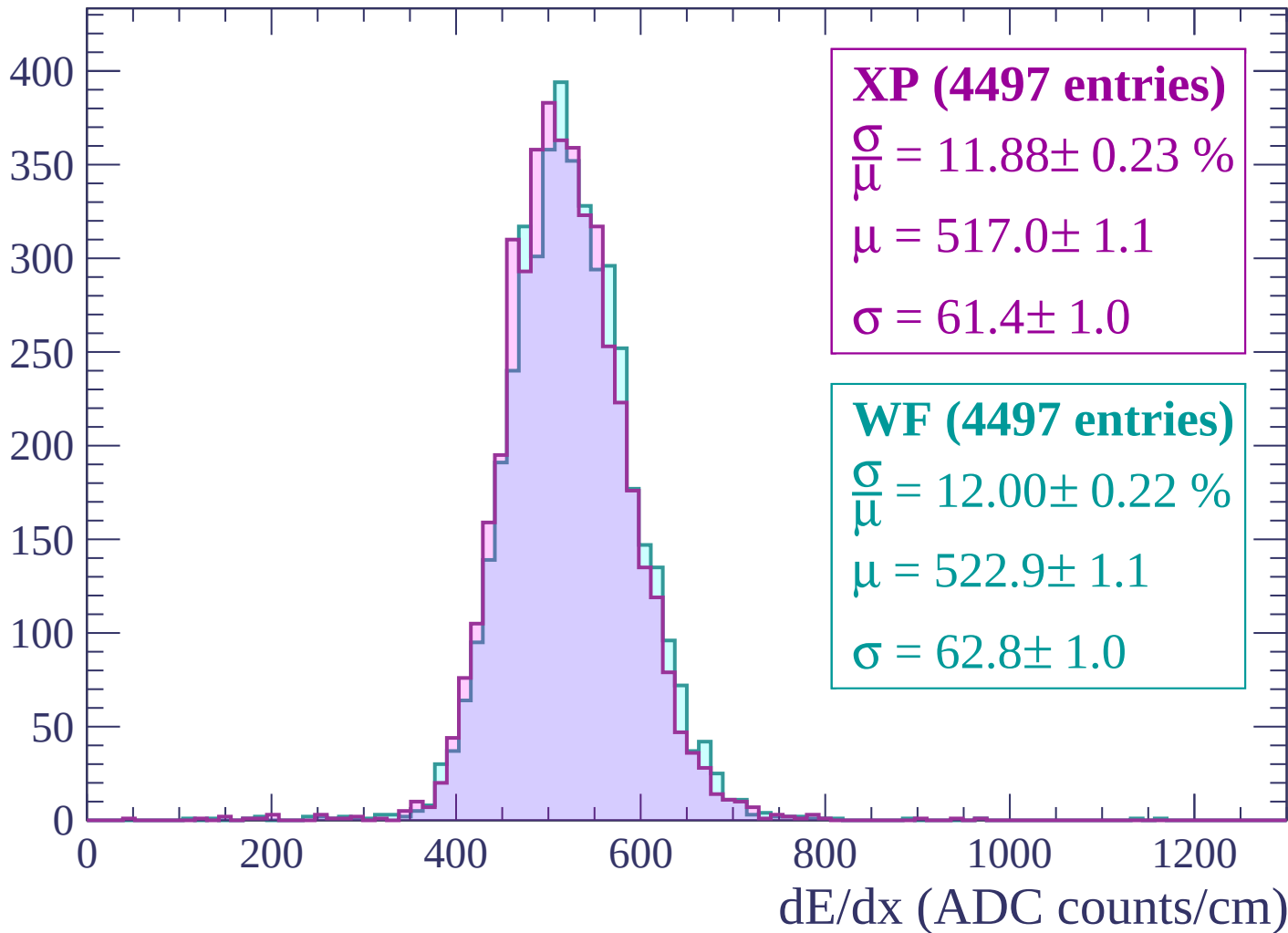
Energy loss | $-2820 < p < -2760$

Count



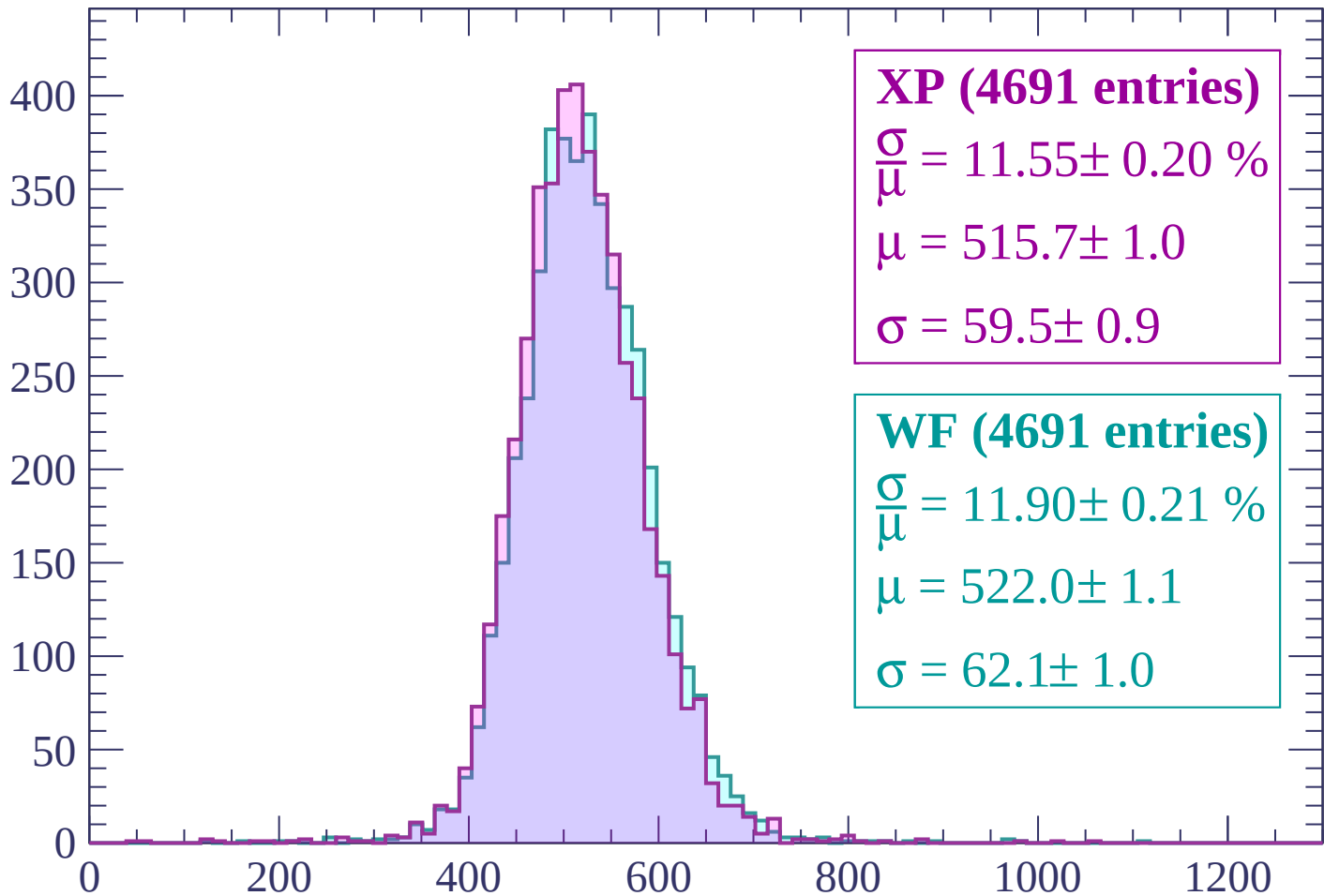
Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count



XP (4691 entries)

$$\frac{\sigma}{\mu} = 11.55 \pm 0.20 \%$$

$$\mu = 515.7 \pm 1.0$$

$$\sigma = 59.5 \pm 0.9$$

WF (4691 entries)

$$\frac{\sigma}{\mu} = 11.90 \pm 0.21 \%$$

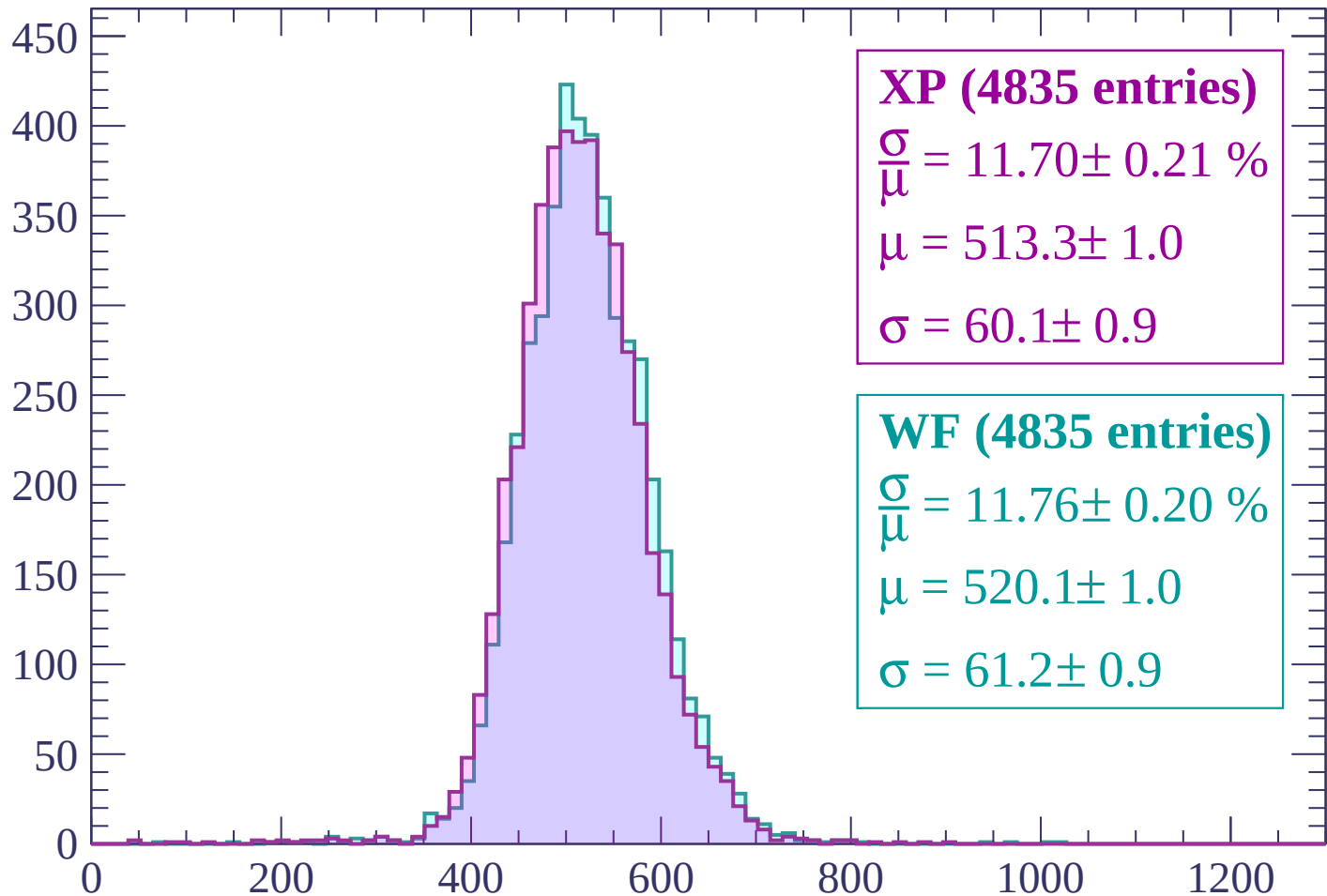
$$\mu = 522.0 \pm 1.1$$

$$\sigma = 62.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-2640 < p < -2580$

Count



XP (4835 entries)

$$\frac{\sigma}{\mu} = 11.70 \pm 0.21 \%$$

$$\mu = 513.3 \pm 1.0$$

$$\sigma = 60.1 \pm 0.9$$

WF (4835 entries)

$$\frac{\sigma}{\mu} = 11.76 \pm 0.20 \%$$

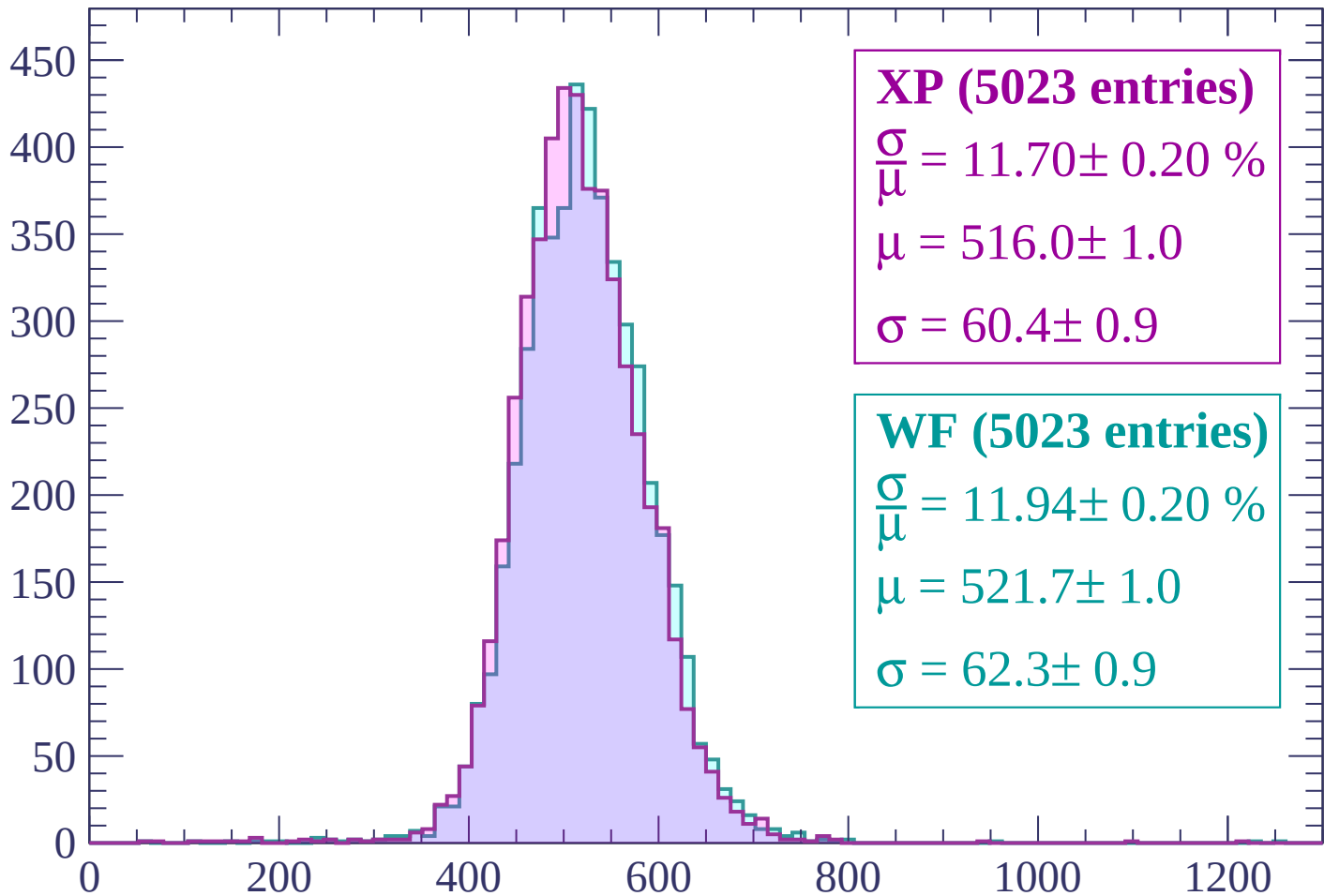
$$\mu = 520.1 \pm 1.0$$

$$\sigma = 61.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -2580 < p < -2520

Count



XP (5023 entries)

$$\frac{\sigma}{\mu} = 11.70 \pm 0.20 \%$$

$$\mu = 516.0 \pm 1.0$$

$$\sigma = 60.4 \pm 0.9$$

WF (5023 entries)

$$\frac{\sigma}{\mu} = 11.94 \pm 0.20 \%$$

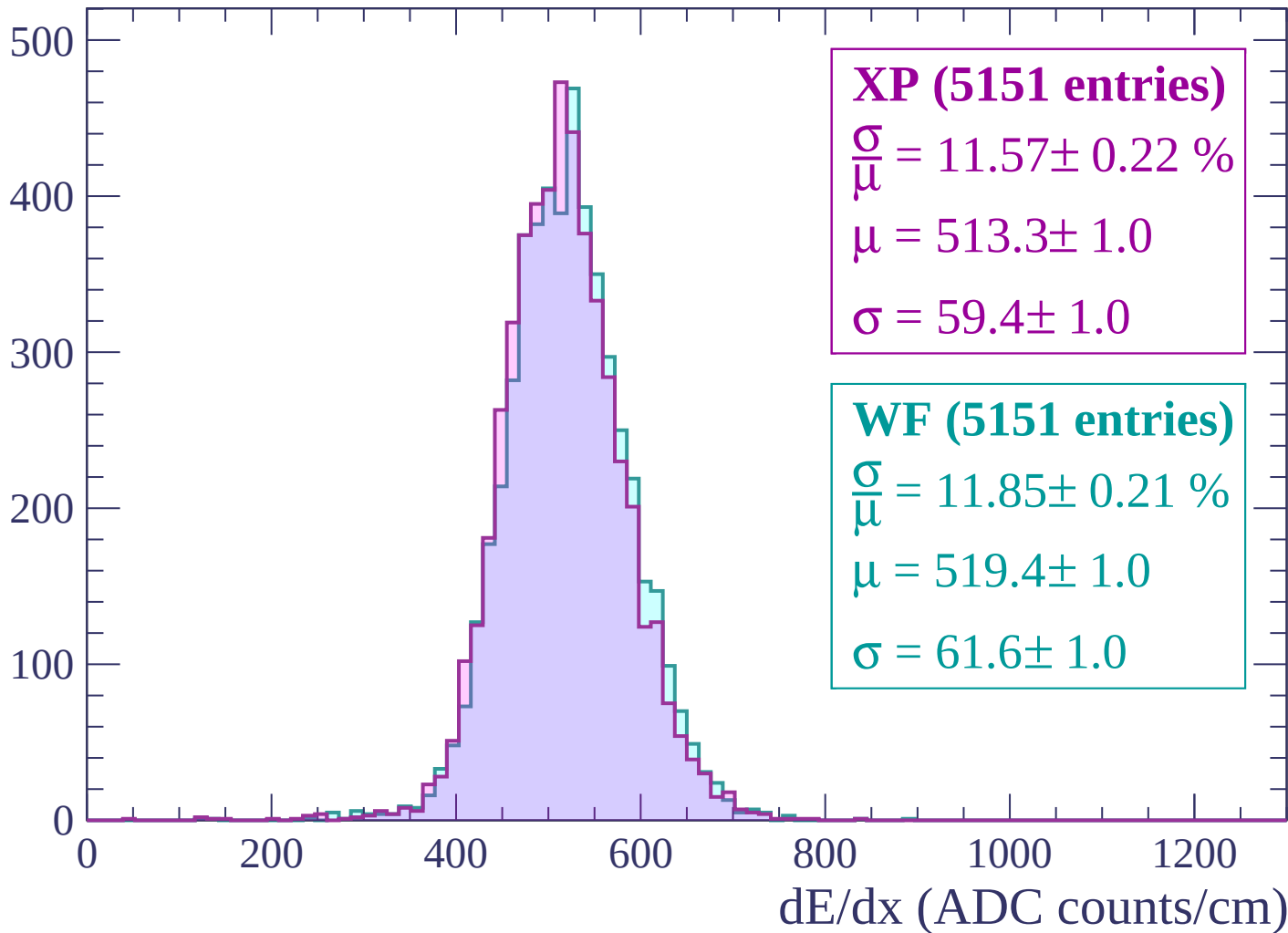
$$\mu = 521.7 \pm 1.0$$

$$\sigma = 62.3 \pm 0.9$$

dE/dx (ADC counts/cm)

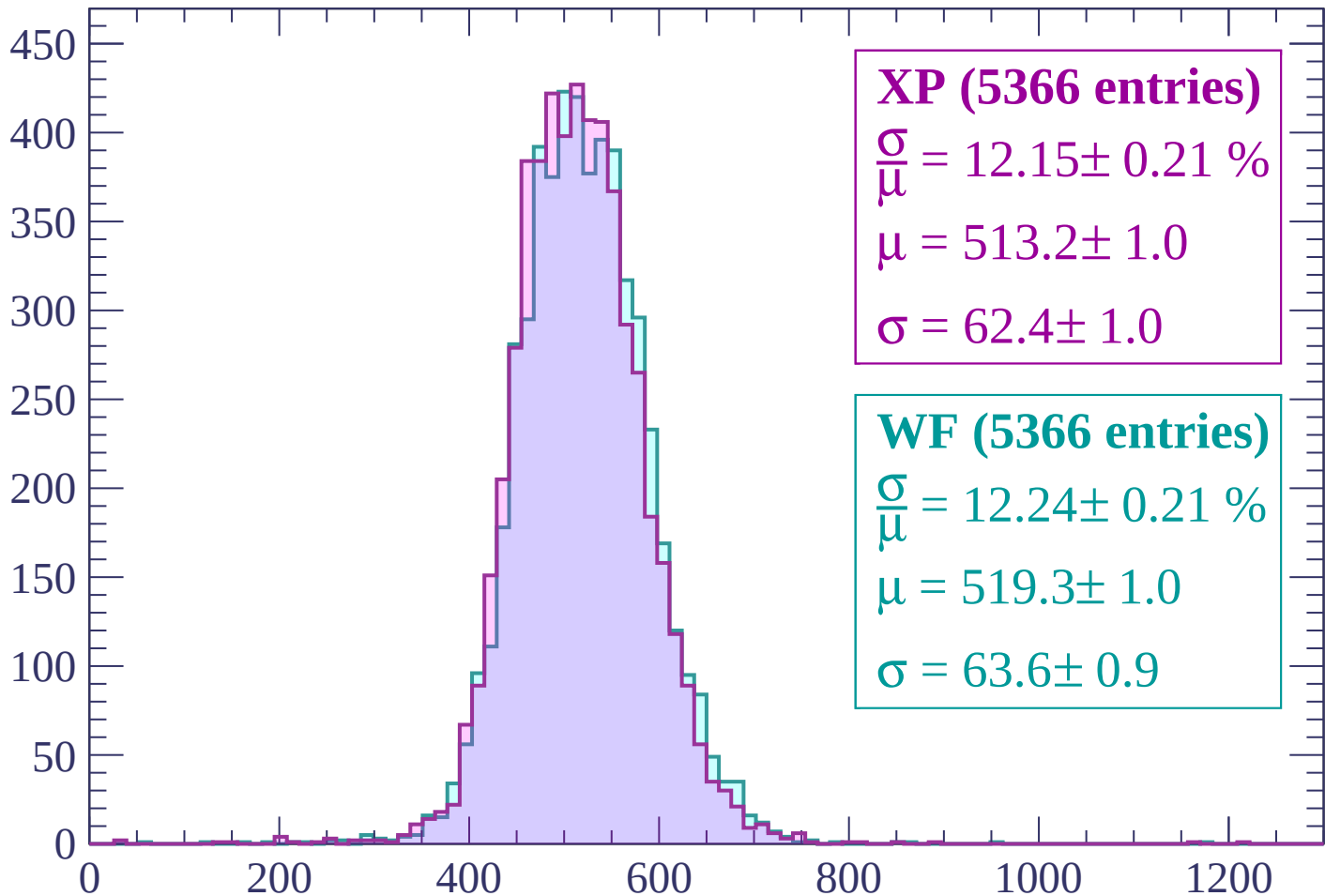
Energy loss | -2520 < p < -2460

Count



Energy loss | $-2460 < p < -2400$

Count



XP (5366 entries)

$$\frac{\sigma}{\mu} = 12.15 \pm 0.21 \%$$

$$\mu = 513.2 \pm 1.0$$

$$\sigma = 62.4 \pm 1.0$$

WF (5366 entries)

$$\frac{\sigma}{\mu} = 12.24 \pm 0.21 \%$$

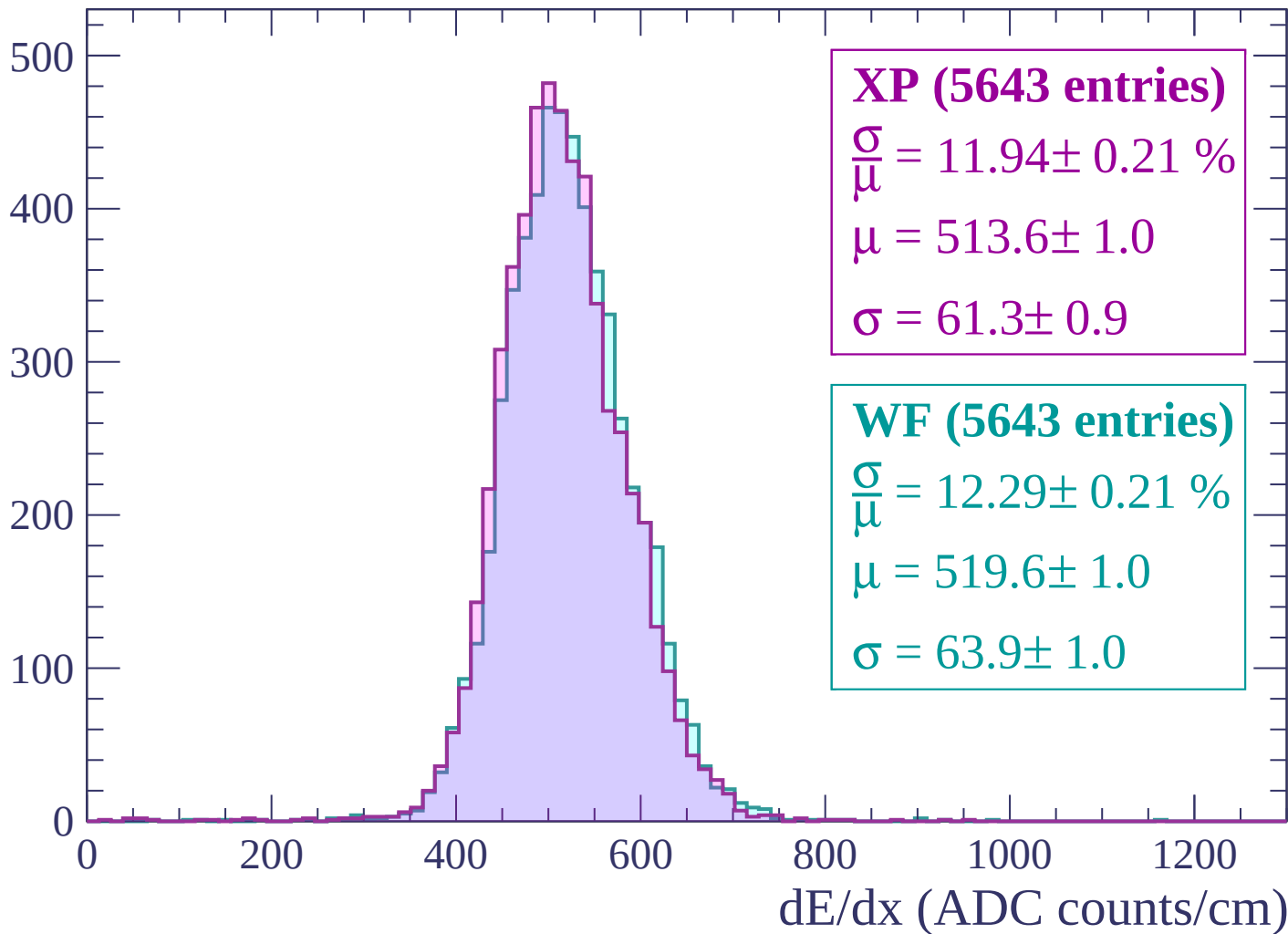
$$\mu = 519.3 \pm 1.0$$

$$\sigma = 63.6 \pm 0.9$$

dE/dx (ADC counts/cm)

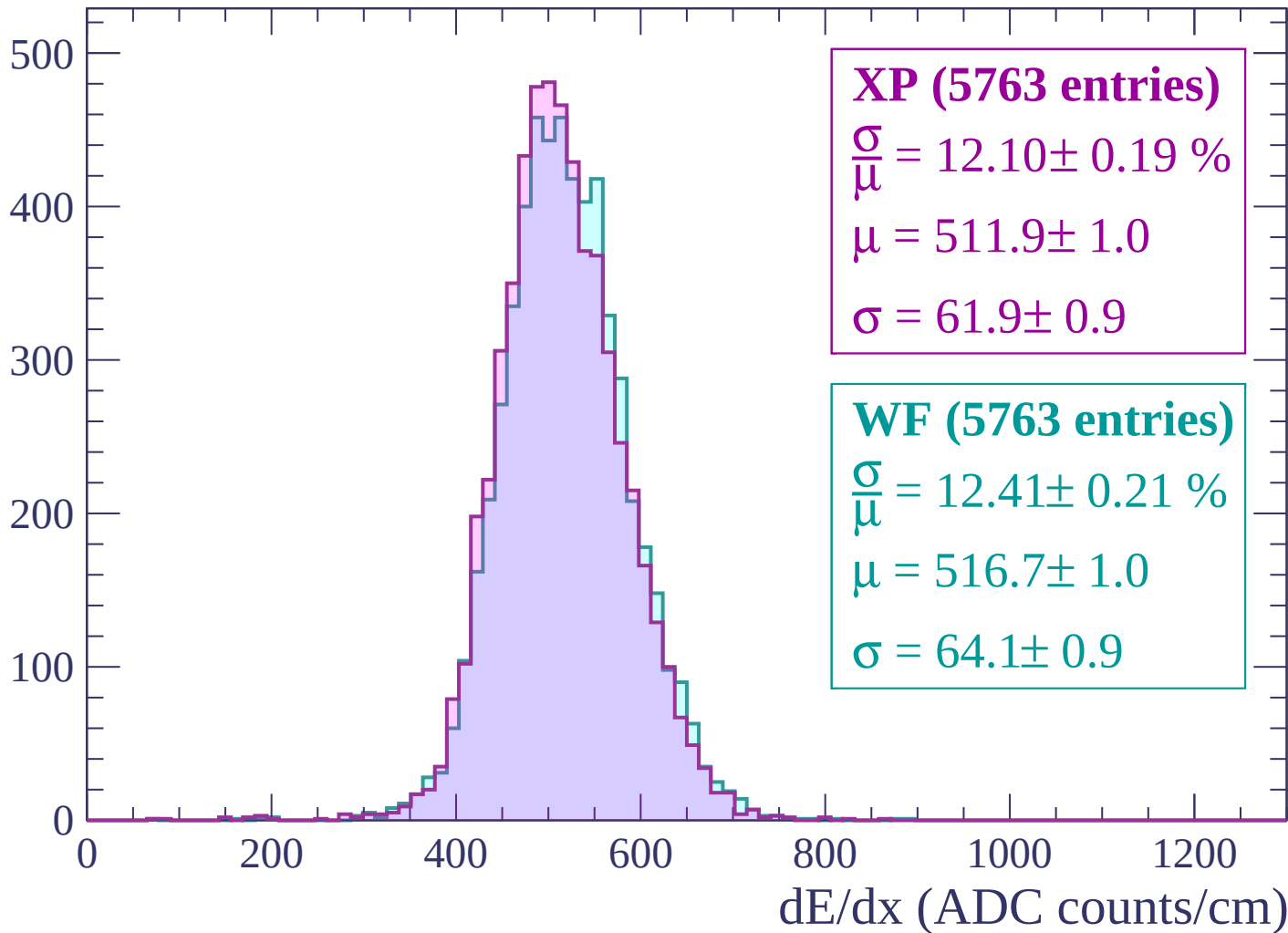
Energy loss | -2400 < p < -2340

Count



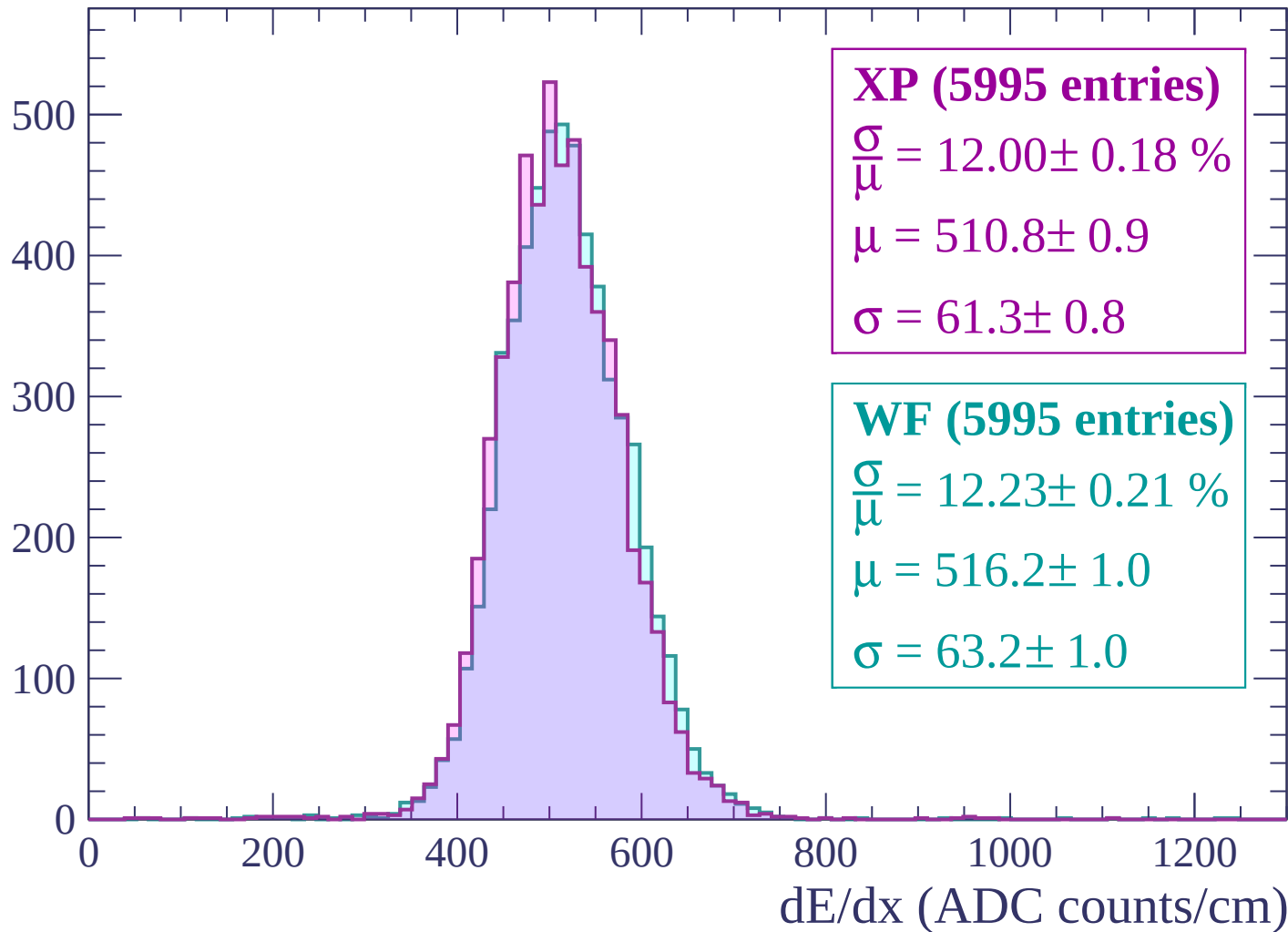
Energy loss | $-2340 < p < -2280$

Count



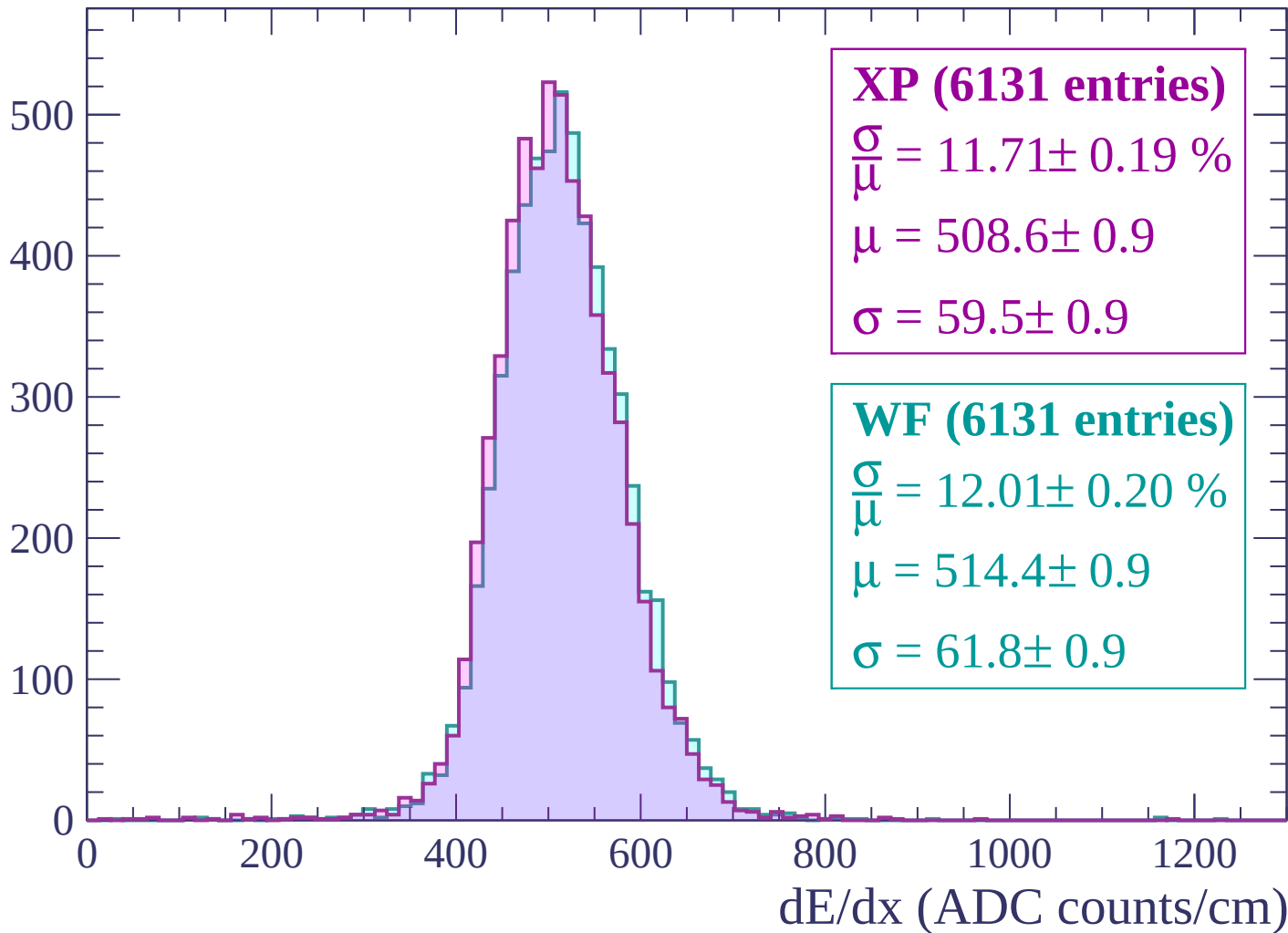
Energy loss | $-2280 < p < -2220$

Count



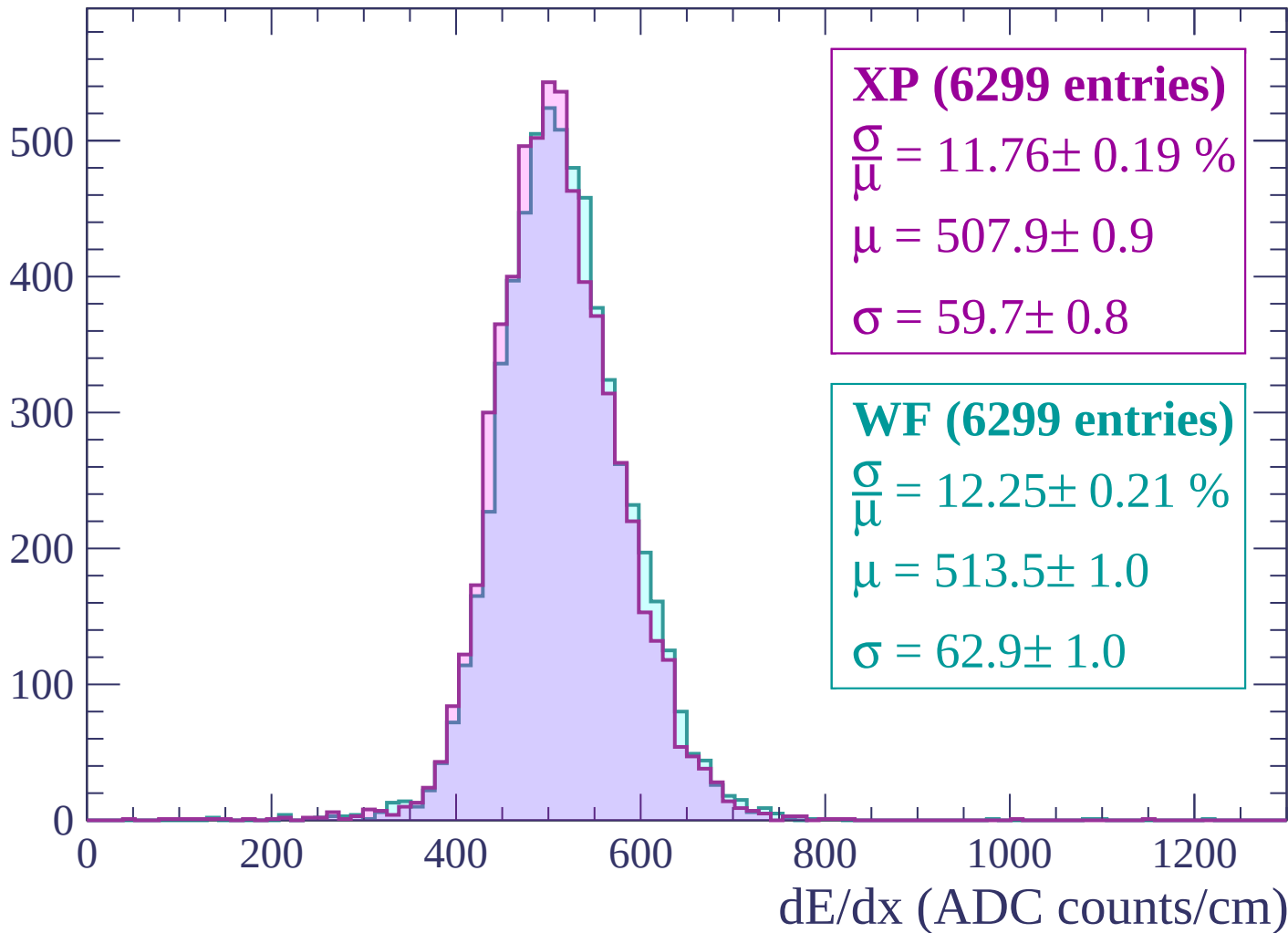
Energy loss | -2220 < p < -2160

Count



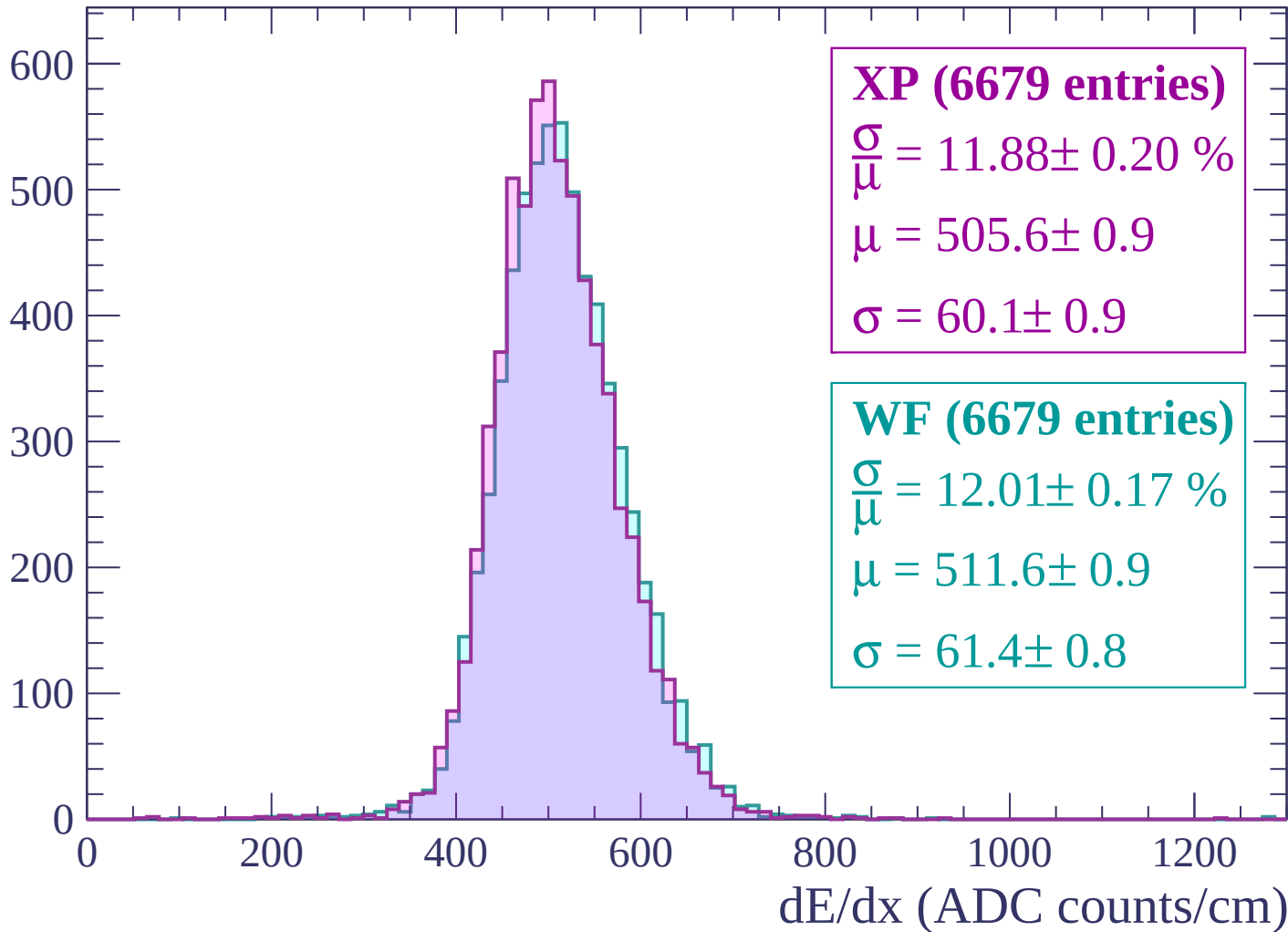
Energy loss | $-2160 < p < -2100$

Count



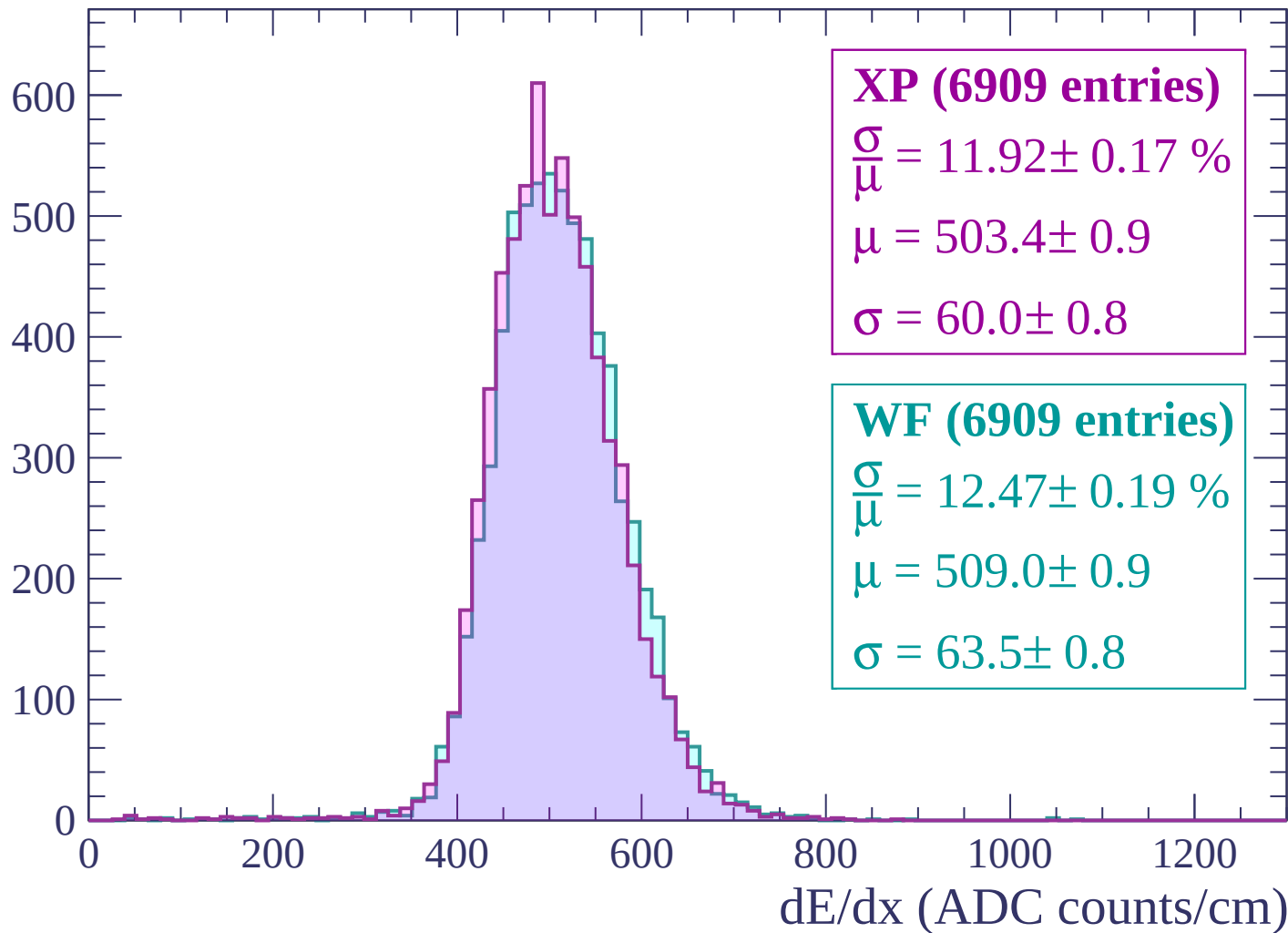
Energy loss | -2100 < p < -2040

Count



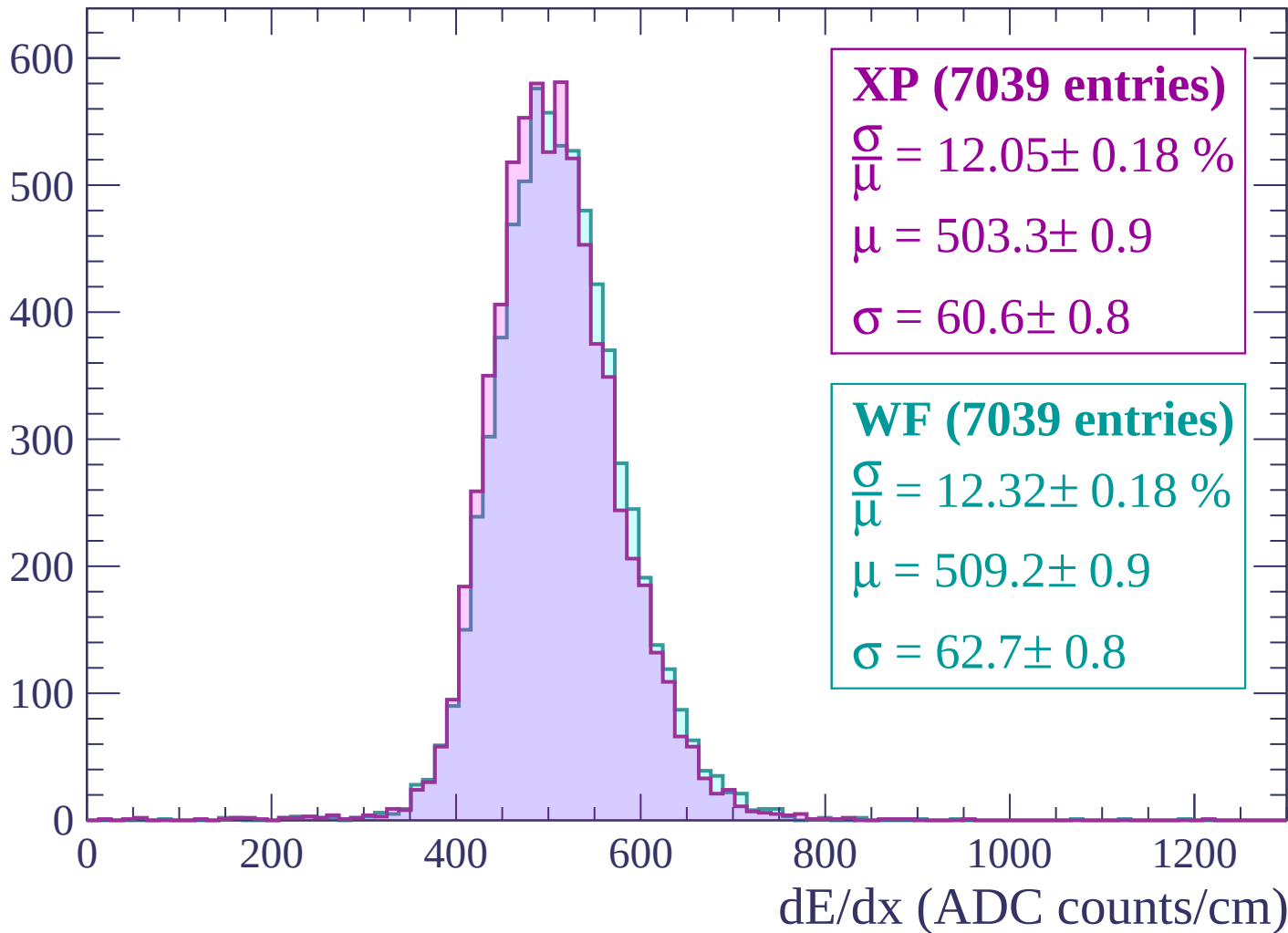
Energy loss | $-2040 < p < -1980$

Count



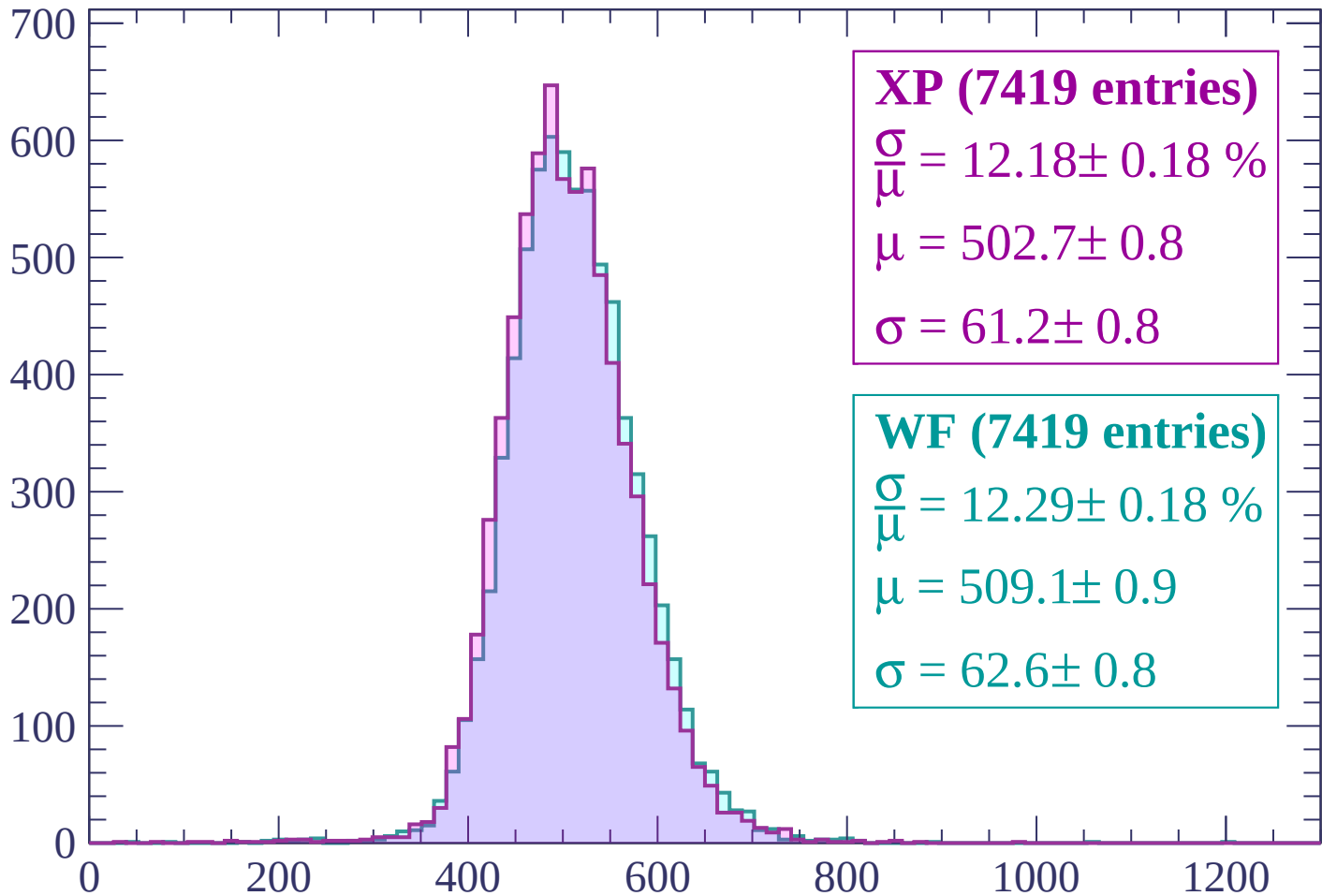
Energy loss | $-1980 < p < -1920$

Count



Energy loss | -1920 < p < -1860

Count



XP (7419 entries)

$$\frac{\sigma}{\mu} = 12.18 \pm 0.18 \%$$

$$\mu = 502.7 \pm 0.8$$

$$\sigma = 61.2 \pm 0.8$$

WF (7419 entries)

$$\frac{\sigma}{\mu} = 12.29 \pm 0.18 \%$$

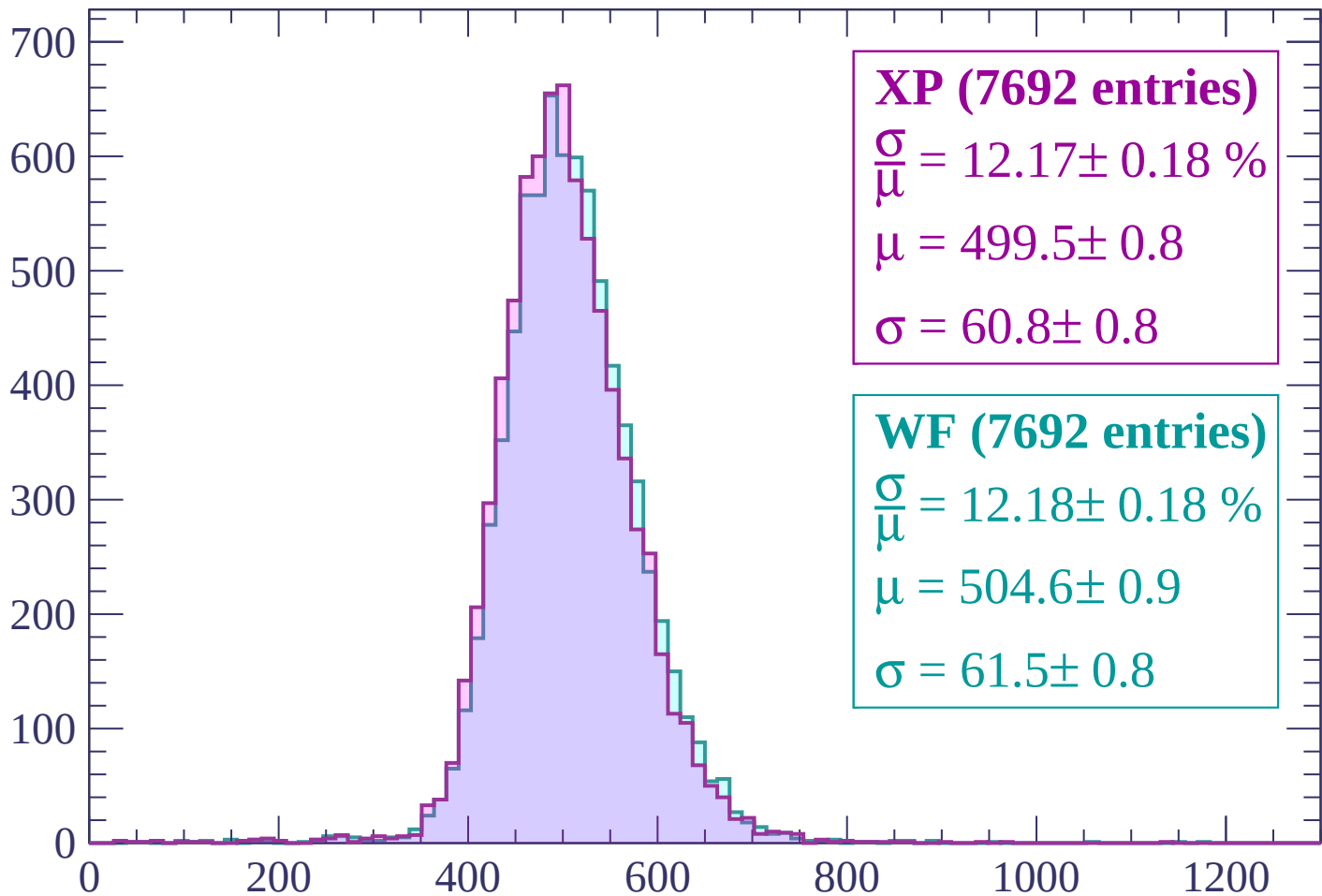
$$\mu = 509.1 \pm 0.9$$

$$\sigma = 62.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1860 < p < -1800

Count



XP (7692 entries)

$$\frac{\sigma}{\mu} = 12.17 \pm 0.18 \%$$

$$\mu = 499.5 \pm 0.8$$

$$\sigma = 60.8 \pm 0.8$$

WF (7692 entries)

$$\frac{\sigma}{\mu} = 12.18 \pm 0.18 \%$$

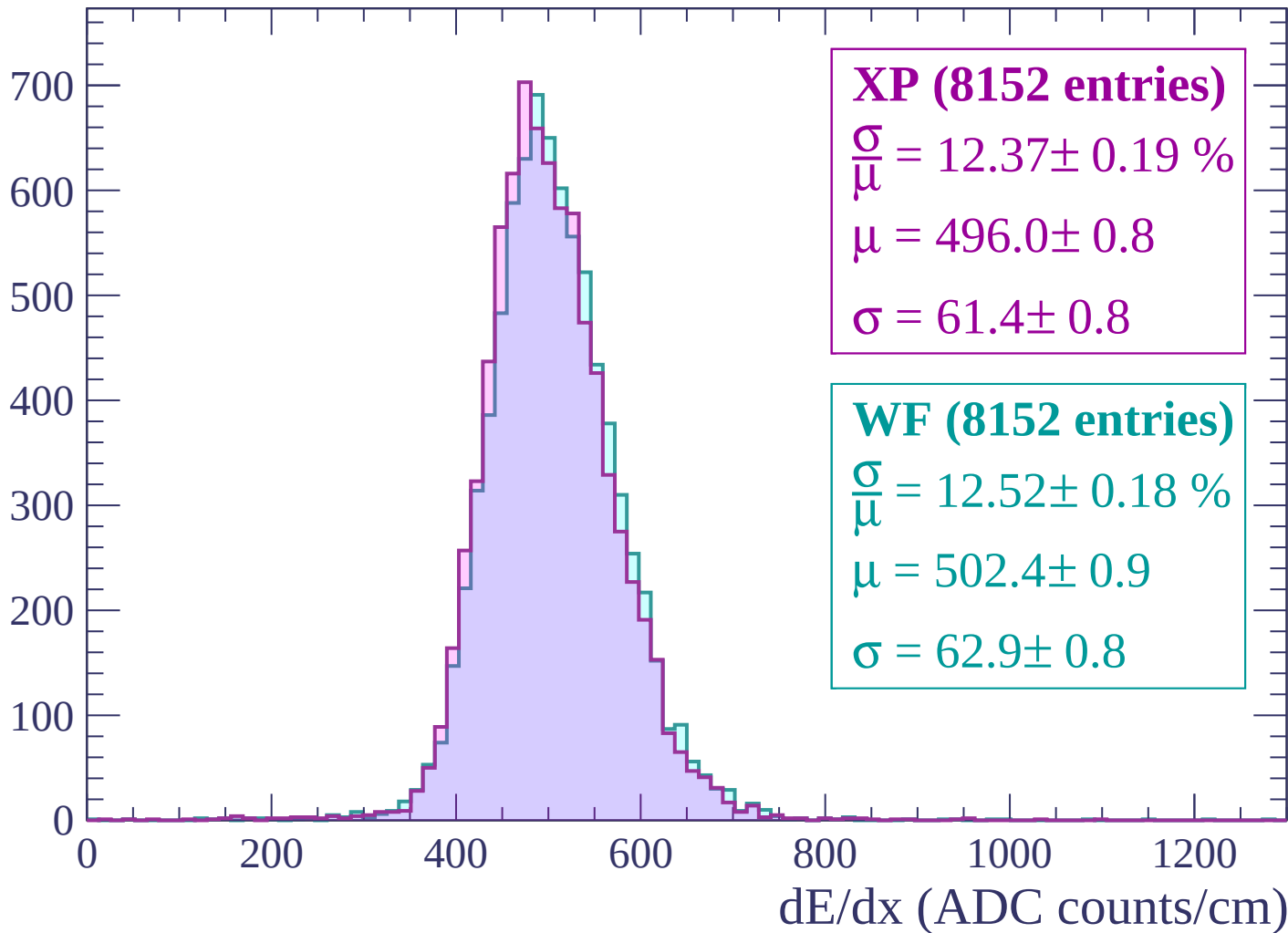
$$\mu = 504.6 \pm 0.9$$

$$\sigma = 61.5 \pm 0.8$$

dE/dx (ADC counts/cm)

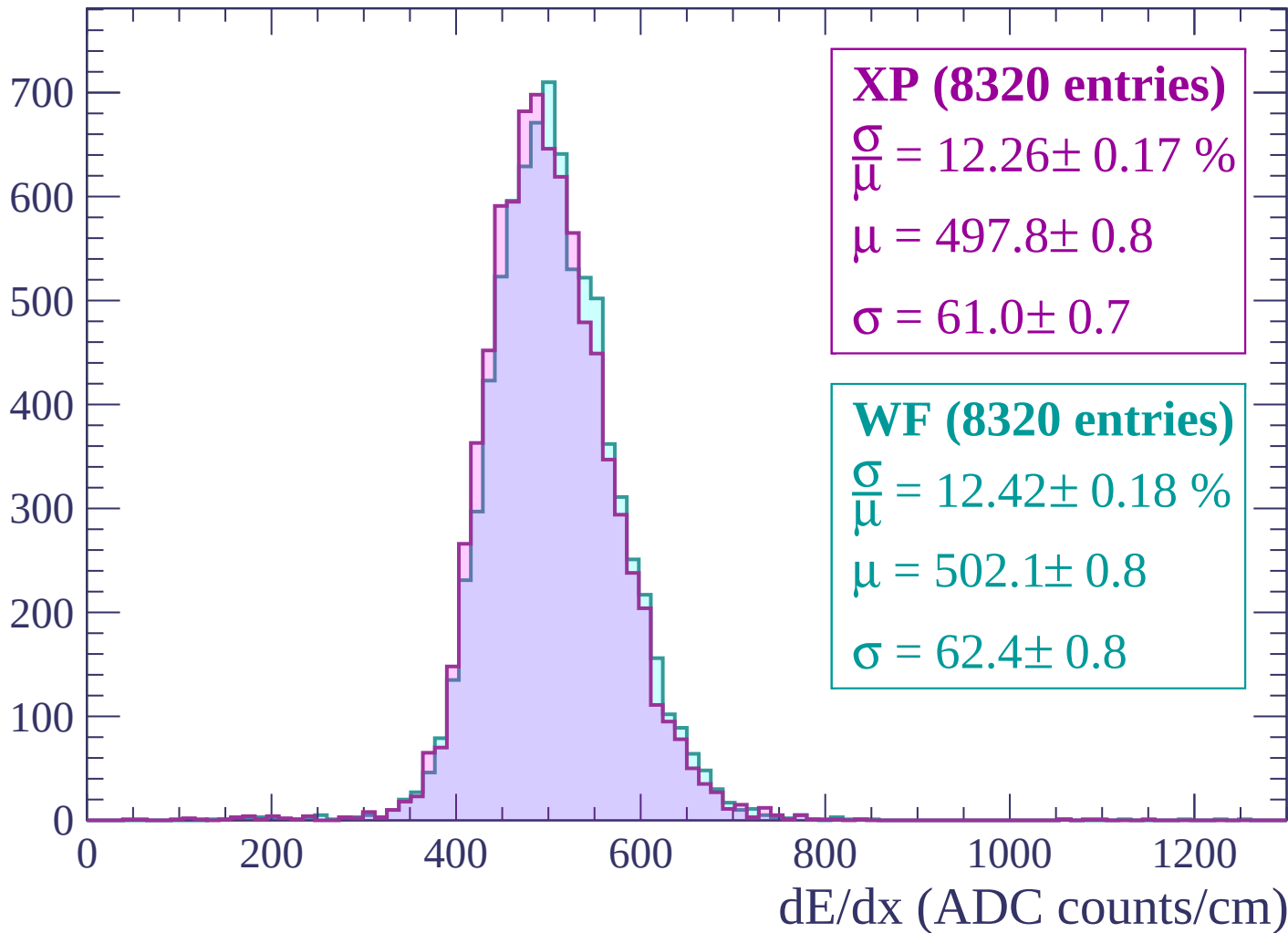
Energy loss | -1800 < p < -1740

Count



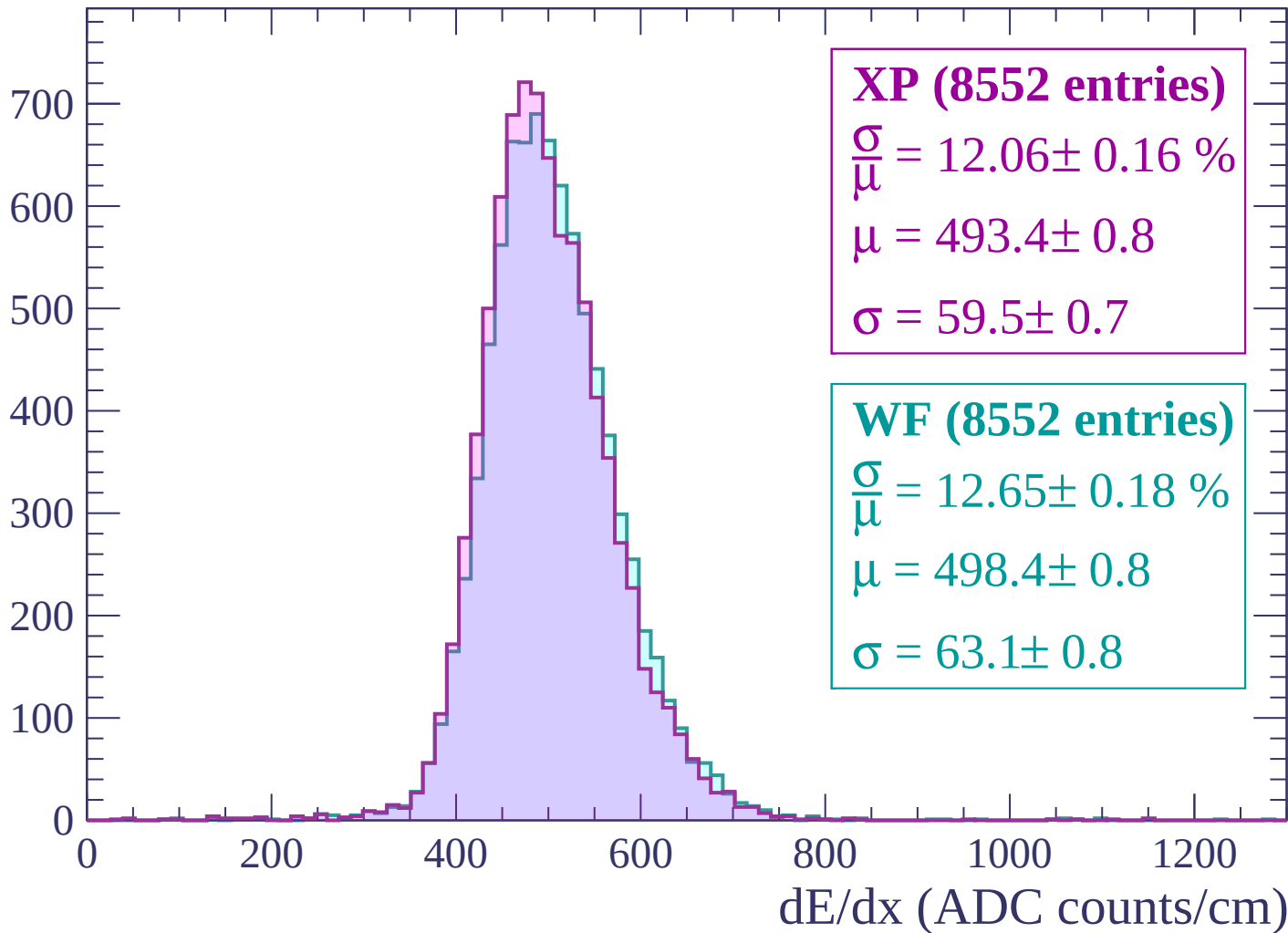
Energy loss | -1740 < p < -1680

Count

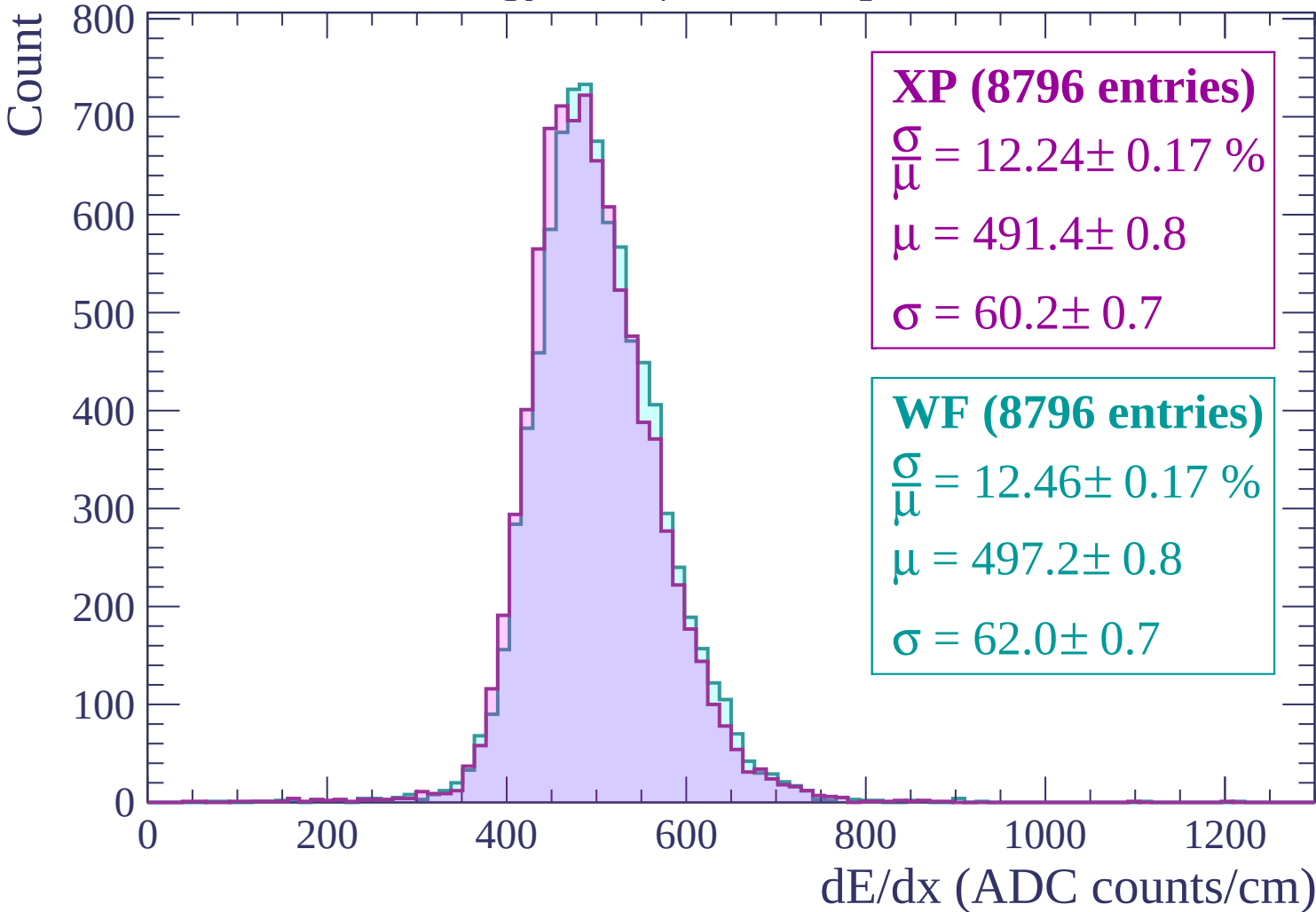


Energy loss | $-1680 < p < -1620$

Count

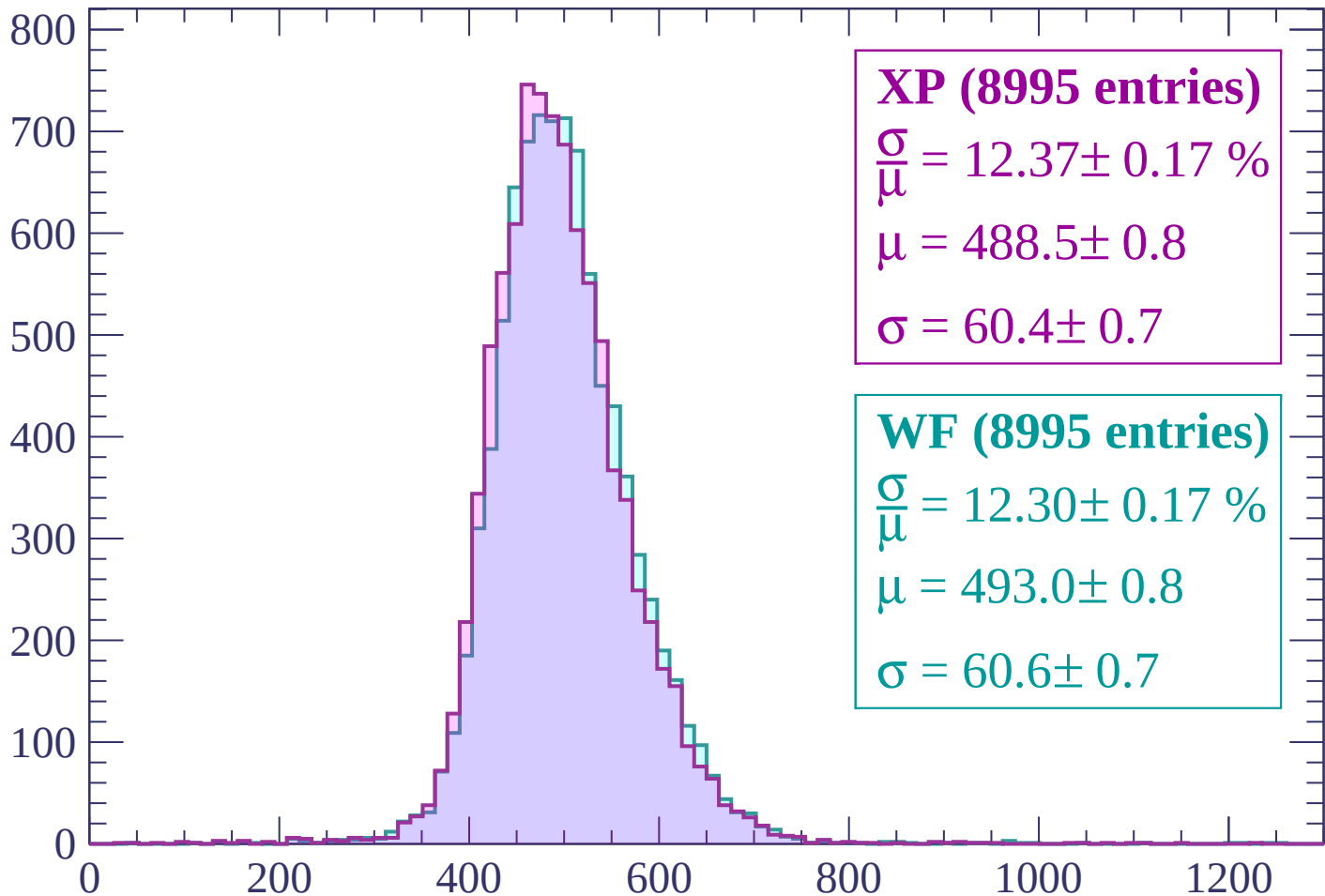


Energy loss | -1620 < p < -1560



Energy loss | -1560 < p < -1500

Count



XP (8995 entries)

$$\frac{\sigma}{\mu} = 12.37 \pm 0.17 \%$$

$$\mu = 488.5 \pm 0.8$$

$$\sigma = 60.4 \pm 0.7$$

WF (8995 entries)

$$\frac{\sigma}{\mu} = 12.30 \pm 0.17 \%$$

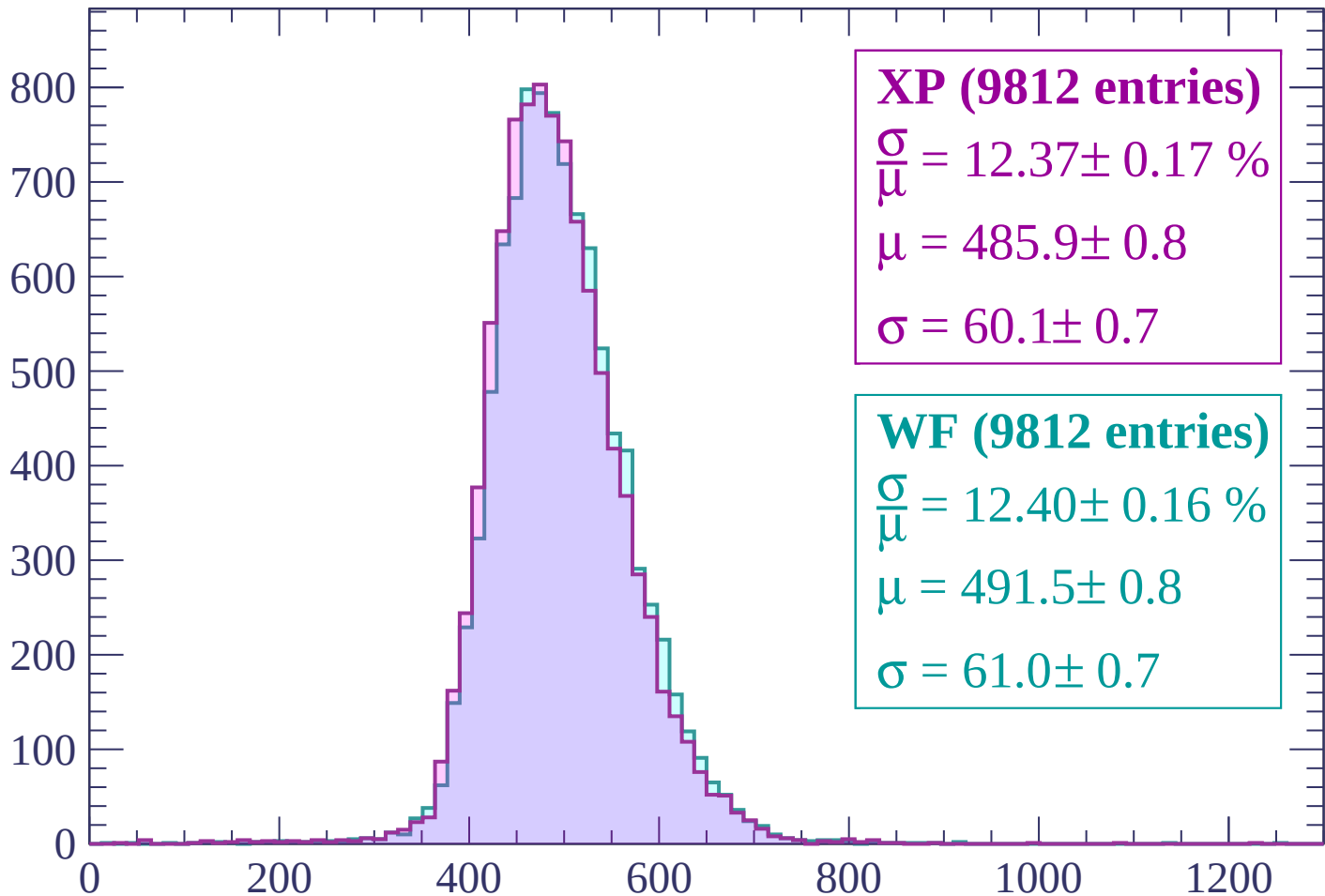
$$\mu = 493.0 \pm 0.8$$

$$\sigma = 60.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1500 < p < -1440

Count



XP (9812 entries)

$$\frac{\sigma}{\mu} = 12.37 \pm 0.17 \%$$

$$\mu = 485.9 \pm 0.8$$

$$\sigma = 60.1 \pm 0.7$$

WF (9812 entries)

$$\frac{\sigma}{\mu} = 12.40 \pm 0.16 \%$$

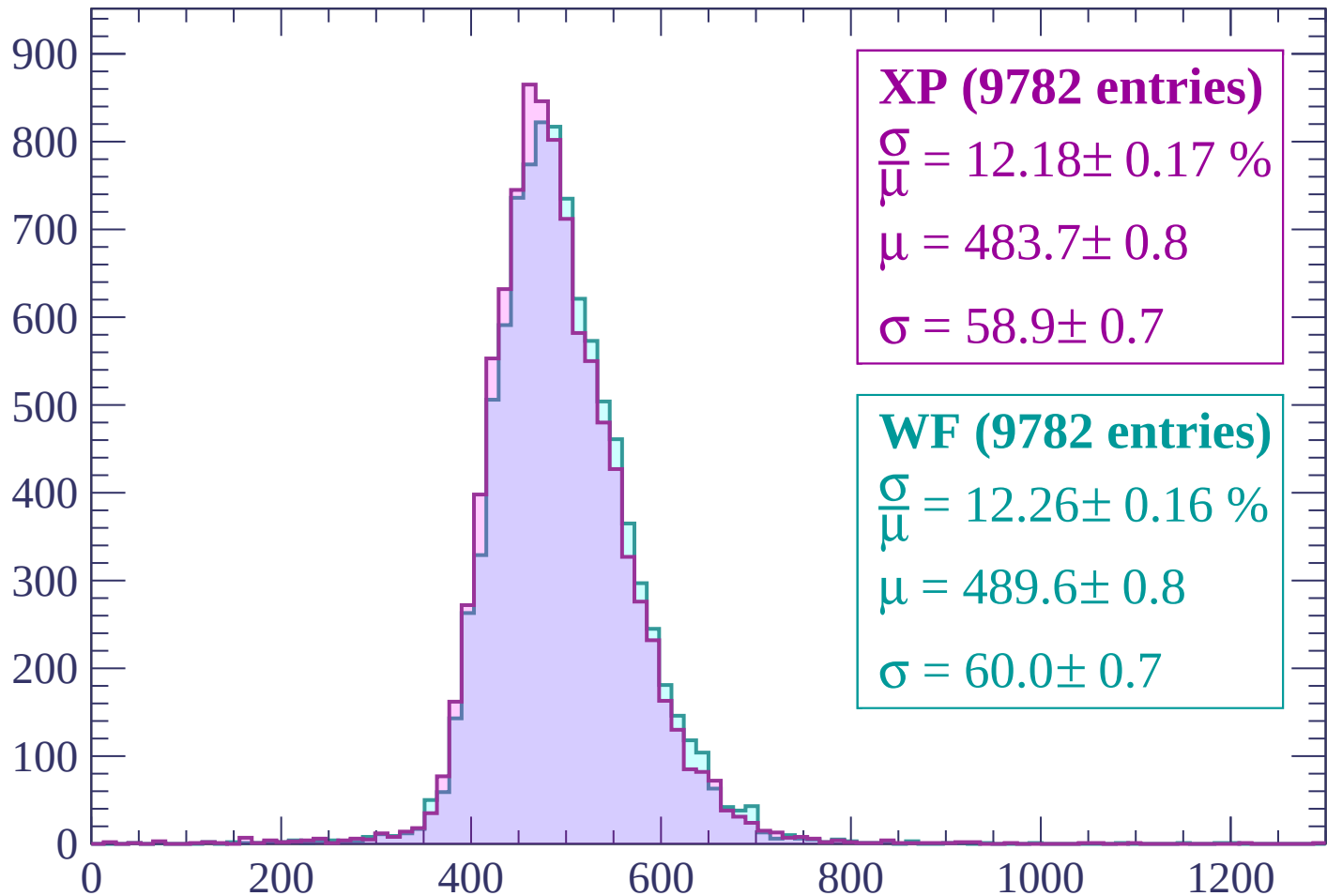
$$\mu = 491.5 \pm 0.8$$

$$\sigma = 61.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1380$

Count



XP (9782 entries)

$$\frac{\sigma}{\mu} = 12.18 \pm 0.17 \%$$

$$\mu = 483.7 \pm 0.8$$

$$\sigma = 58.9 \pm 0.7$$

WF (9782 entries)

$$\frac{\sigma}{\mu} = 12.26 \pm 0.16 \%$$

$$\mu = 489.6 \pm 0.8$$

$$\sigma = 60.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1380 < p < -1320$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (10474 entries)

$$\frac{\sigma}{\mu} = 12.45 \pm 0.17 \%$$

$$\mu = 478.7 \pm 0.7$$

$$\sigma = 59.6 \pm 0.7$$

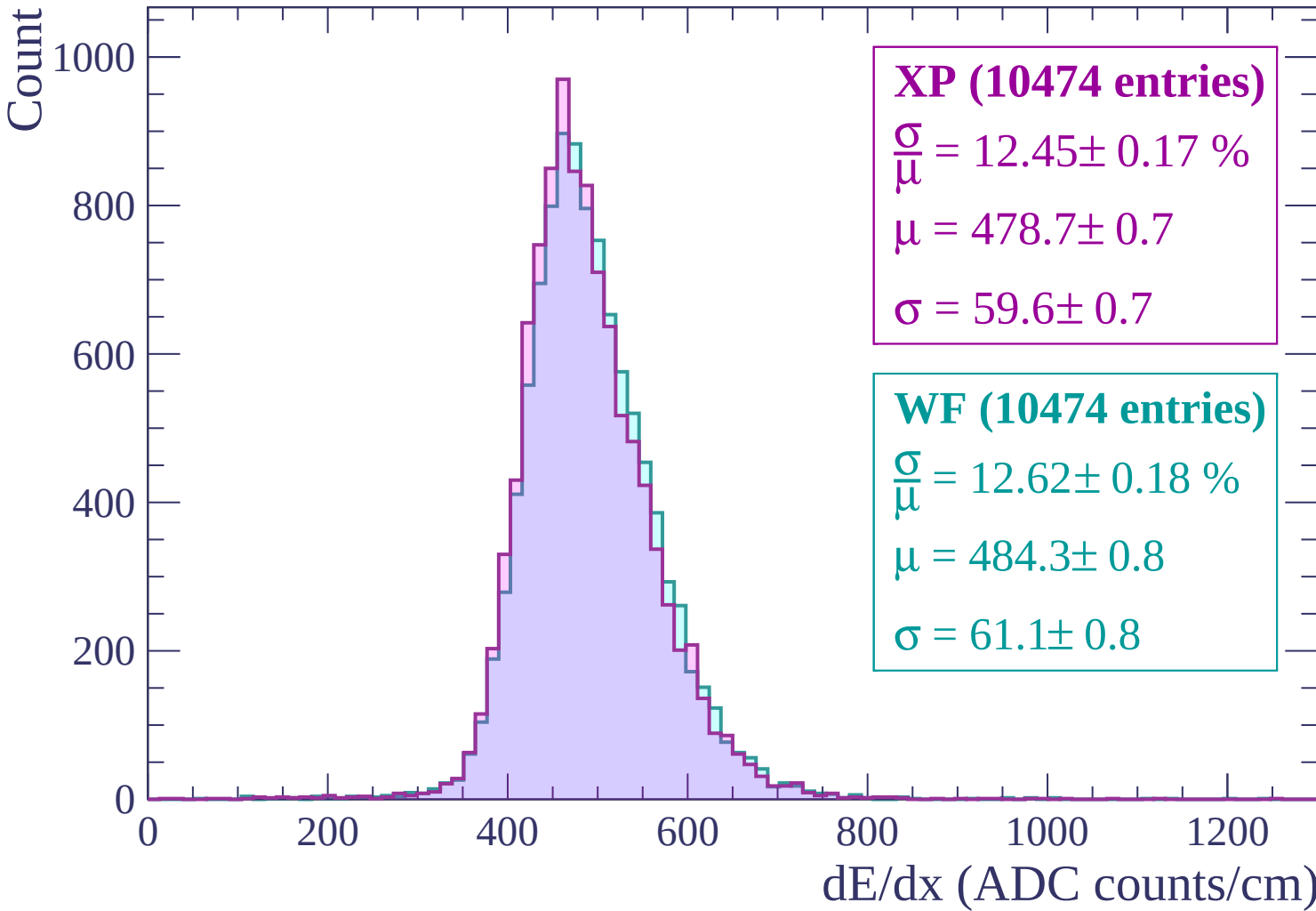
WF (10474 entries)

$$\frac{\sigma}{\mu} = 12.62 \pm 0.18 \%$$

$$\mu = 484.3 \pm 0.8$$

$$\sigma = 61.1 \pm 0.8$$

dE/dx (ADC counts/cm)



Energy loss | -1320 < p < -1260

Count

1000

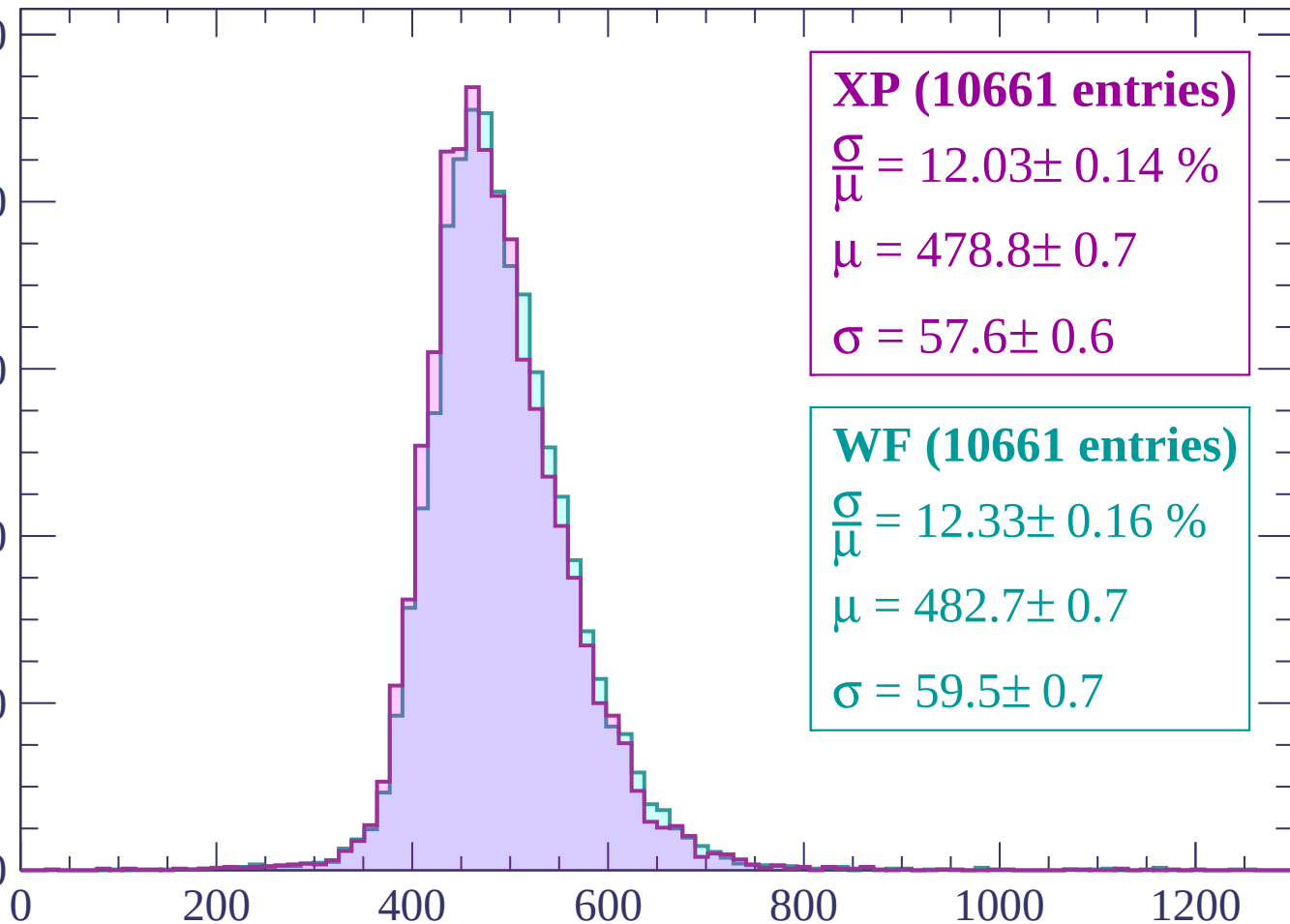
800

600

400

200

0



XP (10661 entries)

$\frac{\sigma}{\mu} = 12.03 \pm 0.14 \%$

$\mu = 478.8 \pm 0.7$

$\sigma = 57.6 \pm 0.6$

WF (10661 entries)

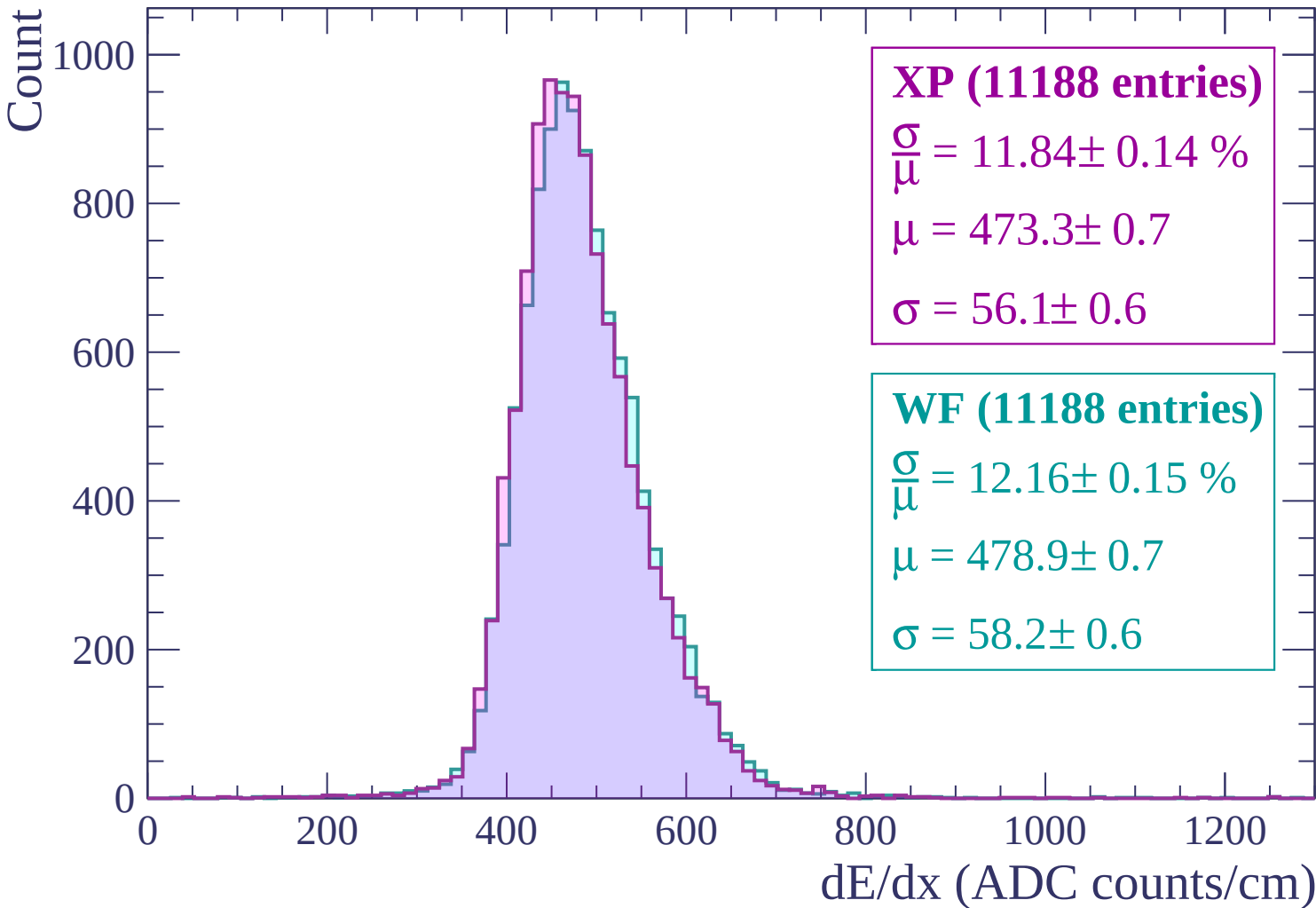
$\frac{\sigma}{\mu} = 12.33 \pm 0.16 \%$

$\mu = 482.7 \pm 0.7$

$\sigma = 59.5 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | -1260 < p < -1200



Energy loss | -1200 < p < -1140

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (11502 entries)

$$\frac{\sigma}{\mu} = 11.81 \pm 0.17 \%$$

$$\mu = 467.1 \pm 0.7$$

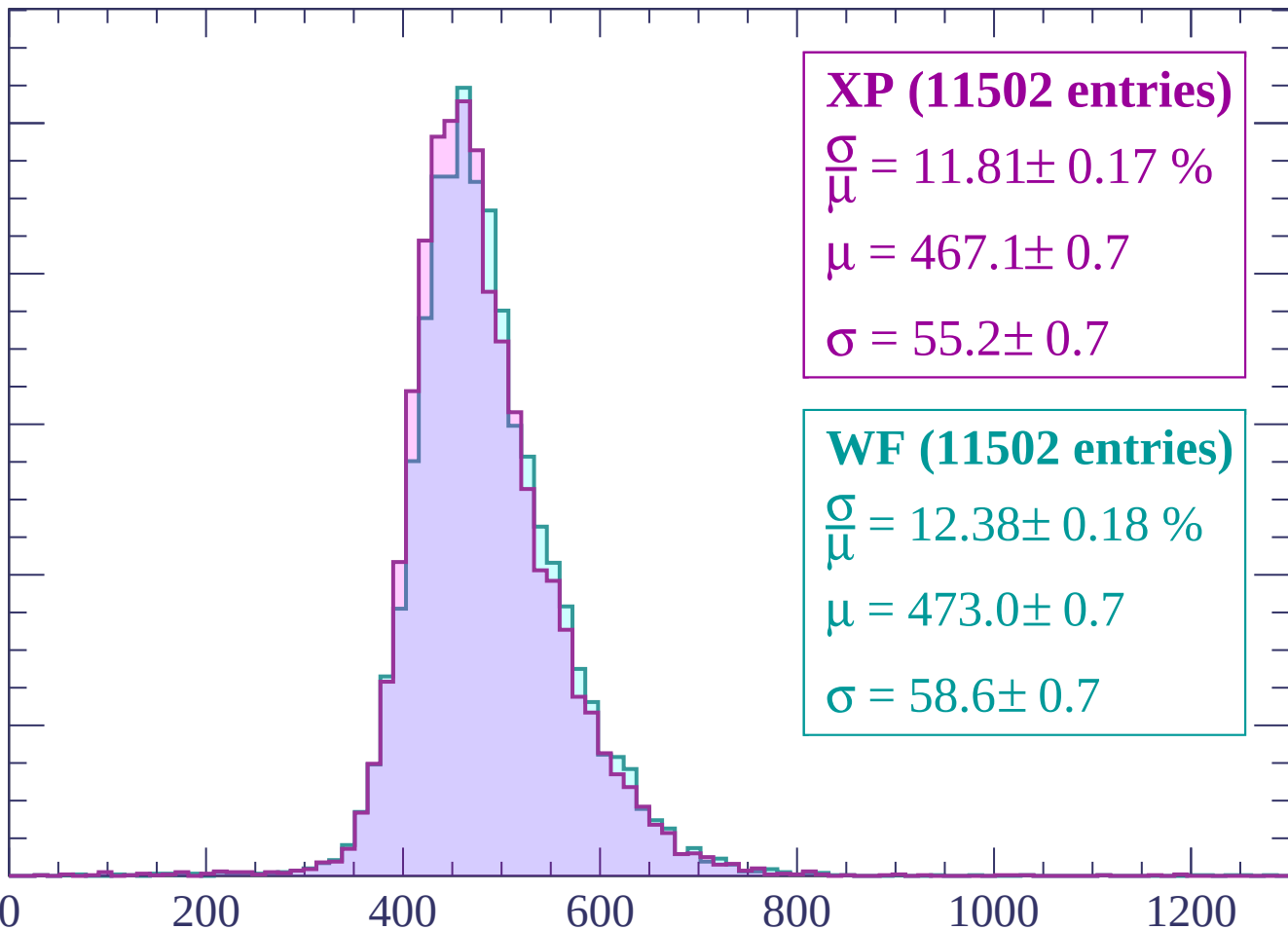
$$\sigma = 55.2 \pm 0.7$$

WF (11502 entries)

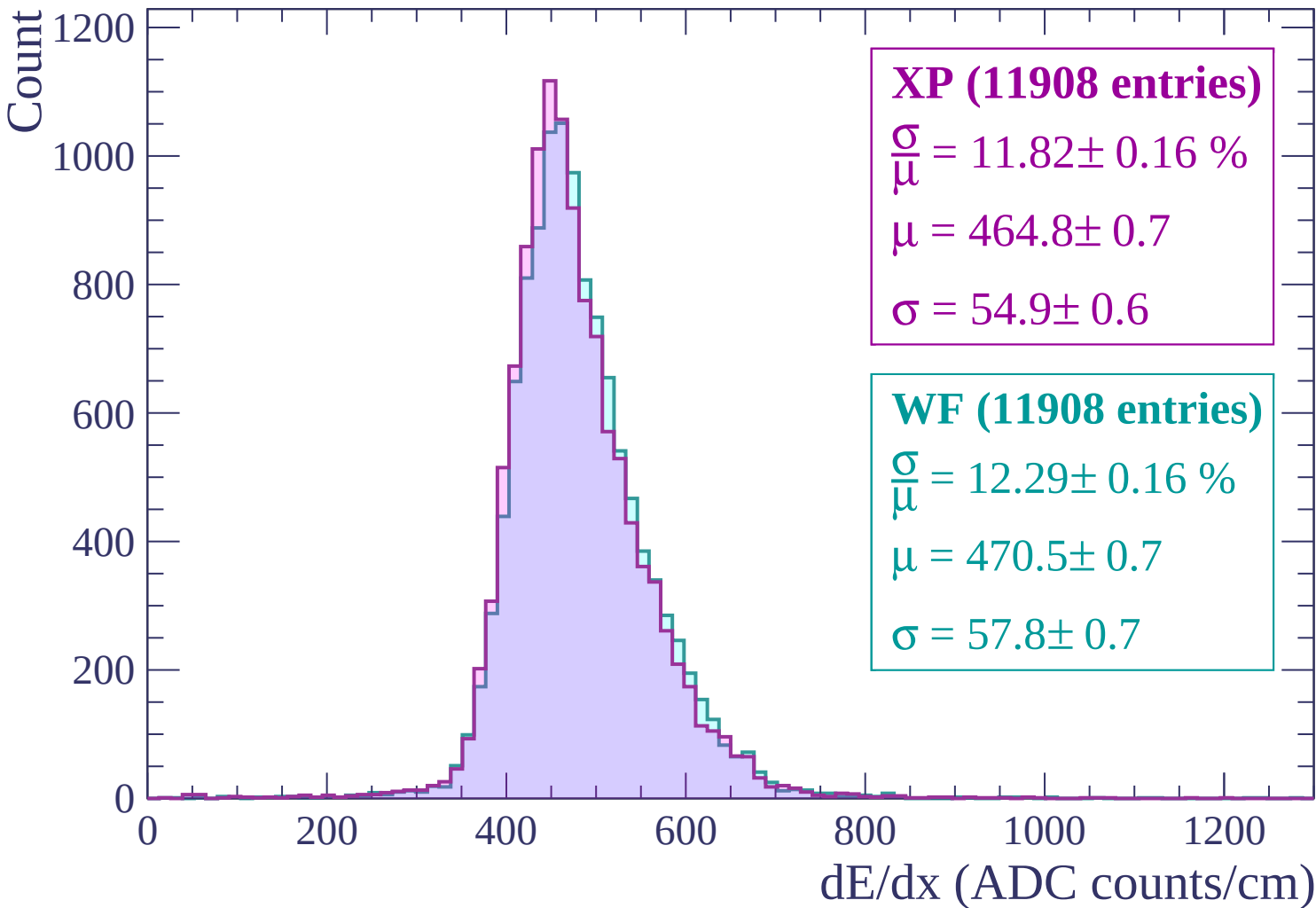
$$\frac{\sigma}{\mu} = 12.38 \pm 0.18 \%$$

$$\mu = 473.0 \pm 0.7$$

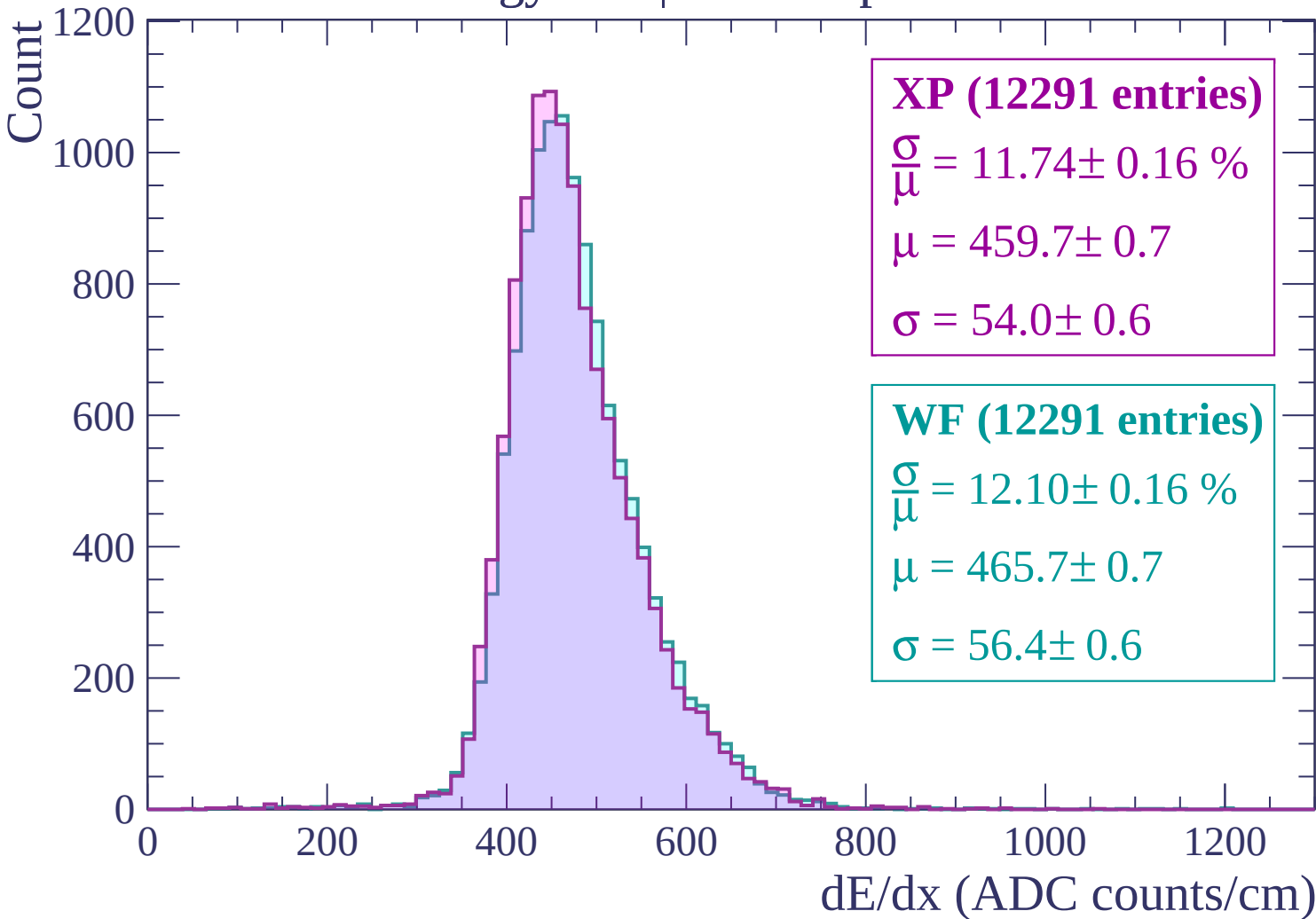
$$\sigma = 58.6 \pm 0.7$$



Energy loss | -1140 < p < -1080



Energy loss | -1080 < p < -1020



Energy loss | $-1020 < p < -960$

Count

1200
1000
800
600
400
200
0

XP (12677 entries)

$$\frac{\sigma}{\mu} = 11.17 \pm 0.15 \%$$

$$\mu = 454.2 \pm 0.6$$

$$\sigma = 50.7 \pm 0.6$$

WF (12677 entries)

$$\frac{\sigma}{\mu} = 11.76 \pm 0.16 \%$$

$$\mu = 459.3 \pm 0.6$$

$$\sigma = 54.0 \pm 0.6$$

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count

1200
1000
800
600
400
200
0

XP (13368 entries)

$\frac{\sigma}{\mu} = 11.63 \pm 0.13 \%$

$\mu = 452.9 \pm 0.6$

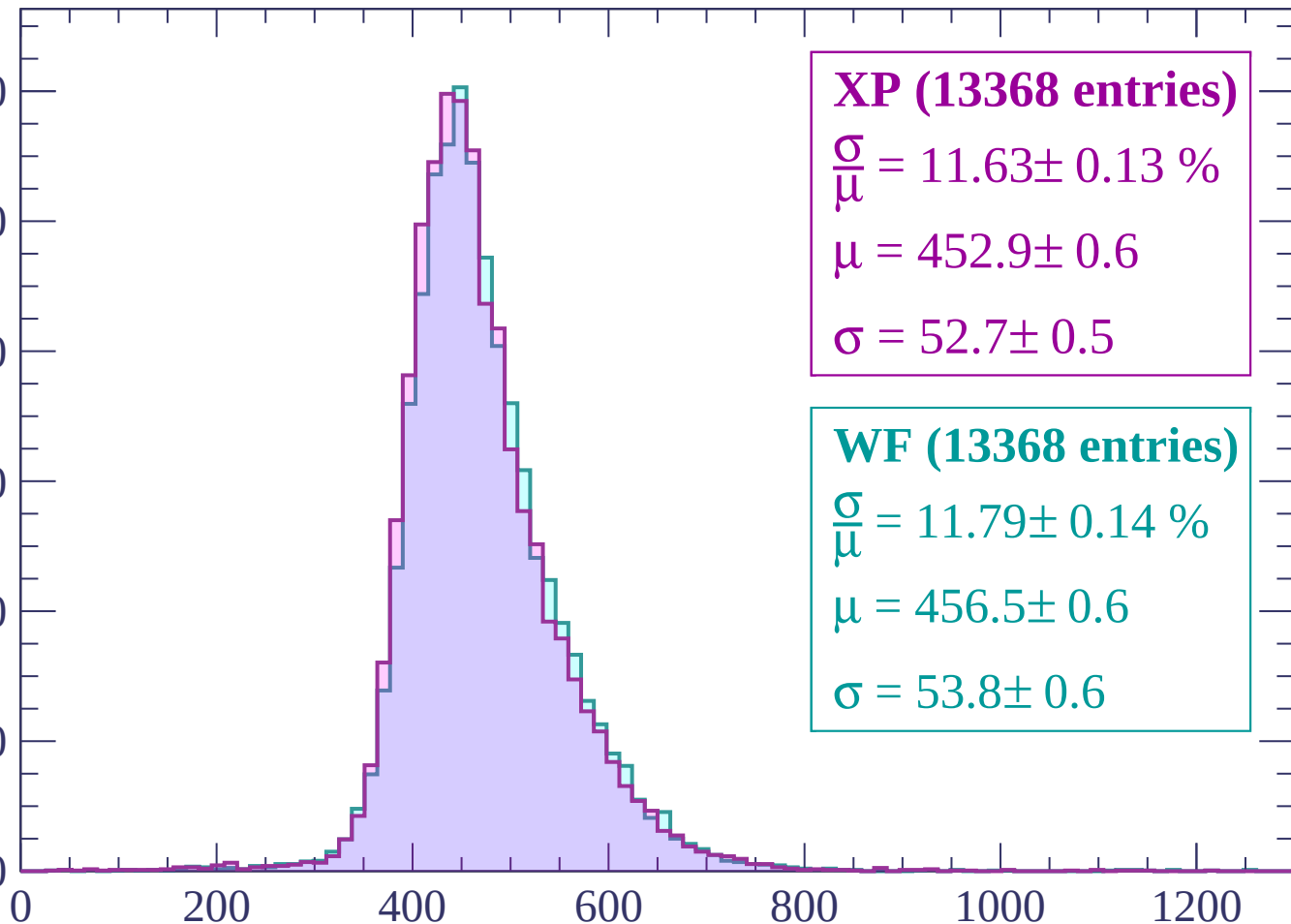
$\sigma = 52.7 \pm 0.5$

WF (13368 entries)

$\frac{\sigma}{\mu} = 11.79 \pm 0.14 \%$

$\mu = 456.5 \pm 0.6$

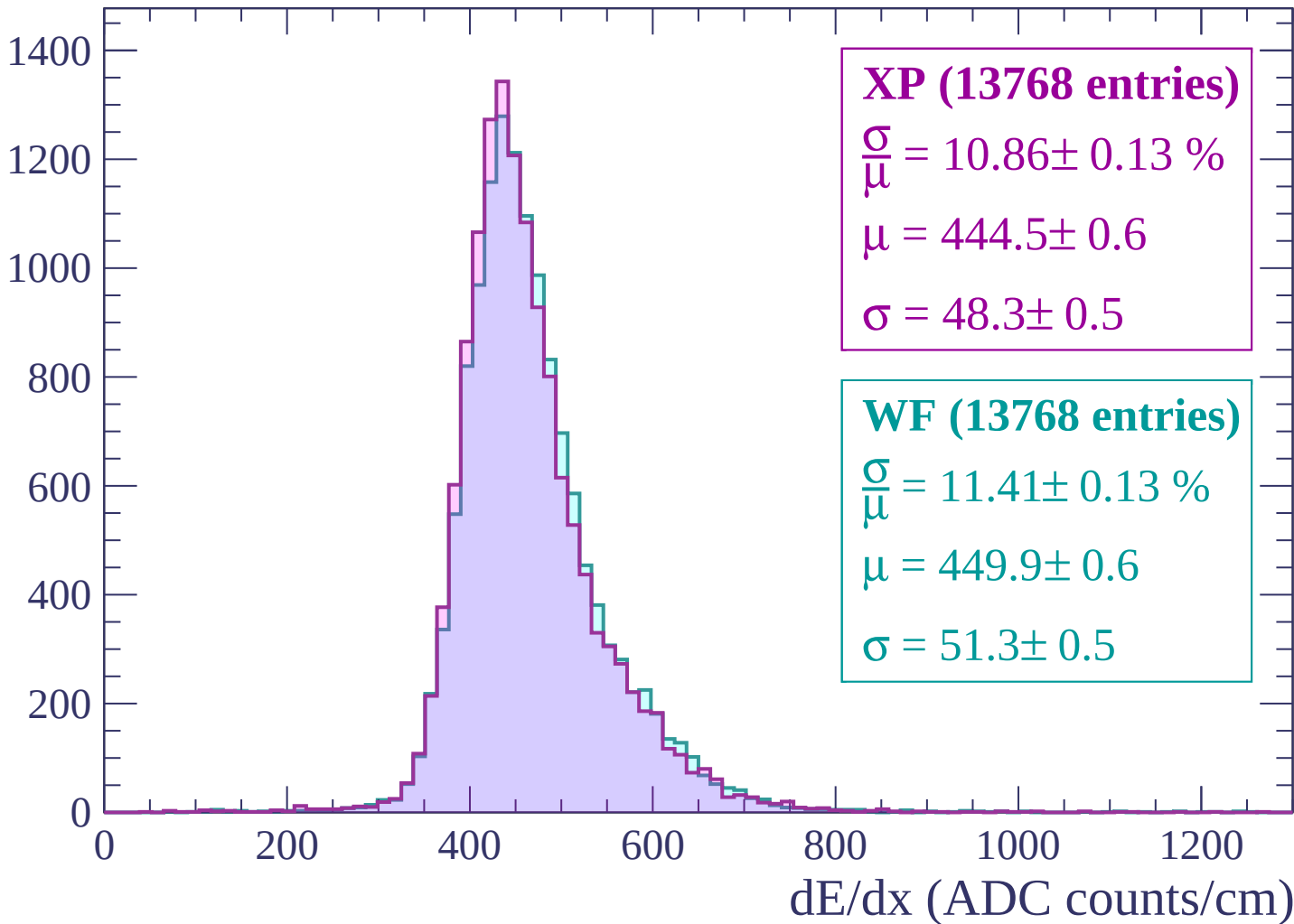
$\sigma = 53.8 \pm 0.6$



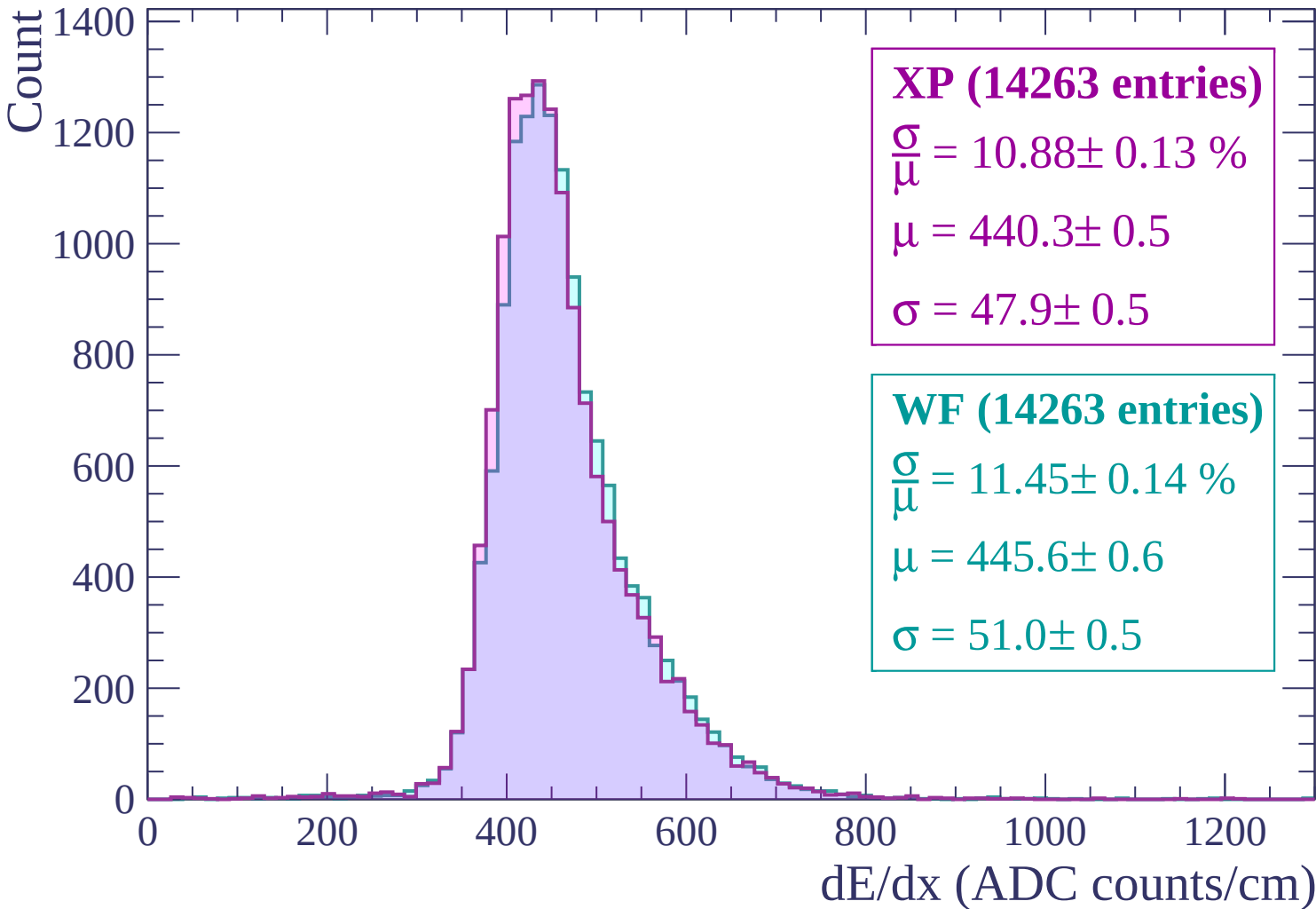
dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



Energy loss | $-840 < p < -780$



Energy loss | $-780 < p < -720$

Count

1400
1200
1000
800
600
400
200
0

XP (14332 entries)

$\frac{\sigma}{\mu} = 11.09 \pm 0.14 \%$

$\mu = 436.1 \pm 0.5$

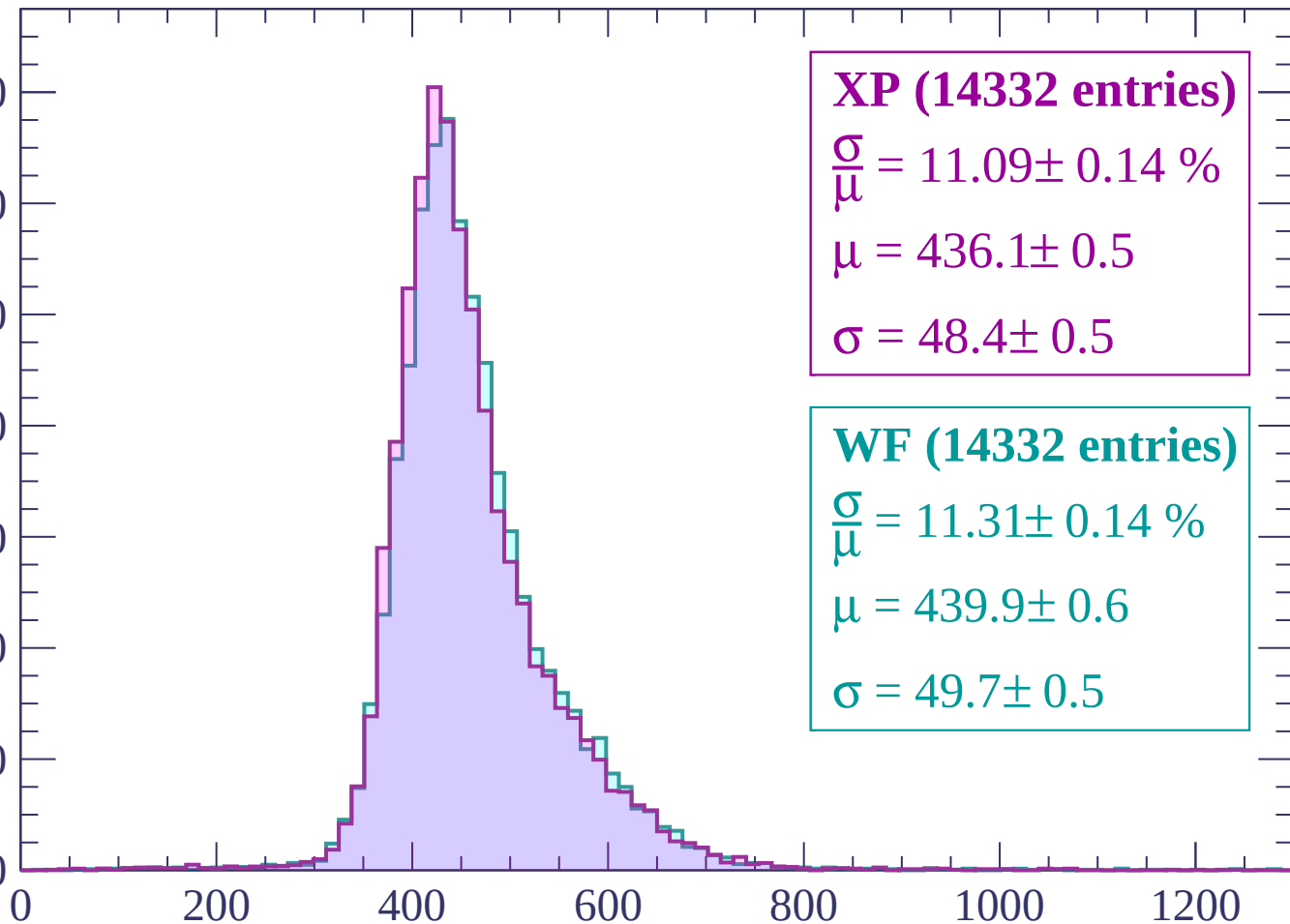
$\sigma = 48.4 \pm 0.5$

WF (14332 entries)

$\frac{\sigma}{\mu} = 11.31 \pm 0.14 \%$

$\mu = 439.9 \pm 0.6$

$\sigma = 49.7 \pm 0.5$



dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count

1400
1200
1000
800
600
400
200
0

XP (14961 entries)

$$\frac{\sigma}{\mu} = 10.78 \pm 0.13 \%$$

$$\mu = 429.5 \pm 0.5$$

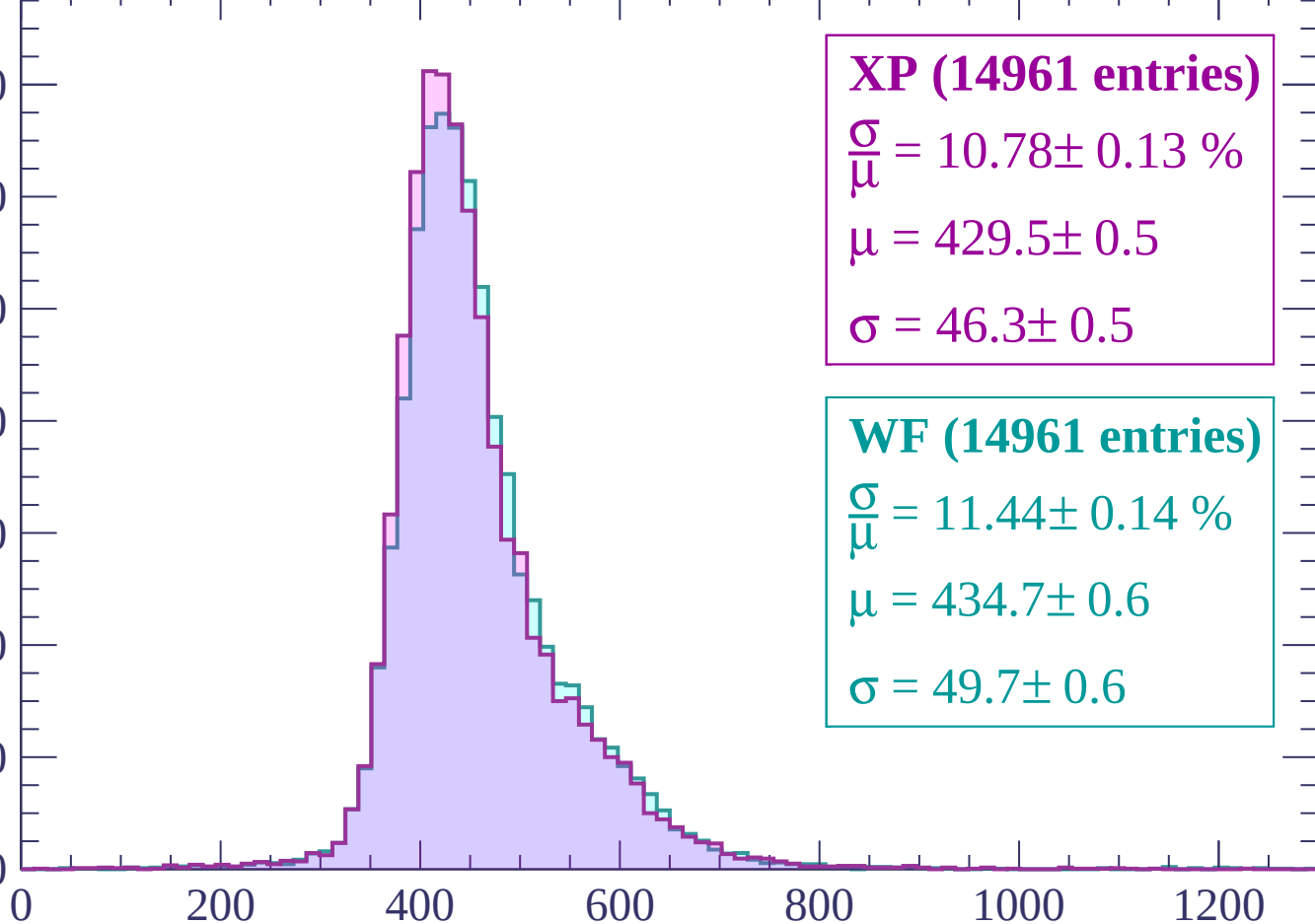
$$\sigma = 46.3 \pm 0.5$$

WF (14961 entries)

$$\frac{\sigma}{\mu} = 11.44 \pm 0.14 \%$$

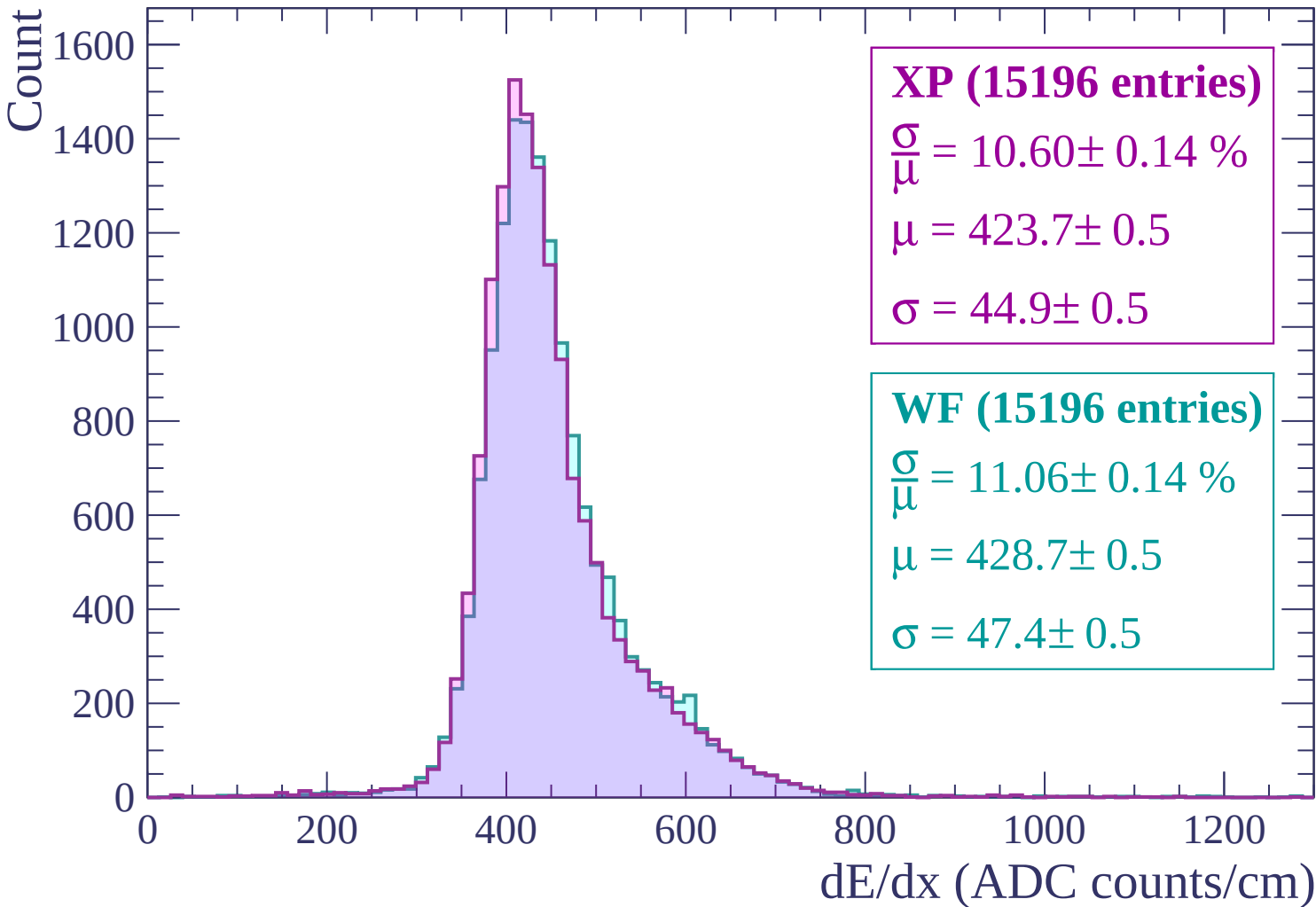
$$\mu = 434.7 \pm 0.6$$

$$\sigma = 49.7 \pm 0.6$$



dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$



Energy loss | $-600 < p < -540$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (15531 entries)

$$\frac{\sigma}{\mu} = 10.17 \pm 0.11 \%$$

$$\mu = 419.1 \pm 0.4$$

$$\sigma = 42.6 \pm 0.4$$

WF (15531 entries)

$$\frac{\sigma}{\mu} = 10.51 \pm 0.13 \%$$

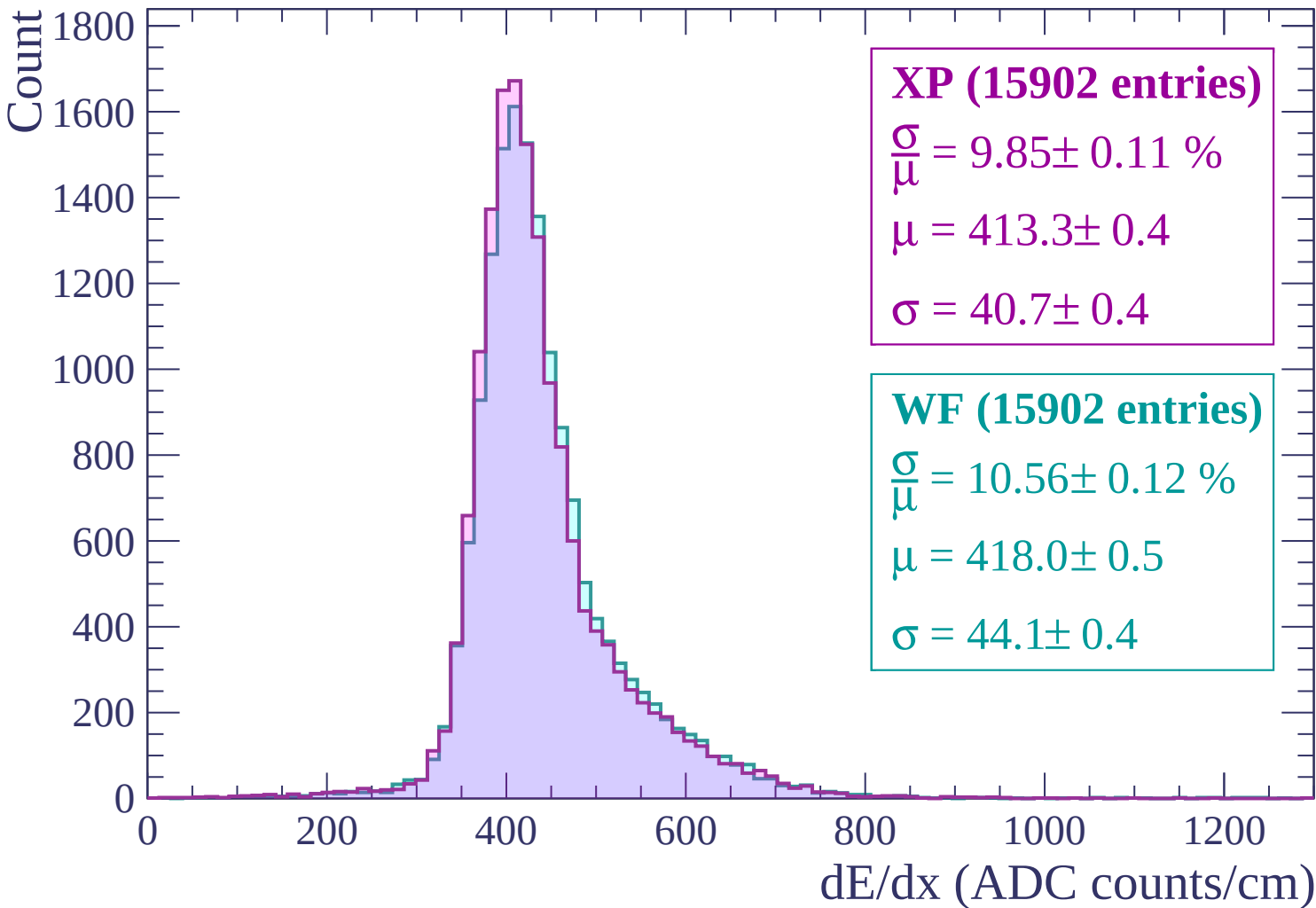
$$\mu = 422.2 \pm 0.5$$

$$\sigma = 44.4 \pm 0.5$$

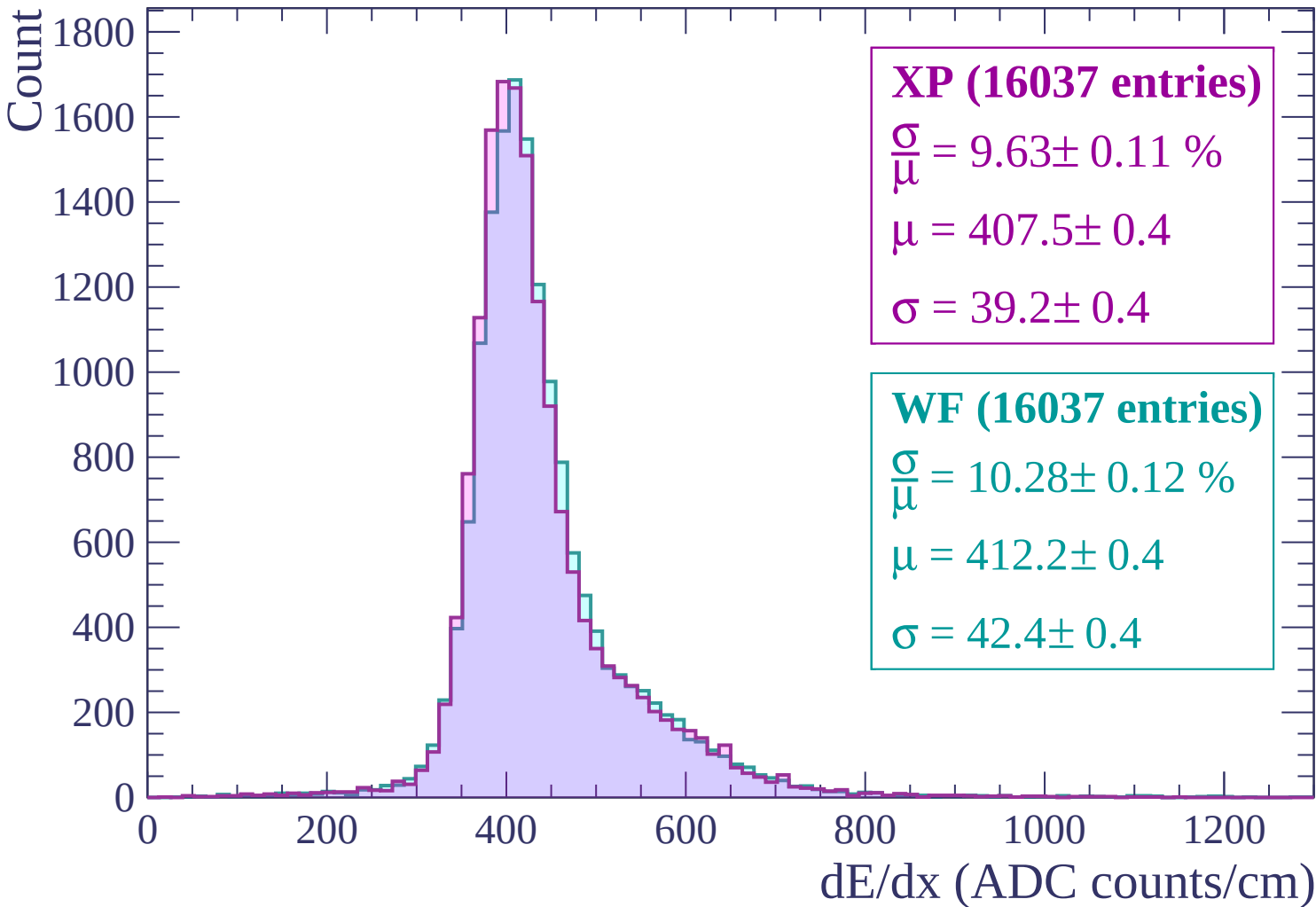
0 200 400 600 800 1000 1200

dE/dx (ADC counts/cm)

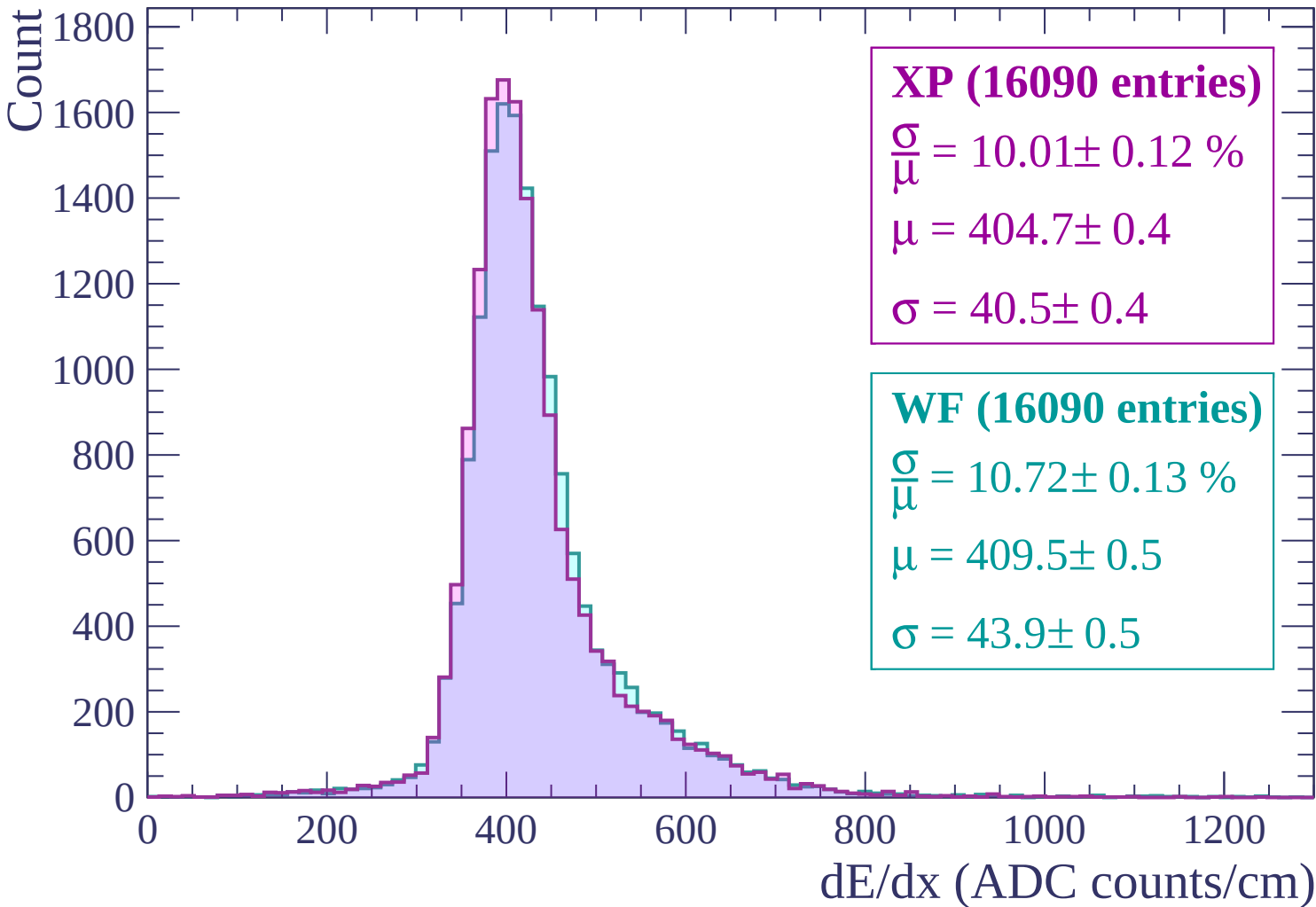
Energy loss | $-540 < p < -480$



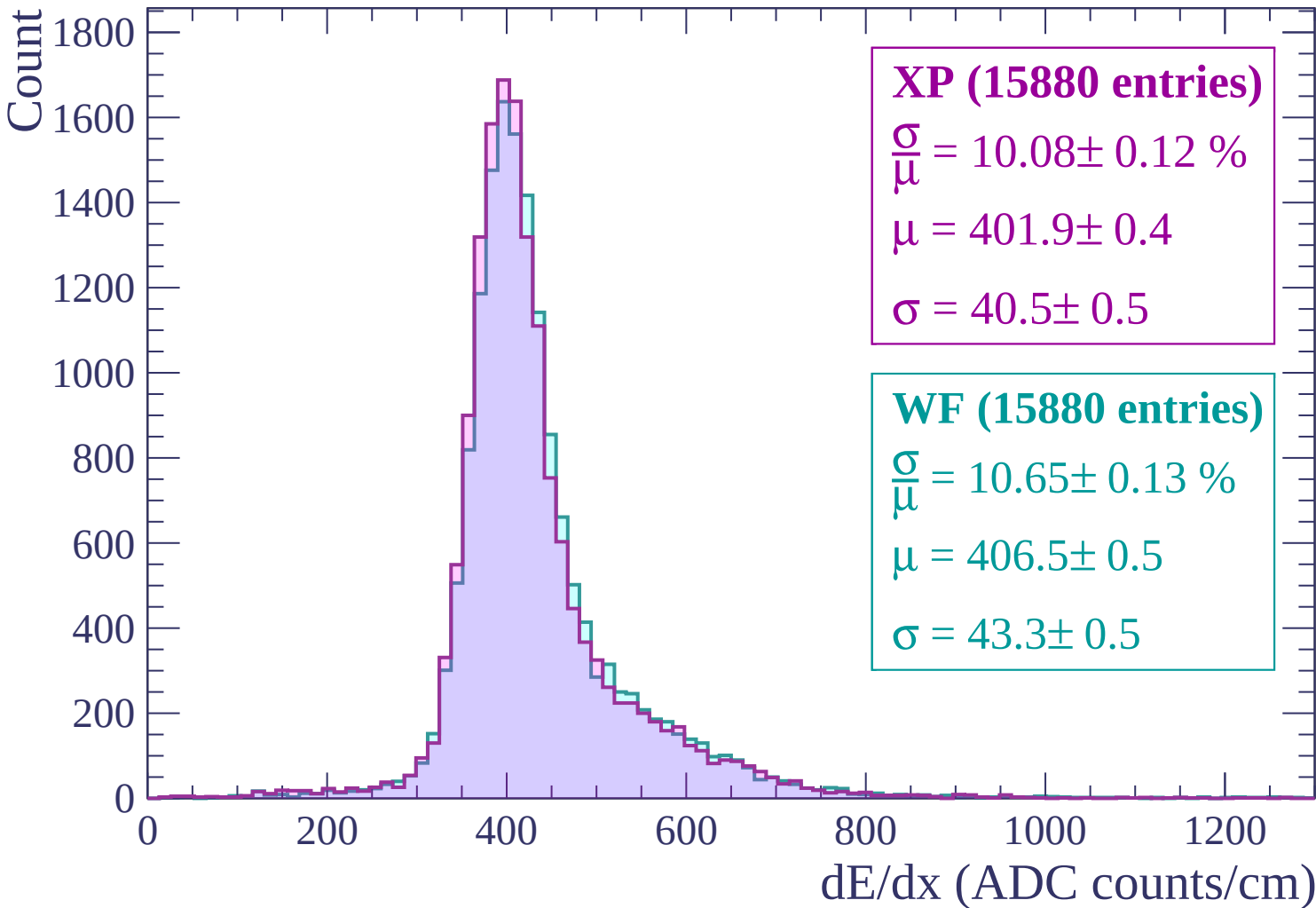
Energy loss | $-480 < p < -420$



Energy loss | $-420 < p < -360$



Energy loss | -360 < p < -300



Energy loss | $-300 < p < -240$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (15335 entries)

$$\frac{\sigma}{\mu} = 10.18 \pm 0.13 \%$$

$$\mu = 403.5 \pm 0.5$$

$$\sigma = 41.1 \pm 0.5$$

WF (15335 entries)

$$\frac{\sigma}{\mu} = 10.64 \pm 0.13 \%$$

$$\mu = 407.2 \pm 0.5$$

$$\sigma = 43.3 \pm 0.5$$

0 200 400 600 800 1000 1200

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (13431 entries)

$\frac{\sigma}{\mu} = 11.25 \pm 0.15 \%$

$\mu = 414.4 \pm 0.5$

$\sigma = 46.6 \pm 0.6$

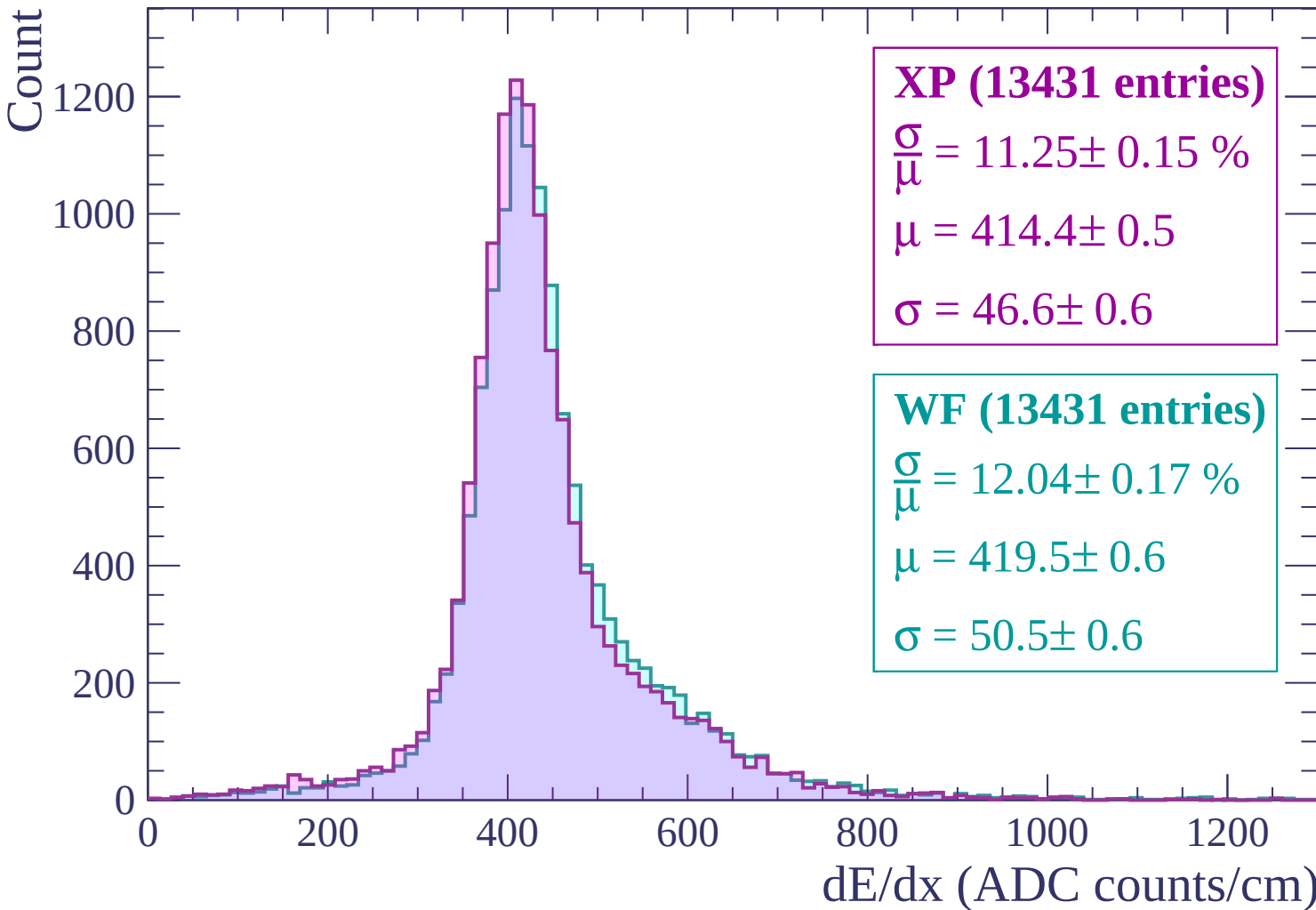
WF (13431 entries)

$\frac{\sigma}{\mu} = 12.04 \pm 0.17 \%$

$\mu = 419.5 \pm 0.6$

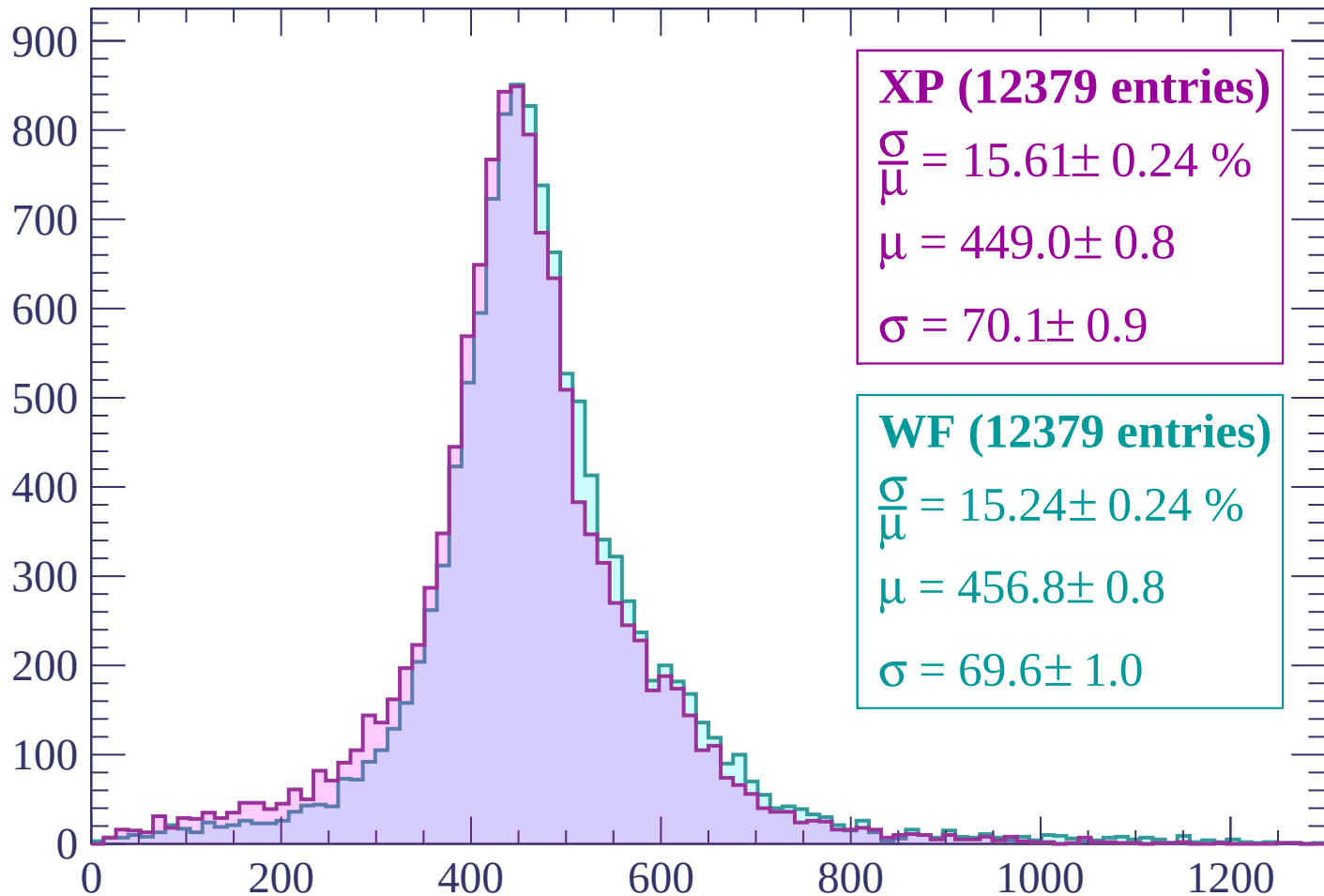
$\sigma = 50.5 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | -180 < p < -120

Count



XP (12379 entries)

$$\frac{\sigma}{\mu} = 15.61 \pm 0.24 \%$$

$$\mu = 449.0 \pm 0.8$$

$$\sigma = 70.1 \pm 0.9$$

WF (12379 entries)

$$\frac{\sigma}{\mu} = 15.24 \pm 0.24 \%$$

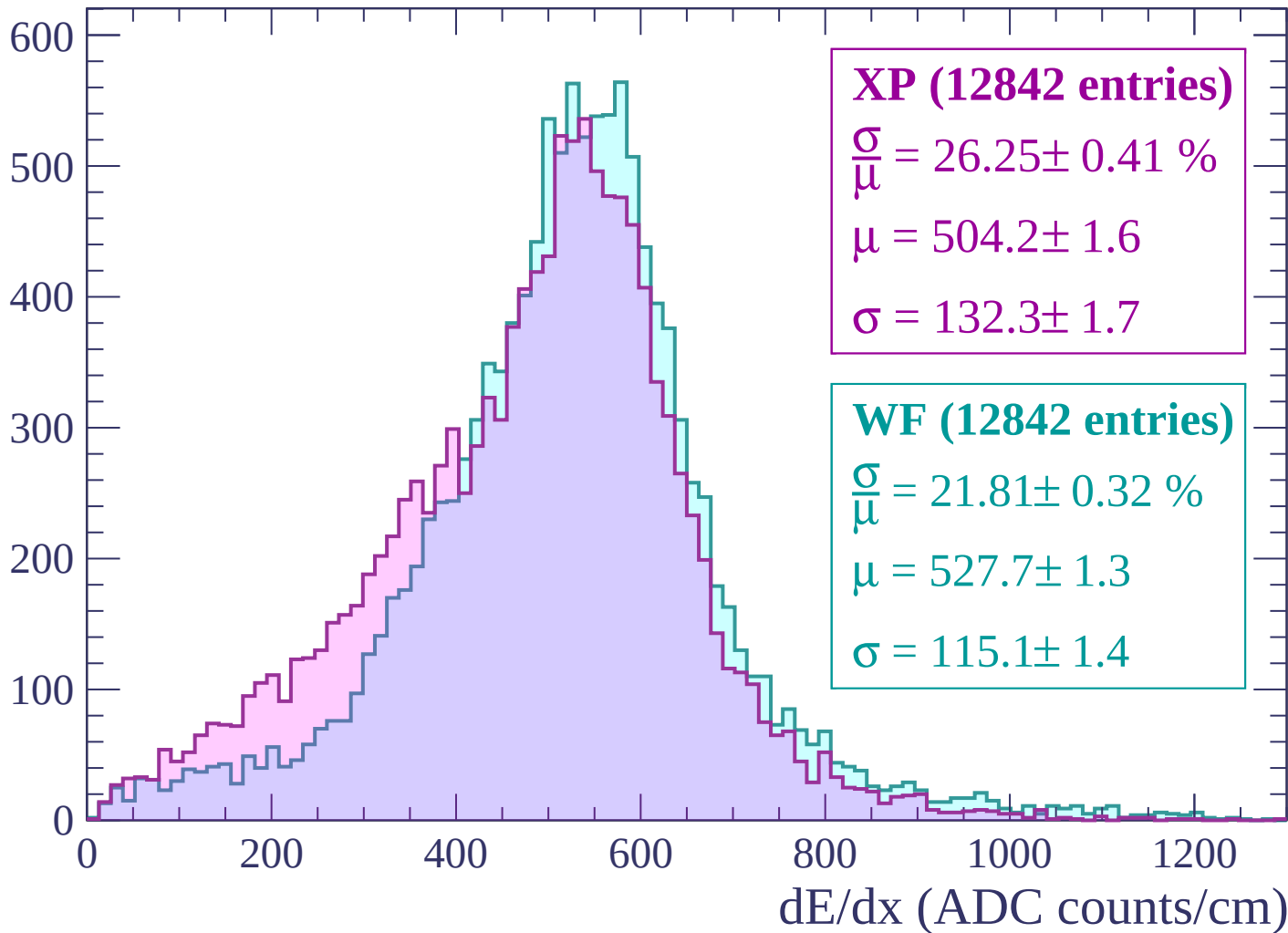
$$\mu = 456.8 \pm 0.8$$

$$\sigma = 69.6 \pm 1.0$$

dE/dx (ADC counts/cm)

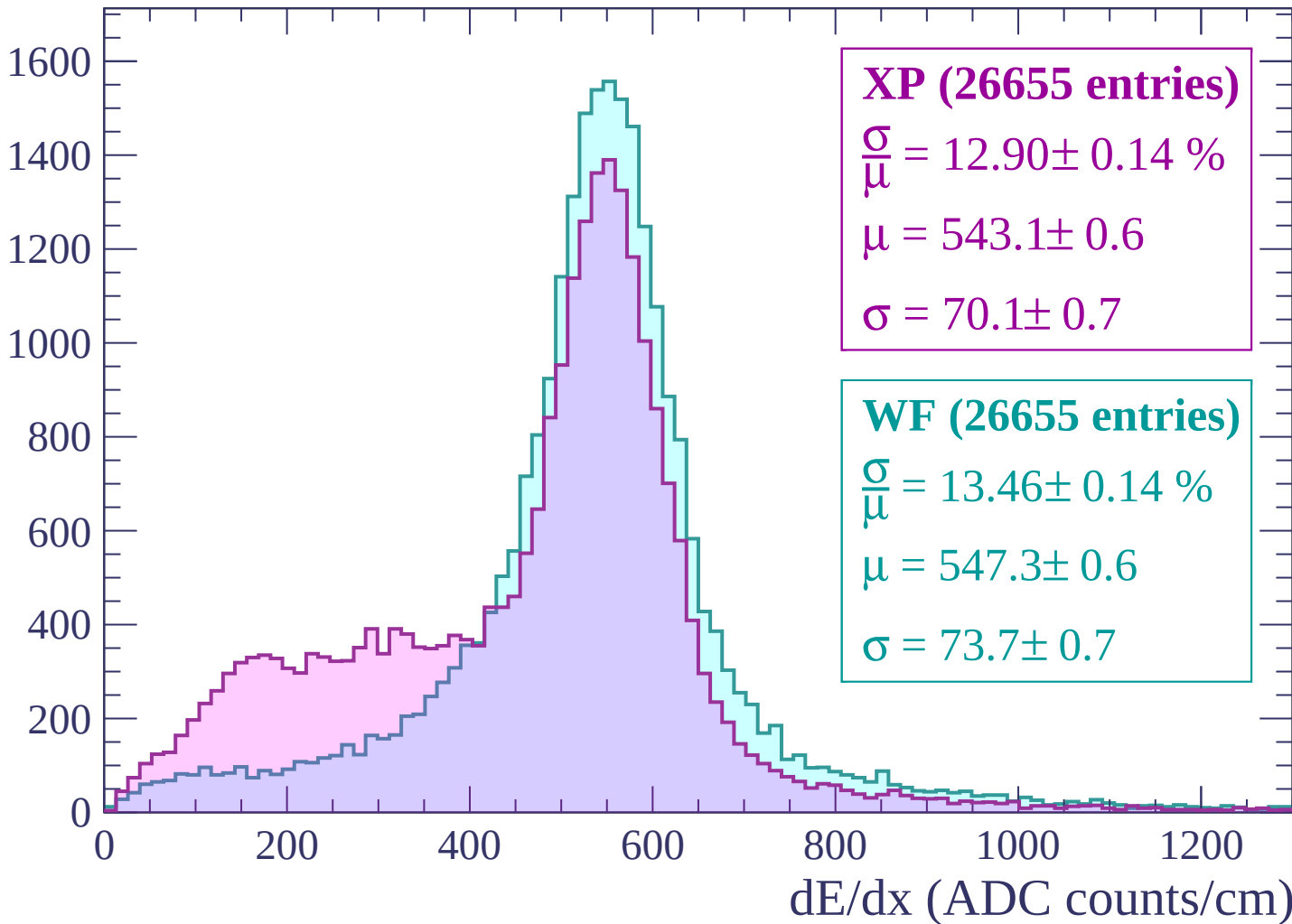
Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

Count



Energy loss | $0 < p < 60$

Count

4000
3500
3000
2500
2000
1500
1000
500
0

XP (68192 entries)

$\frac{\sigma}{\mu} = 18.23 \pm 0.13 \%$

$\mu = 484.3 \pm 0.5$

$\sigma = 88.3 \pm 0.5$

WF (68192 entries)

$\frac{\sigma}{\mu} = 15.85 \pm 0.10 \%$

$\mu = 500.3 \pm 0.4$

$\sigma = 79.3 \pm 0.4$

0

200

400

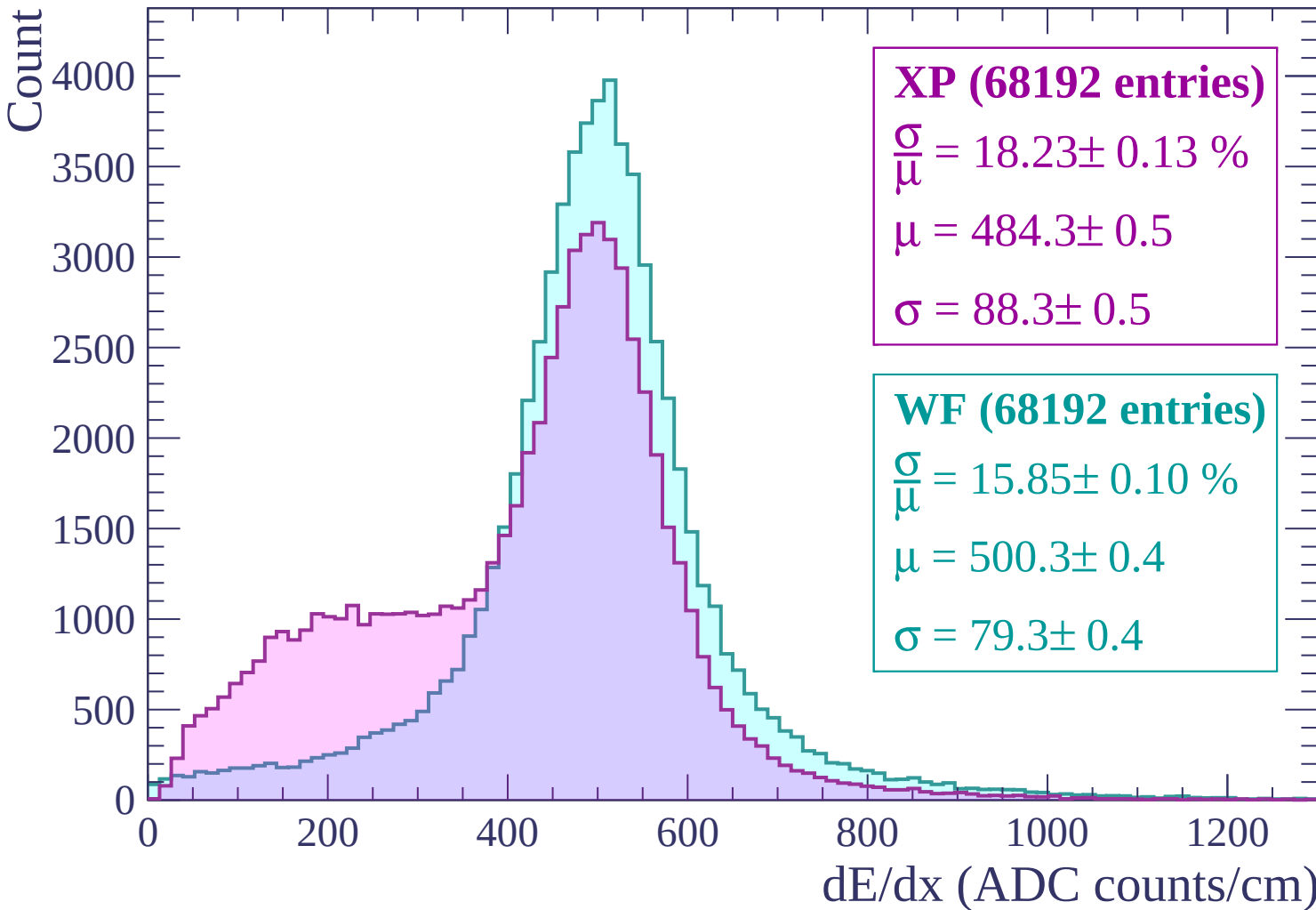
600

800

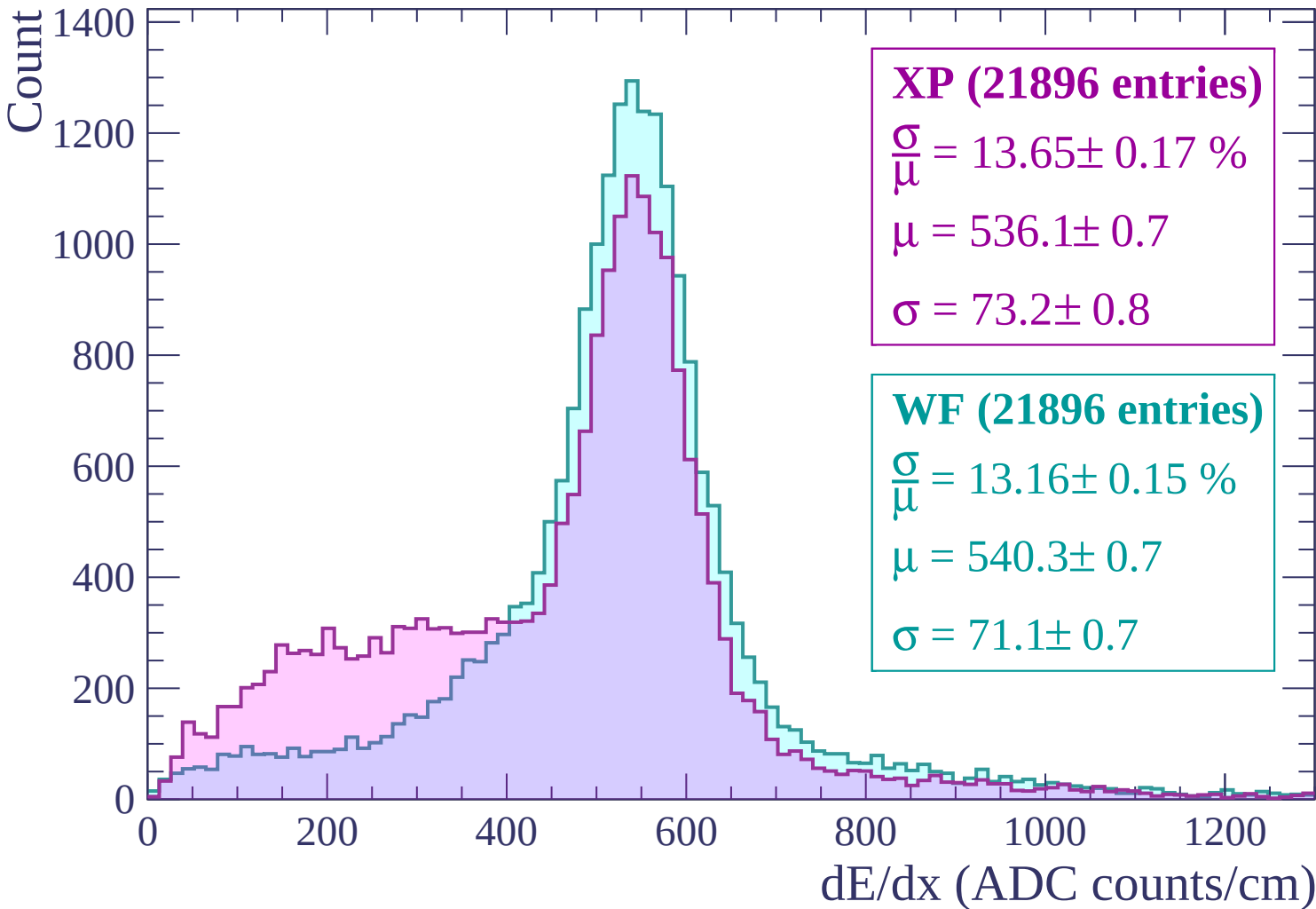
1000

1200

dE/dx (ADC counts/cm)

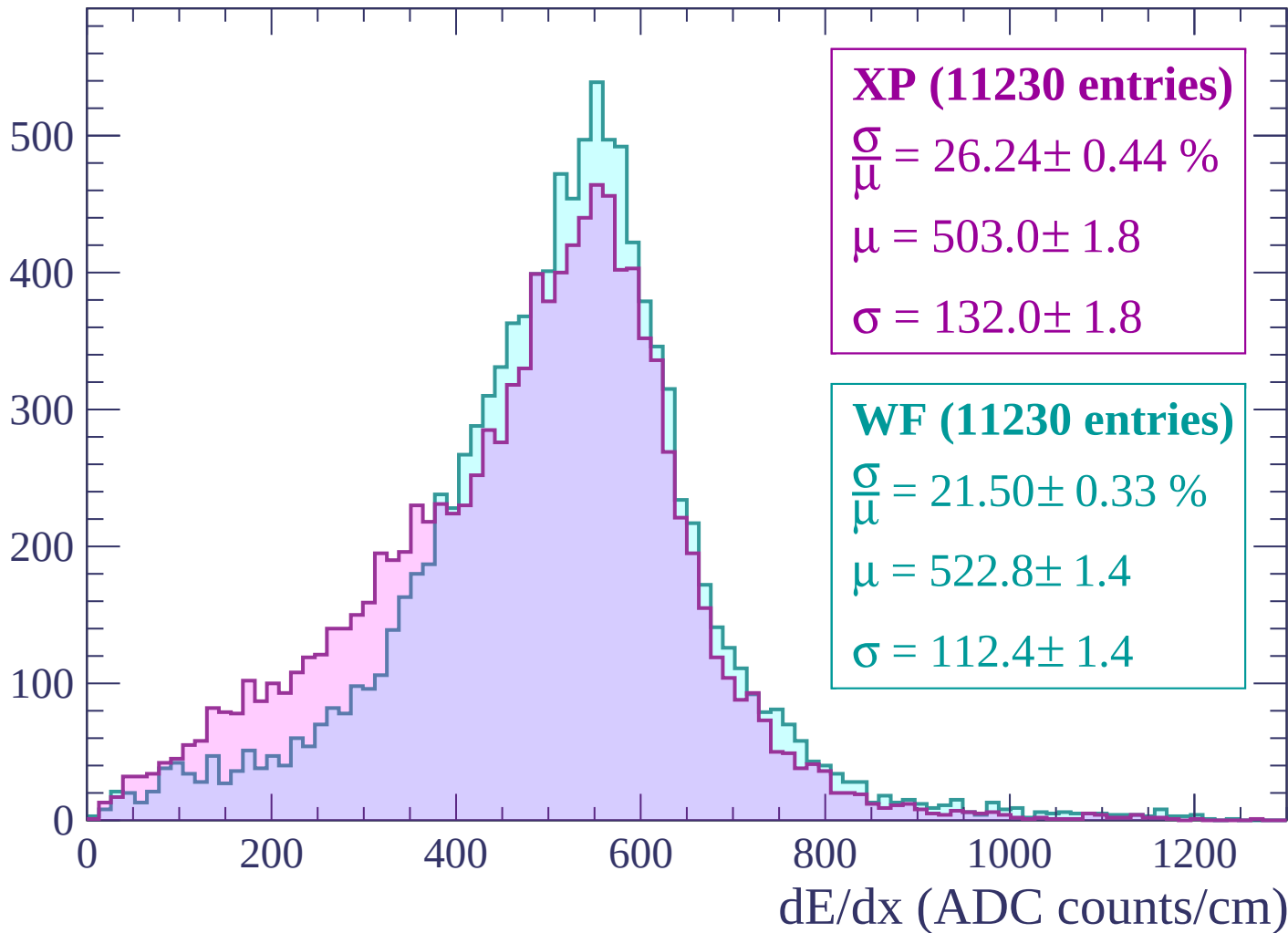


Energy loss | $60 < p < 120$



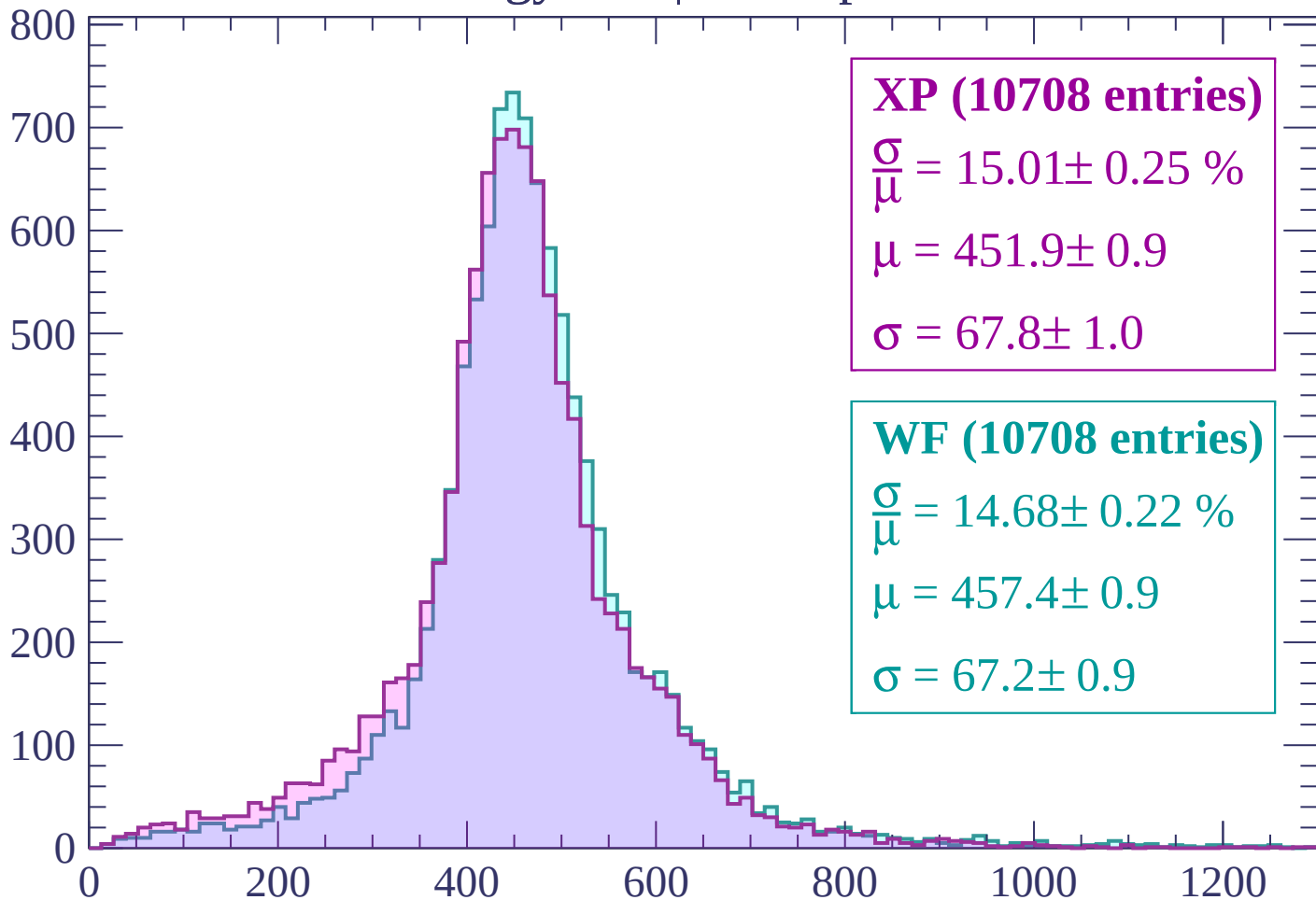
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (10708 entries)

$$\frac{\sigma}{\mu} = 15.01 \pm 0.25 \%$$

$$\mu = 451.9 \pm 0.9$$

$$\sigma = 67.8 \pm 1.0$$

WF (10708 entries)

$$\frac{\sigma}{\mu} = 14.68 \pm 0.22 \%$$

$$\mu = 457.4 \pm 0.9$$

$$\sigma = 67.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (11697 entries)

$\frac{\sigma}{\mu} = 11.66 \pm 0.16 \%$

$\mu = 413.7 \pm 0.6$

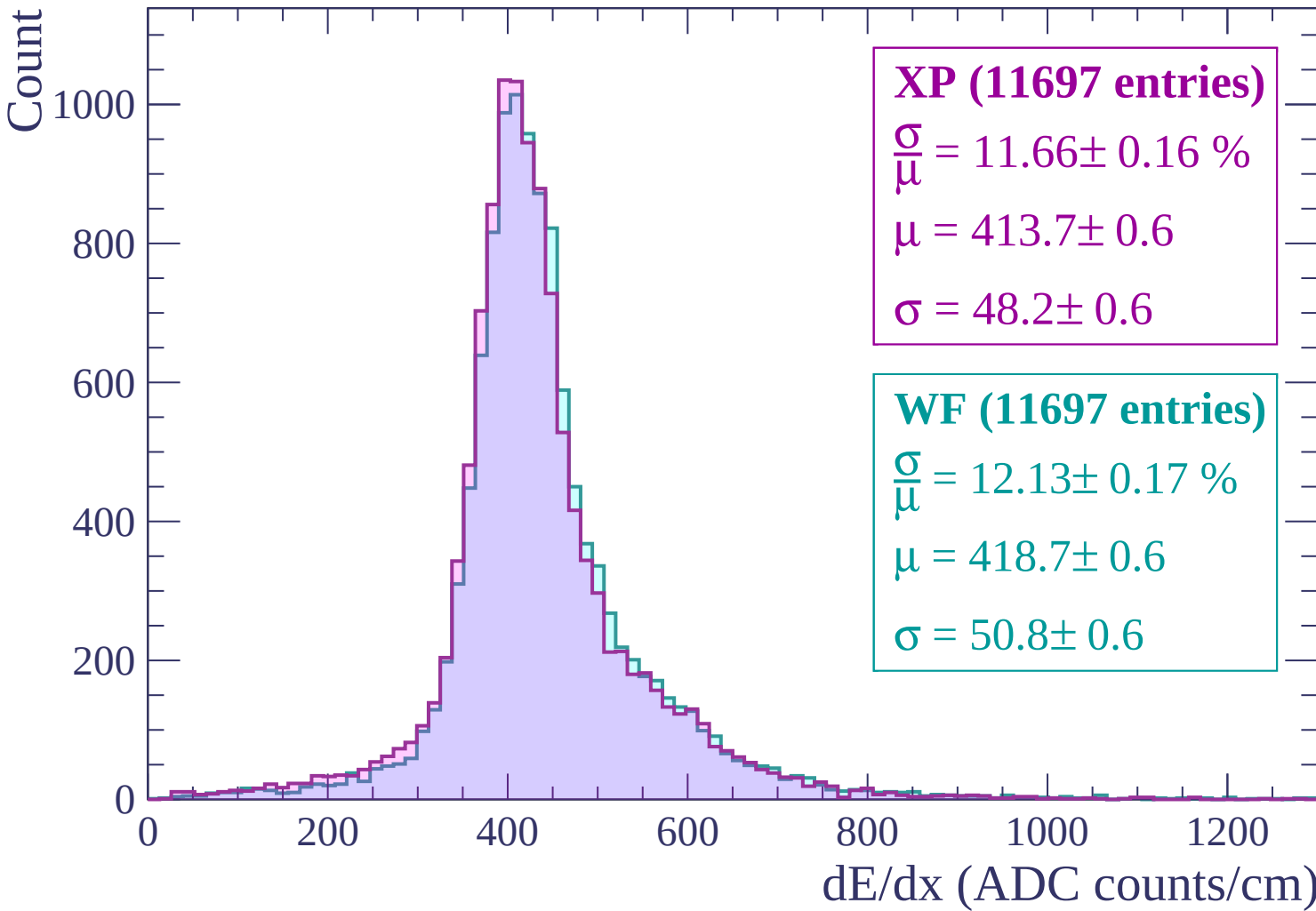
$\sigma = 48.2 \pm 0.6$

WF (11697 entries)

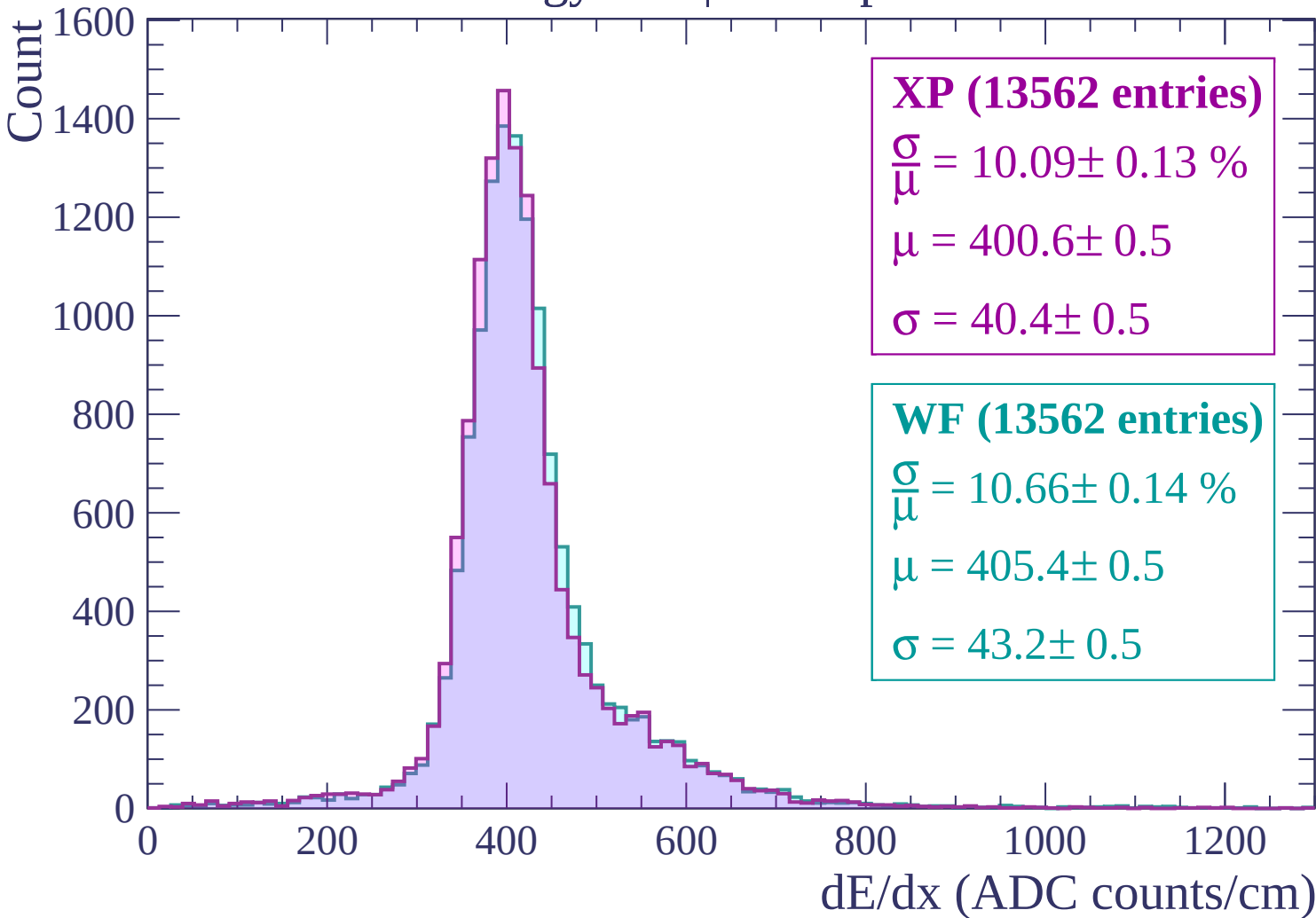
$\frac{\sigma}{\mu} = 12.13 \pm 0.17 \%$

$\mu = 418.7 \pm 0.6$

$\sigma = 50.8 \pm 0.6$



Energy loss | $300 < p < 360$



Energy loss | $360 < p < 420$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (13906 entries)

$$\frac{\sigma}{\mu} = 9.53 \pm 0.13 \%$$

$$\mu = 396.5 \pm 0.4$$

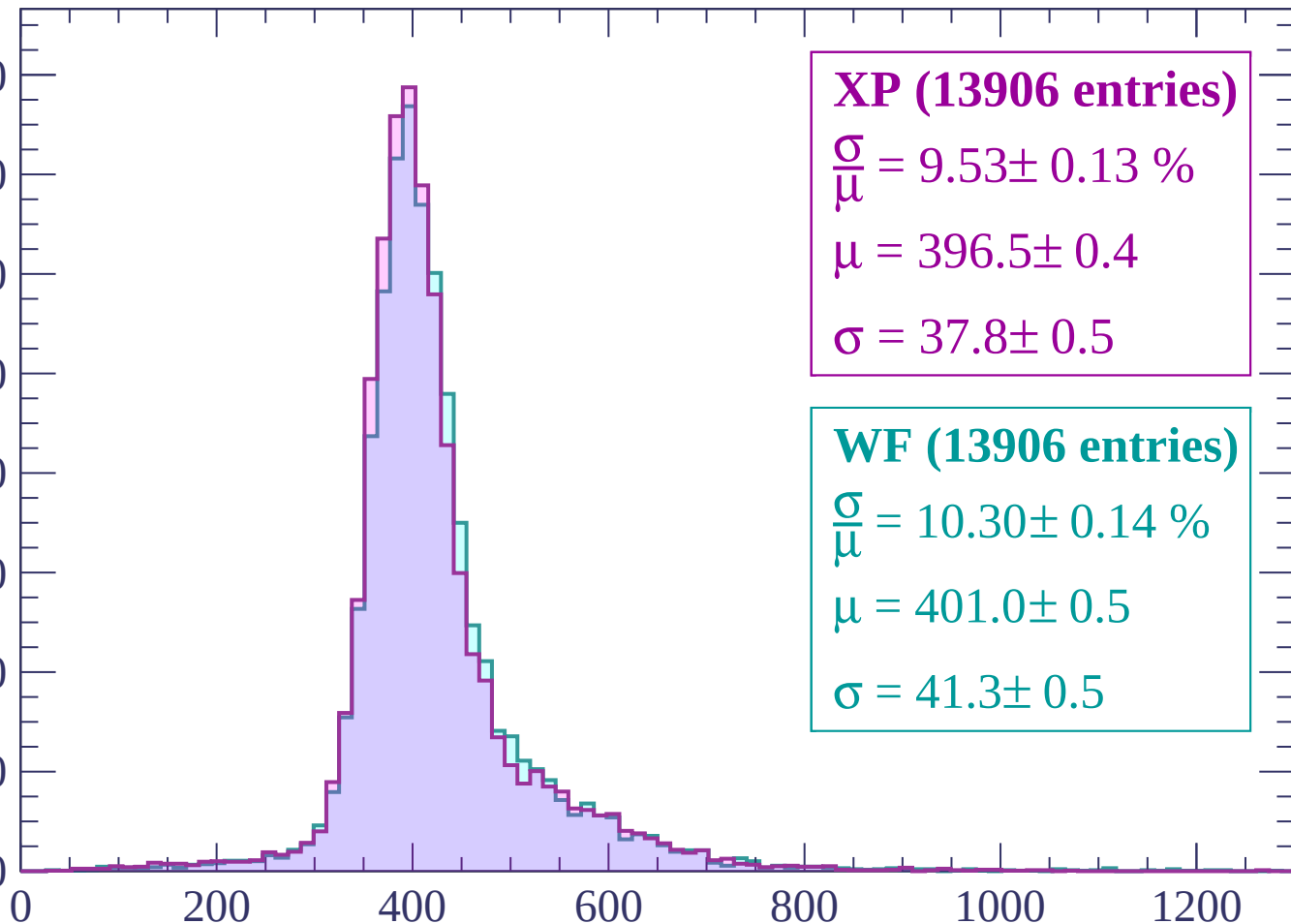
$$\sigma = 37.8 \pm 0.5$$

WF (13906 entries)

$$\frac{\sigma}{\mu} = 10.30 \pm 0.14 \%$$

$$\mu = 401.0 \pm 0.5$$

$$\sigma = 41.3 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (13486 entries)

$$\frac{\sigma}{\mu} = 9.79 \pm 0.12 \%$$

$$\mu = 399.5 \pm 0.4$$

$$\sigma = 39.1 \pm 0.4$$

WF (13486 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.13 \%$$

$$\mu = 403.4 \pm 0.5$$

$$\sigma = 40.3 \pm 0.5$$

0

200

400

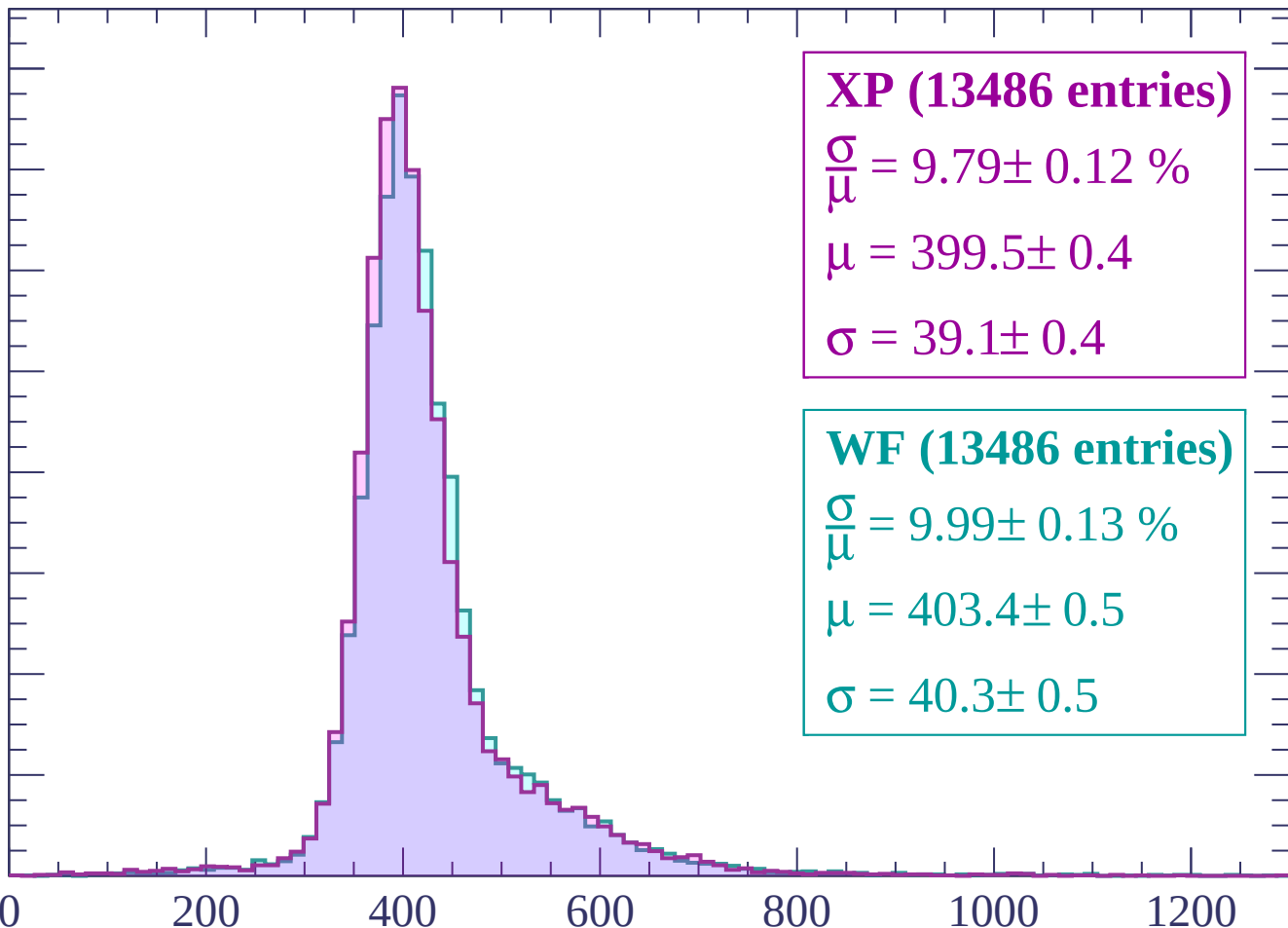
600

800

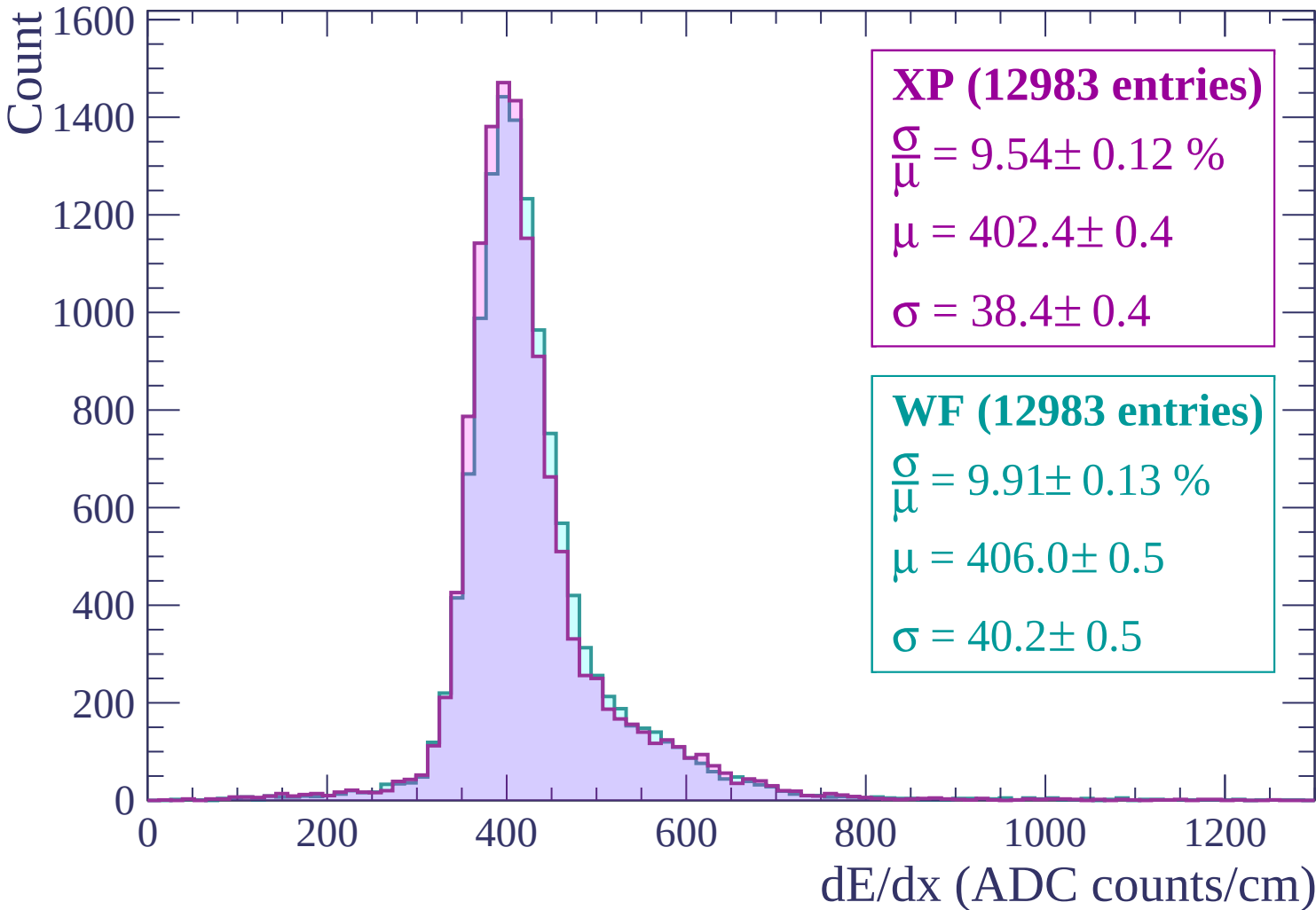
1000

1200

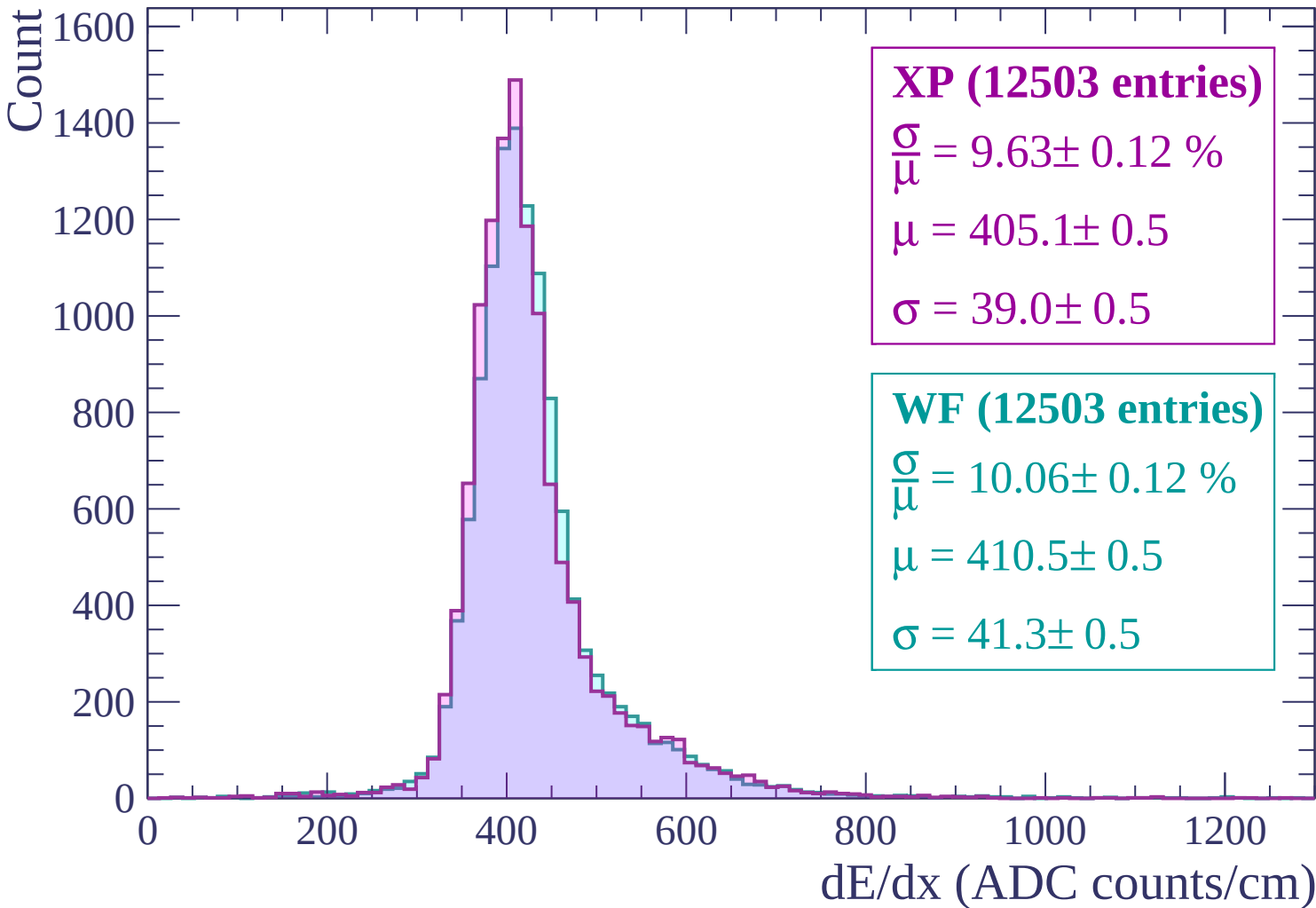
dE/dx (ADC counts/cm)



Energy loss | $480 < p < 540$



Energy loss | $540 < p < 600$



Energy loss | $600 < p < 660$

Count

1400
1200
1000
800
600
400
200
0

XP (11967 entries)

$$\frac{\sigma}{\mu} = 9.52 \pm 0.14 \%$$

$$\mu = 408.1 \pm 0.5$$

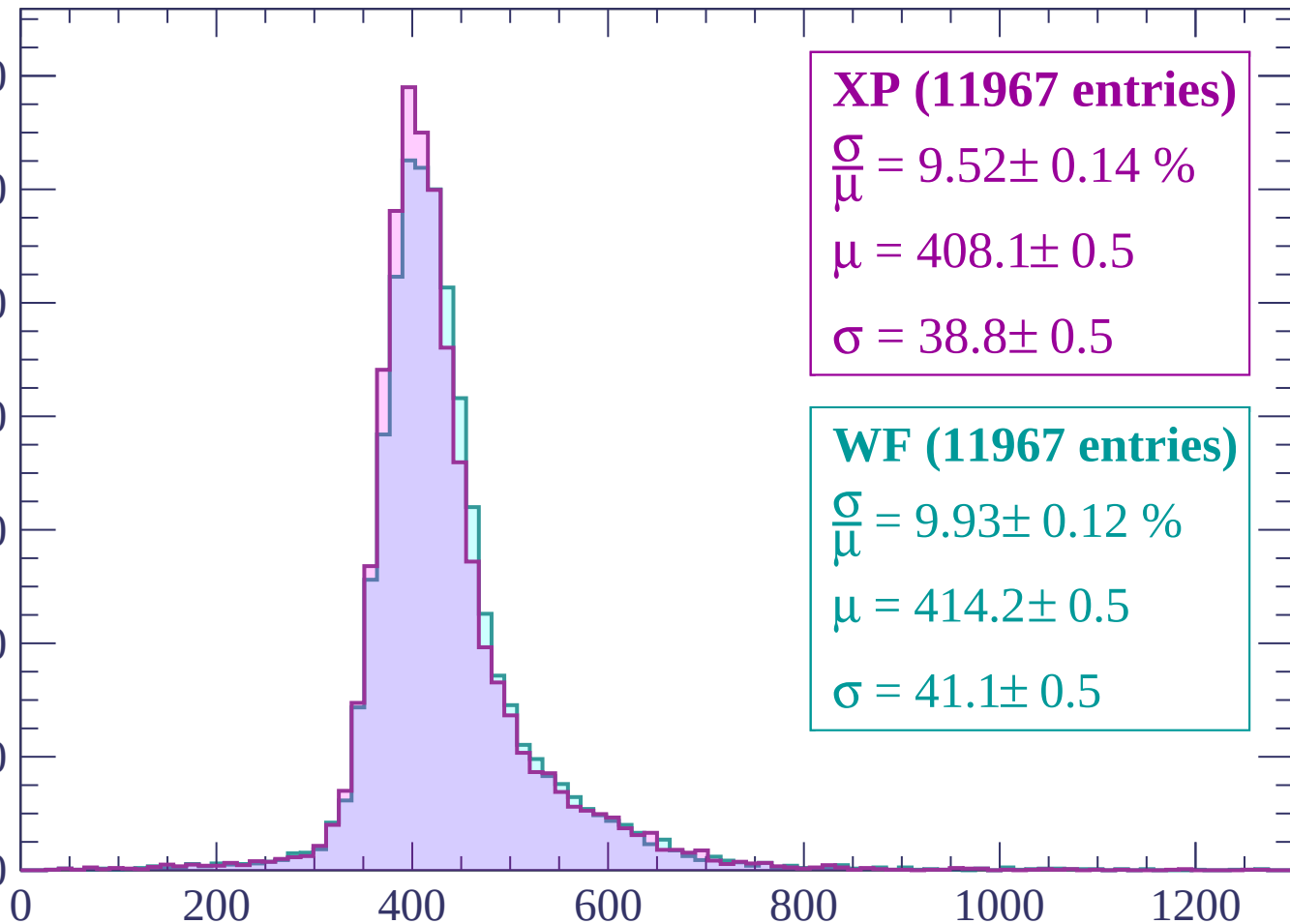
$$\sigma = 38.8 \pm 0.5$$

WF (11967 entries)

$$\frac{\sigma}{\mu} = 9.93 \pm 0.12 \%$$

$$\mu = 414.2 \pm 0.5$$

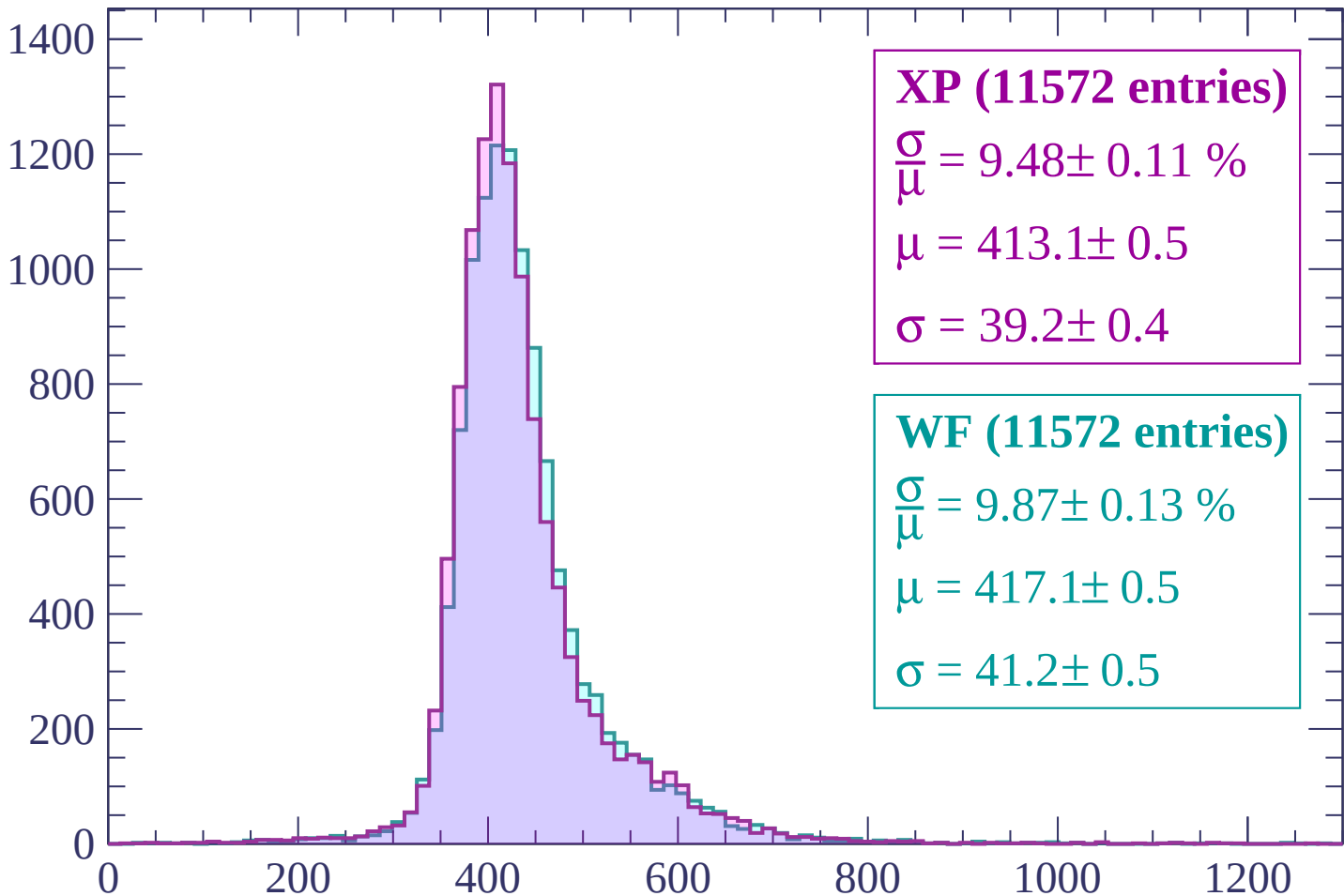
$$\sigma = 41.1 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (11572 entries)

$$\frac{\sigma}{\mu} = 9.48 \pm 0.11 \%$$

$$\mu = 413.1 \pm 0.5$$

$$\sigma = 39.2 \pm 0.4$$

WF (11572 entries)

$$\frac{\sigma}{\mu} = 9.87 \pm 0.13 \%$$

$$\mu = 417.1 \pm 0.5$$

$$\sigma = 41.2 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count

1200

1000

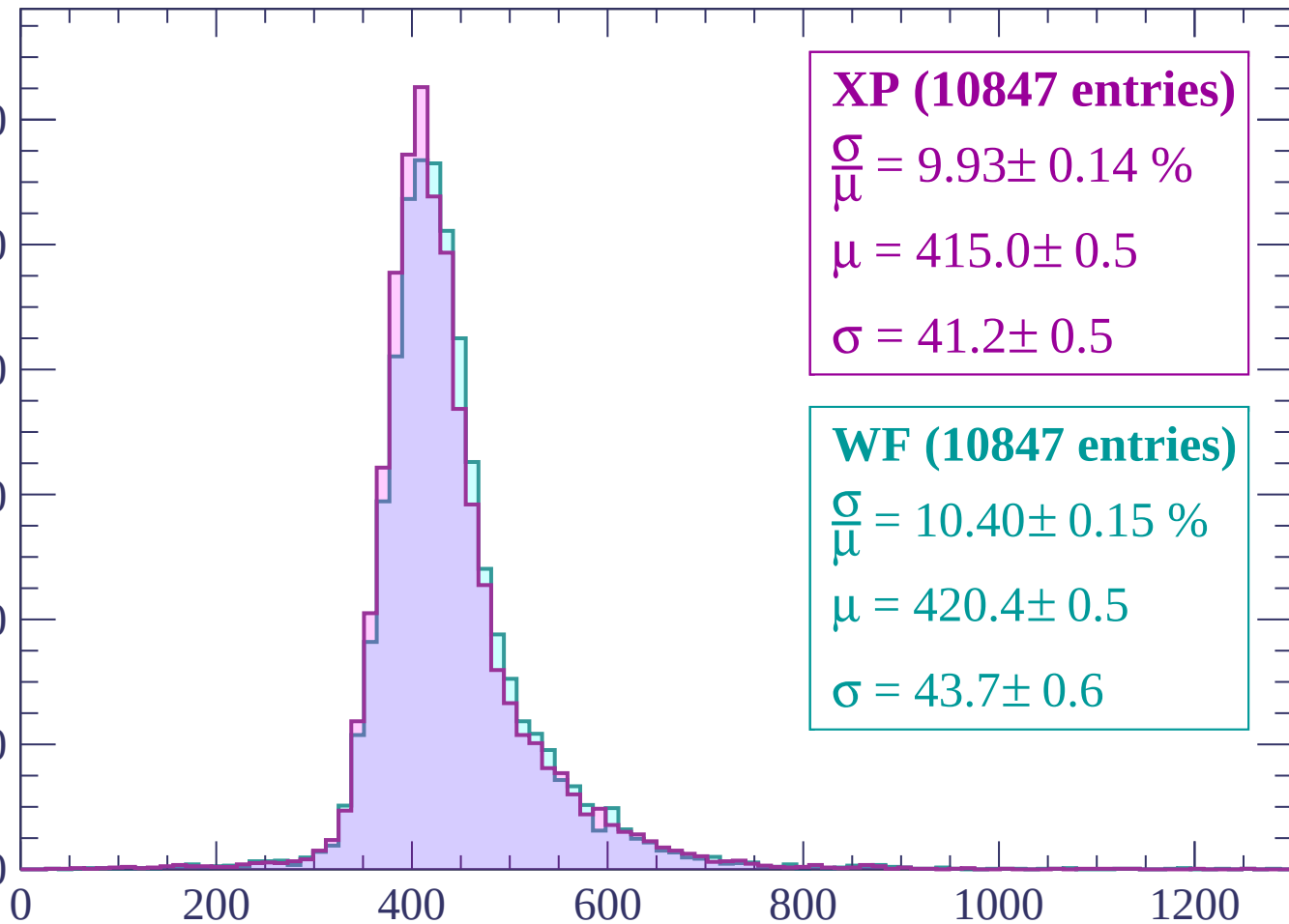
800

600

400

200

0



XP (10847 entries)

$\frac{\sigma}{\mu} = 9.93 \pm 0.14 \%$

$\mu = 415.0 \pm 0.5$

$\sigma = 41.2 \pm 0.5$

WF (10847 entries)

$\frac{\sigma}{\mu} = 10.40 \pm 0.15 \%$

$\mu = 420.4 \pm 0.5$

$\sigma = 43.7 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (10449 entries)

$\frac{\sigma}{\mu} = 9.42 \pm 0.13 \%$

$\mu = 419.0 \pm 0.5$

$\sigma = 39.5 \pm 0.5$

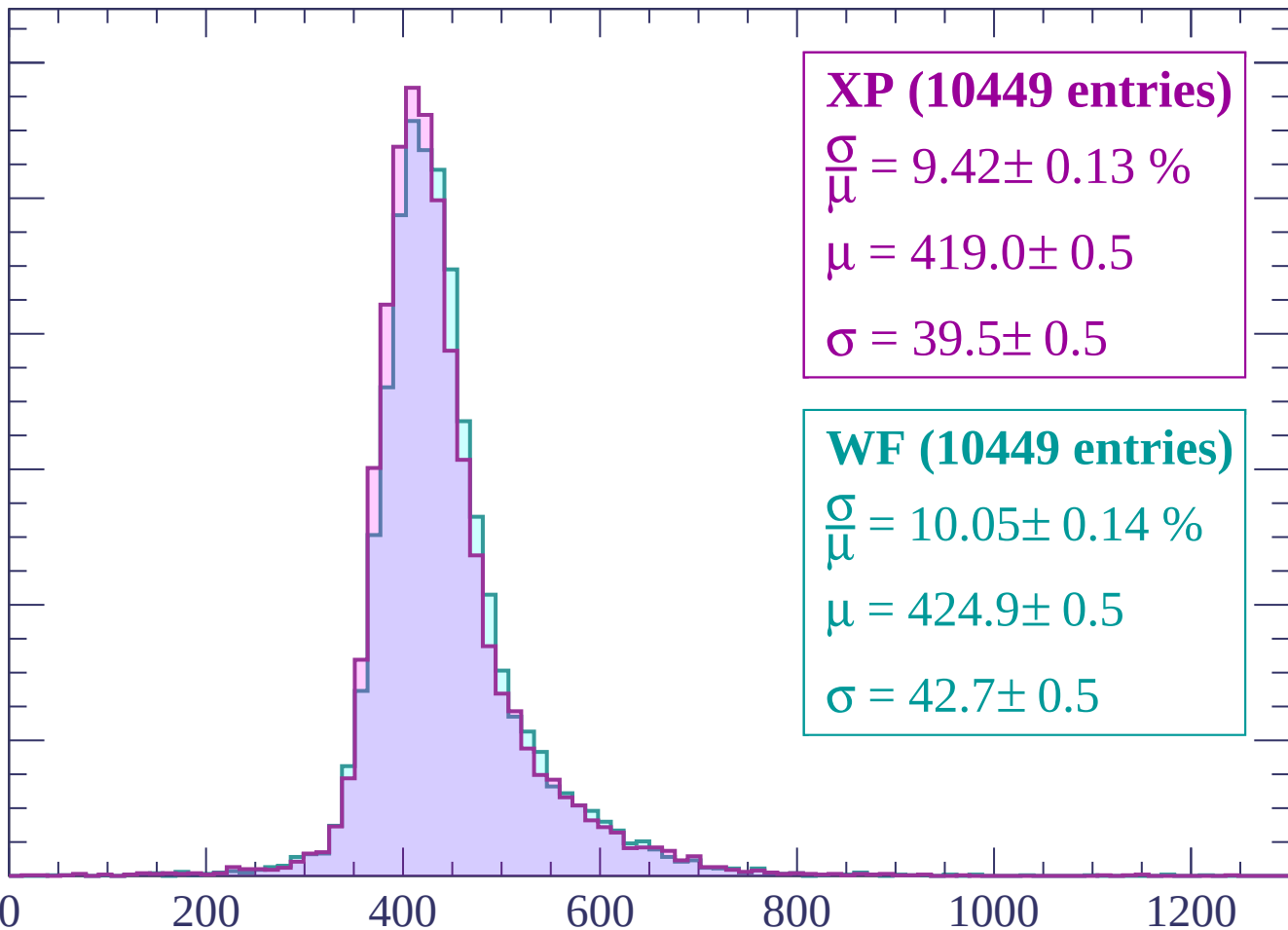
WF (10449 entries)

$\frac{\sigma}{\mu} = 10.05 \pm 0.14 \%$

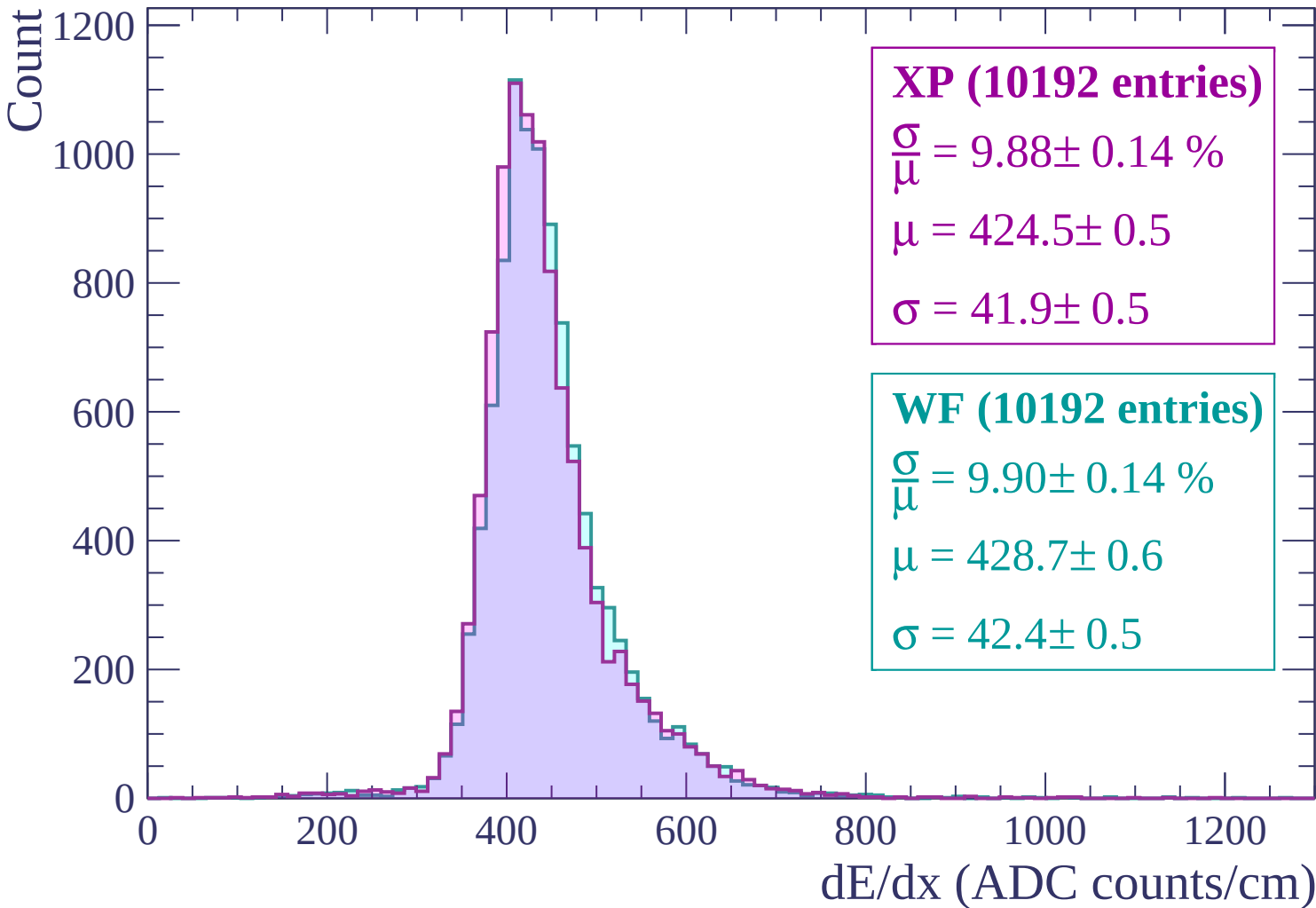
$\mu = 424.9 \pm 0.5$

$\sigma = 42.7 \pm 0.5$

dE/dx (ADC counts/cm)



Energy loss | $840 < p < 900$



Energy loss | $900 < p < 960$

Count

1200
1000
800
600
400
200
0

XP (9861 entries)

$$\frac{\sigma}{\mu} = 9.87 \pm 0.16 \%$$

$$\mu = 426.8 \pm 0.6$$

$$\sigma = 42.1 \pm 0.6$$

WF (9861 entries)

$$\frac{\sigma}{\mu} = 10.29 \pm 0.14 \%$$

$$\mu = 433.7 \pm 0.6$$

$$\sigma = 44.6 \pm 0.5$$

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (9367 entries)

$$\frac{\sigma}{\mu} = 9.39 \pm 0.13 \%$$

$$\mu = 429.4 \pm 0.6$$

$$\sigma = 40.3 \pm 0.5$$

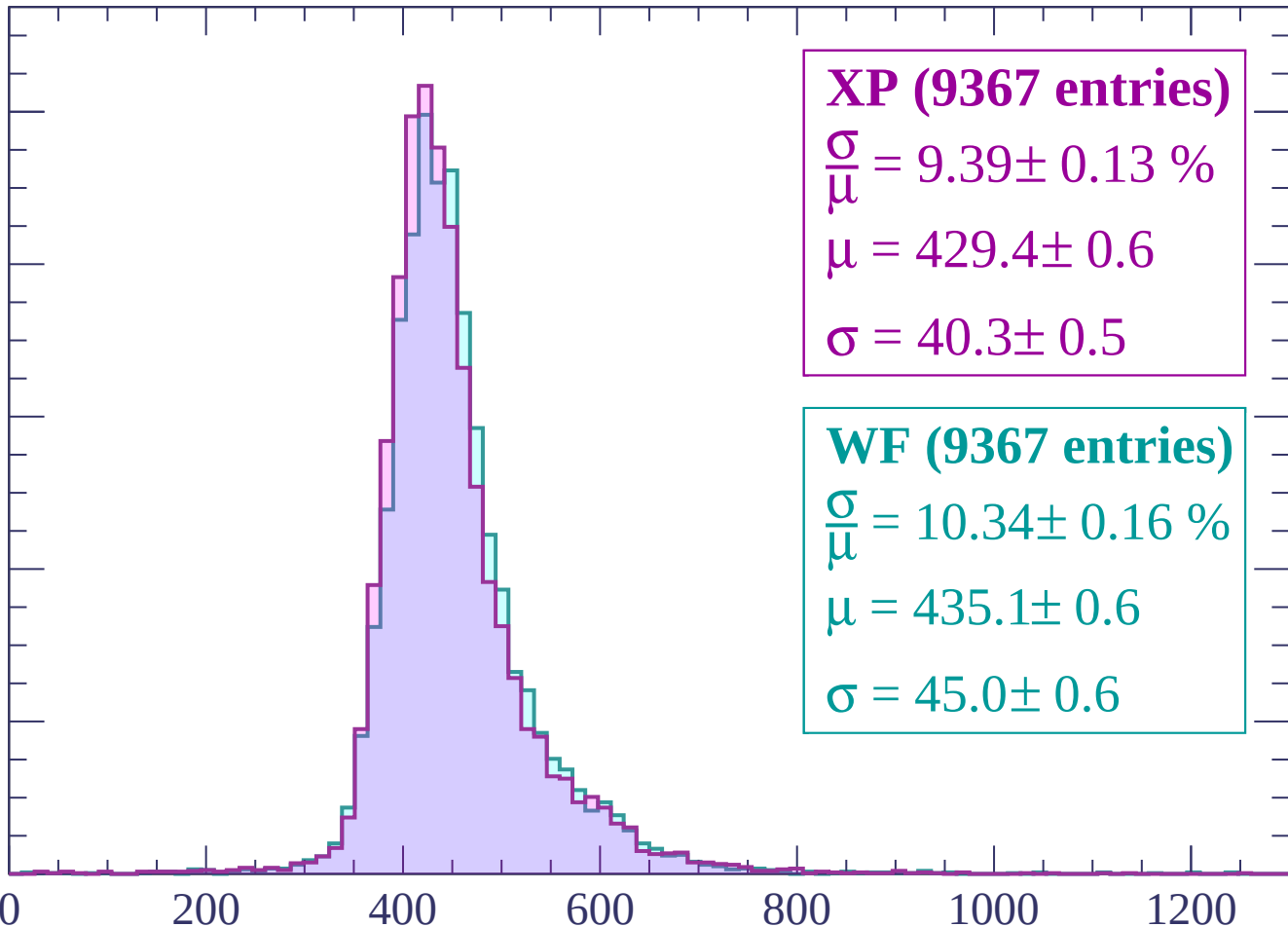
WF (9367 entries)

$$\frac{\sigma}{\mu} = 10.34 \pm 0.16 \%$$

$$\mu = 435.1 \pm 0.6$$

$$\sigma = 45.0 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss | $1020 < p < 1080$

Count

1000

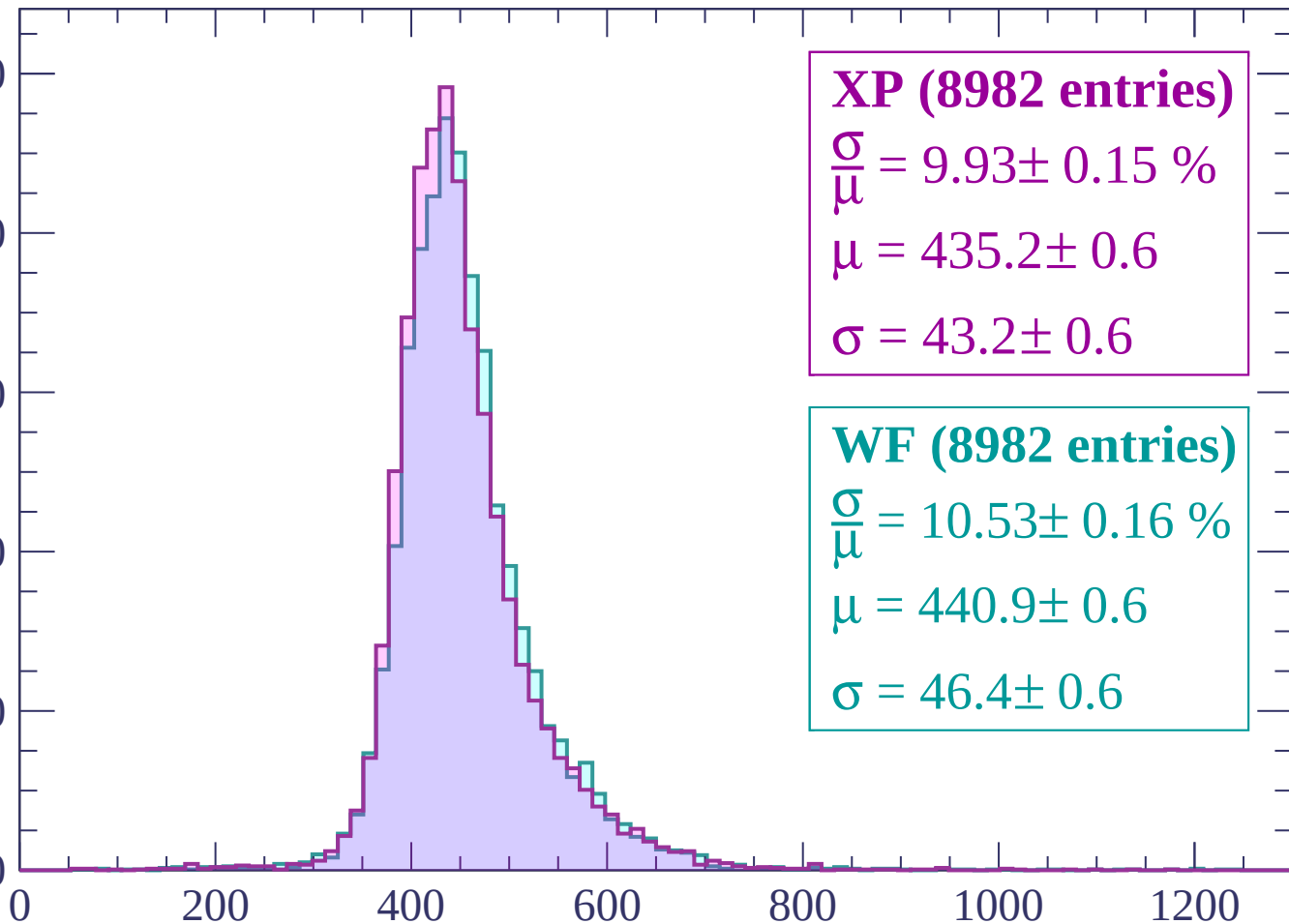
800

600

400

200

0



XP (8982 entries)

$\frac{\sigma}{\mu} = 9.93 \pm 0.15 \%$

$\mu = 435.2 \pm 0.6$

$\sigma = 43.2 \pm 0.6$

WF (8982 entries)

$\frac{\sigma}{\mu} = 10.53 \pm 0.16 \%$

$\mu = 440.9 \pm 0.6$

$\sigma = 46.4 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (8784 entries)

$\frac{\sigma}{\mu} = 9.92 \pm 0.15 \%$

$\mu = 436.7 \pm 0.6$

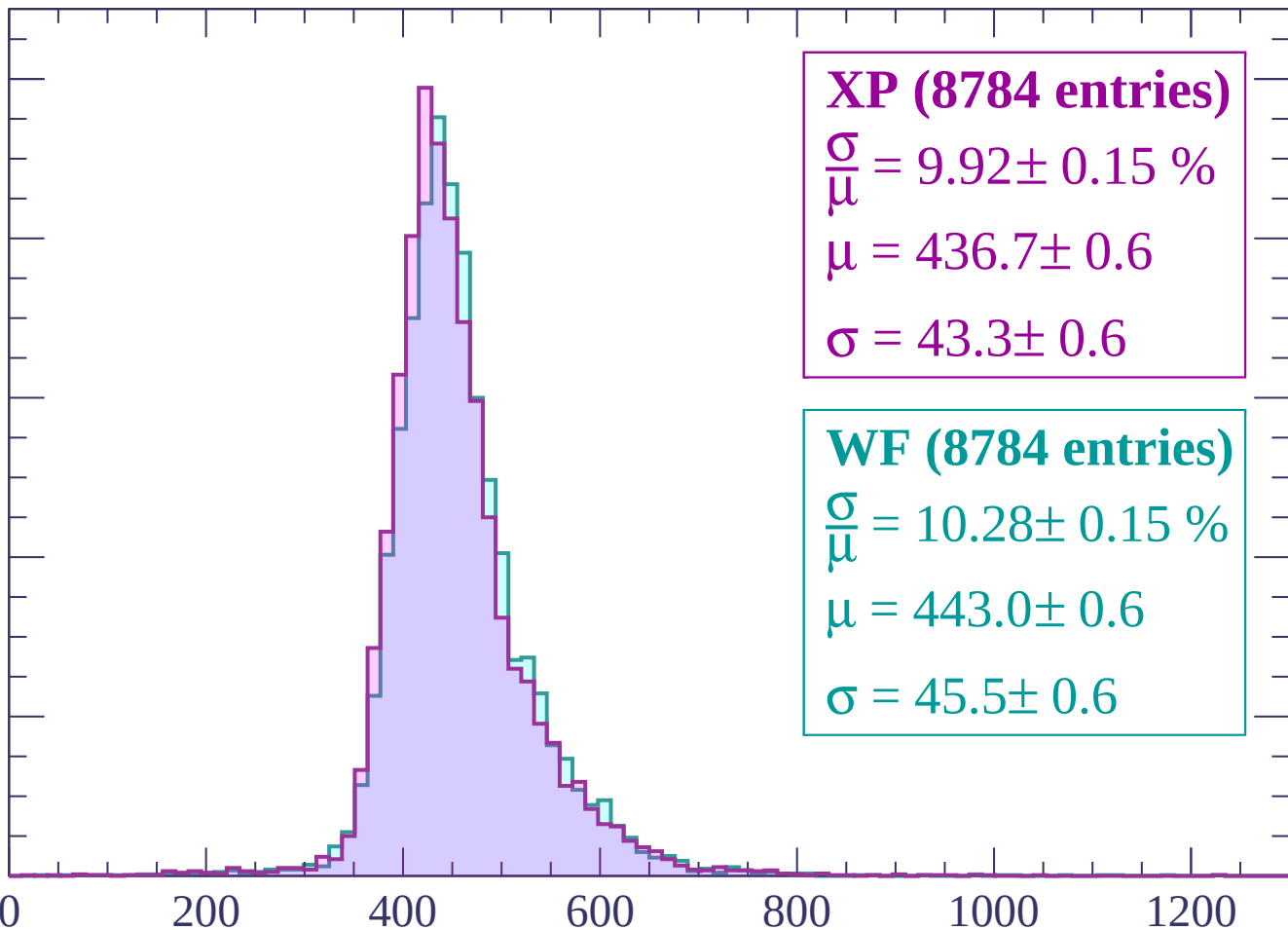
$\sigma = 43.3 \pm 0.6$

WF (8784 entries)

$\frac{\sigma}{\mu} = 10.28 \pm 0.15 \%$

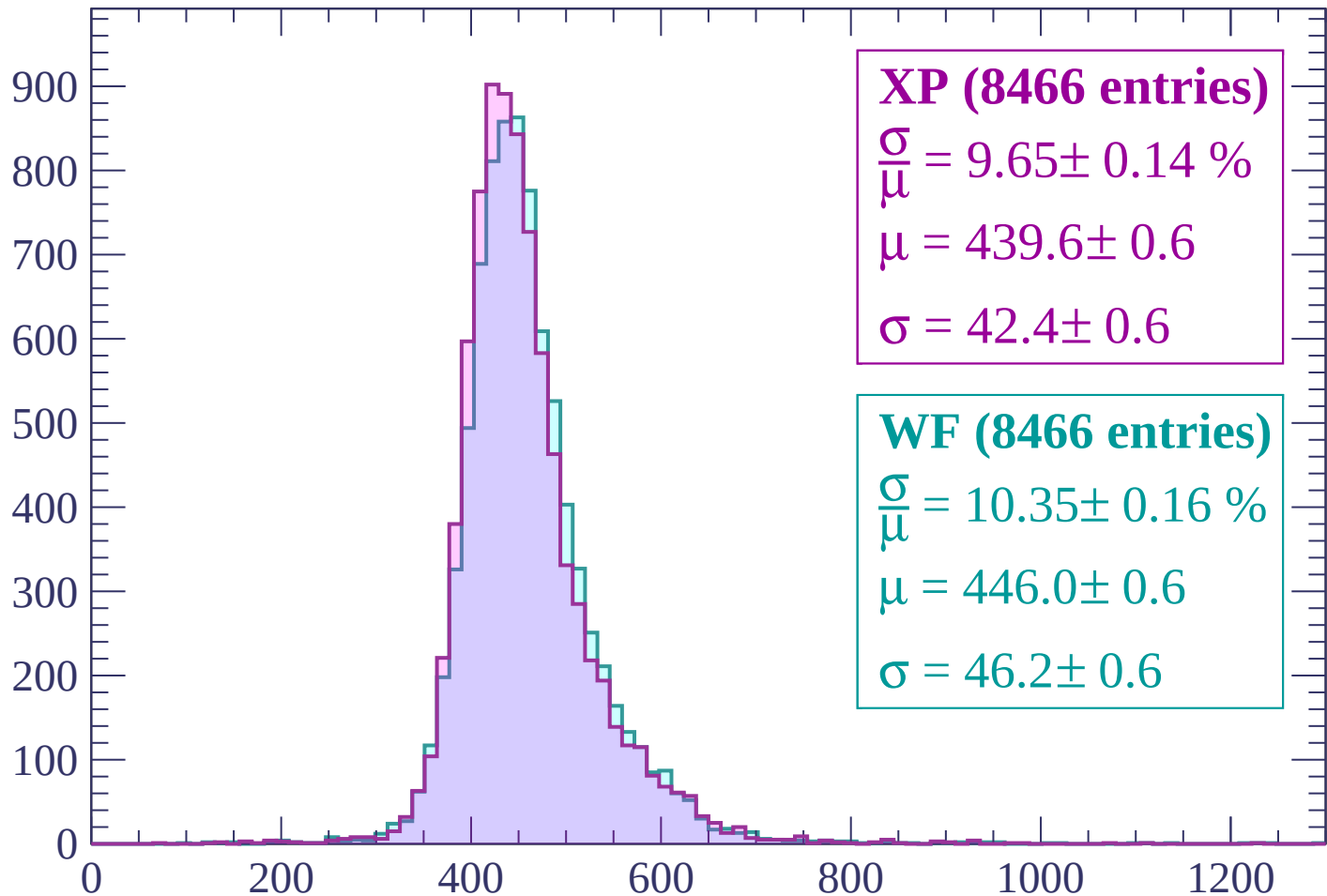
$\mu = 443.0 \pm 0.6$

$\sigma = 45.5 \pm 0.6$



Energy loss | $1140 < p < 1200$

Count



XP (8466 entries)

$$\frac{\sigma}{\mu} = 9.65 \pm 0.14 \%$$

$$\mu = 439.6 \pm 0.6$$

$$\sigma = 42.4 \pm 0.6$$

WF (8466 entries)

$$\frac{\sigma}{\mu} = 10.35 \pm 0.16 \%$$

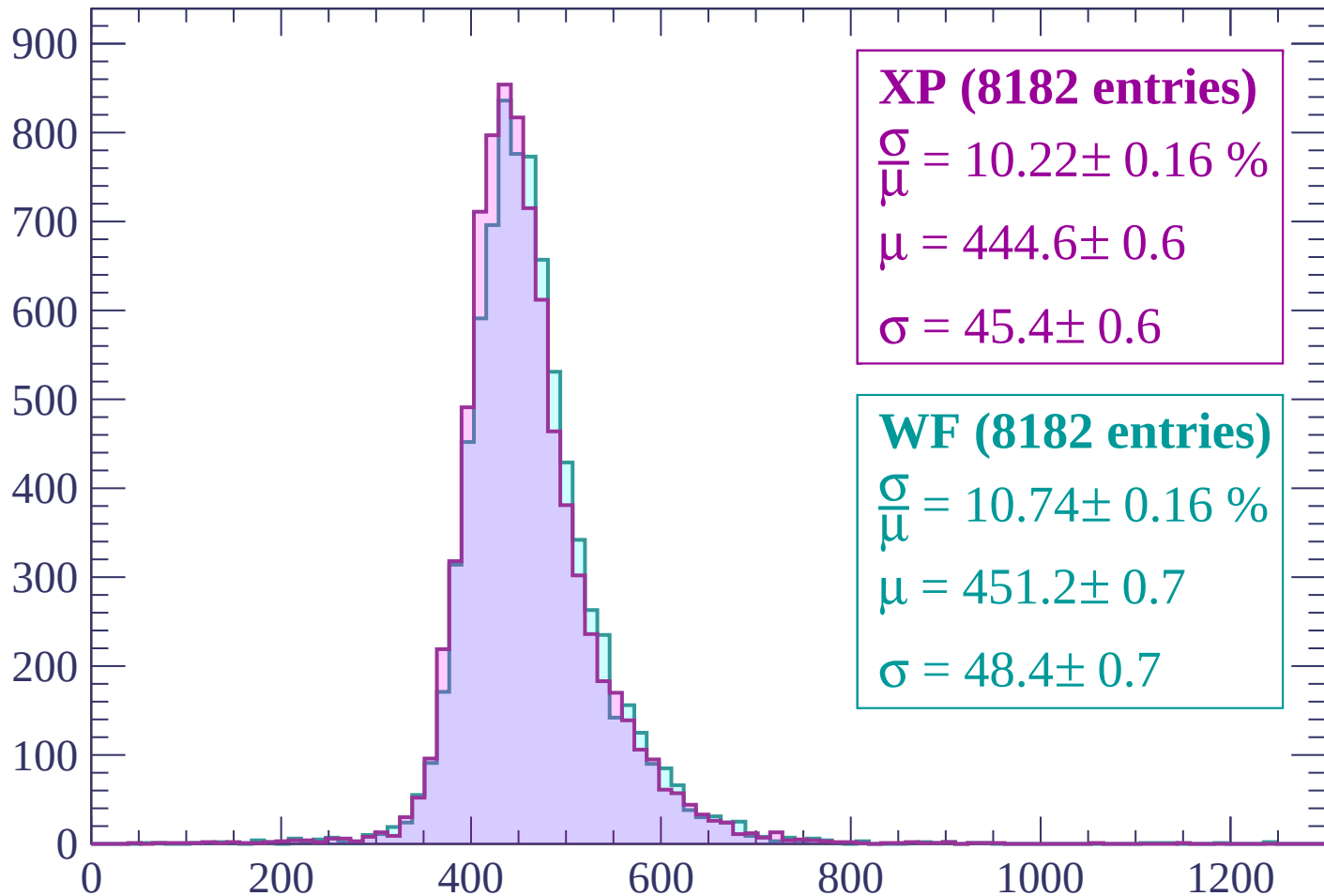
$$\mu = 446.0 \pm 0.6$$

$$\sigma = 46.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

Count



XP (8182 entries)

$$\frac{\sigma}{\mu} = 10.22 \pm 0.16 \%$$

$$\mu = 444.6 \pm 0.6$$

$$\sigma = 45.4 \pm 0.6$$

WF (8182 entries)

$$\frac{\sigma}{\mu} = 10.74 \pm 0.16 \%$$

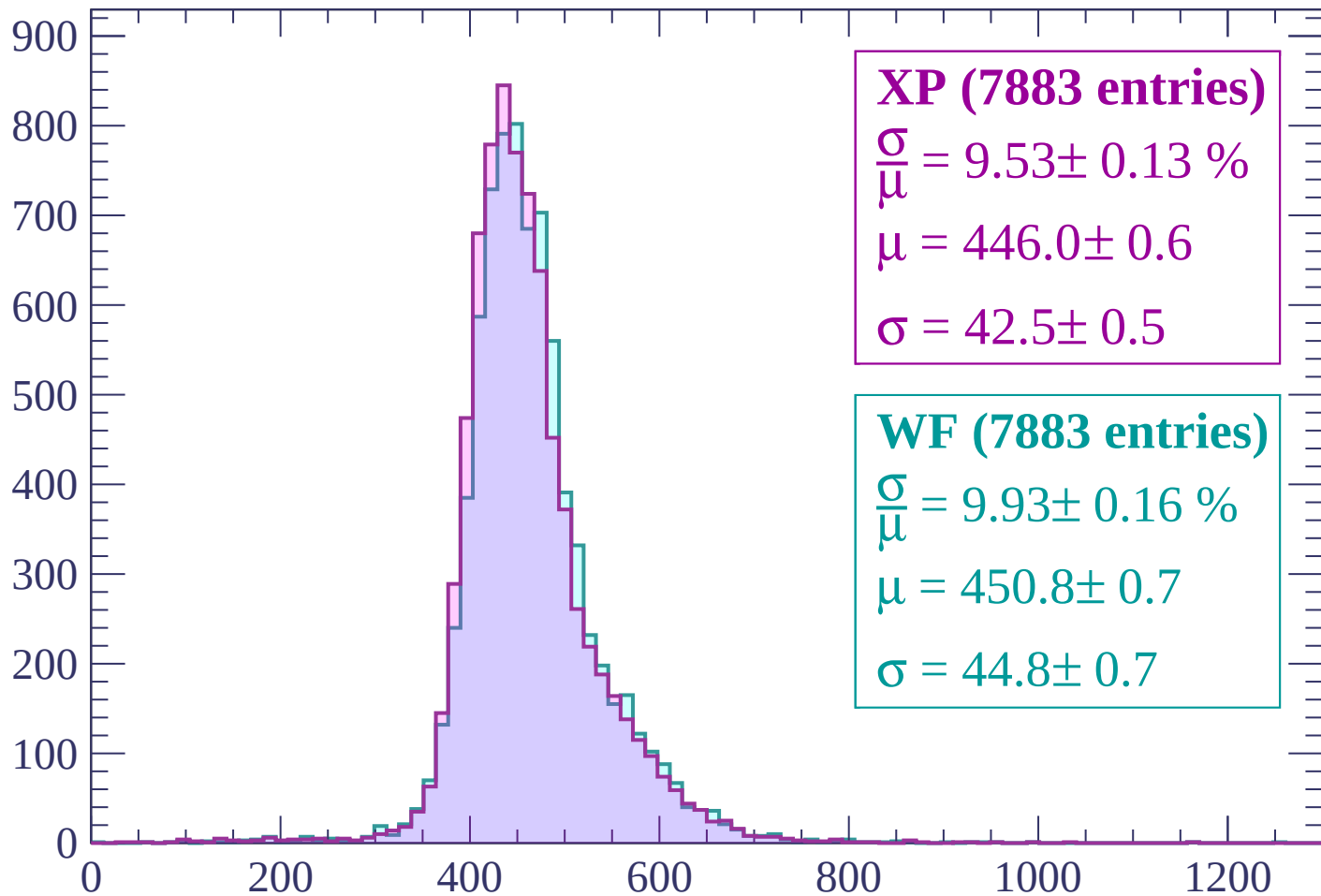
$$\mu = 451.2 \pm 0.7$$

$$\sigma = 48.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (7883 entries)

$\frac{\sigma}{\mu} = 9.53 \pm 0.13 \%$

$\mu = 446.0 \pm 0.6$

$\sigma = 42.5 \pm 0.5$

WF (7883 entries)

$\frac{\sigma}{\mu} = 9.93 \pm 0.16 \%$

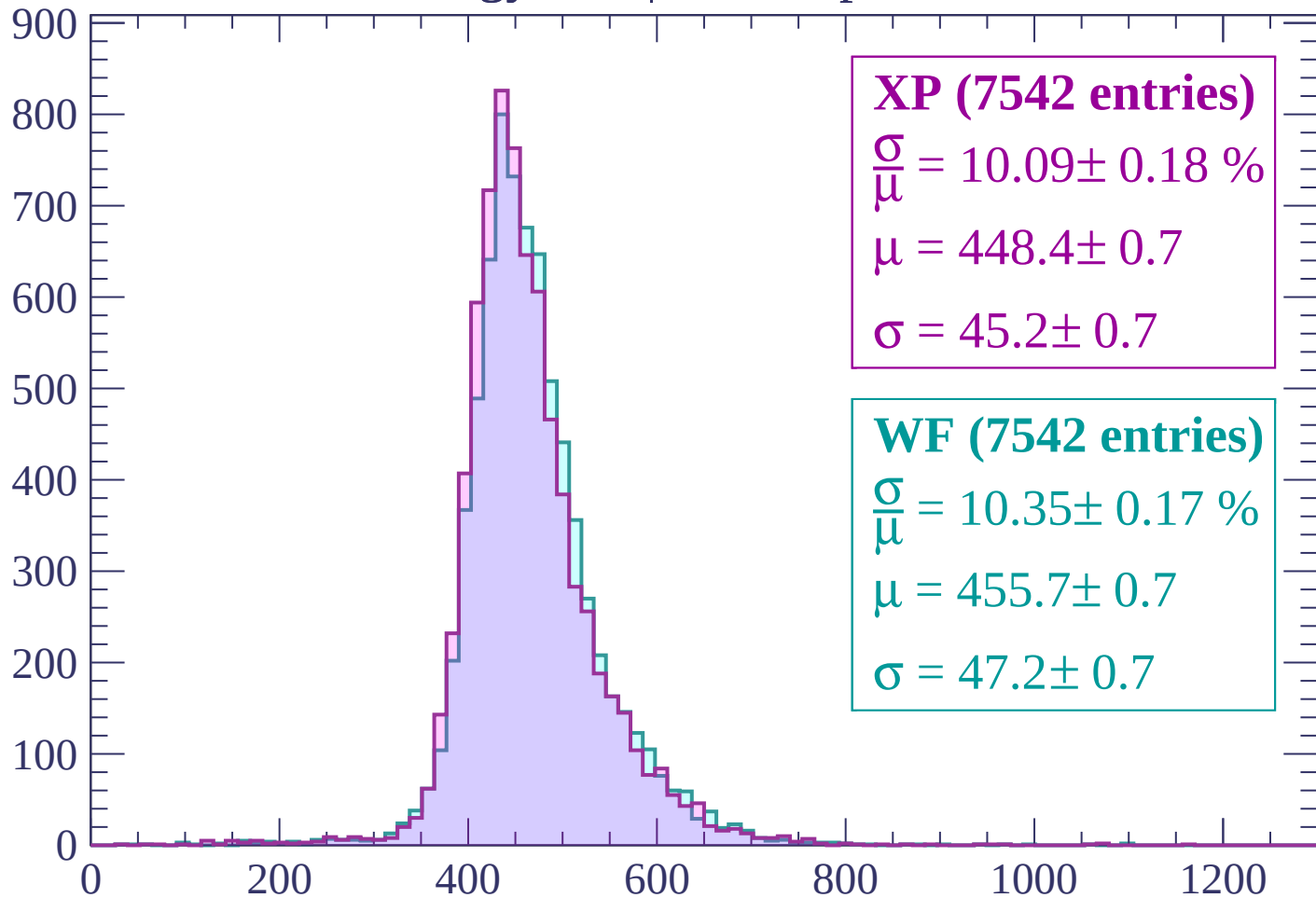
$\mu = 450.8 \pm 0.7$

$\sigma = 44.8 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (7542 entries)

$\frac{\sigma}{\mu} = 10.09 \pm 0.18 \%$

$\mu = 448.4 \pm 0.7$

$\sigma = 45.2 \pm 0.7$

WF (7542 entries)

$\frac{\sigma}{\mu} = 10.35 \pm 0.17 \%$

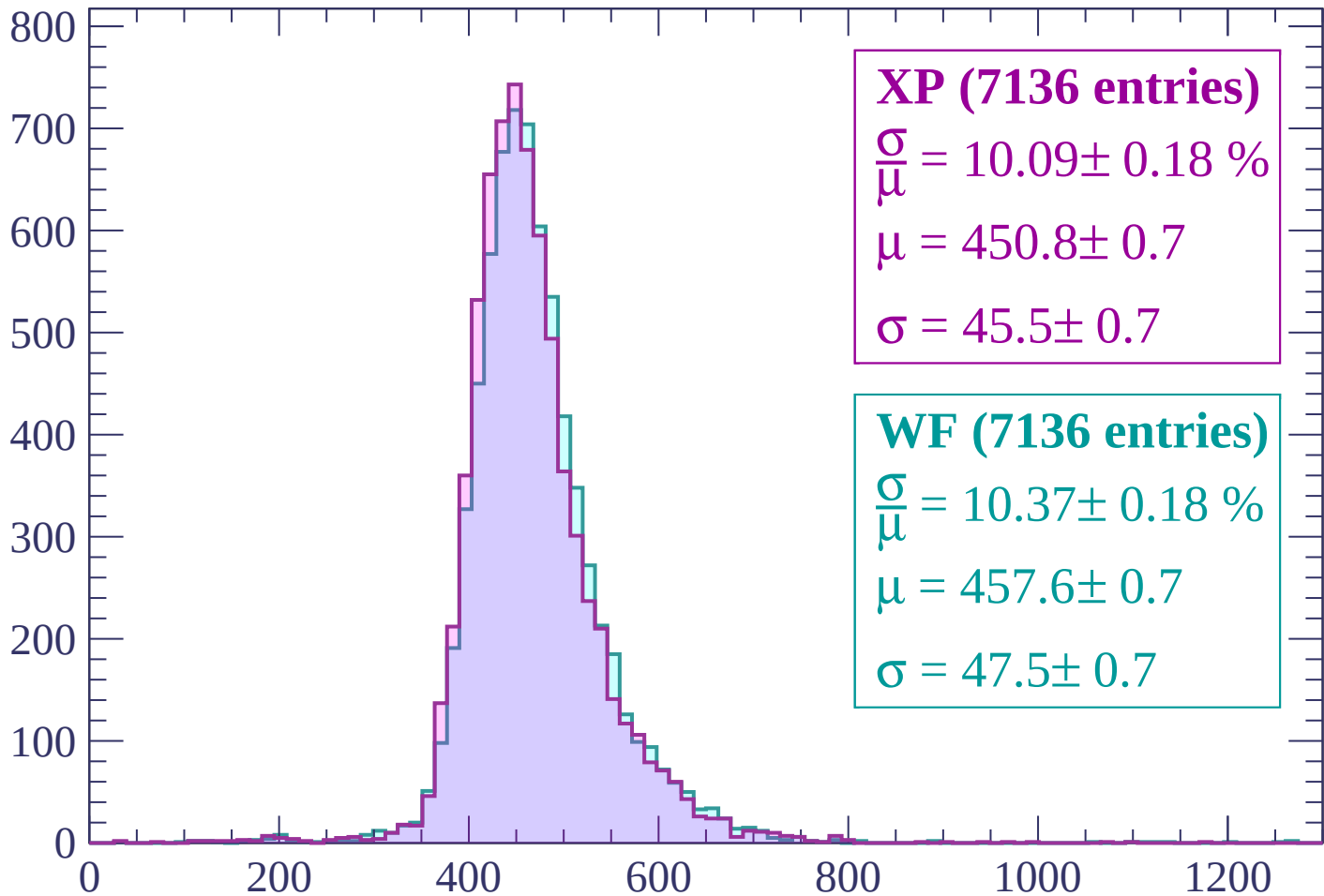
$\mu = 455.7 \pm 0.7$

$\sigma = 47.2 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP (7136 entries)

$$\frac{\sigma}{\mu} = 10.09 \pm 0.18 \%$$

$$\mu = 450.8 \pm 0.7$$

$$\sigma = 45.5 \pm 0.7$$

WF (7136 entries)

$$\frac{\sigma}{\mu} = 10.37 \pm 0.18 \%$$

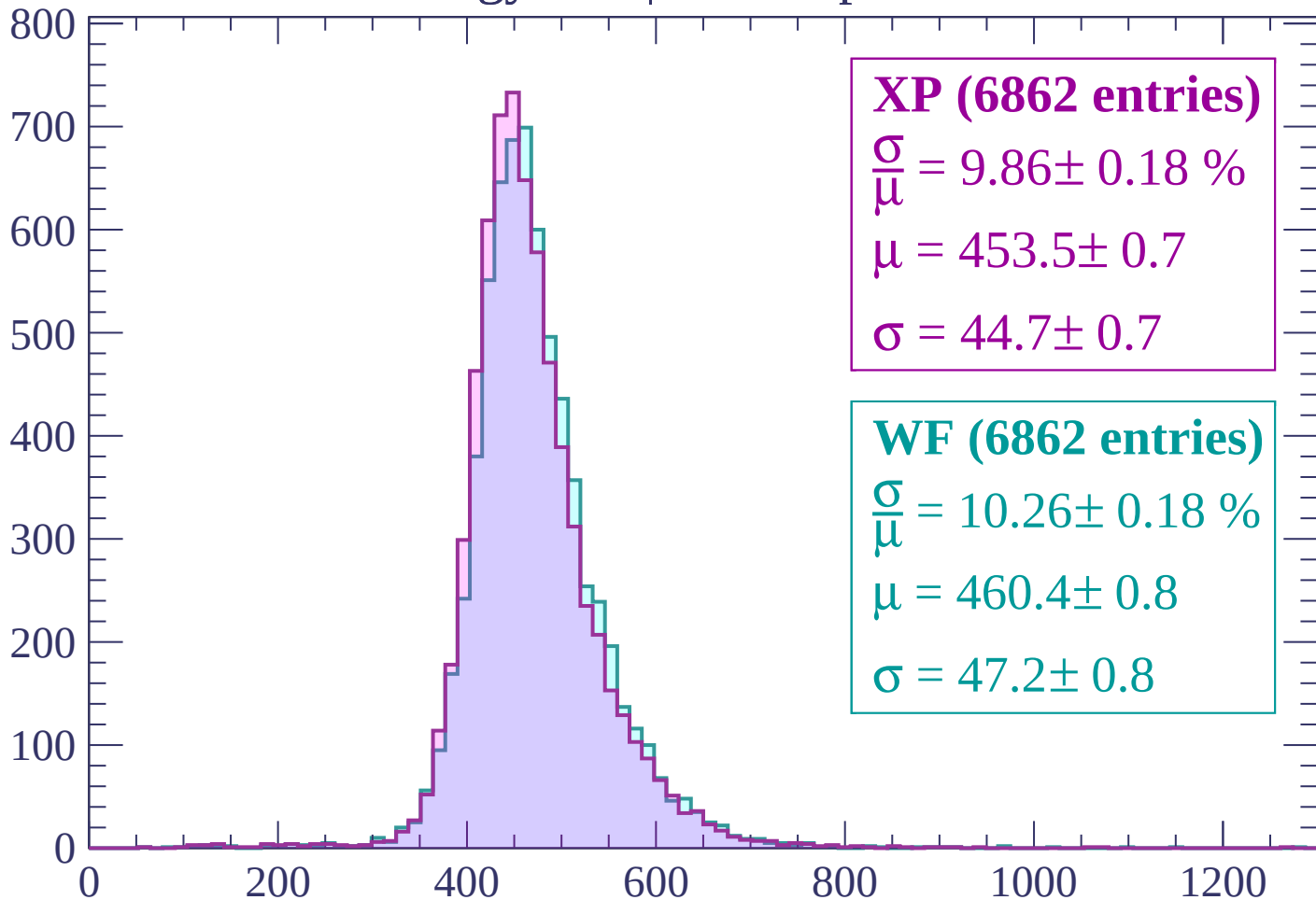
$$\mu = 457.6 \pm 0.7$$

$$\sigma = 47.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP (6862 entries)

$$\frac{\sigma}{\mu} = 9.86 \pm 0.18 \%$$

$$\mu = 453.5 \pm 0.7$$

$$\sigma = 44.7 \pm 0.7$$

WF (6862 entries)

$$\frac{\sigma}{\mu} = 10.26 \pm 0.18 \%$$

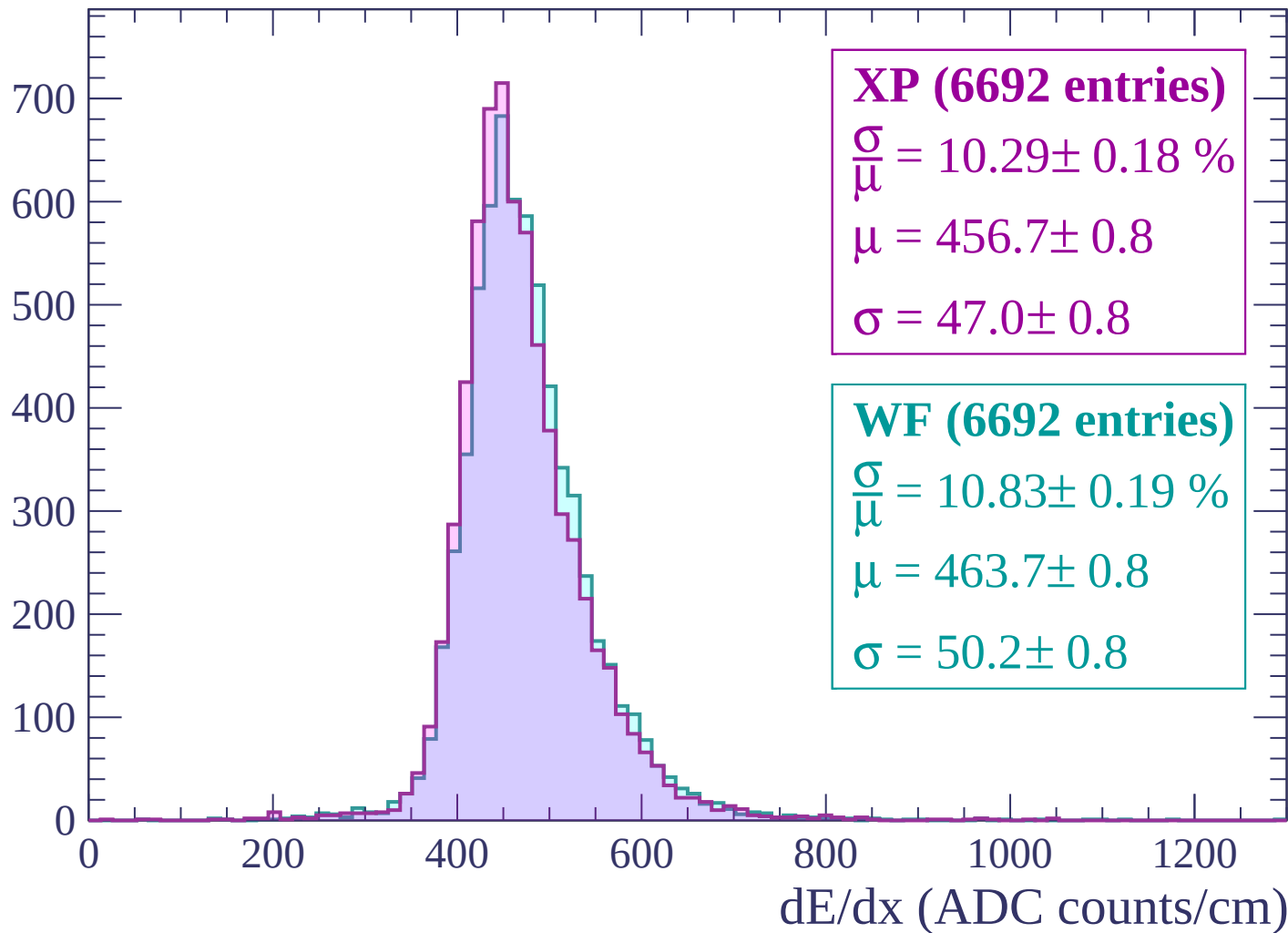
$$\mu = 460.4 \pm 0.8$$

$$\sigma = 47.2 \pm 0.8$$

dE/dx (ADC counts/cm)

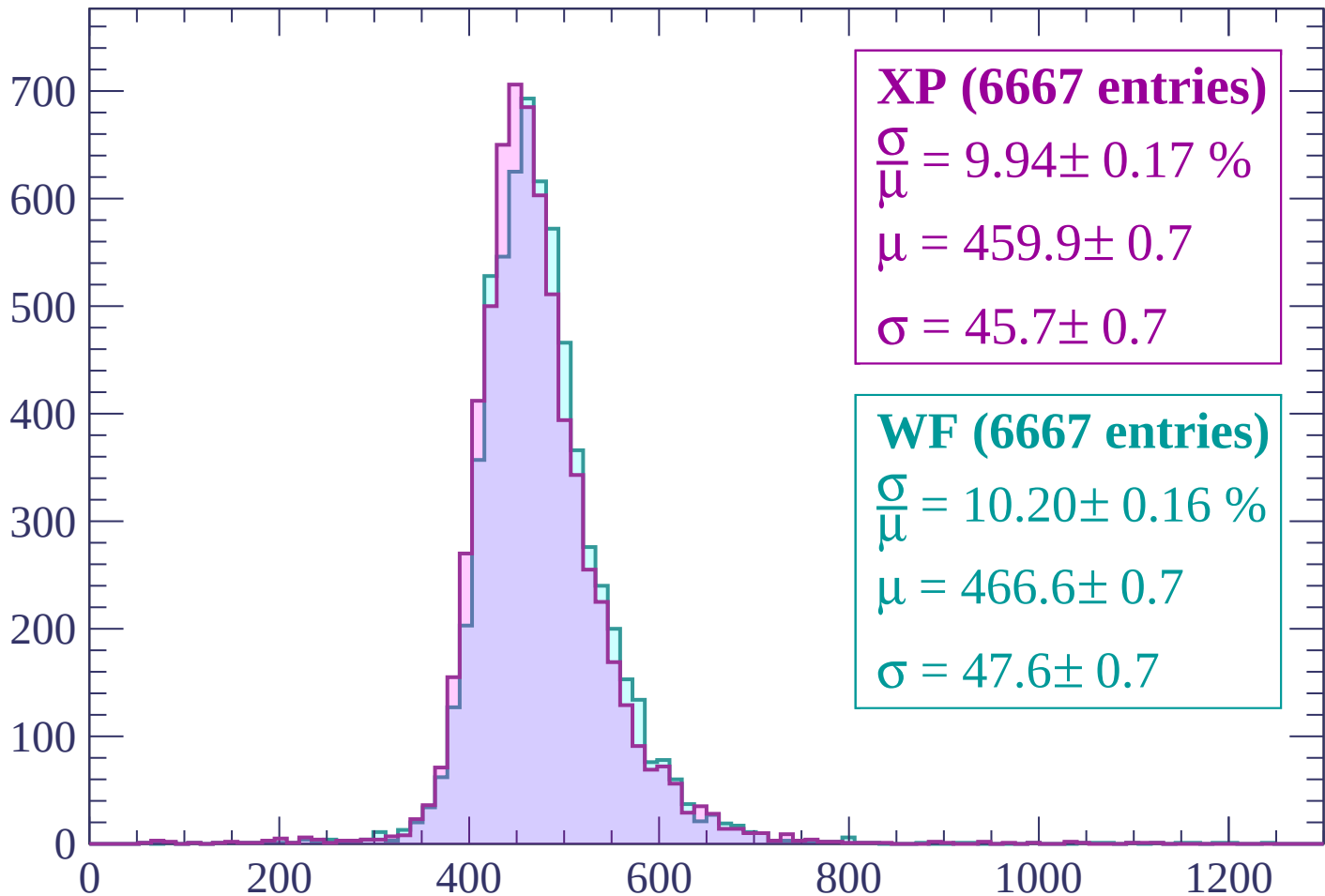
Energy loss | $1500 < p < 1560$

Count



Energy loss | $1560 < p < 1620$

Count



XP (6667 entries)

$$\frac{\sigma}{\mu} = 9.94 \pm 0.17 \%$$

$$\mu = 459.9 \pm 0.7$$

$$\sigma = 45.7 \pm 0.7$$

WF (6667 entries)

$$\frac{\sigma}{\mu} = 10.20 \pm 0.16 \%$$

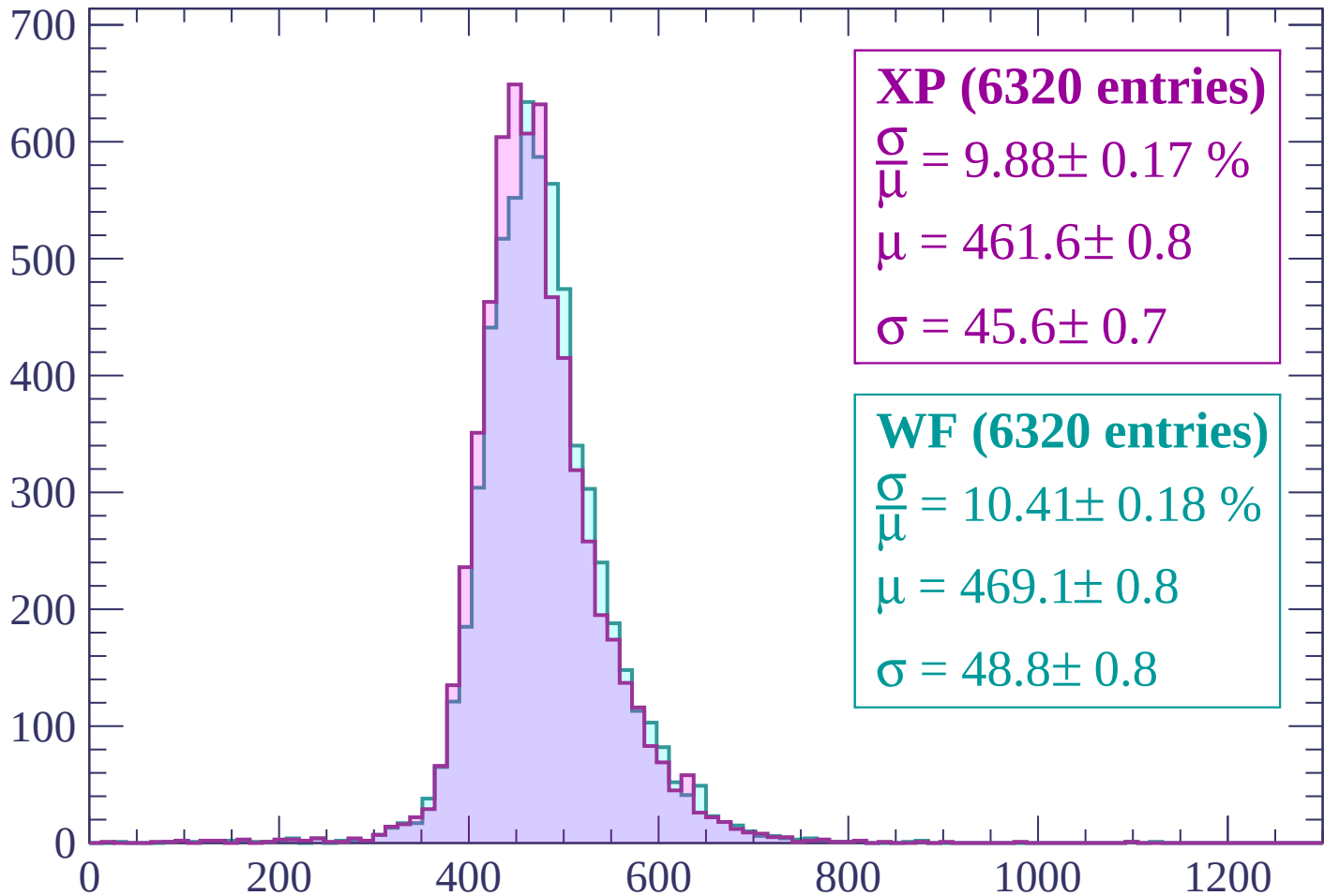
$$\mu = 466.6 \pm 0.7$$

$$\sigma = 47.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP (6320 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.17 \%$$

$$\mu = 461.6 \pm 0.8$$

$$\sigma = 45.6 \pm 0.7$$

WF (6320 entries)

$$\frac{\sigma}{\mu} = 10.41 \pm 0.18 \%$$

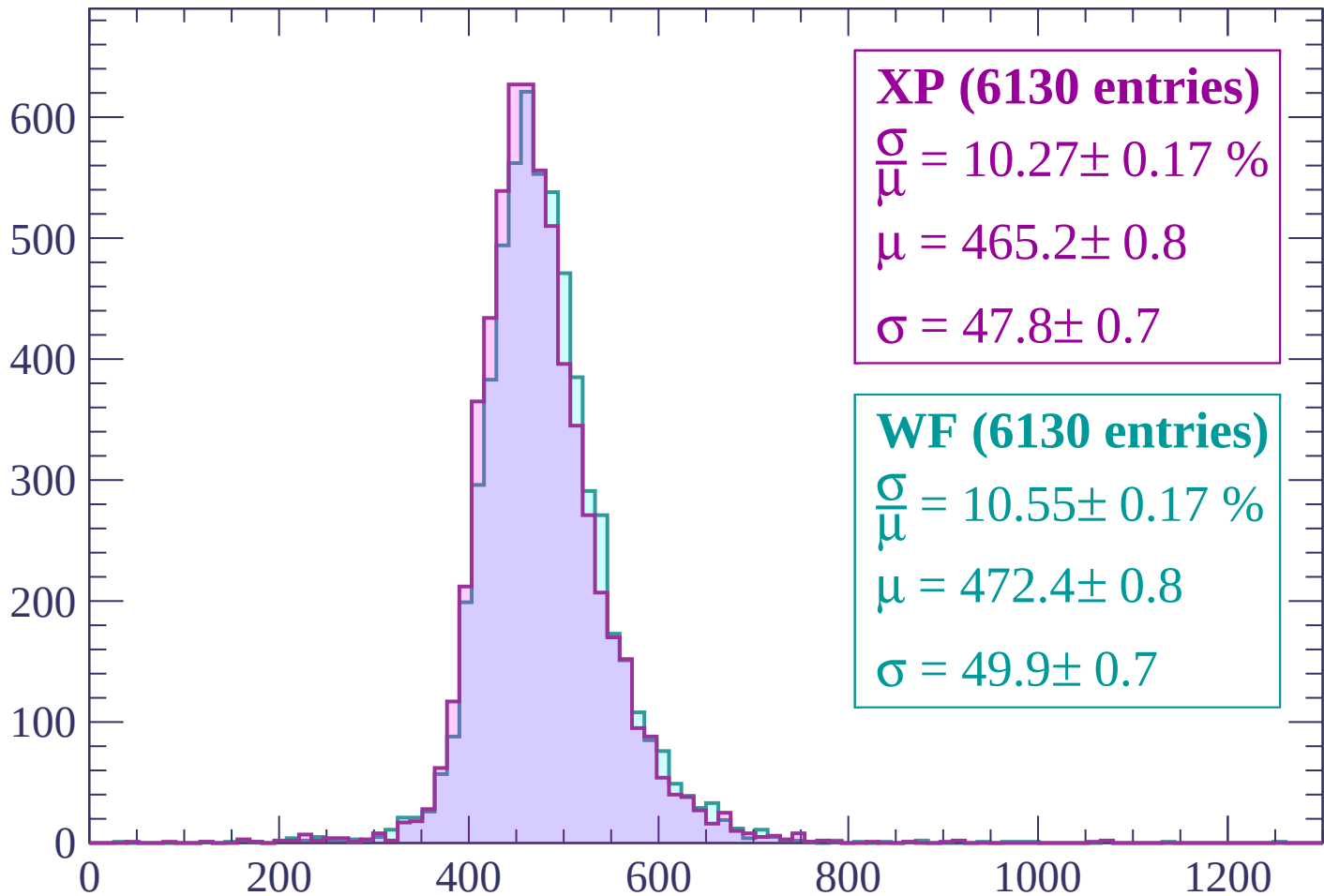
$$\mu = 469.1 \pm 0.8$$

$$\sigma = 48.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (6130 entries)

$$\frac{\sigma}{\mu} = 10.27 \pm 0.17 \%$$

$$\mu = 465.2 \pm 0.8$$

$$\sigma = 47.8 \pm 0.7$$

WF (6130 entries)

$$\frac{\sigma}{\mu} = 10.55 \pm 0.17 \%$$

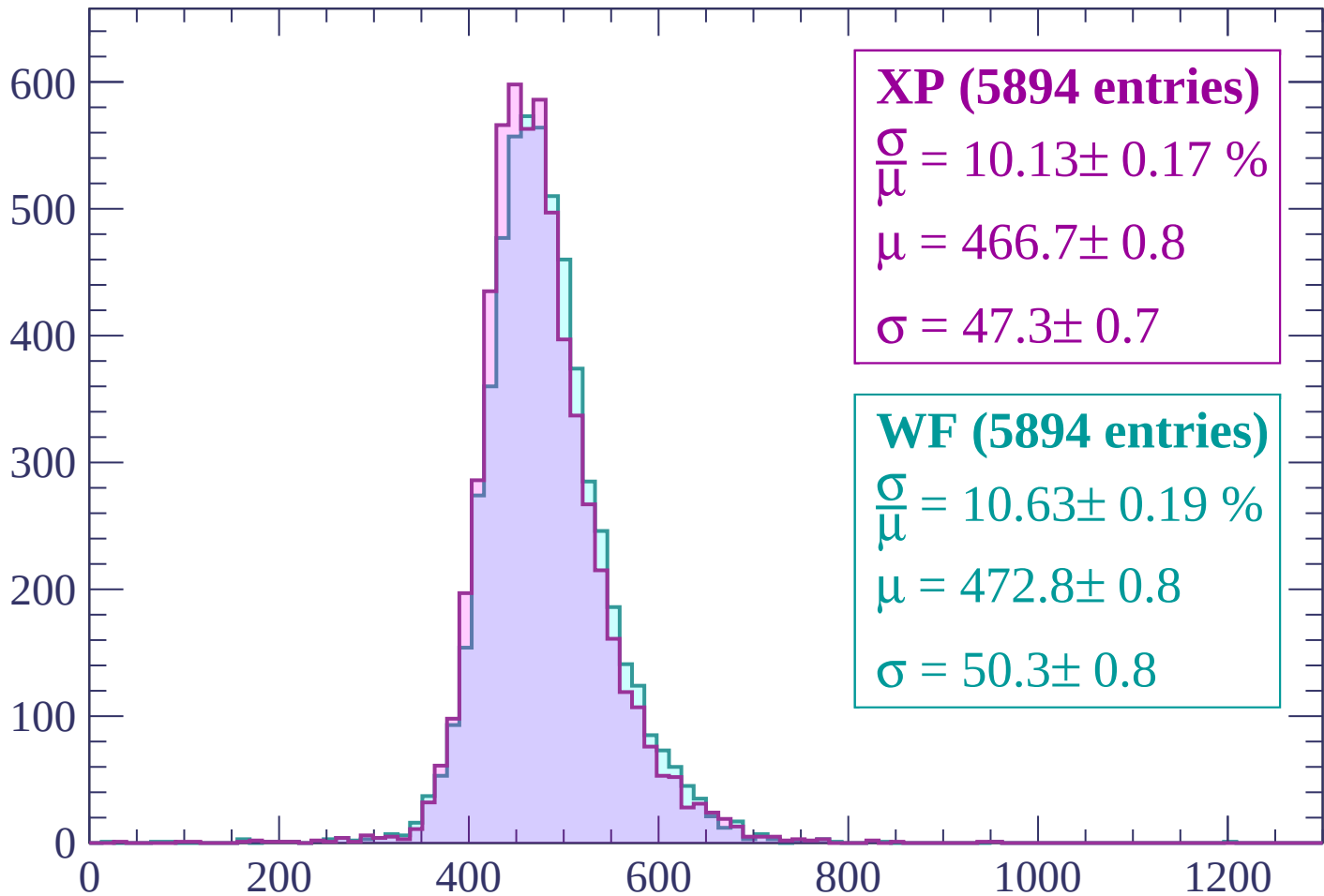
$$\mu = 472.4 \pm 0.8$$

$$\sigma = 49.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (5894 entries)

$$\frac{\sigma}{\mu} = 10.13 \pm 0.17 \%$$

$$\mu = 466.7 \pm 0.8$$

$$\sigma = 47.3 \pm 0.7$$

WF (5894 entries)

$$\frac{\sigma}{\mu} = 10.63 \pm 0.19 \%$$

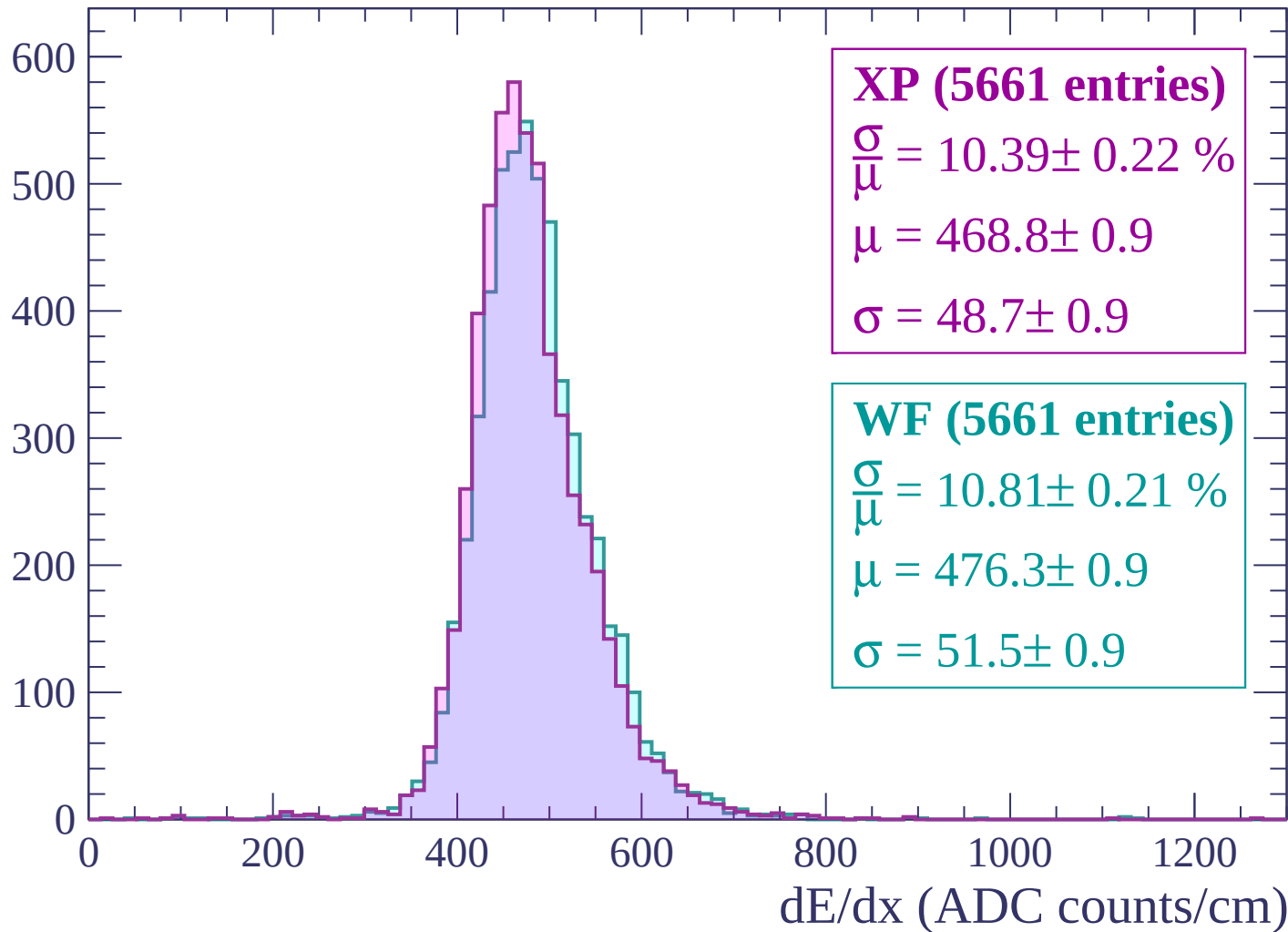
$$\mu = 472.8 \pm 0.8$$

$$\sigma = 50.3 \pm 0.8$$

dE/dx (ADC counts/cm)

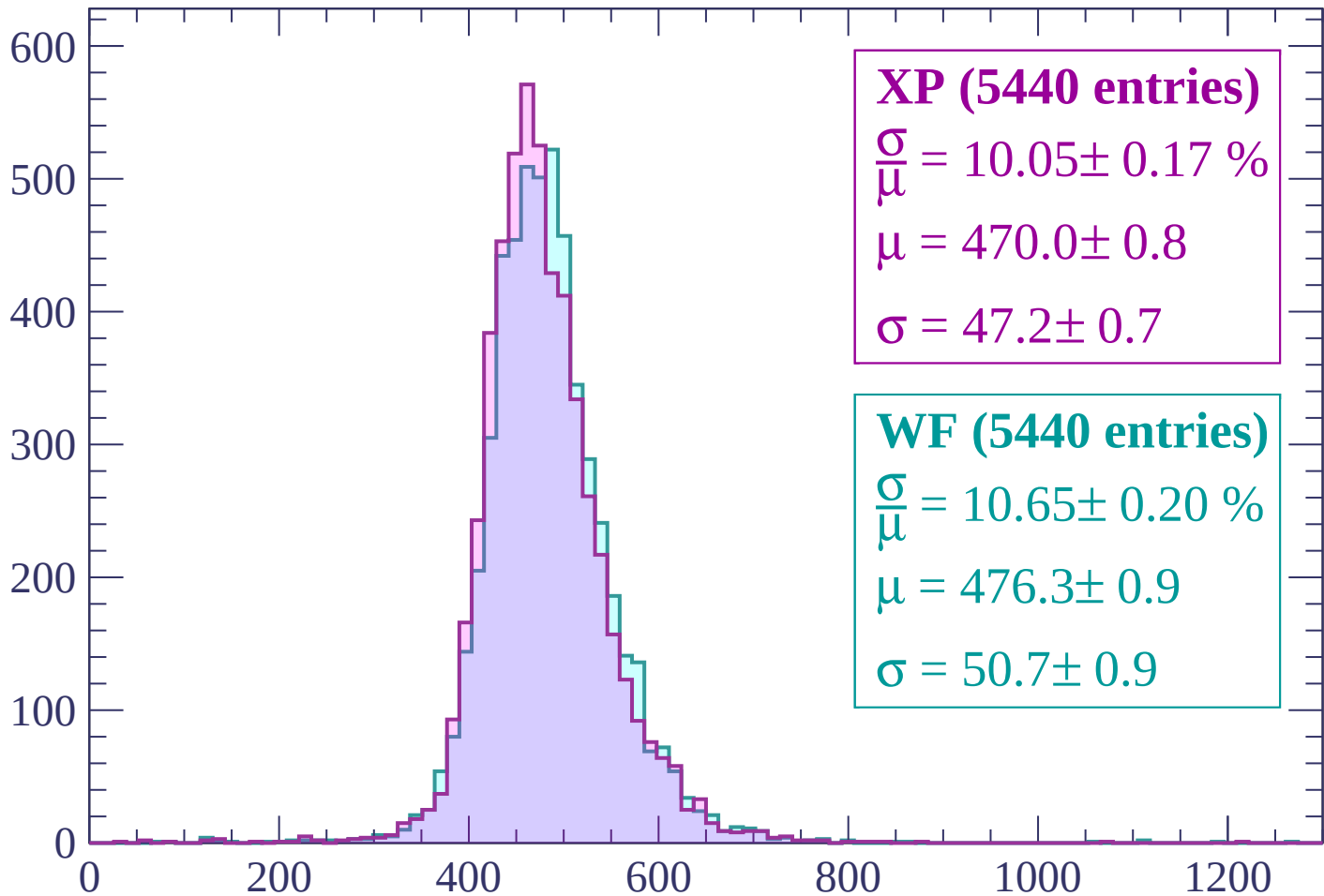
Energy loss | $1800 < p < 1860$

Count



Energy loss | $1860 < p < 1920$

Count



XP (5440 entries)

$\frac{\sigma}{\mu} = 10.05 \pm 0.17 \%$

$\mu = 470.0 \pm 0.8$

$\sigma = 47.2 \pm 0.7$

WF (5440 entries)

$\frac{\sigma}{\mu} = 10.65 \pm 0.20 \%$

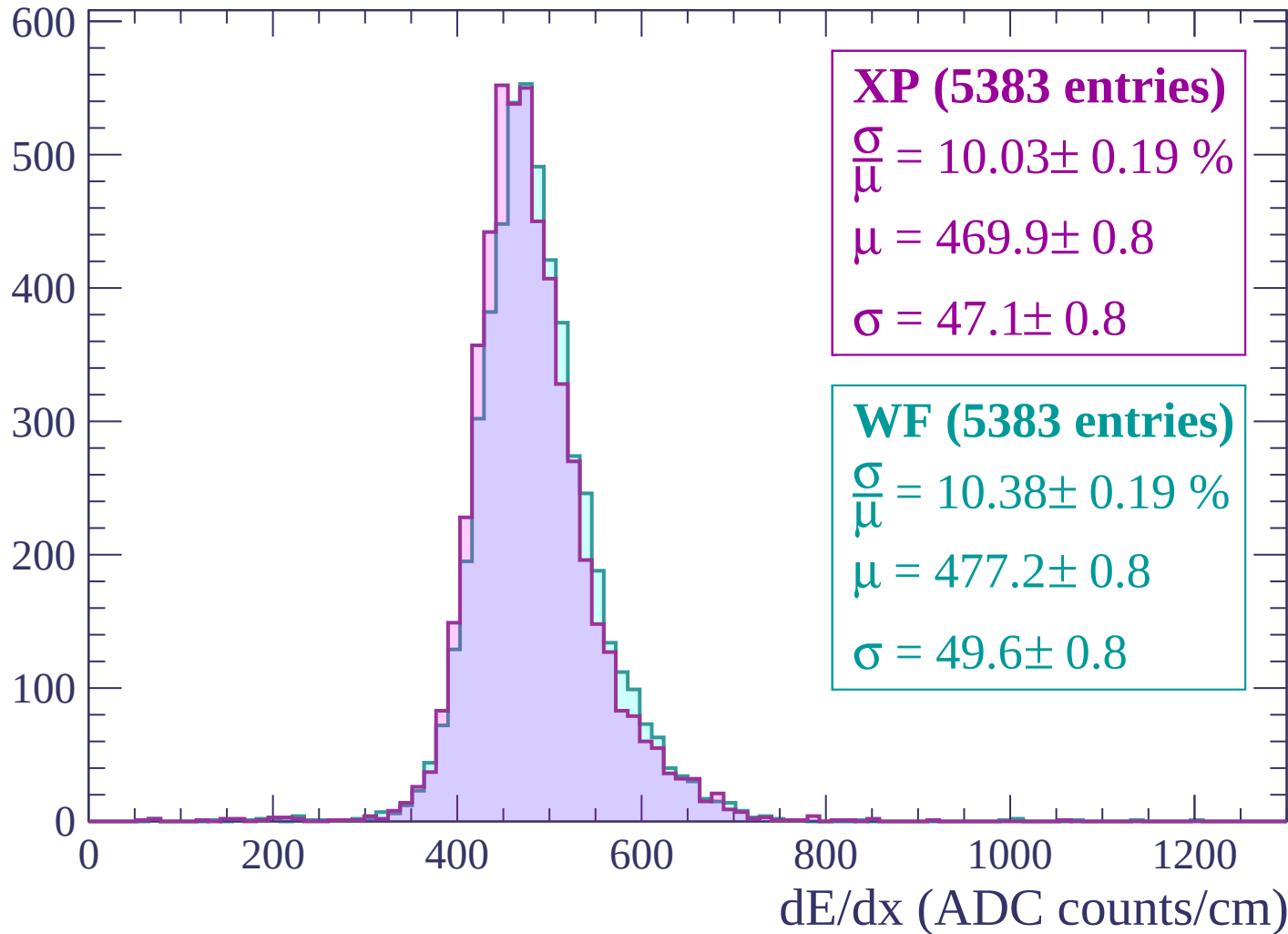
$\mu = 476.3 \pm 0.9$

$\sigma = 50.7 \pm 0.9$

dE/dx (ADC counts/cm)

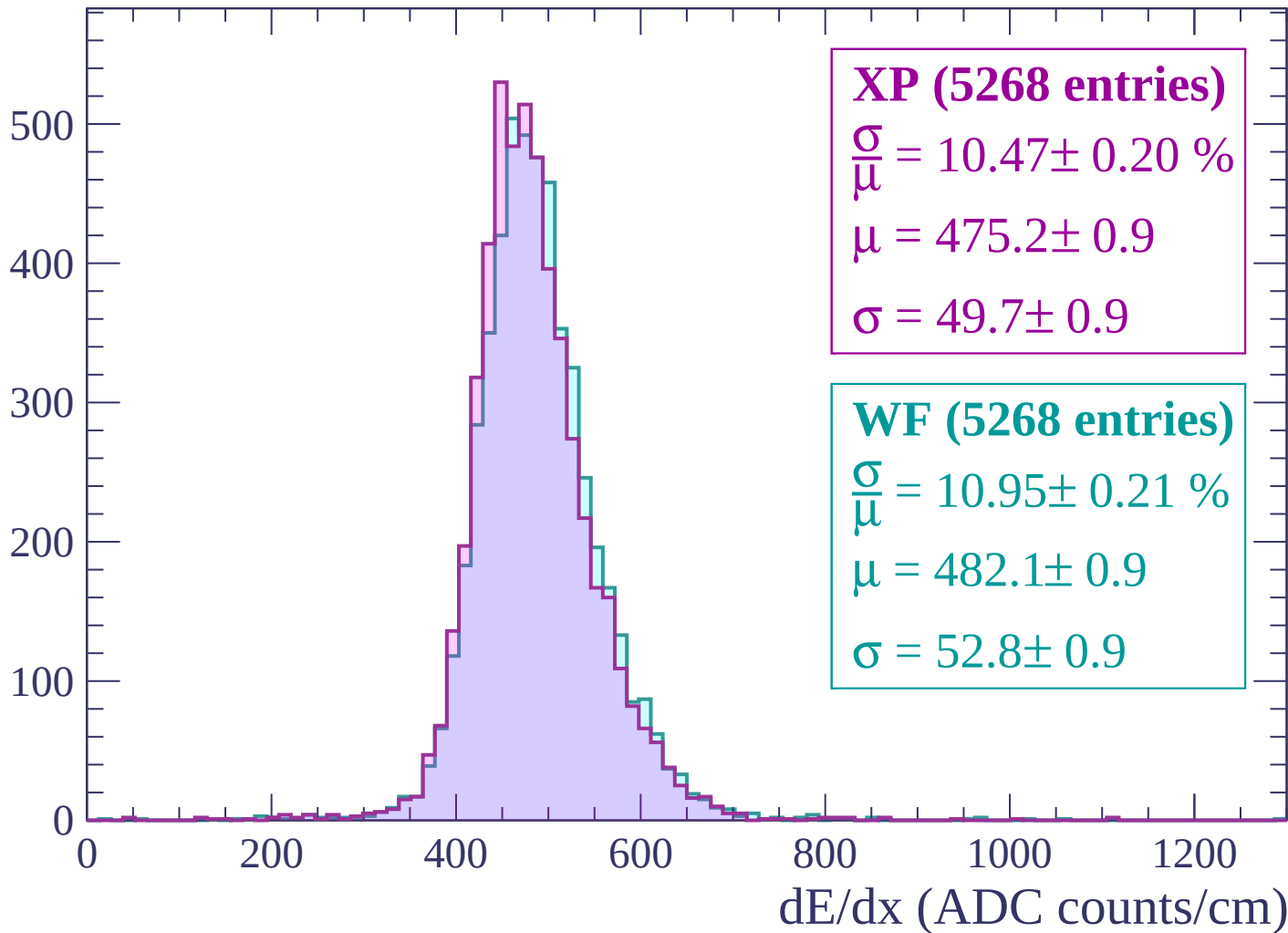
Energy loss | $1920 < p < 1980$

Count



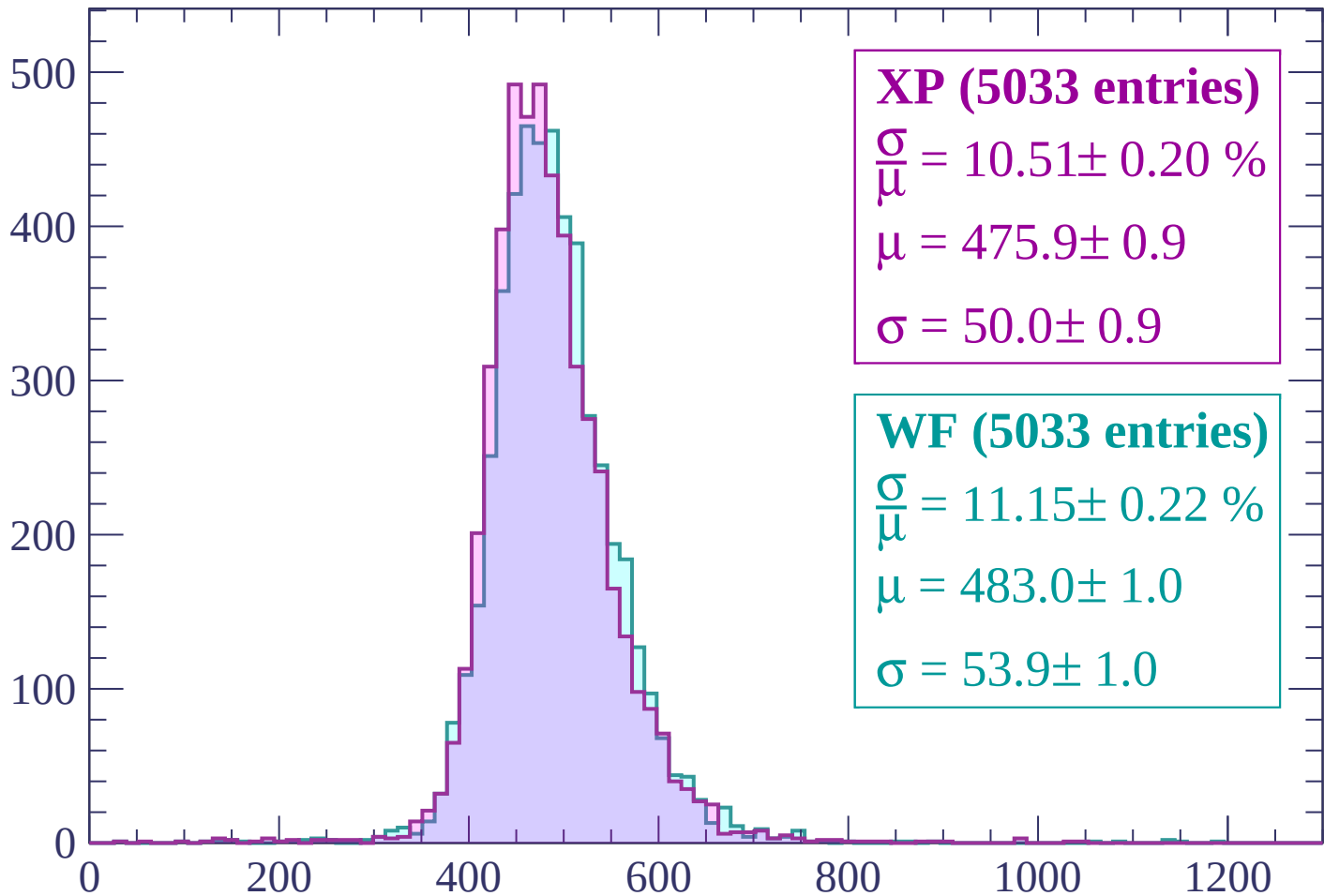
Energy loss | $1980 < p < 2040$

Count



Energy loss | $2040 < p < 2100$

Count



XP (5033 entries)

$$\frac{\sigma}{\mu} = 10.51 \pm 0.20 \%$$

$$\mu = 475.9 \pm 0.9$$

$$\sigma = 50.0 \pm 0.9$$

WF (5033 entries)

$$\frac{\sigma}{\mu} = 11.15 \pm 0.22 \%$$

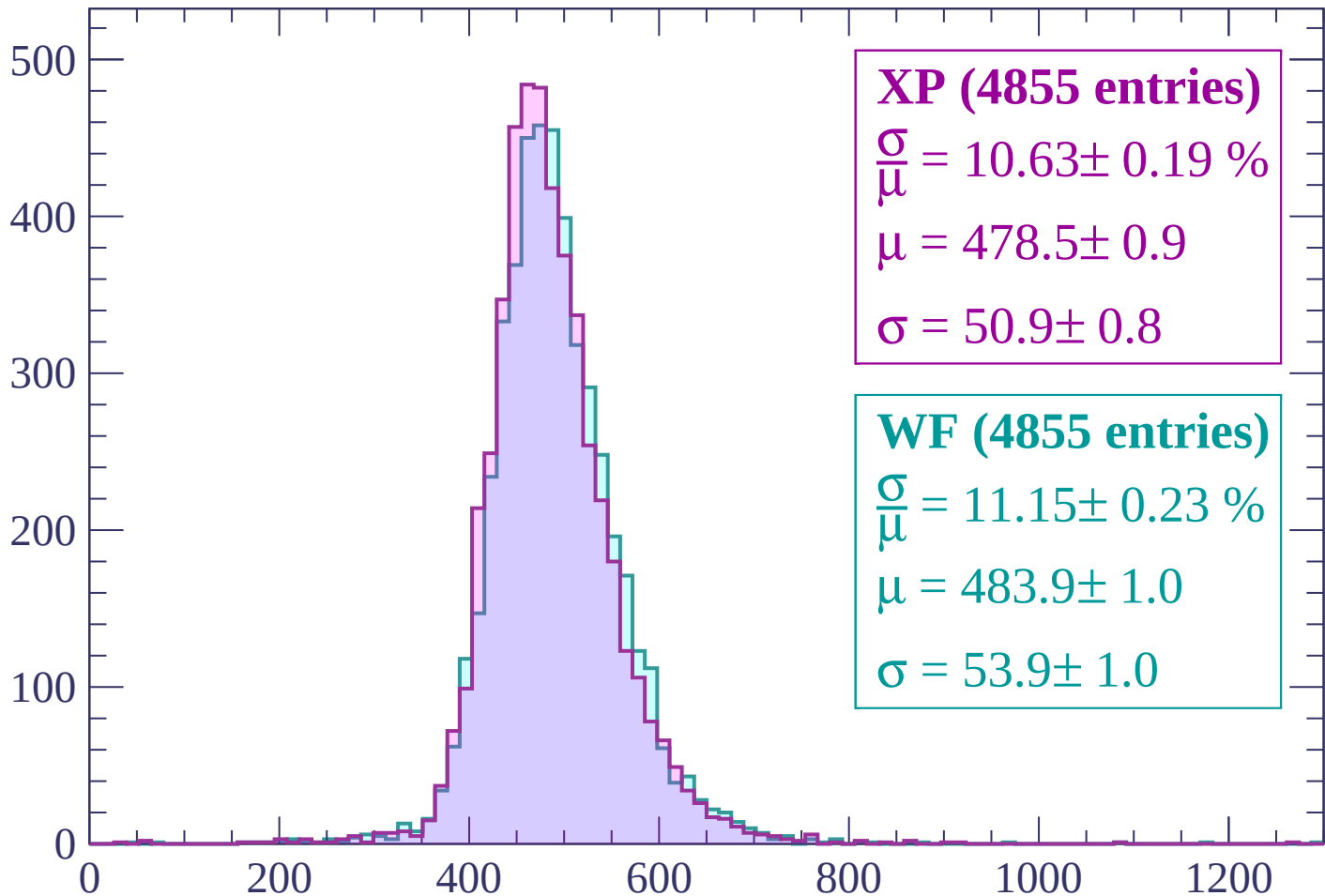
$$\mu = 483.0 \pm 1.0$$

$$\sigma = 53.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP (4855 entries)

$\frac{\sigma}{\mu} = 10.63 \pm 0.19 \%$

$\mu = 478.5 \pm 0.9$

$\sigma = 50.9 \pm 0.8$

WF (4855 entries)

$\frac{\sigma}{\mu} = 11.15 \pm 0.23 \%$

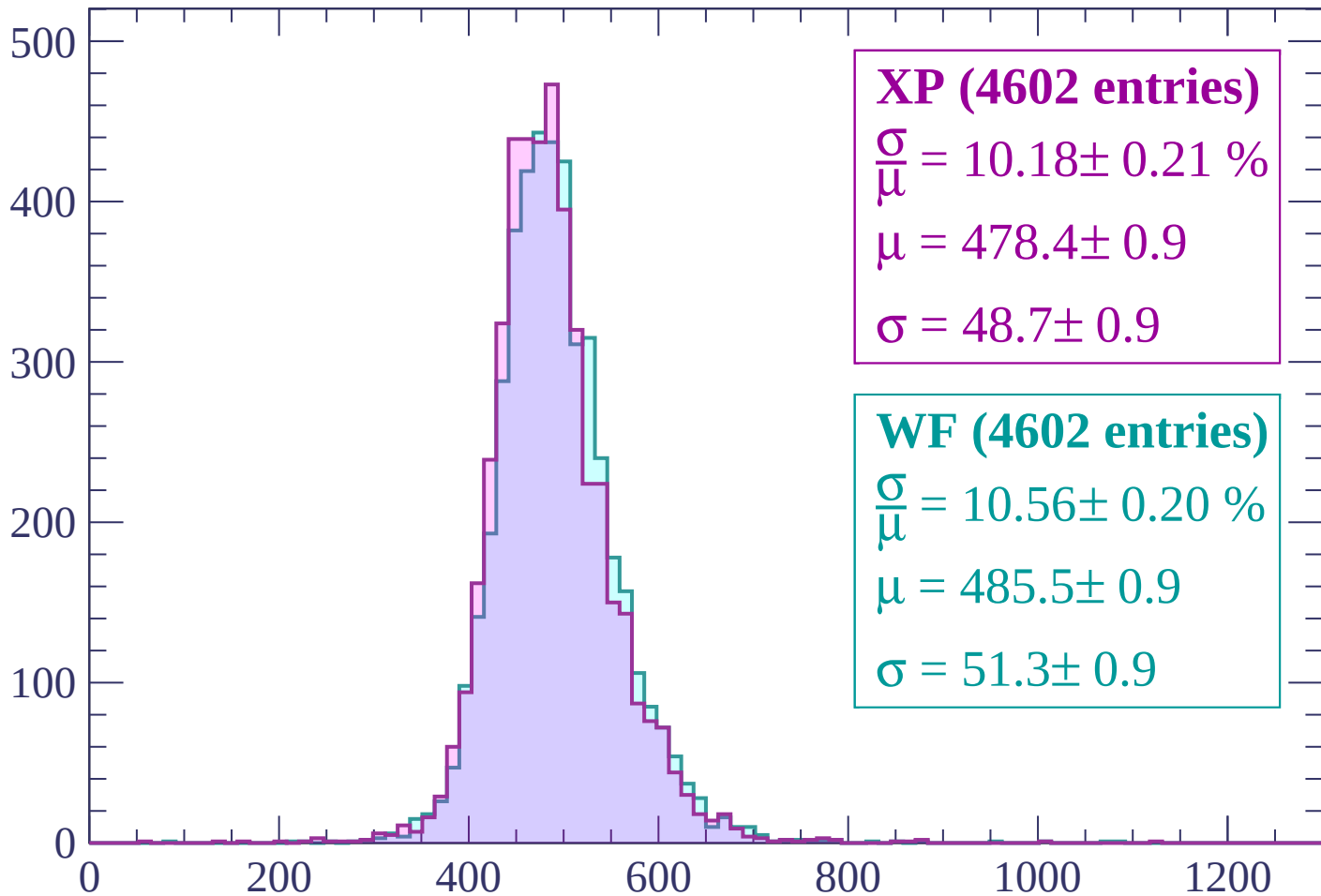
$\mu = 483.9 \pm 1.0$

$\sigma = 53.9 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP (4602 entries)

$\frac{\sigma}{\mu} = 10.18 \pm 0.21 \%$

$\mu = 478.4 \pm 0.9$

$\sigma = 48.7 \pm 0.9$

WF (4602 entries)

$\frac{\sigma}{\mu} = 10.56 \pm 0.20 \%$

$\mu = 485.5 \pm 0.9$

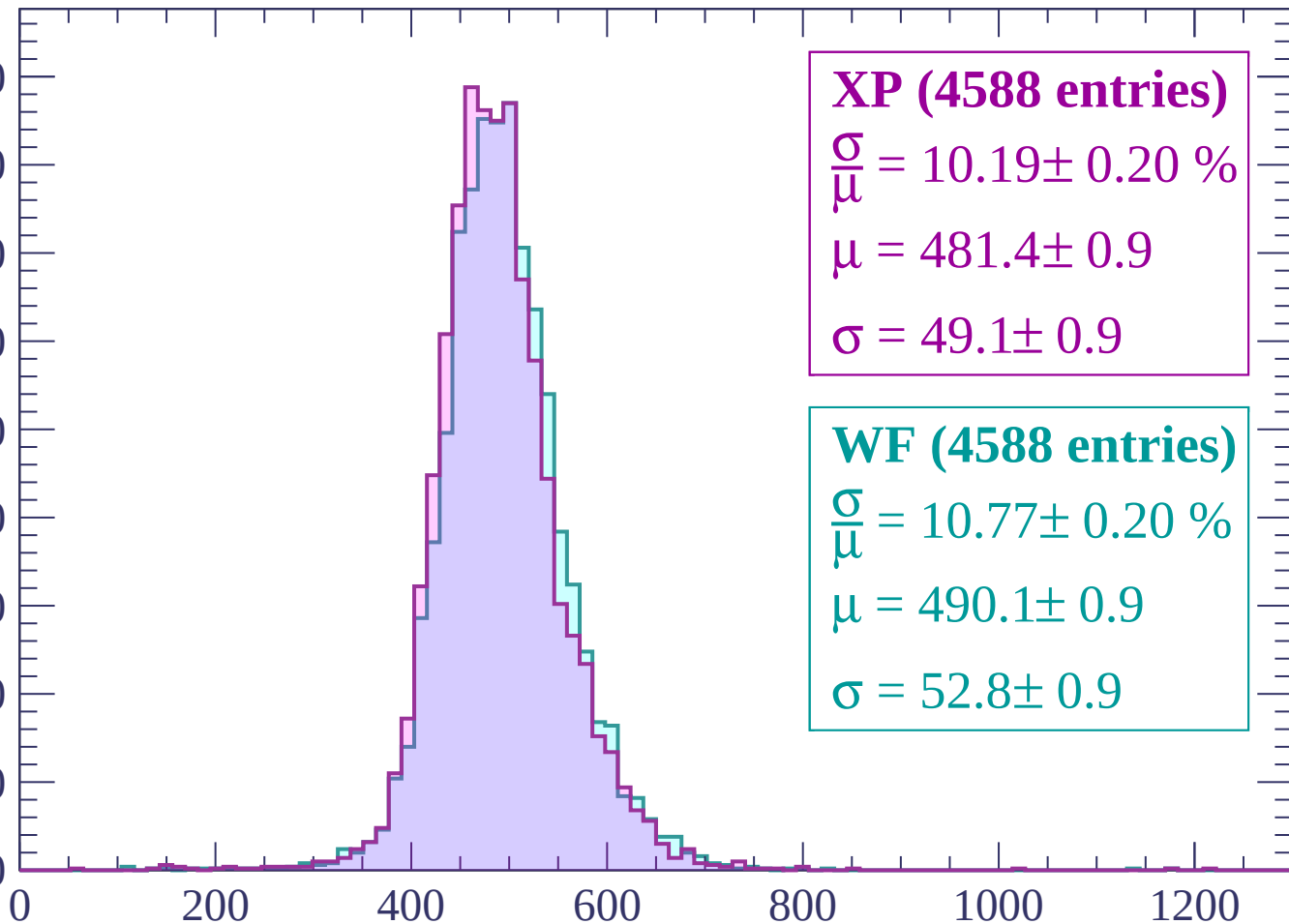
$\sigma = 51.3 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count

450
400
350
300
250
200
150
100
50
0



XP (4588 entries)

$\frac{\sigma}{\mu} = 10.19 \pm 0.20 \%$

$\mu = 481.4 \pm 0.9$

$\sigma = 49.1 \pm 0.9$

WF (4588 entries)

$\frac{\sigma}{\mu} = 10.77 \pm 0.20 \%$

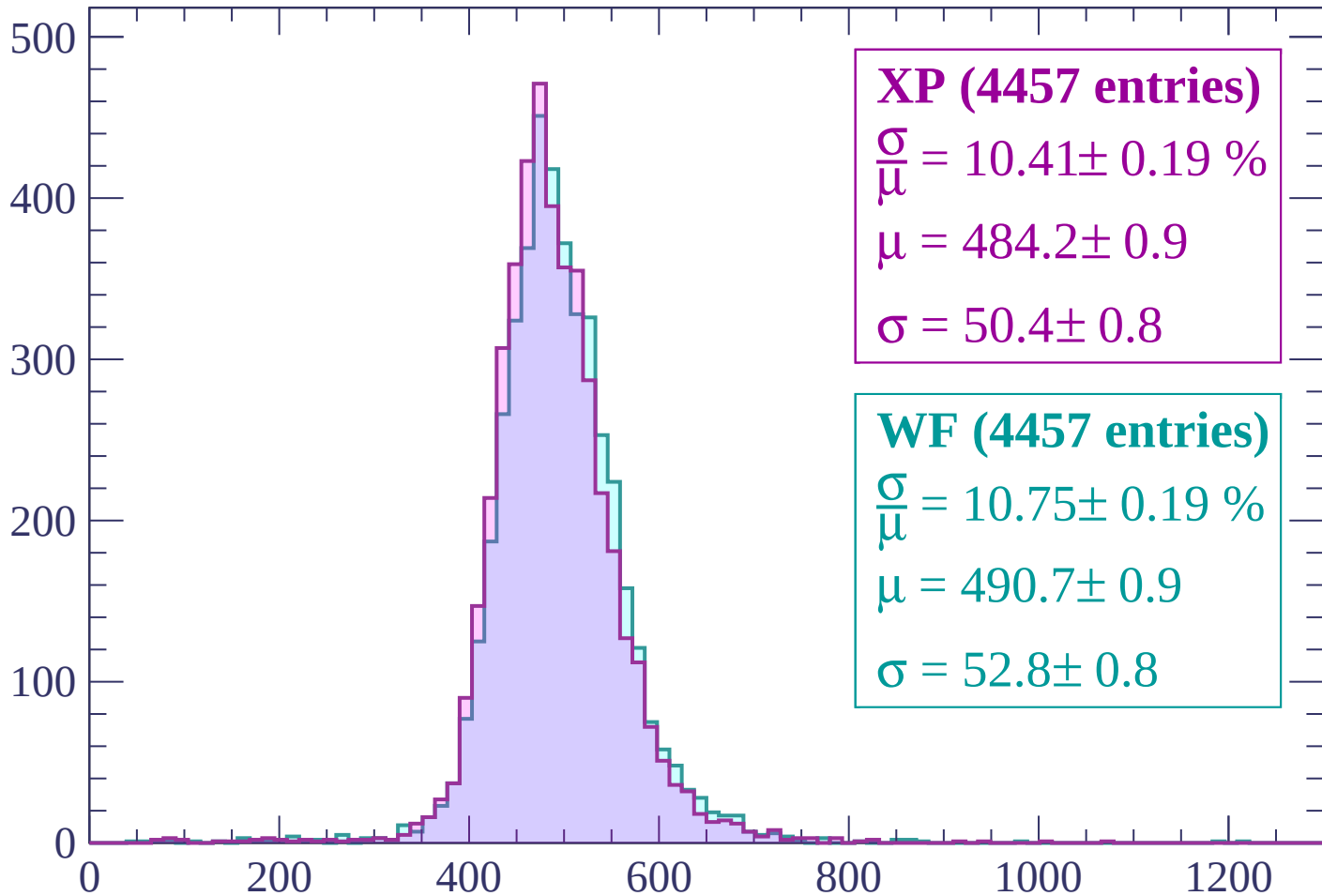
$\mu = 490.1 \pm 0.9$

$\sigma = 52.8 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

Count



XP (4457 entries)

$\frac{\sigma}{\mu} = 10.41 \pm 0.19 \%$

$\mu = 484.2 \pm 0.9$

$\sigma = 50.4 \pm 0.8$

WF (4457 entries)

$\frac{\sigma}{\mu} = 10.75 \pm 0.19 \%$

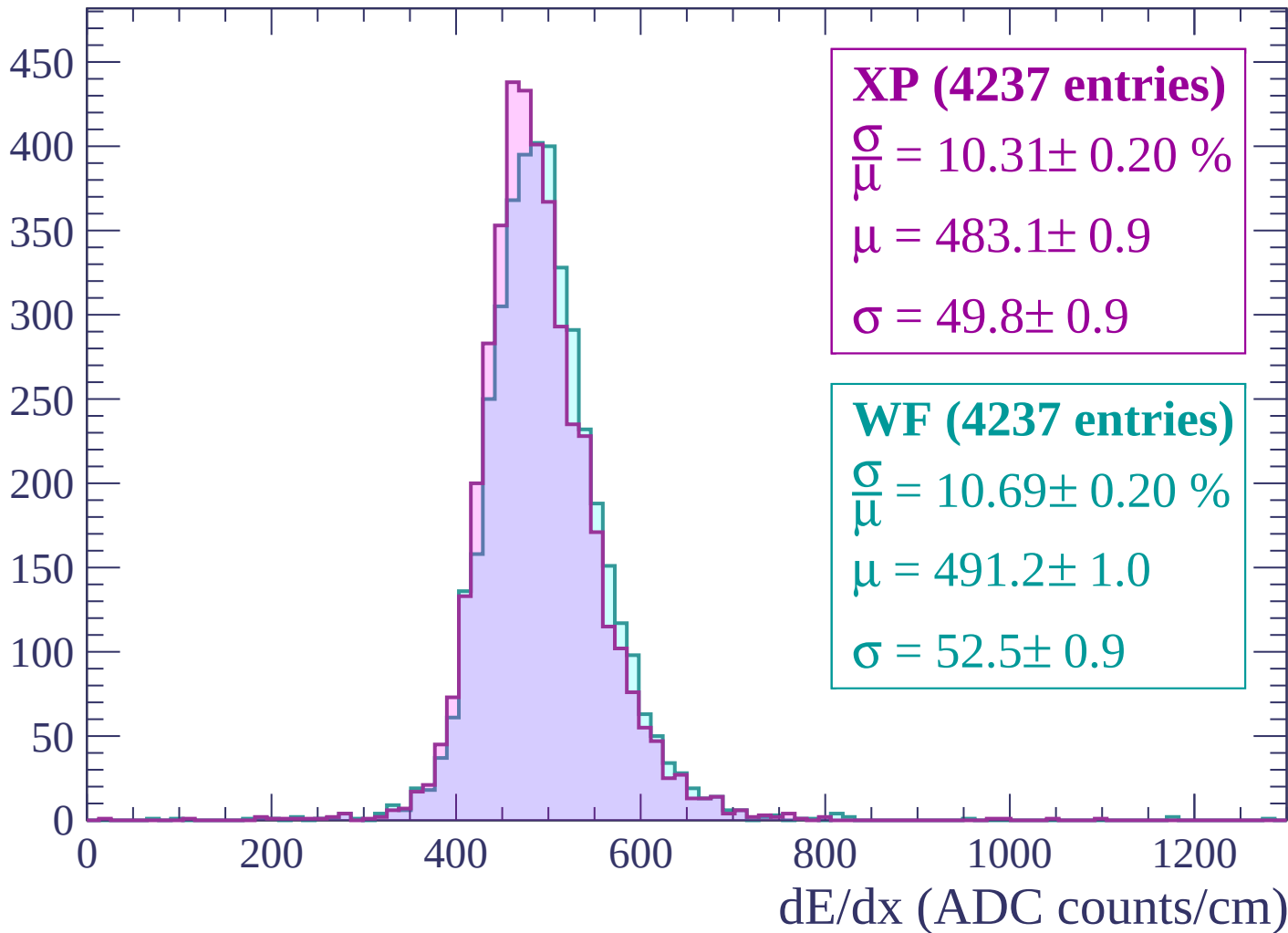
$\mu = 490.7 \pm 0.9$

$\sigma = 52.8 \pm 0.8$

dE/dx (ADC counts/cm)

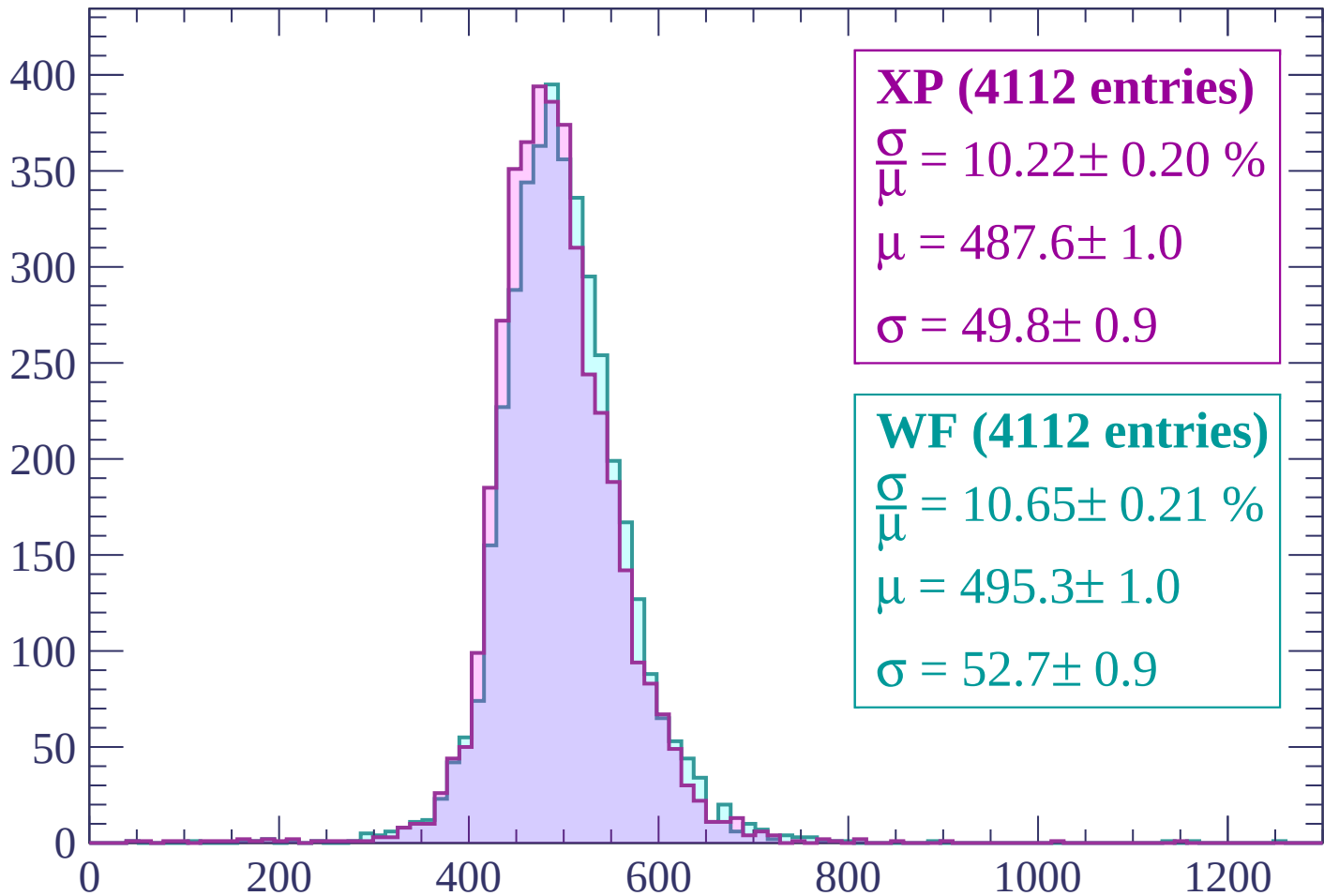
Energy loss | $2340 < p < 2400$

Count



Energy loss | $2400 < p < 2460$

Count



XP (4112 entries)

$\frac{\sigma}{\mu} = 10.22 \pm 0.20 \%$

$\mu = 487.6 \pm 1.0$

$\sigma = 49.8 \pm 0.9$

WF (4112 entries)

$\frac{\sigma}{\mu} = 10.65 \pm 0.21 \%$

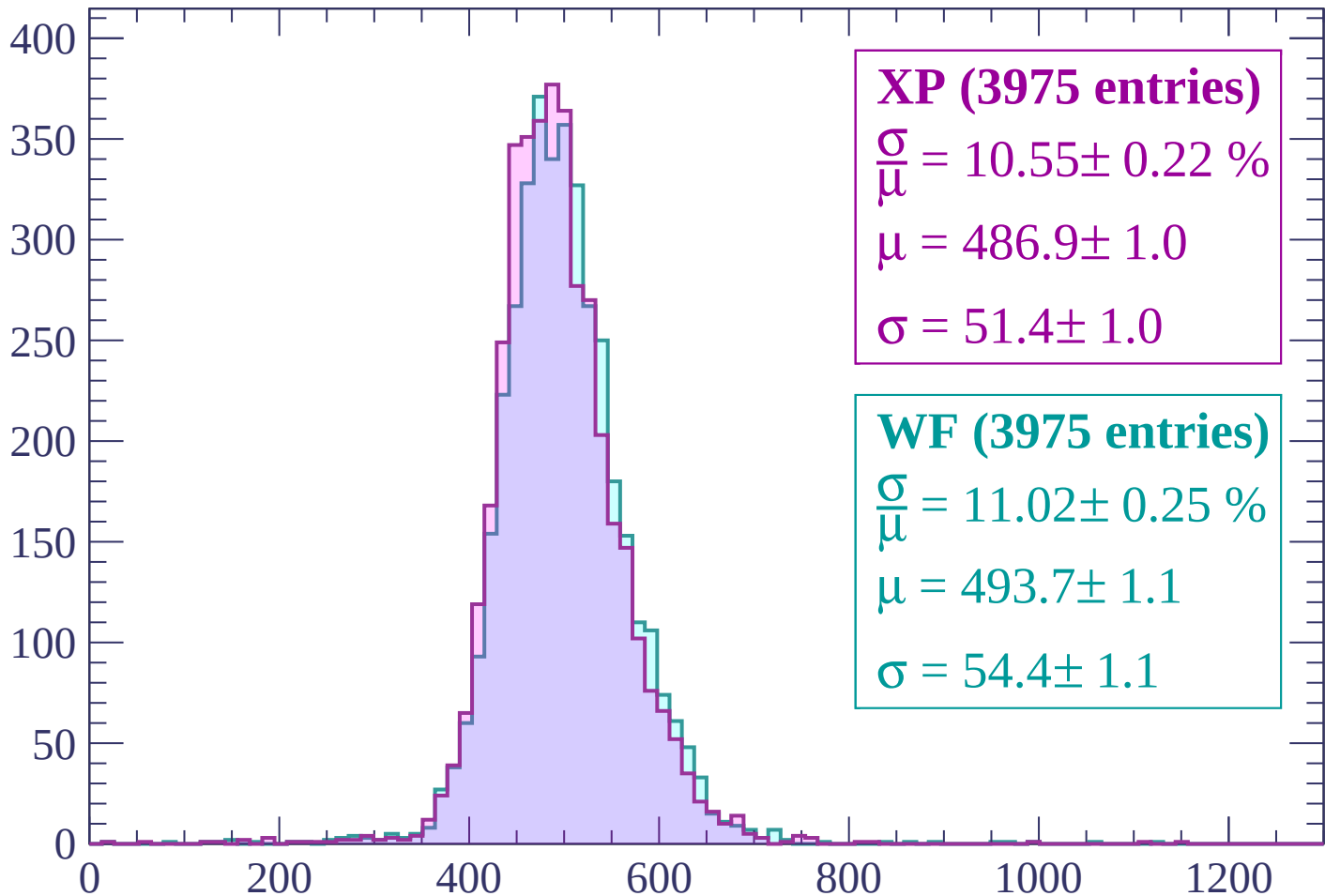
$\mu = 495.3 \pm 1.0$

$\sigma = 52.7 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $2460 < p < 2520$

Count



XP (3975 entries)

$\frac{\sigma}{\mu} = 10.55 \pm 0.22 \%$

$\mu = 486.9 \pm 1.0$

$\sigma = 51.4 \pm 1.0$

WF (3975 entries)

$\frac{\sigma}{\mu} = 11.02 \pm 0.25 \%$

$\mu = 493.7 \pm 1.1$

$\sigma = 54.4 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | $2520 < p < 2580$

Count

400
350
300
250
200
150
100
50
0

XP (3921 entries)

$\frac{\sigma}{\mu} = 10.60 \pm 0.24 \%$

$\mu = 489.4 \pm 1.0$

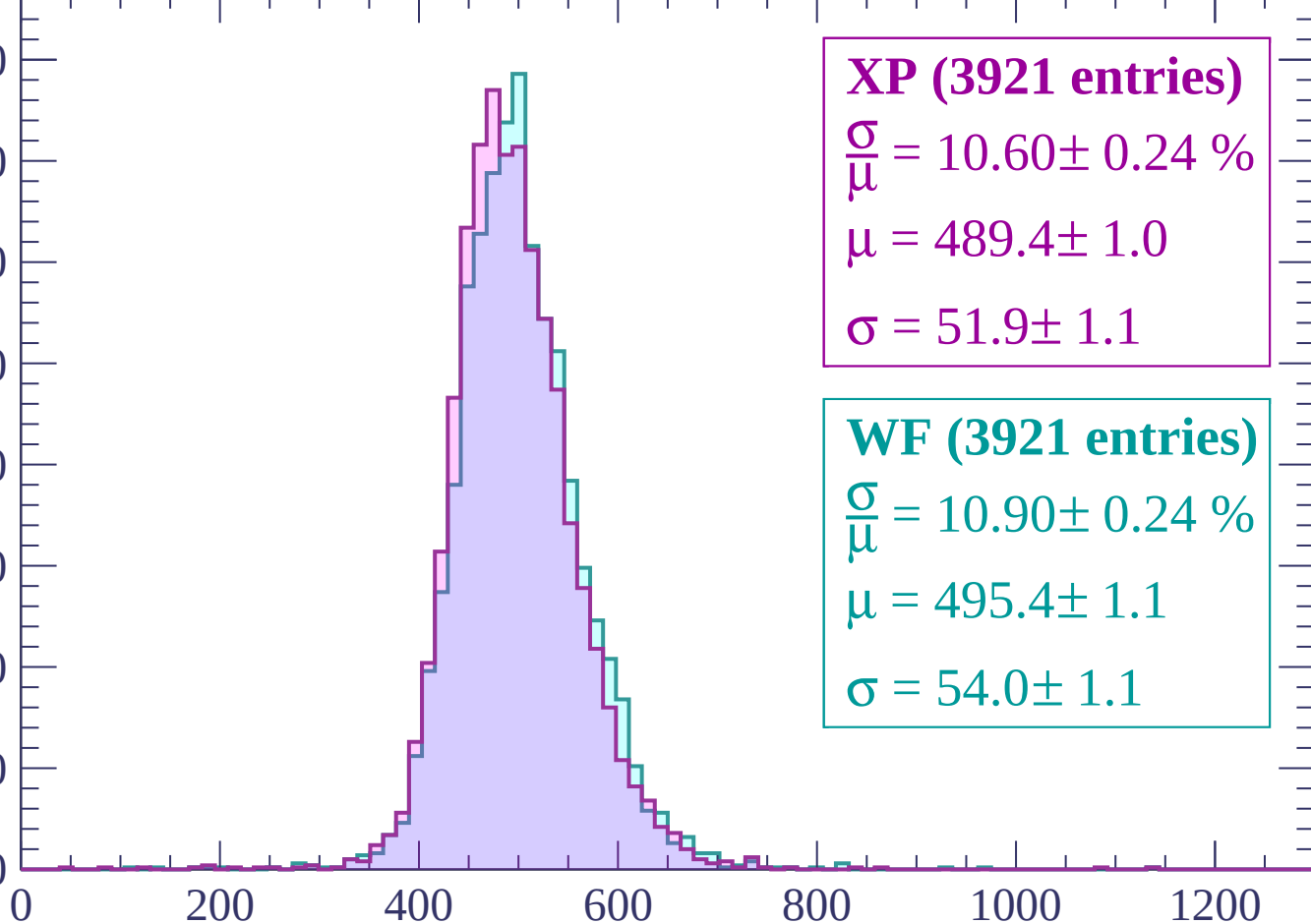
$\sigma = 51.9 \pm 1.1$

WF (3921 entries)

$\frac{\sigma}{\mu} = 10.90 \pm 0.24 \%$

$\mu = 495.4 \pm 1.1$

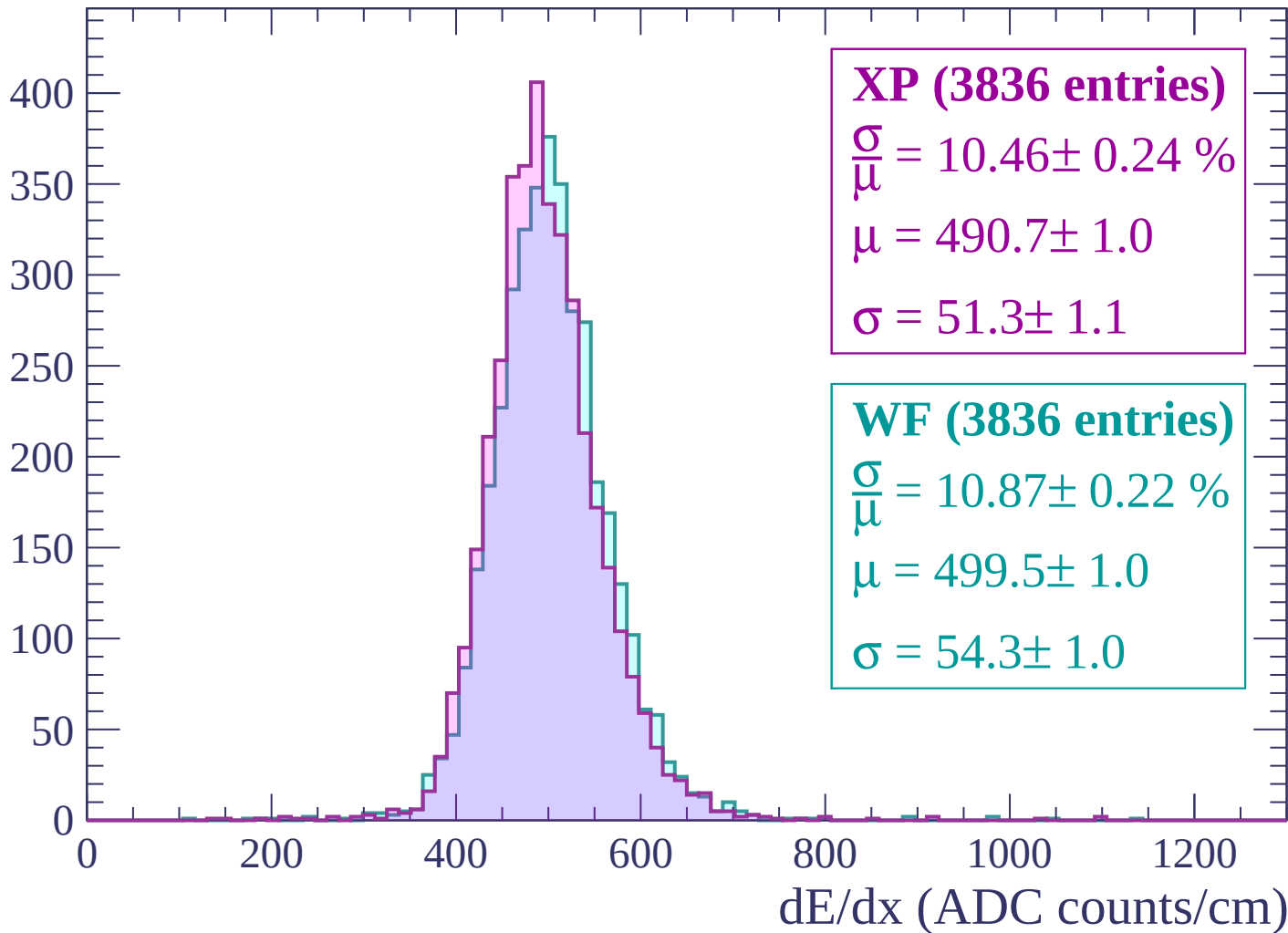
$\sigma = 54.0 \pm 1.1$



dE/dx (ADC counts/cm)

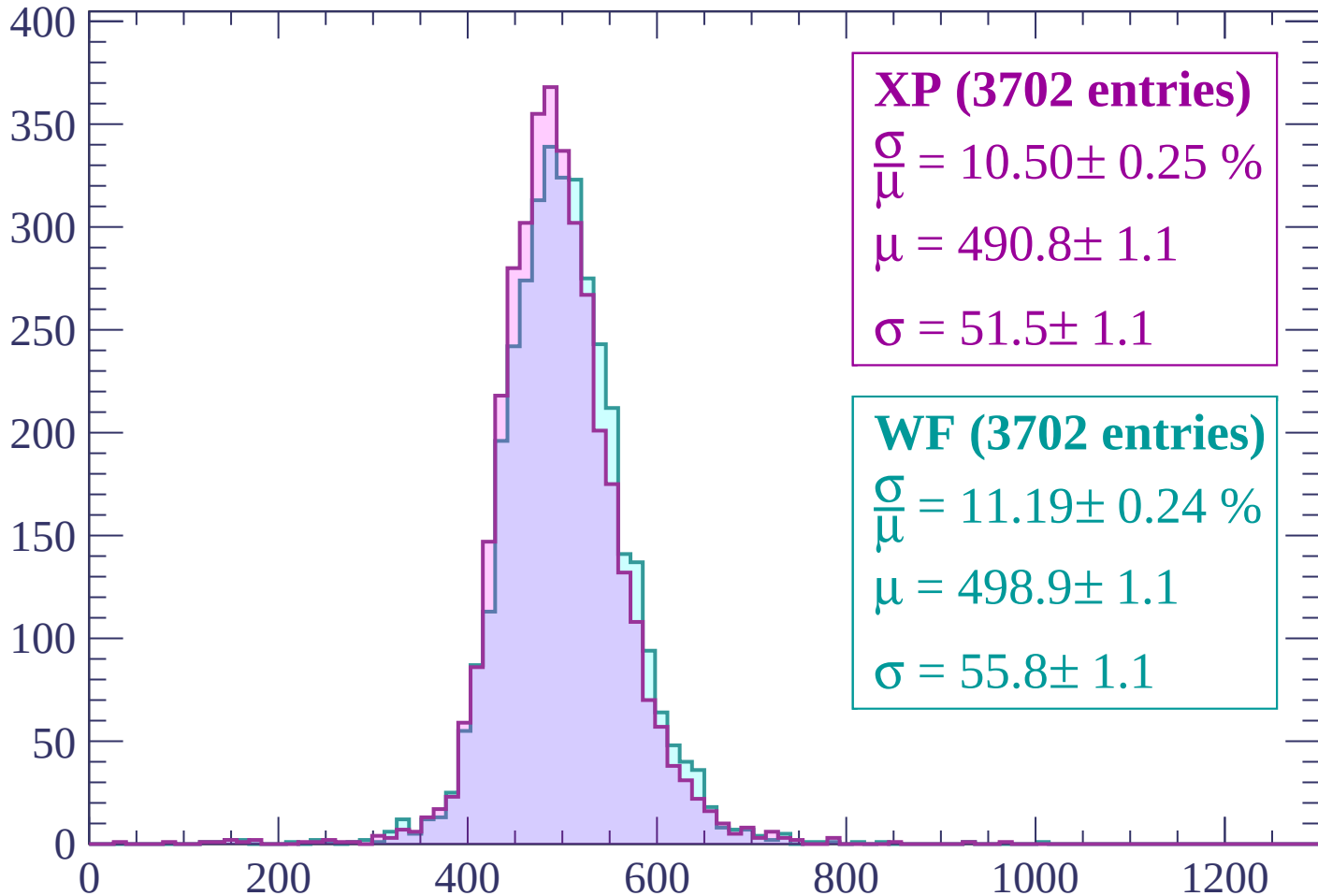
Energy loss | $2580 < p < 2640$

Count



Energy loss | $2640 < p < 2700$

Count



XP (3702 entries)

$\frac{\sigma}{\mu} = 10.50 \pm 0.25 \%$

$\mu = 490.8 \pm 1.1$

$\sigma = 51.5 \pm 1.1$

WF (3702 entries)

$\frac{\sigma}{\mu} = 11.19 \pm 0.24 \%$

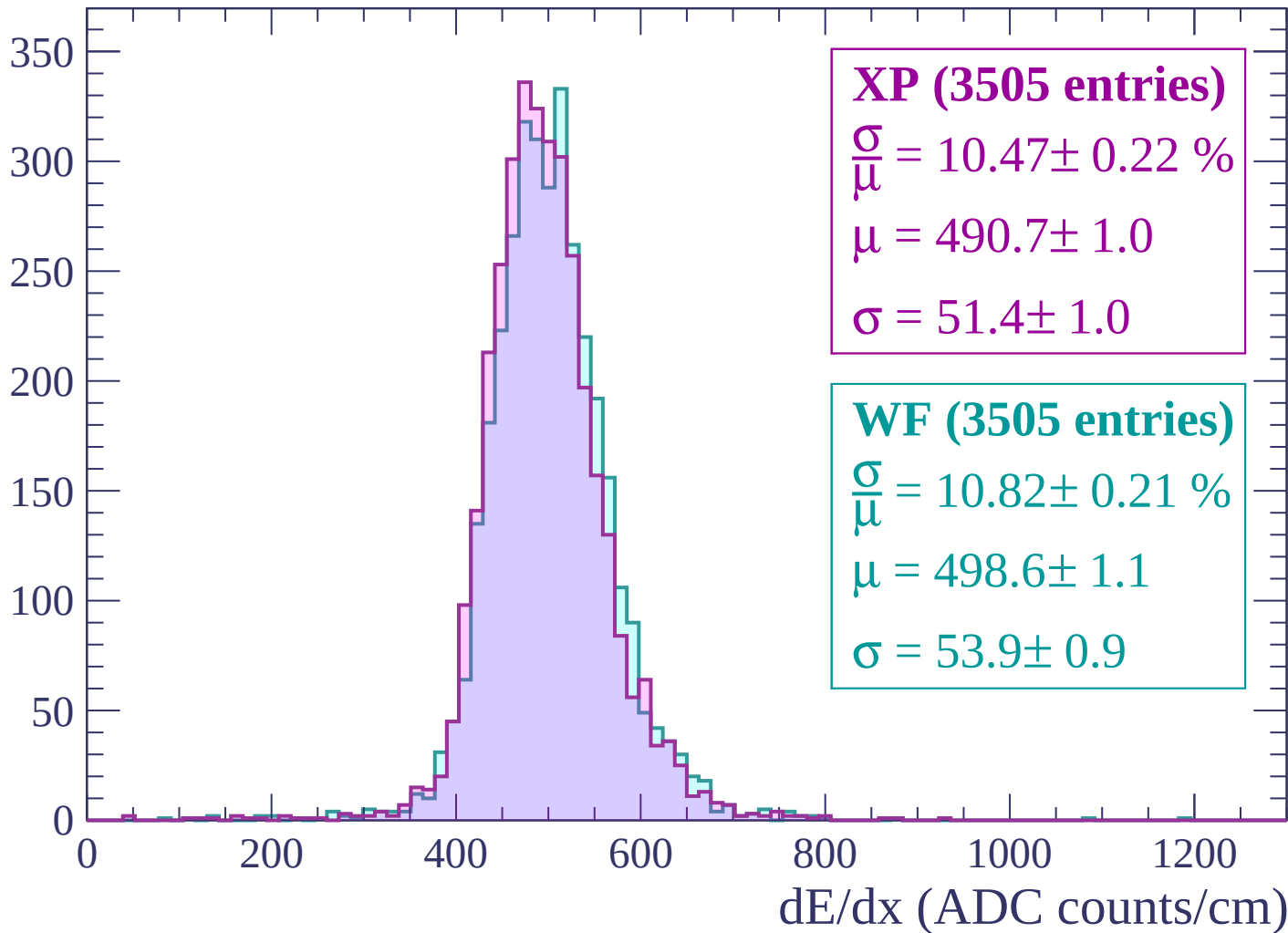
$\mu = 498.9 \pm 1.1$

$\sigma = 55.8 \pm 1.1$

dE/dx (ADC counts/cm)

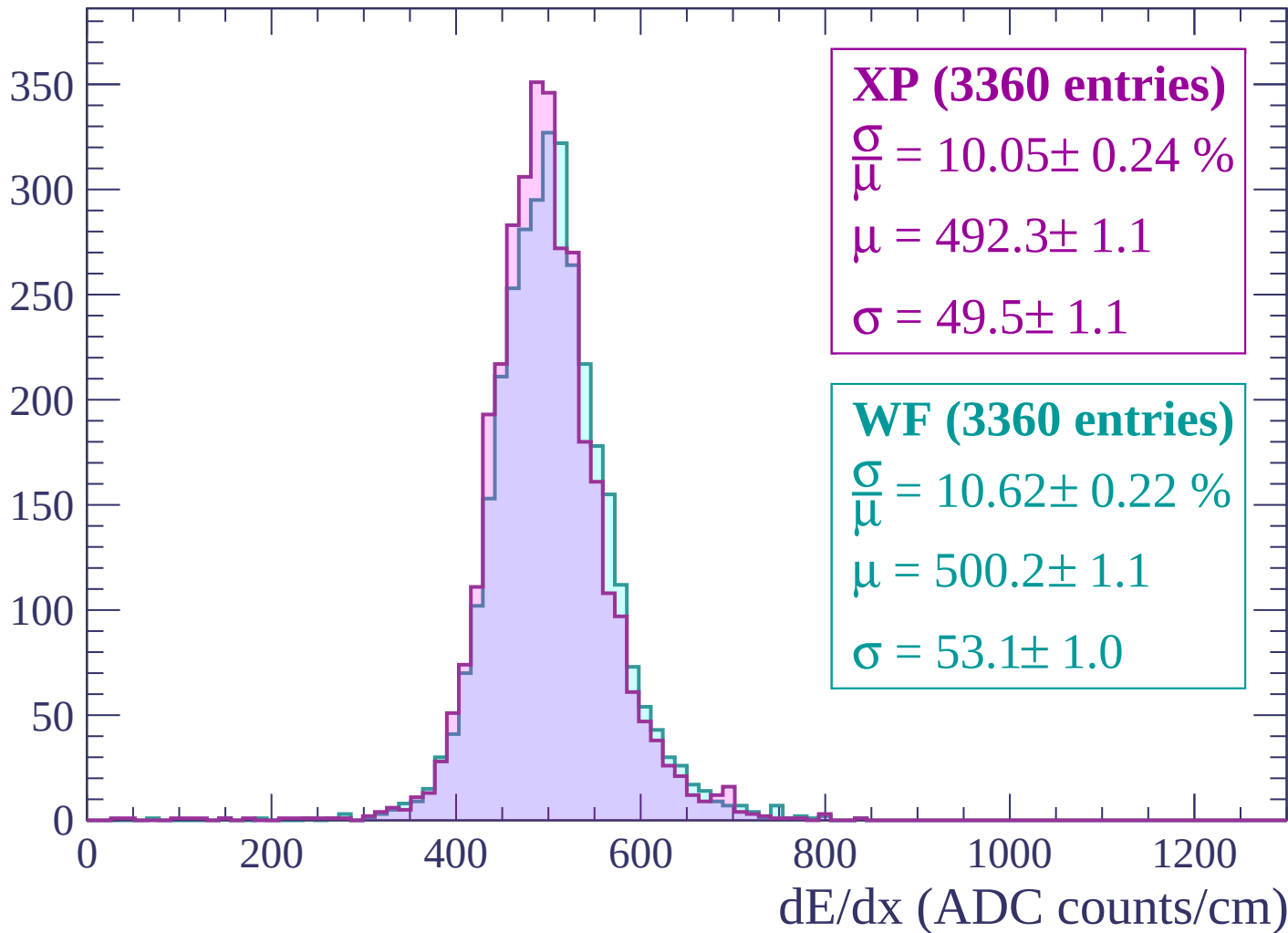
Energy loss | $2700 < p < 2760$

Count



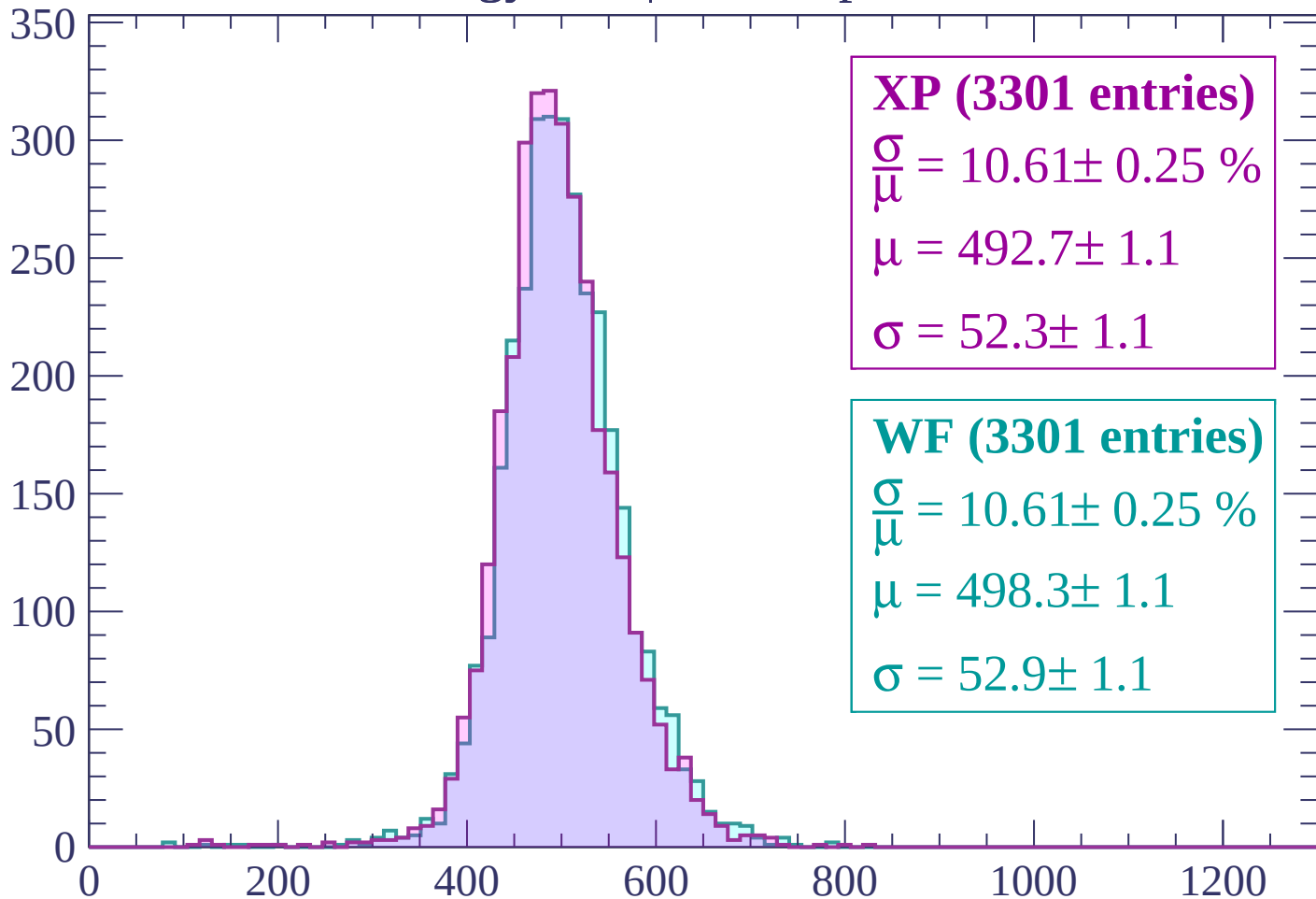
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP (3301 entries)

$\frac{\sigma}{\mu} = 10.61 \pm 0.25 \%$

$\mu = 492.7 \pm 1.1$

$\sigma = 52.3 \pm 1.1$

WF (3301 entries)

$\frac{\sigma}{\mu} = 10.61 \pm 0.25 \%$

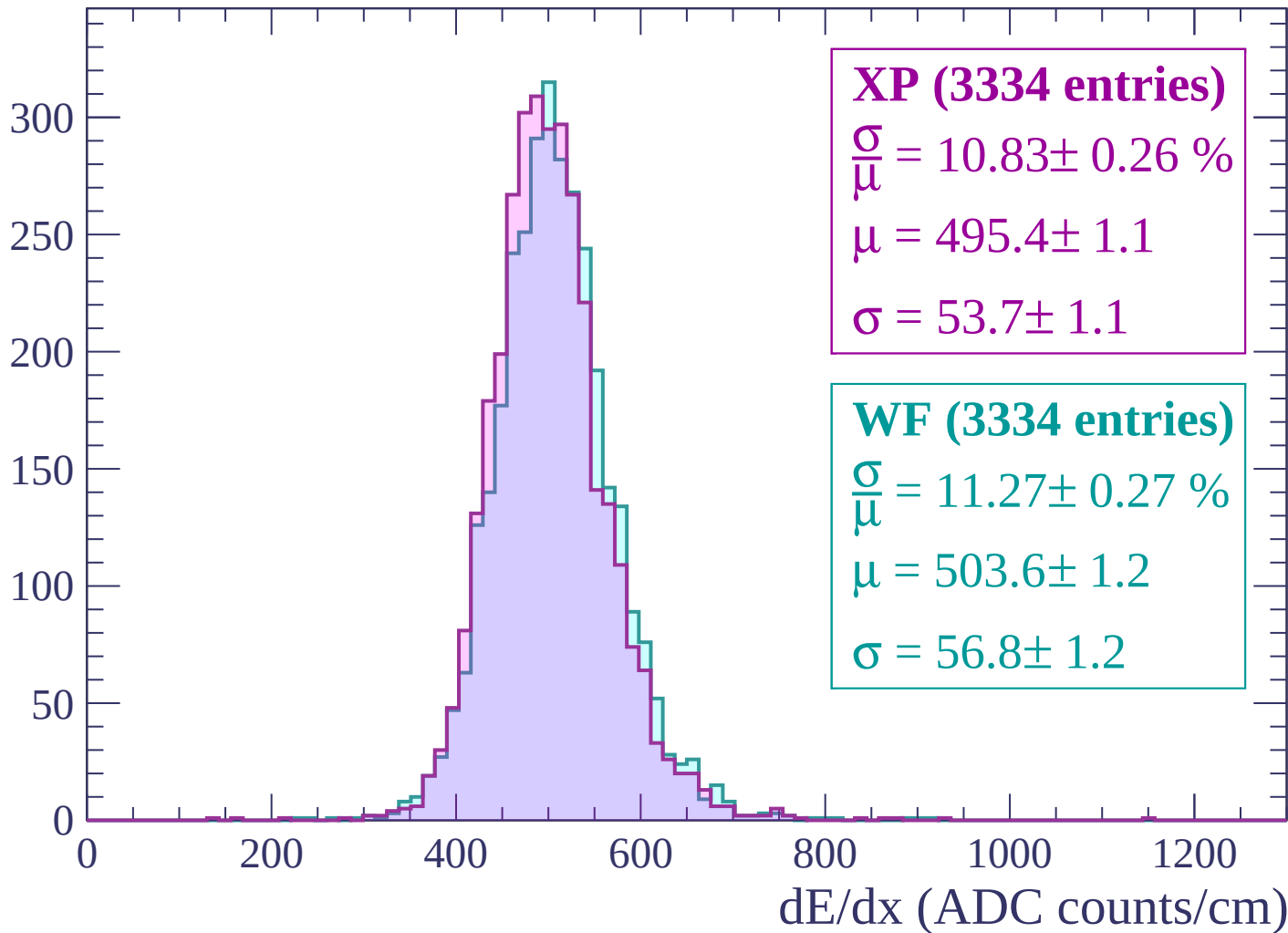
$\mu = 498.3 \pm 1.1$

$\sigma = 52.9 \pm 1.1$

dE/dx (ADC counts/cm)

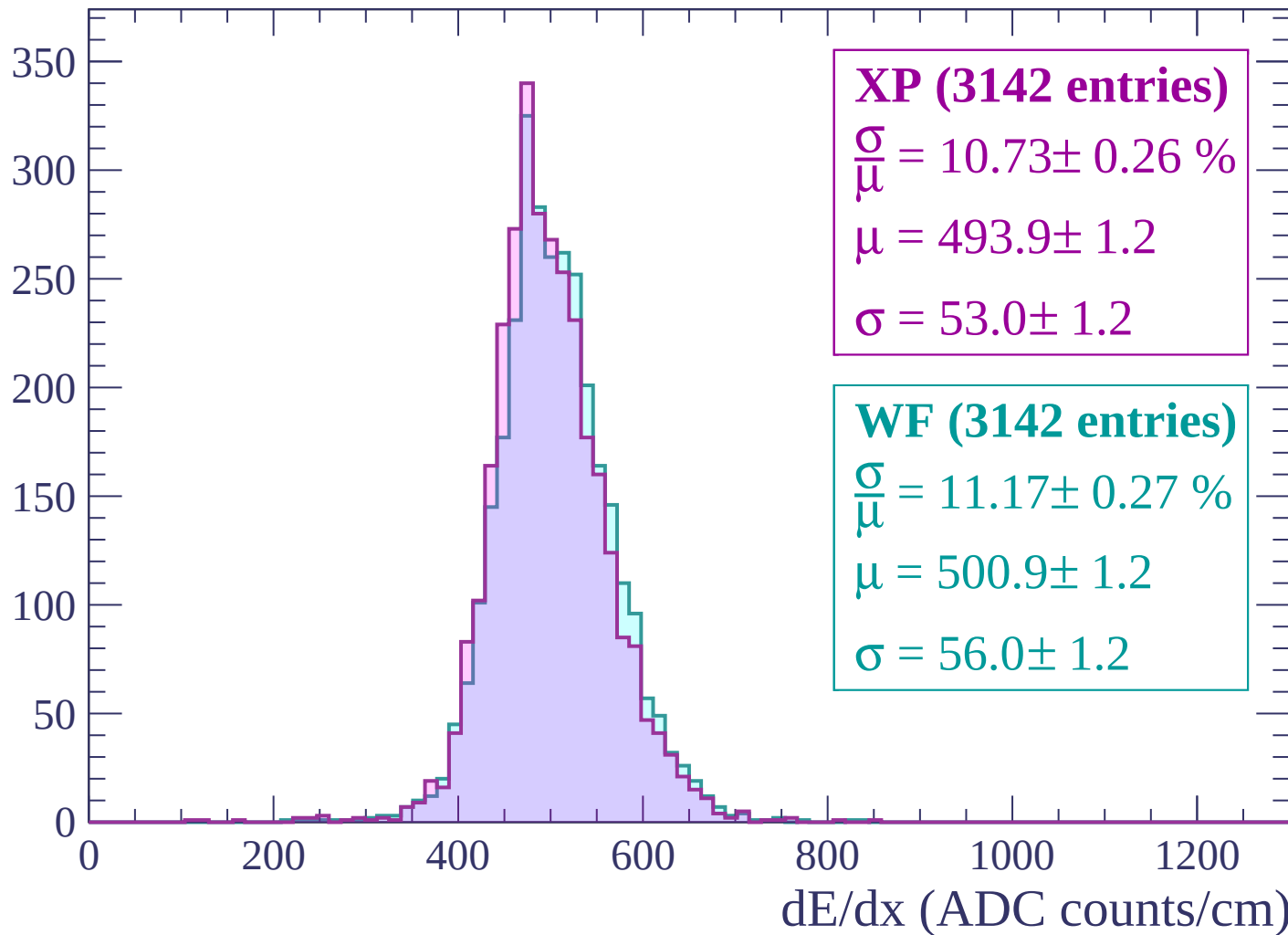
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count

